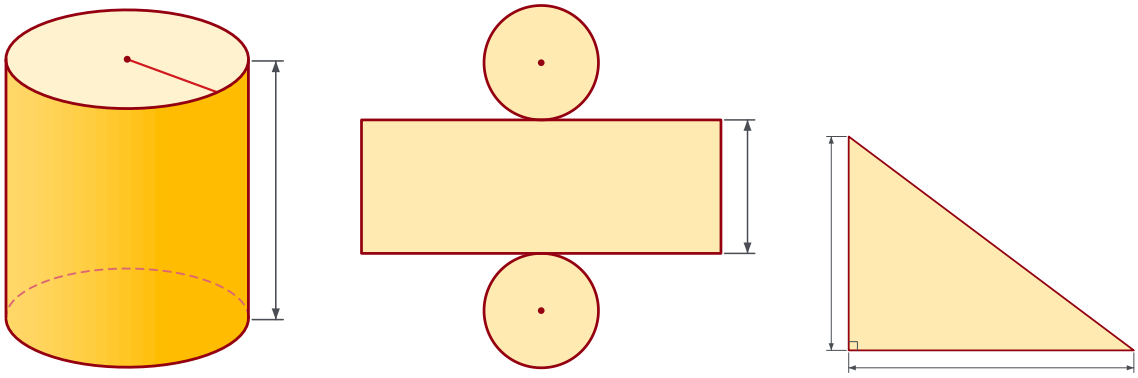
ACCESS INDIGENOUS SKILLS4SUCCESS

Trades Numeracy Textbook

Whole numbers to trigonometry, with key-by-key calculator steps for the TI-30Xa and the Sharp EL-531XT. For fabrication, pipe trades, millwright and carpentry pre-apprentices.



Skills for Success: Numeracy
Metric and imperial

About this book

This textbook covers the mathematics a pre-apprentice uses every day in the shop: whole numbers, fractions and decimals, percent and ratio, measurement and unit conversion, formulas and how to rearrange them, perimeter, area, volume, right triangles and trigonometry.

Each chapter opens with what you will learn and the key words to watch for. The ideas are explained in plain language, then shown in worked examples. Short Try it sets let you practise straight away. Each chapter ends with Shape Canvas worksheets that use the same drawings and practice ideas as the ACCESS Shape Canvas app.

Two calculators are used throughout: the Texas Instruments TI-30Xa and the Sharp EL-531XT. Every calculator box shows the keys for both, side by side, and what each display reads after the last key.

Sizes are given in metric first and imperial second. Answers to every Try it set and worksheet are in the separate Answer Key. The companion Exercise Book has more practice for each chapter.

The book serves four trades: fabrication, pipe trades, millwright and carpentry. Carpentry sits alongside the other three. Its examples and worksheets cover framing, stairs, sheet goods and concrete, and its code values are quoted from the BC Building Code. The page Using the BC Building Code explains how those quotes work.

Appendix A, Kitchen Math, follows Chapter 13. It covers recipe scaling, cooking measures and food yield percent.

HOW TO READ A SIZE

In this book 1,250 mm means one thousand two hundred fifty millimetres. $5' \; 3\frac{1}{2}''$ means five feet three and one half inches. The symbol $\approx$ means “about equal to” and is used when an answer is rounded.

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How to use this book

Worked examples

Examples are numbered by chapter. Read the question, try it yourself, then check each step.

Example 1.8 Counting bolts

A job needs 24 brackets. Each bracket is held by 4 bolts. How many bolts are needed?

SOLUTION

1. 24 groups of 4: 24 × 4.
2. 20 × 4 = 80 and 4 × 4 = 16.
   80 + 16 = 96.

Answer **96 bolts**

Calculator boxes

The left column shows the TI-30Xa keys, the right column the Sharp EL-531XT keys. Grey keys are numbers, white keys are operations and functions, and a gold key is the second-function key (2nd or 2ndF) pressed before the key after it. The display shows what that calculator reads after the last key.

CALCULATOR

TI-30Xa	Sharp EL-531XT
π × 152.4 =	π × 152.4 =
Display 478.7787204	Display 478.7787204

Rounded to two decimal places: **478.78**

Try it

Quick practice at the end of a section. Do these before you move on. Answers are in the Answer Key.

Boxes

TIP
A shortcut or a good habit.

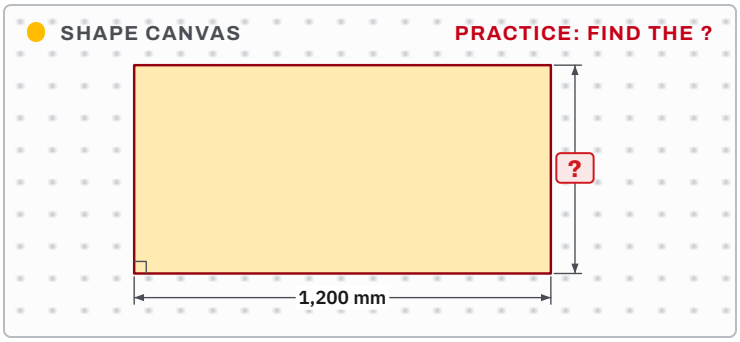
WATCH OUT
A common mistake and how to avoid it.

ON THE JOB
How the idea shows up in the shop.

KEY IDEA
The one thing to remember from a section.

Shape Canvas worksheets

The worksheets use the drawings and practice ideas of the ACCESS Shape Canvas app. A red ? marks a size that is hidden: find it. Dashed lines **break a part down** into simple shapes. A solid shown beside its flat pattern is **unfolded** into its net. Callouts point to the part a question is about. Show your work in the working box and write the answer on the answer line.



Using the BC Building Code

The carpentry pages of this book quote the **British Columbia Building Code 2024**, Book I: General, written **BCBC 2024** in this book. The code is set out in three Divisions. Division A covers compliance, objectives and functional statements. Division B holds the acceptable solutions, which are the technical requirements. Division C holds the administrative provisions.

Part 9 of Division B is titled Housing and Small Buildings. It applies to buildings of 3 storeys or less in building height, with a building area of not more than 600 m², used for residential occupancies, business and personal services occupancies, mercantile occupancies, or medium- and low-hazard industrial occupancies (Division A, Sentence 1.3.3.3.(1), page 1-12). The framing, stair, footing and floor values in this book all come from Part 9.

How a code number is built

Level	Number	For example
Part	9	Housing and Small Buildings
Section	9.8.	Stairs, Ramps, Landings, Handrails and Guards
Subsection	9.8.4.	Step Dimensions
Article	9.8.4.1.	Dimensions for Risers
Sentence	9.8.4.1.(1)	The numbered sentence (1) inside the Article
Clause	9.8.4.1.(1)(a)	A lettered part of a Sentence, where the code uses one
Subclause	9.8.4.1.(1)(a)(i)	A part of a Clause, where the code uses one

The first number is the Part, the second the Section, the third the Subsection and the fourth the Article. The requirements themselves are in the Sentences, the numbers in brackets. A Sentence can be broken into Clauses and Subclauses.

How this book quotes the code

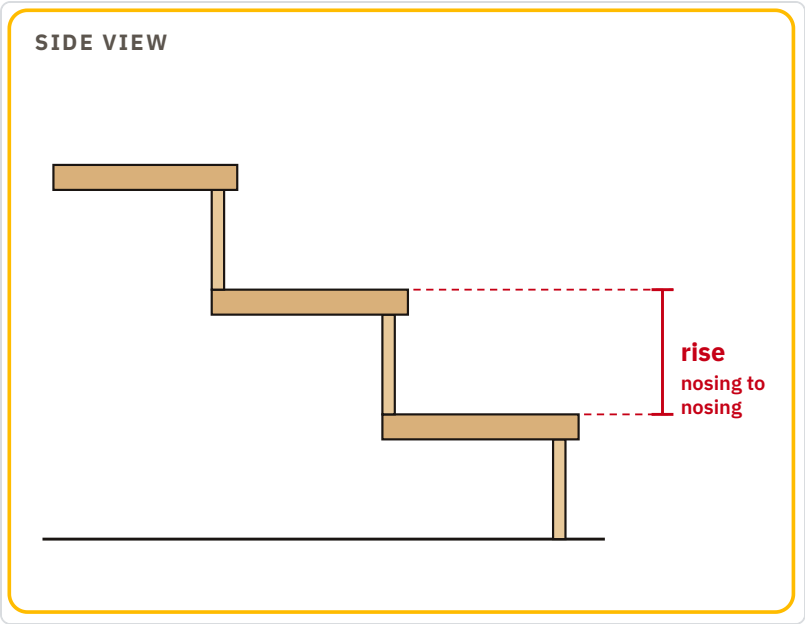
Every code value in this book names the Article and Sentence it comes from and the page of the code it is printed on, like this: **BCBC 2024, Division B, Article 9.8.4.1. (page 9-30)**. Page 9-30 is the page number printed in the code itself, not a page of this book.

Values are quoted at the level the code states them. The stair rise limits are given in millimetres, so the book quotes millimetres. When the book shows the same value in inches, that is a conversion for the shop, not a code value.

A QUOTE, READ FROM THE CODE TEXT

Except for stairs serving areas only used as service rooms or service spaces, the rise, measured as the vertical nosing-to-nosing distance, must comply with Table 9.8.4.1. For a private stair the table gives a maximum rise of 200 mm and a minimum rise of 125 mm. Source: BCBC 2024, Division B, Sentence 9.8.4.1.(1) and Table 9.8.4.1. (page 9-30).

The plates are teaching drawings



The side view panel of the ACCESS teaching plate for Article 9.8.4.1. The plate shows the idea. The code text sets the values.

The drawings with a BCBC source line under them are **ACCESS teaching plates**. Each one shows a Part 9 Article. They were drawn from the code text to help you see the math, but a plate is not the code. The book prints the drawing part of a plate and writes the requirement under it, with each value quoted from the code text. **The code text governs.** If a plate and the code ever seem to disagree, follow the code text.

The full set of 922 plates is in the **bcbc** folder of the ACCESS Indigenous Skills4Success Numeracy site, the same site as the Numeracy Hub. There is one folder for each Section of Part 9 and one file for each Article, named with the Article number and title, for example “9.8.4.1 Dimensions for Risers”.

> **CHECK THE CODE THAT APPLIES TO YOUR JOB**
>
> BCBC 2024 took effect on March 8, 2024. The order that set it up lets some work carry on under the 2018 code, for example work under a building permit applied for before that date. Before you rely on a value on the job, check which code applies to your project.

CHAPTER

1

Whole Numbers

Every job in the shop starts with whole numbers. You count bolts and brackets, read sizes off a drawing, add up a cut list and work out how many pieces you can get from one length of stock.

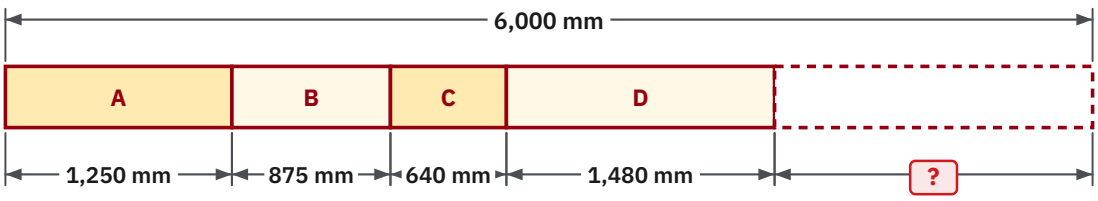
This chapter reviews place value, rounding and estimating, and the four operations: adding, subtracting, multiplying and dividing. Each method is shown by hand first and then on the TI-30Xa and the Sharp EL-531XT, so you can check your work two ways.

In this chapter you will

- Read and write whole numbers using place value
- Round whole numbers and use rounding to estimate
- Add and subtract lengths on a cut list
- Multiply and divide to find totals, pieces and spacing
- Work out averages and multi-step shop problems
- Check answers with an estimate and with your calculator

Key words

- whole number
- digit
- place value
- rounding
- estimate
- sum and difference
- factors and product
- quotient and remainder
- kerf
- average



1.1 Whole numbers and place value

A **whole number** is a number with no fraction or decimal part: 0, 1, 2, 3 and so on. Whole numbers are written with ten **digits**, 0 to 9.

The value of a digit depends on where it sits in the number. This is called **place value**. Each place is worth ten times the place to its right.

Number	Ten thousands	Thousands	Hundreds	Tens	Ones
6,096		6	0	9	6
12,450	1	2	4	5	0
305			3	0	5

Table 1.1 Place value of three numbers.

A zero holds a place. In 305 the zero shows there are no tens. Without it the number would read 35, which is a very different length.

Example 1.1 Reading a size off a drawing

A drawing gives the length of a beam as 12,450 mm. Write the number in words and say what the 2 is worth.

SOLUTION

1. Split the number at the comma: 12 thousand and 450.
2. Read each part: twelve thousand, four hundred fifty.
3. The 2 is in the thousands place, so it is worth 2 × 1,000 = 2,000.

Answer **Twelve thousand four hundred fifty millimetres. The 2 is worth 2,000.**

COMMAS AND SPACES

This book puts a comma between each group of three digits in numbers with four or more digits, for example 12,450. You may also see a small space used instead, as in 12 450. Both mean the same number. The conversion table at the back of this book groups long decimals with spaces, as in 0.453 592 37.

TRY IT 1.1

1. Write 3,048 in words.
2. Write in digits: seven thousand six hundred two.
3. What is the 5 worth in 25,410?
4. What is the 9 worth in 6,096?
5. Which is longer, 2,450 mm or 2,405 mm?
6. Write in digits: twenty thousand fifty.

Answers are in the Answer Key.

1.2 Rounding

Rounding replaces a number with a nearby number that is easier to work with. You round to a place: the nearest ten, the nearest hundred or the nearest thousand.

1. Find the digit in the place you are rounding to.
2. Look at the digit just to its right.
3. If that digit is 5 or more, add one to the rounding digit. If it is 4 or less, leave the rounding digit as it is.
4. Change every digit to the right of the rounding place to zero.

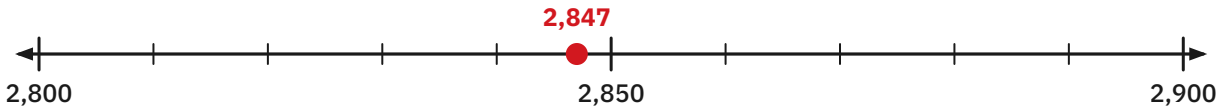


Figure 1.1 2,847 is closer to 2,800 than to 2,900, so it rounds to 2,800 to the nearest hundred.

Example 1.2 Rounding a length

Round 2,847 mm to the nearest ten and to the nearest hundred.

SOLUTION

1. Nearest ten: the tens digit is 4. The digit to its right is 7, which is 5 or more, so the 4 becomes 5: 2,850 mm.
2. Nearest hundred: the hundreds digit is 8. The digit to its right is 4, which is 4 or less, so the 8 stays: 2,800 mm.

Answer **2,850 mm to the nearest ten, 2,800 mm to the nearest hundred.**

Example 1.3 Rounding up past a nine

Round 6,960 to the nearest hundred.

SOLUTION

1. The hundreds digit is 9. The digit to its right is 6, so the 9 goes up by one.
2. Nine plus one is ten, so write 0 in the hundreds place and carry 1 into the thousands: 6 thousands become 7 thousands.

Answer **7,000**

ROUND AT THE END

Do the whole calculation with the exact numbers and round only the final answer. Rounding each step adds small errors together and the result can be a cut that is too short.

TRY IT 1.2

1. Round 1,374 to the nearest ten.
2. Round 1,375 to the nearest ten.
3. Round 6,096 to the nearest hundred.
4. Round 6,096 to the nearest thousand.
5. Round 49,950 to the nearest hundred.
6. Round 845 to the nearest hundred.

Answers are in the Answer Key.

1.3 Estimating

An **estimate** is a quick answer made with rounded numbers. It tells you roughly what the exact answer should be. If your calculator gives an answer far from your estimate, a key was probably missed or pressed twice.

Example 1.4 Estimating a cut list

A cut list has four pieces: 1,250 mm, 875 mm, 640 mm, 1,480 mm. Estimate the total length.

SOLUTION

1. Round each length to the nearest hundred: 1,300, 900, 600, 1,500.
2. Add the rounded numbers: 1,300 + 900 + 600 + 1,500 = 4,300.

Answer **About 4,300 mm. The exact total, found in Example 1.5, is 4,245 mm, which is close to the estimate.**

KEY IDEA

Estimate first, calculate second, compare third. An estimate does not replace the exact answer. It catches mistakes before they reach the saw.

1.4 Adding

Adding puts amounts together. The answer to an addition is the **sum**. To add by hand, line the numbers up by place value, ones under ones and tens under tens, and add each column from the right. When a column adds to 10 or more, write the ones digit and carry the tens digit to the next column.

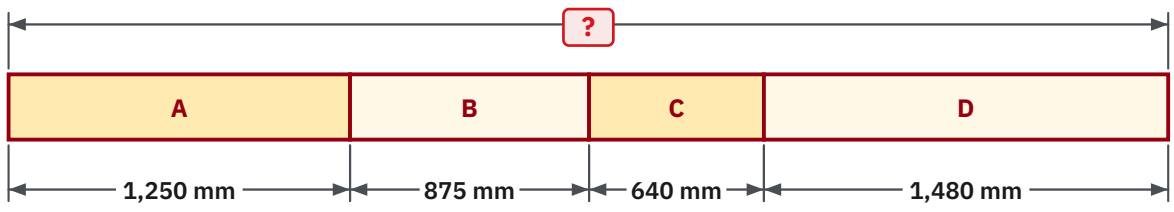


Figure 1.2 The cut list from Example 1.4. The total length is hidden.

Example 1.5 Total length of a cut list

Find the exact total length of the four pieces: 1,250 mm, 875 mm, 640 mm, 1,480 mm.

SOLUTION

1. Ones: 0 + 5 + 0 + 0 = 5.
2. Tens: 5 + 7 + 4 + 8 = 24. Write 4, carry 2.
3. Hundreds: 2 + 8 + 6 + 4 + 2 (carried) = 22. Write 2, carry 2.
4. Thousands: 1 + 1 + 2 (carried) = 4.

CALCULATOR

TI-30Xa

1250 + 875 + 640 + 1480 =

Display 4245.

Sharp EL-531XT

1250 + 875 + 640 + 1480 =

Display 4’245.

The TI-30Xa shows 4245. The Sharp EL-531XT shows 4’245, with a small mark between the thousands and the hundreds. The total is **4,245 mm**.

Answer **4,245 mm, close to the estimate of 4,300 mm.**

The TI-30Xa shows no separator between groups of digits. Read its display from the right in groups of three to put the commas back: 4245 is 4,245. The Sharp EL-531XT marks each group of three with a small raised mark, as in 4’245.

TRY IT 1.4

1. 2,450 + 1,375
2. 608 + 1,295 + 47
3. Pieces of 915, 915 and 1,220 mm. Total length?
4. 12,450 + 3,048 + 9,960

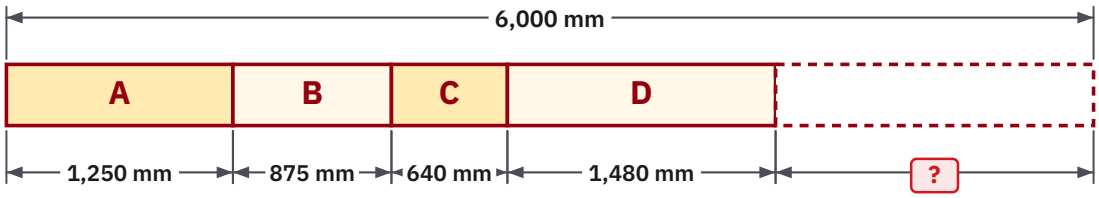
Answers are in the Answer Key.

1.5 Subtracting

Subtracting takes one amount away from another. The answer is the **difference**. Line the numbers up by place value with the larger number on top. When the top digit in a column is smaller than the bottom digit, borrow 1 from the next column to the left. That 1 is worth 10 in the column you are working on.

Example 1.6 What is left of the bar?

The four pieces from Example 1.5 (4,245 mm in total) are cut from a 6,000 mm bar. Ignore the width of the saw cuts. How long is the offcut?



SOLUTION

1. Write 6,000 on top and 4,245 below it.
2. Ones: 0 − 5 needs a borrow. The tens and hundreds are 0, so borrow through them: 6,000 becomes 5 thousands, 9 hundreds, 9 tens and 10 ones.
3. 10 − 5 = 5, 9 − 4 = 5, 9 − 2 = 7, 5 − 4 = 1.

CALCULATOR

TI-30Xa

6000 − 4245 =

Display 1755.

Sharp EL-531XT

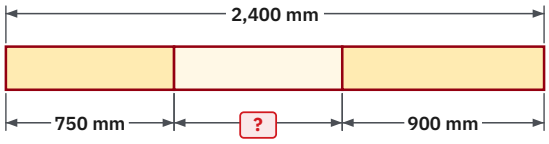
6000 − 4245 =

Display 1'755.

Answer **1,755 mm**

Example 1.7 Finding a hidden dimension

A drawing gives the overall length of a part as 2,400 mm and two of its three sections as 750 mm and 900 mm. Find the middle section.



SOLUTION

1. The three sections add up to the overall length.
2. Add the sections you know: 750 + 900 = 1,650.
3. Subtract from the overall length: 2,400 − 1,650 = 750.

CALCULATOR

TI-30Xa

2400 − 750 − 900 =

Display 750.

Sharp EL-531XT

2400 − 750 − 900 =

Display 750.

Answer **The middle section is 750 mm.**

CHECK A SUBTRACTION BY ADDING

Add the answer back to the number you took away. You should get the number you started with: 1,755 + 4,245 = 6,000.

TRY IT 1.5

1. 5,000 − 1,875
2. 12,450 − 9,960
3. A 3,600 mm bar has 2,475 mm cut off. How much is left?
4. Overall 1,800 mm. Sections of 600 mm and ? mm. Find the ?

Answers are in the Answer Key.

1.6 Multiplying

Multiplying is a fast way to add the same number many times. 24 × 4 means 24 groups of 4. The numbers you multiply are **factors**, and the answer is the **product**.

Example 1.8 Counting bolts

A job needs 24 brackets. Each bracket is held by 4 bolts. How many bolts are needed?

SOLUTION

1. 24 groups of 4: 24 × 4.
2. 20 × 4 = 80 and 4 × 4 = 16.
3. 80 + 16 = 96.

CALCULATOR

TI-30Xa

24 × 4 =

Display 96.

Sharp EL-531XT

24 × 4 =

Display 96.

Answer **96 bolts**

Example 1.9 Multiplying by a two-digit number

You need 36 pieces, each 125 mm long. What total length of stock do the pieces use? Ignore the saw cuts.

SOLUTION

1. Split 36 into 30 + 6.
2. 125 × 6 = 750.
3. 125 × 30 = 3,750.
4. Add the two parts: 750 + 3,750 = 4,500.

CALCULATOR

TI-30Xa

36 × 125 =

Display 4500.

Sharp EL-531XT

36 × 125 =

Display 4'500.

Answer **4,500 mm**

Splitting a number into parts works well in your head. On paper, most people multiply in columns, one digit of the bottom number at a time. Section 1.10 shows the column method step by step, for numbers up to three digits by three digits, with a times table.

SAW CUTS TAKE MATERIAL

A saw blade turns a strip of metal into chips. The width of that strip is the **kerf**. When you cut many pieces, the kerf adds up. With a kerf of 3 mm, 12 cuts remove 12 × 3 = 36 mm of stock. The problems in this book give the kerf whenever it counts.

TRY IT 1.6

1. 15 × 8
2. 48 × 25
3. 125 × 64
4. 18 pieces at 450 mm. Total length?
5. 7 cuts with a 3 mm kerf. Stock lost?
6. 6 rows of 14 bolts. How many bolts?

Answers are in the Answer Key.

1.7 Dividing

Dividing splits an amount into equal groups or equal parts. In 6,000 ÷ 450, the number being divided (6,000) is the dividend, the number you divide by (450) is the divisor, and the answer is the **quotient**. When the division does not come out even, the amount left over is the **remainder**.

Example 1.10 Pieces from one bar

How many 450 mm pieces can you cut from a 6,000 mm bar? Ignore the saw cuts. How long is the offcut?

SOLUTION

1. Divide: 6,000 ÷ 450. The calculator shows 13.33333333.
2. Only whole pieces count, so you get 13 pieces.
3. The 13 pieces use 13 × 450 = 5,850 mm.
4. The offcut is the remainder: 6,000 − 5,850 = 150 mm.

CALCULATOR

TI-30Xa

Pieces

6000 ÷ 450 =

Display 13.33333333

Offcut

6000 − 13 × 450 =

Display 150.

Sharp EL-531XT

Pieces

6000 ÷ 450 =

Display 13.33333333

Offcut

6000 − 13 × 450 =

Display 150.

Answer **13 pieces, with a 150 mm offcut.**

WHOLE PIECES ONLY

The calculator shows 13.33333333 pieces. Only whole pieces count, so the answer is 13. Even 13.9 would still be 13 pieces, because the 14th piece does not fit. Think about what the answer means before you round.

Example 1.11 Long division by hand

Divide 4,500 by 36.

SOLUTION

1. 36 does not go into 4. It goes into 45 once: 1 × 36 = 36, and 45 − 36 = 9.
2. Bring down the 0 to make 90. 36 goes into 90 twice: 2 × 36 = 72, and 90 − 72 = 18.
3. Bring down the last 0 to make 180. 36 goes into 180 five times: 5 × 36 = 180, remainder 0.
4. Read the answer from the top: 125.

CALCULATOR

TI-30Xa

4500 ÷ 36 =

Display 125.

Sharp EL-531XT

4500 ÷ 36 =

Display 125.

Answer **125. This undoes Example 1.9: 36 × 125 = 4,500.**

Equal spacing is a division problem with one twist. When there is a hole at each end of a straight row, the number of spaces is one less than the number of holes.

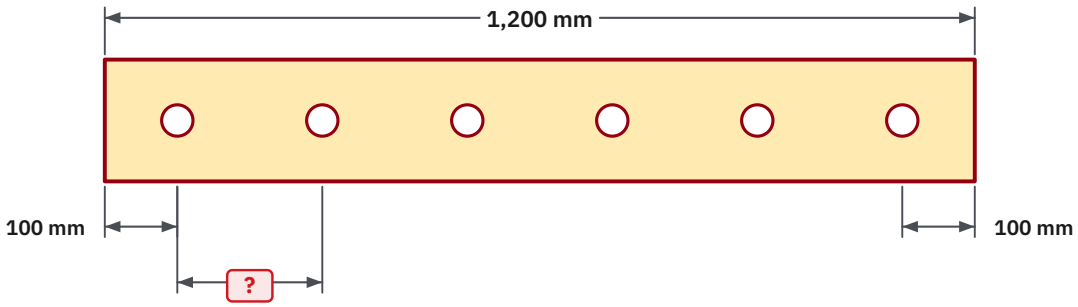


Figure 1.3 Six holes in a row, with the end holes 100 mm from the ends of a 1,200 mm plate.

Example 1.12 Equal hole spacing

Six holes are drilled in a straight row along a 1,200 mm plate. The first and last holes are 100 mm from the ends. The holes are evenly spaced. How far apart are the holes, centre to centre?

SOLUTION

1. Distance from the first hole to the last hole: 1,200 − 100 − 100 = 1,000 mm.
2. Six holes make 6 − 1 = 5 spaces.
3. Spacing: 1,000 ÷ 5 = 200 mm.

CALCULATOR

TI-30Xa

(1200 − 2 × 100) ÷ 5 =

Display 200.

Sharp EL-531XT

(1200 − 2 × 100) ÷ 5 =

Display 200.

Answer **200 mm centre to centre.**

KEY IDEA

In a straight row with a hole at each end, spaces = holes − 1. Draw a quick sketch and count the gaps if you are unsure.

Carpenters use the same idea the other way round: studs = spaces + 1. Section 1.12 counts studs and joists along a wall or floor.

TRY IT 1.7

1. 144 ÷ 12
2. 5,850 ÷ 450
3. 1,000 ÷ 8
4. How many 380 mm pieces from a 3,000 mm bar? Offcut?
5. 96 bolts shared equally among 24 brackets. Bolts each?
6. 6 holes evenly spaced, end holes 50 mm from the ends of a 650 mm bar. Spacing?

Answers are in the Answer Key.

1.8 Averages and multi-step problems

The **average** of a set of numbers is one number that represents the whole set. Add the numbers, then divide by how many numbers there are.

Example 1.13 Average welds per day

Over five days a welder completes 38, 42, 45, 40 and 35 joints. What is the average number of joints per day?

SOLUTION

1. Add: 38 + 42 + 45 + 40 + 35 = 200.
2. There are 5 days, so divide by 5: 200 ÷ 5 = 40.

CALCULATOR

TI-30Xa

(38 + 42 + 45 + 40 + 35) ÷ 5 =

Display 40.

Sharp EL-531XT

(38 + 42 + 45 + 40 + 35) ÷ 5 =

Display 40.

The brackets make the calculator add first and divide second. Chapter 3 explains why.

Answer **40 joints per day.**

The average is one way to describe a set of numbers. Two more are the median and the mode. The **median** is the middle number once the set is put in order. For the welder, 35, 38, 40, 42, 45 has 40 in the middle, so the median is 40 joints. The **mode** is the number that comes up most often. Each number of joints comes up once, so this set has no mode.

MEAN, MEDIAN AND MODE

Mean (average): add them up and divide by how many. Median: put them in order and take the middle one. Mode: the one that comes up most often. Section 1.11 works through all three with shop tables and graphs.

Many shop problems take more than one step. Write down what you know, decide which operation each step needs, and check the final answer against an estimate.

Example 1.14 Ordering brackets

A job has 3 units. Each unit needs 14 brackets, and you want 6 spares. Brackets come in boxes of 10. How many boxes do you order?

SOLUTION

1. Brackets for the units: 3 × 14 = 42.
2. Add the spares: 42 + 6 = 48 brackets.
3. Boxes: 48 ÷ 10 = 4.8.
4. 4 boxes hold only 40 brackets, which is not enough. Round up to 5 boxes.

CALCULATOR

TI-30Xa

Brackets

3 × 14 + 6 =

Display 48.

Boxes

48 ÷ 10 =

Display 4.8

Sharp EL-531XT

Brackets

3 × 14 + 6 =

Display 48.

Boxes

48 ÷ 10 =

Display 4.8

Answer **5 boxes (50 brackets).**

ROUNDING UP OR ROUNDING DOWN

When you are counting pieces you can cut, round down: a part piece is scrap. When you are counting boxes, lengths or sheets you need to buy, round up: a part box still has to be bought.

1.9 Time on the job

Shifts, breaks, timesheets and pay all run on the clock. The amount of time that passes between a start time and an end time is the **elapsed time**. Time does not count in tens: 60 minutes make an hour, so you cannot subtract clock times the way you subtract lengths.

Reading a clock

A clock face has two hands. The short hand points to the hour. The long hand points to the minutes. The numbers 1 to 12 mark the hours for the short hand. For the long hand, each number is 5 minutes more than the one before: the 1 means 5 minutes, the 2 means 10 minutes, and the 9 means 45 minutes. The small marks between the numbers are single minutes.

When the long hand is past the 12, the short hand has moved past the hour it just reached. At 7:15 it sits a quarter of the way from the 7 to the 8.

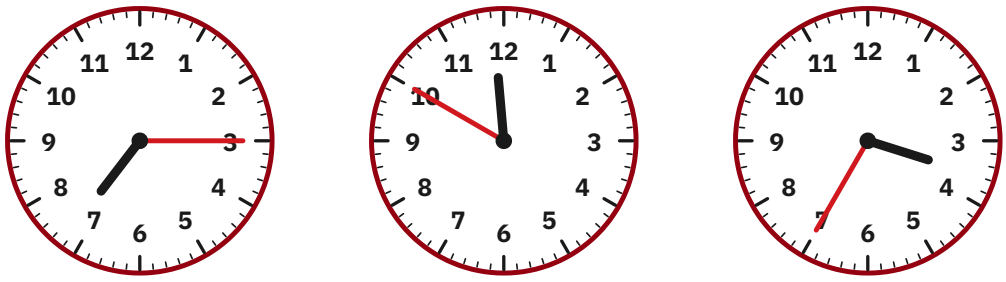


Figure 1.4 Three clocks. The short hand shows the hour and the long red hand shows the minutes.

Example 1.15 Reading three clocks

Read clocks A, B and C in Figure 1.4.

SOLUTION

1. A: the short hand is just past the 7, so the hour is 7. The long hand is on the 3: 3 × 5 = 15 minutes. A reads 7:15.
2. B: the short hand is near the 12 but has not reached it, so the hour is still 11. The long hand is on the 10: 10 × 5 = 50 minutes. B reads 11:50.
3. C: the short hand is past the 3. The long hand is on the 7: 7 × 5 = 35 minutes. C reads 3:35.

Answer **A is 7:15, B is 11:50 and C is 3:35.**

Elapsed time

To find how long a job took, count up from the start time in three steps: up to the next whole hour, then whole hours, then the minutes that are left. This works across noon too.

Example 1.16 A job that runs past noon

A pour starts at 7:45 a.m. and finishes at 3:20 p.m. How long does it take?

SOLUTION

1. Up to the next whole hour: 7:45 a.m. to 8:00 a.m. is 15 minutes.
2. Whole hours: 8:00 a.m. to 3:00 p.m. is 7 hours: 8:00 to 12:00 is 4 hours and 12:00 to 3:00 is 3 more.
3. The minutes that are left: 3:00 p.m. to 3:20 p.m. is 20 minutes.
4. Add: 7 h + 15 min + 20 min = 7 h 35 min.

Answer **7 h 35 min.**

A clock face does not say morning or afternoon. Times from midnight to noon are a.m. and times from noon to midnight are p.m. Many sites, schedules and phones use the 24-hour clock instead, which counts the hours from 00 to 23 after midnight. For a p.m. time from 1:00 p.m. on, add 12 to the hour: 3:35 p.m. is 15:35.

12-hour clock	24-hour clock
7:15 a.m.	07:15
11:50 a.m.	11:50
12:00 p.m.	12:00
1:30 p.m.	13:30
3:35 p.m.	15:35
11:05 p.m.	23:05

Table 1.2 The same times on the 12-hour and the 24-hour clock. 12:00 p.m. is noon.

Decimal hours and timesheets

Timesheets and pay use decimal hours. To change minutes to a decimal part of an hour, divide by 60: 45 minutes is 45 ÷ 60 = 0.75 hours. To change a decimal part of an hour back to minutes, multiply by 60. Chapter 5 covers decimals in full.

CALCULATOR Minutes to decimal hours

TI-30Xa

45 min

45 ÷ 60 =

Display 0.75

24 min

24 ÷ 60 =

Display 0.4

Sharp EL-531XT

45 min

45 ÷ 60 =

Display 0.75

24 min

24 ÷ 60 =

Display 0.4

Minutes	Decimal hours
6	0.1
12	0.2
15	0.25
18	0.3
24	0.4
30	0.5
36	0.6
42	0.7
45	0.75
48	0.8
54	0.9

Table 1.3 Minutes that change to an exact decimal of an hour (minutes ÷ 60).

Example 1.17 A week on the timesheet

A pre-apprentice works the week below. Each day has a 30 minute unpaid lunch. Find the hours for each day, the total for the week in hours and minutes and in decimal hours, and the pay before deductions at $27.80 an hour.

SOLUTION

1. For each day, find the time on site by counting up from the start to the finish. Monday: 7:00 a.m. to 3:30 p.m. is 8 h 30 min.
2. Take off the 30 minute lunch: 8 h 30 min − 30 min = 8 h.
3. Add the paid time for the five days: 8 h + 8 h 45 min + 8 h 15 min + 7 h 30 min + 6 h 15 min = 38 h 45 min.
4. In decimal hours: 45 ÷ 60 = 0.75, so 38 h 45 min = 38.75 hours.
5. Pay: 38.75 × 27.80 = $1,077.25.

Day	Start	Finish	Time on site	Paid time	Decimal hours
Monday	7:00 a.m.	3:30 p.m.	8 h 30 min	8 h	8
Tuesday	7:00 a.m.	4:15 p.m.	9 h 15 min	8 h 45 min	8.75
Wednesday	6:45 a.m.	3:30 p.m.	8 h 45 min	8 h 15 min	8.25
Thursday	7:00 a.m.	3:00 p.m.	8 h	7 h 30 min	7.5
Friday	7:00 a.m.	1:45 p.m.	6 h 45 min	6 h 15 min	6.25
Week				**38 h 45 min**	**38.75**

Table 1.4 The timesheet for Example 1.17.

CALCULATOR

TI-30Xa

38.75 × 27.8 =

Display 1077.25

Sharp EL-531XT

38.75 × 27.8 =

Display 1’077.25

The display shows 1077.25 on the TI-30Xa. Money is written with two decimal places: $1,077.25.

Answer **38 h 45 min, or 38.75 hours. Pay before deductions: $1,077.25.**

Example 1.18 Decimal hours back to minutes

A job log shows 6.4 hours. How many hours and minutes is that?

SOLUTION

1. The whole number is the hours: 6 hours.
2. Multiply the decimal part by 60: 0.4 × 60 = 24 minutes.

CALCULATOR

TI-30Xa

0.4 × 60 =

Display 24.

Sharp EL-531XT

0.4 × 60 =

Display 24.

Answer **6 h 24 min.**

6.4 HOURS IS NOT 6 HOURS 40 MINUTES

The digits after the point are tenths of an hour, not minutes. 6.4 hours is 6 hours 24 minutes. Always multiply the decimal part by 60.

TRY IT 1.9

1. Write 2:45 p.m. on the 24-hour clock.
2. Write 19:20 on the 12-hour clock.
3. How long is it from 6:50 a.m. to 9:15 a.m.?
4. How long is it from 10:40 a.m. to 2:05 p.m.?
5. Change 7 h 30 min to decimal hours.
6. Change 8 h 12 min to decimal hours.
7. Change 3.75 hours to hours and minutes.
8. A shift runs 7:00 a.m. to 3:30 p.m. with a 30 minute unpaid lunch. How many paid hours?

Answers are in the Answer Key.

1.10 Multiplying in columns

To multiply by a number with two or more digits on paper, work in columns. Write the larger number on top and line the digits up on the right. Multiply the top number by each digit of the bottom number in turn, starting with the ones. Each answer is a partial product. Then add the partial products.

The digits of the bottom number are worth different amounts. In 26, the 2 is worth 20. So when you multiply by the tens digit, the answer is ten times bigger, and you write a zero in the ones column first to hold the place. When you multiply by the hundreds digit, write two zeros. These zeros hold the places. They are shown in red below.

```
   3 4 7
   × 2 6
 -------
   2 0 8 2     347 × 6
+ 6 9 4 0      347 × 20
 -------
   9 0 2 2     product
```

Figure 1.5 347 × 26 in columns. The red zero holds the ones place in the second row.

Example 1.19 Three digits by two digits

A job needs 26 pieces of flat bar, each 347 mm long. Ignore the saw cuts. Find the total length by multiplying 347 × 26 in columns.

SOLUTION

1. Multiply 347 by the ones digit, 6: 347 × 6 = 2,082.
2. Multiply by the tens digit, 2, which is worth 20. Write a zero to hold the place, then 347 × 2 = 694 goes in front: 6,940.
3. Add the partial products: 2,082 + 6,940 = 9,022.
4. Check with an estimate: 350 × 30 = 10,500. The answer is close.

CALCULATOR

TI-30Xa

347 × 26 =

Display 9022.

Sharp EL-531XT

347 × 26 =

Display 9′022.

Answer 9,022 mm

```
   4 0 8
 × 2 3 5
---------
   2 0 4 0      408 × 5
 1 2 2 4 0      408 × 30
+ 8 1 6 0 0     408 × 200
---------
 9 5 8 8 0      product
```

Figure 1.6 408 × 235 in columns. Each row moves one place to the left, and the red zeros hold the places.

Example 1.20 Three digits by three digits

Multiply 408 × 235 in columns.

SOLUTION

1. Multiply 408 by the ones digit, 5: 408 × 5 = 2,040.
2. Multiply by the tens digit, 3, which is worth 30. Write a zero to hold the place, then 408 × 3 = 1,224 goes in front: 12,240.
3. Multiply by the hundreds digit, 2, which is worth 200. Write 2 zeros to hold the place, then 408 × 2 = 816 goes in front: 81,600.
4. Add the partial products: 2,040 + 12,240 + 81,600 = 95,880.
5. Estimate: 400 × 200 = 80,000 and 400 × 250 = 100,000. The answer sits between them.

CALCULATOR

TI-30Xa	Sharp EL-531XT
408 × 235 =	408 × 235 =
Display 95880.	Display 95’880.

Answer 95,880

DO NOT DROP THE ZEROS THAT HOLD THE PLACES

Without the zeros, the three rows of 408 × 235 would add to 2,040 + 1,224 + 816 = 4,080, far below the estimate of about 80,000. An estimate catches this mistake every time.

Column multiplication goes faster when you know the times table. Find a product where a row meets a column: 7 × 8 is in row 7 under column 8.

×	1	2	3	4	5	6	7	8	9	10	11	12
1	1	2	3	4	5	6	7	8	9	10	11	12
2	2	4	6	8	10	12	14	16	18	20	22	24
3	3	6	9	12	15	18	21	24	27	30	33	36
4	4	8	12	16	20	24	28	32	36	40	44	48
5	5	10	15	20	25	30	35	40	45	50	55	60
6	6	12	18	24	30	36	42	48	54	60	66	72
7	7	14	21	28	35	42	49	56	63	70	77	84
8	8	16	24	32	40	48	56	64	72	80	88	96
9	9	18	27	36	45	54	63	72	81	90	99	108
10	10	20	30	40	50	60	70	80	90	100	110	120
11	11	22	33	44	55	66	77	88	99	110	121	132
12	12	24	36	48	60	72	84	96	108	120	132	144

Table 1.5 Times table from 1 × 1 to 12 × 12.

TRY IT 1.10

1. 56 × 34
2. 125 × 48
3. 307 × 24
4. 214 × 136
5. 605 × 408
6. 36 bolts at 165 g each. Total mass in grams?

Answers are in the Answer Key.

1.11 Median, mode, tables and graphs

Section 1.8 found the average, also called the mean. The median and the mode describe a set of numbers in two other ways. Put the numbers in order first. That makes the median and the mode easy to find.

Example 1.21 The median of an odd number of values

Find the median of the welder's joints in Example 1.13: 38, 42, 45, 40, 35.

SOLUTION

1. Put them in order: 35, 38, 40, 42, 45.
2. There are 5 values, so the middle one is the 3rd: 40.

Answer **The median is 40 joints. Here the mean is also 40.**

Example 1.22 The median and mode of an even number of values

A machine shop logs parts finished each shift: 12, 15, 9, 14, 15, 11. Find the mean, the median and the mode.

SOLUTION

1. Mean: 12 + 15 + 9 + 14 + 15 + 11 = 76, and 76 ÷ 6 = 12.67 to two decimal places.
2. Order: 9, 11, 12, 14, 15, 15. There are 6 values, so there are two middle ones, 12 and 14. The median is halfway between: (12 + 14) ÷ 2 = 13.
3. Mode: 15 comes up twice. Every other value comes up once.

CALCULATOR

TI-30Xa

(12 + 15 + 9 + 14 + 15 + 11) ÷ 6 =

Display 12.66666667

Sharp EL-531XT

(12 + 15 + 9 + 14 + 15 + 11) ÷ 6 =

Display 12.66666667

Answer **Mean 12.67 parts, median 13 parts, mode 15 parts.**

Reading a table

Trade tables put numbers in rows and columns. Read across a row and down a column to find a value. The table below logs the bolts a crew used on each shift of a 7-shift week.

Shift	1	2	3	4	5	6	7
Bolts used	46	52	48	52	55	41	49

Table 1.6 Bolts used on each shift.

Example 1.23 Mean, median and mode from a table

Use Table 1.6. Find the mean, median and mode of the bolts used per shift.

SOLUTION

1. Mean: 46 + 52 + 48 + 52 + 55 + 41 + 49 = 343. 343 ÷ 7 = 49.
2. Median: in order, 41, 46, 48, 49, 52, 52, 55. The 4th of 7 values is the middle: 49.
3. Mode: 52 comes up twice.

CALCULATOR

TI-30Xa

(46 + 52 + 48 + 52 + 55 + 41 + 49) ÷ 7 =

Display 49.

Sharp EL-531XT

(46 + 52 + 48 + 52 + 55 + 41 + 49) ÷ 7 =

Display 49.

Answer **Mean 49, median 49, mode 52 bolts.**

Bar graphs and line graphs

A bar graph shows each value as a bar. Read the value at the top of the bar against the scale on the left. A line graph joins the values with a line, so you can see a change from one point to the next.

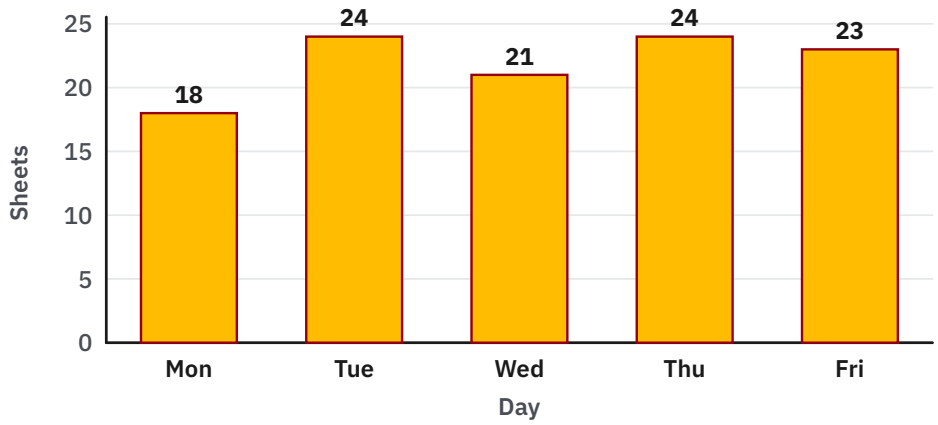


Figure 1.7 Drywall sheets hung by a crew each day.

Example 1.24 Reading a bar graph

Use Figure 1.7. On which day were the most sheets hung? Find the mean, median and mode.

SOLUTION

1. The tallest bars are Tuesday and Thursday, 24 sheets each.
2. Mean: 18 + 24 + 21 + 24 + 23 = 110, and 110 ÷ 5 = 22.
3. Median: in order, 18, 21, 23, 24, 24. The middle value is 23.
4. Mode: 24 comes up twice.

Answer **Tuesday and Thursday. Mean 22, median 23, mode 24 sheets.**

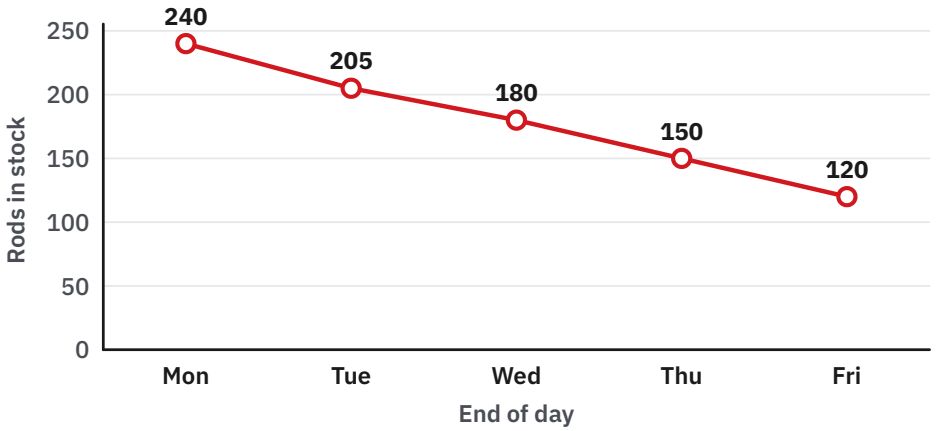


Figure 1.8 Welding rods left in stock at the end of each day.

Example 1.25 Reading a line graph

Use Figure 1.8. How many rods were used from Monday night to Friday night? What is the mean number used per day over those 4 days, and what is the mode?

SOLUTION

1. Used each day: 240 − 205 = 35, 205 − 180 = 25, 180 − 150 = 30, 150 − 120 = 30.
2. Total used: 240 − 120 = 120 rods.
3. Mean: 120 ÷ 4 = 30 rods per day.
4. Mode: 30 comes up twice in 35, 25, 30, 30.

Answer **120 rods. Mean 30 rods per day, mode 30.**

TRY IT 1.11

1. Find the median of 14, 9, 22, 17, 11.
2. Find the median of 8, 3, 10, 6.
3. Find the mode of 5, 7, 5, 9, 7, 5.
4. Find the mean, median and mode of 30, 25, 30, 35, 40.
5. In Figure 1.7, how many more sheets were hung on Tuesday than on Monday?
6. In Figure 1.8, on which day were the most rods used?

Answers are in the Answer Key.

1.12 Carpentry: studs, joists and cut lists

In a wood-frame wall, the upright pieces are studs. They stand between a bottom plate and a top plate. In a floor, the joists lie side by side. Studs and joists are set out at an equal spacing measured from the centre of one to the centre of the next. Carpenters call this the spacing **on centre**, written o.c.

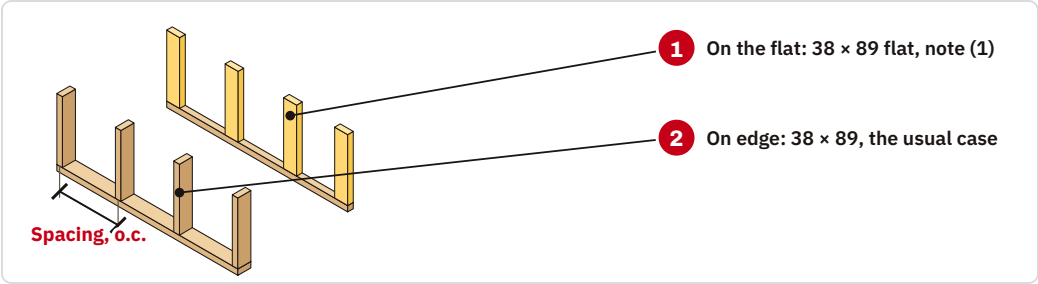


Figure 1.9 Studs on edge and on the flat along a bottom plate. The spacing is measured on centre.

Source: BCBC 2024, Division B, Article 9.23.10.1. (page 9-182)

The BC Building Code sets the largest stud spacing allowed. Article 9.23.10.1, Sentence (1) says the size and spacing of studs shall conform to Table 9.23.10.1 (page 9-182). Part of the table for exterior walls is below. The spacings in the code are in millimetres.

Exterior wall supports	Minimum stud size, mm	Maximum stud spacing, mm	Maximum unsupported height, m
Roof with or without attic storage	38 × 64	400	2.4
	38 × 89	600	3.0
Roof with or without attic storage plus one floor	38 × 89	400	3.0
	38 × 140	600	3.0
Roof with or without attic storage plus 2 floors	38 × 89	300	3.0
	64 × 89	400	3.0
	38 × 140	400	3.6

Table 1.7 Exterior wall rows of Table 9.23.10.1, Size and Spacing of Studs. BCBC 2024, Division B, Article 9.23.10.1. (page 9-182). The full table also lists interior walls.

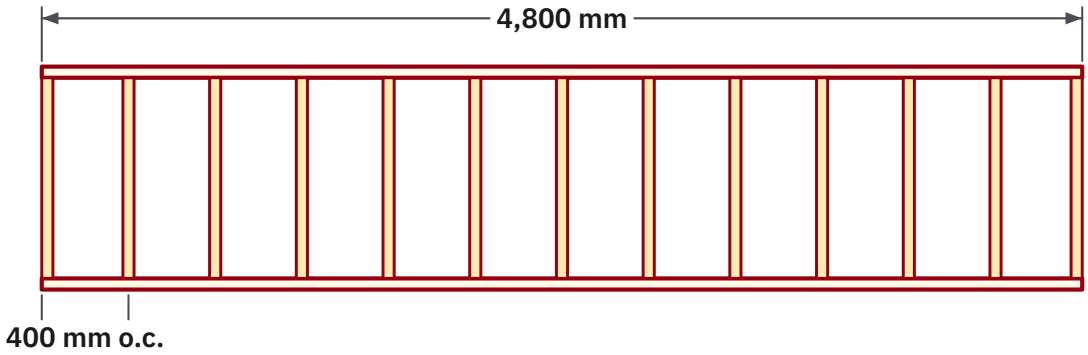


Figure 1.10 A 4,800 mm wall with studs at 400 mm o.c. and a stud at each end. The plates run along the top and bottom.

Example 1.26 Counting studs

A wall is 4,800 mm long. Studs go at 400 mm o.c., starting at one end, with a stud at each end. How many studs are there?

SOLUTION

1. Spaces: 4,800 ÷ 400 = 12.
2. There is a stud at each end, so studs = spaces + 1 = 12 + 1 = 13.
3. Count the studs in Figure 1.10 to check.

CALCULATOR

TI-30Xa

4800 ÷ 400 + 1 =

Display 13.

Sharp EL-531XT

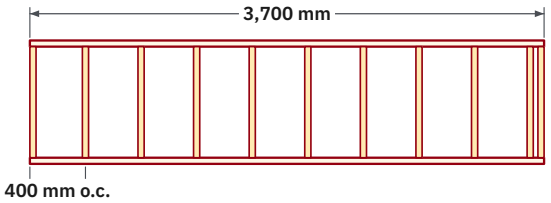
4800 ÷ 400 + 1 =

Display 13.

Answer 13 studs. This count is for a plain wall. Corners, openings and wall ends often need extra studs, which a framing plan shows.

Example 1.27 A length that does not divide evenly

A wall is 3,700 mm long. The studs may be no more than 400 mm o.c. Lay them out at 400 mm from one end, with a stud at each end. How many studs, and how wide is the last space?



SOLUTION

1. Spaces: 3,700 ÷ 400 = 9.25.
2. A space wider than 400 mm is not allowed, so round the spaces up: 10 spaces.
3. Studs: 10 + 1 = 11.
4. The first 9 spaces are 400 mm: 9 × 400 = 3,600 mm. The last space is 3,700 − 3,600 = 100 mm.

CALCULATOR

TI-30Xa

3700 ÷ 400 =

Display 9.25

Sharp EL-531XT

3700 ÷ 400 =

Display 9.25

Answer 11 studs. The last space is 100 mm.

SPACING IS A MAXIMUM

When the spacing is the largest allowed, round the number of spaces up, never down. Rounding down would make every space wider than the limit.

Example 1.28 A cut list for blocking

Blocking fits between studs that are 38 mm thick and set at 400 mm o.c. Each block is 400 − 38 = 362 mm long. You need 12 blocks, cut from boards 2,440 mm long. Each cut takes a 3 mm kerf. How many blocks come from one board, how many boards do you need, and how long is the offcut from each full board?

SOLUTION

1. Each block uses its length plus one saw cut: 362 + 3 = 365 mm.
2. Blocks per board: 2,440 ÷ 365 = 6.68. Only whole blocks count: 6 blocks.
3. Boards: 12 ÷ 6 = 2.00. Round up: 2 boards.
4. Offcut from a board cut into 6 blocks: 2,440 − 6 × 365 = 250 mm.

CALCULATOR

TI-30Xa

Blocks per board

2440 ÷ 365 =

Display 6.684931507

Offcut

2440 − 6 × 365 =

Display 250.

Sharp EL-531XT

Blocks per board

2440 ÷ 365 =

Display 6.684931507

Offcut

2440 − 6 × 365 =

Display 250.

Answer **6 blocks per board, 2 boards, and a 250 mm offcut from each full board.**

Example 1.29 Estimating sheathing

A shed has walls of 9,760 mm, 6,100 mm, 9,760 mm, 6,100 mm. The walls are sheathed with sheets 1,220 mm wide, standing on end, one row high. Ignore openings and corners. Estimate the number of sheets, then find it exactly.

SOLUTION

1. Estimate: round each wall to the nearest thousand:
   10,000 + 6,000 + 10,000 + 6,000 = 32,000 mm. 32,000 ÷ 1,200 is a little under 27.
2. Exact: the walls add to 9,760 + 6,100 + 9,760 + 6,100 = 31,720 mm.
3. 31,720 ÷ 1,220 = 26.

CALCULATOR

TI-30Xa

(9760 + 6100 + 9760 + 6100) ÷ 1220 =

Display 26.

Sharp EL-531XT

(9760 + 6100 + 9760 + 6100) ÷ 1220 =

Display 26.

Answer **26 sheets, close to the estimate.**

TRY IT 1.12

1. A wall is 6,000 mm long with studs at 400 mm o.c. and a stud at each end. How many studs?
2. A wall is 3,000 mm long with studs at 600 mm o.c. and a stud at each end. How many studs?
3. A floor is 5,500 mm long. Joists may be no more than 400 mm o.c., with a joist at each end. How many joists?
4. In Table 1.7, what is the largest spacing for 38 × 89 studs in an exterior wall that supports a roof plus one floor?
5. How many 600 mm blocks come from a 3,660 mm board with a 3 mm kerf on each cut?

Answers are in the Answer Key.

WORKSHEET 1A

Cut lists on the canvas

Each drawing is to scale. Find every red ?

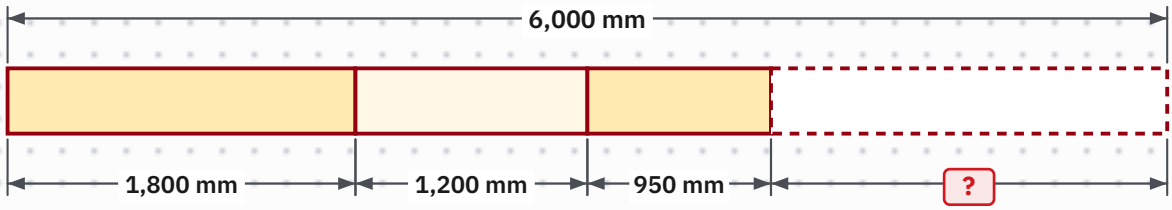
Name _____

Date _____

Work each problem in the working box. Ignore saw cuts unless the problem gives a kerf. Estimate first, then check with your calculator.

1 Three pieces are cut from a 6,000 mm bar. How long is the offcut?

SHAPE CANVAS — PRACTICE: FIND THE ?

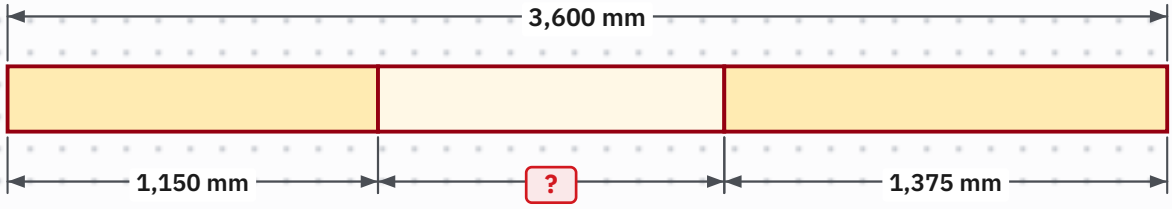


WORKING

Answer _____

2 The overall length is 3,600 mm. Find the middle section.

SHAPE CANVAS — PRACTICE: FIND THE ?

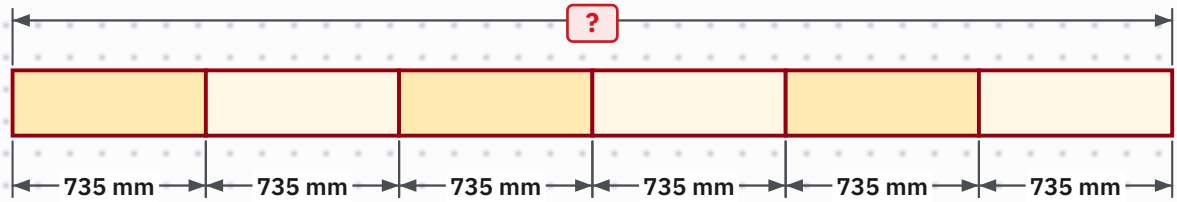


WORKING

Answer _____

3 Six equal pieces of 735 mm. Find the total length.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

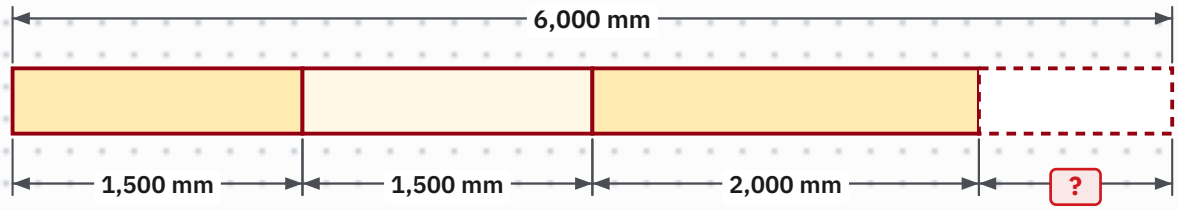
4 How many 380 mm pieces can be cut from a 6,000 mm bar? How long is the offcut?

WORKING

Answer _____

5 Two 1,500 mm pieces and one 2,000 mm piece are cut from a 6,000 mm bar. You make one cut for each piece and the kerf is 3 mm. How long is the offcut?

SHAPE CANVAS — PRACTICE: FIND THE ?



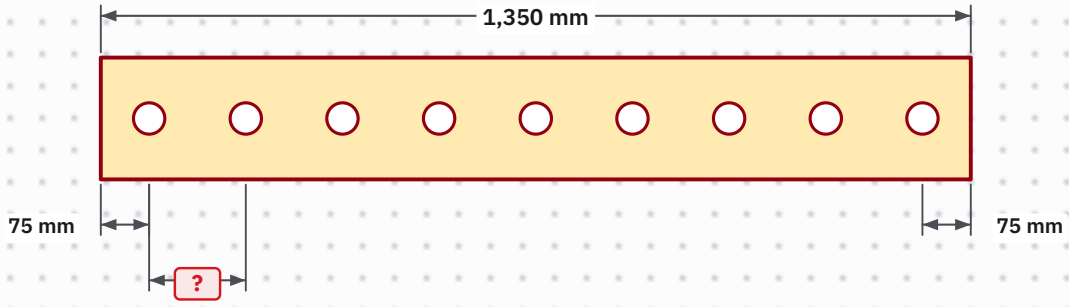
WORKING

Answer _____

6 Nine holes are evenly spaced along a 1,350 mm bar. The end holes are 75 mm from the ends. Find the spacing.

SHAPE CANVAS

PRACTICE: FIND THE ?



WORKING

Answer _____

WORKSHEET 1B

Break it down: plates and parts

Split each part into simple pieces to find the ?

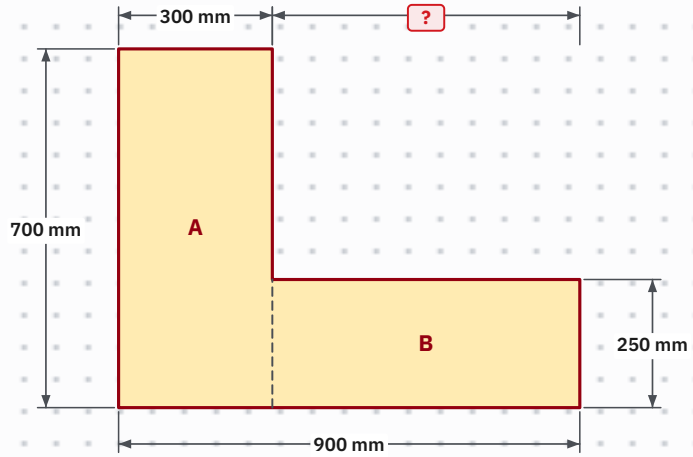
Name _____

Date _____

The dashed line splits the plate into a part A and a part B. Use the sizes you know to find the sizes you do not know.

1 Find the hidden width.

SHAPE CANVAS — PRACTICE: FIND THE ?

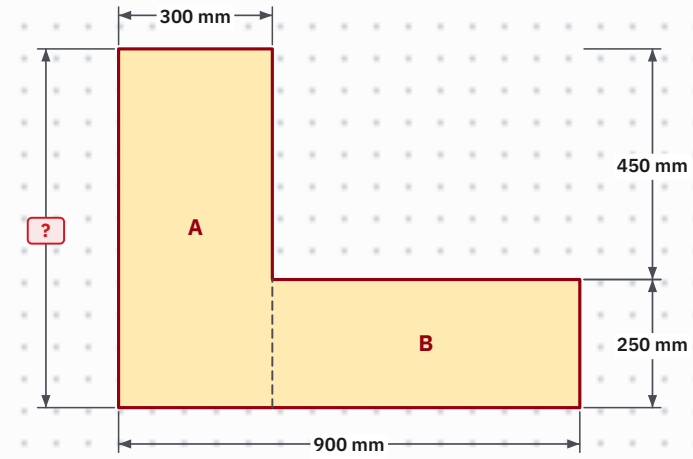


WORKING

Answer _____

2 Find the overall height.

SHAPE CANVAS — PRACTICE: FIND THE ?

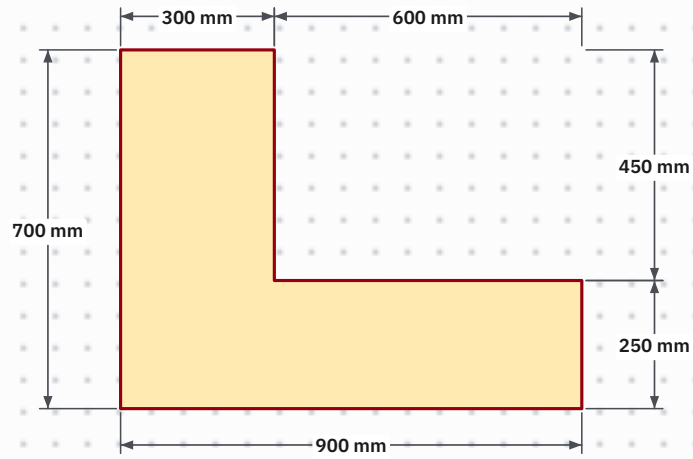


WORKING

Answer _____

3 Find the length of all six edges of the plate, then add them to find the distance around.

SHAPE CANVAS — READ THE DRAWING

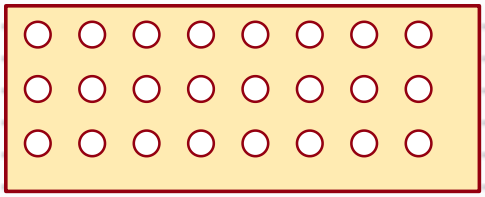


WORKING

Answer _____

4 A plate has holes in equal rows. Count them with one multiplication. Each hole takes 2 minutes to drill. How long does the plate take?

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

5 Estimate 4 × 1,180 by rounding 1,180 to the nearest hundred. Then find the exact answer.

WORKING

Answer _____

6 Four test plates measure 312, 298, 305 and 301 mm across. Find the average.

WORKING

Answer _____

WORKSHEET 1C

Carpentry: studs, joists and cut lists

Walls and floors on the canvas. Find every red ?

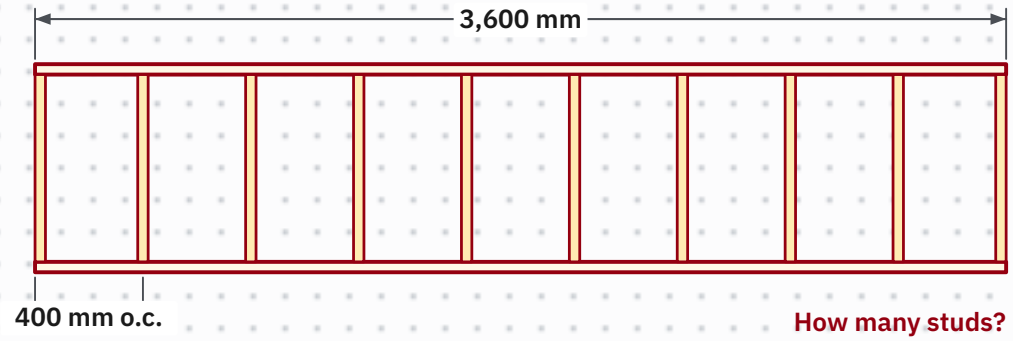
Name _____

Date _____

Every wall has a stud at each end. Spacings are on centre (o.c.). Round the number of spaces up when the spacing is a maximum. Code values come from Table 1.7.

1 Studs go at 400 mm o.c. along a 3,600 mm wall. How many studs are there?

SHAPE CANVAS — READ THE DRAWING

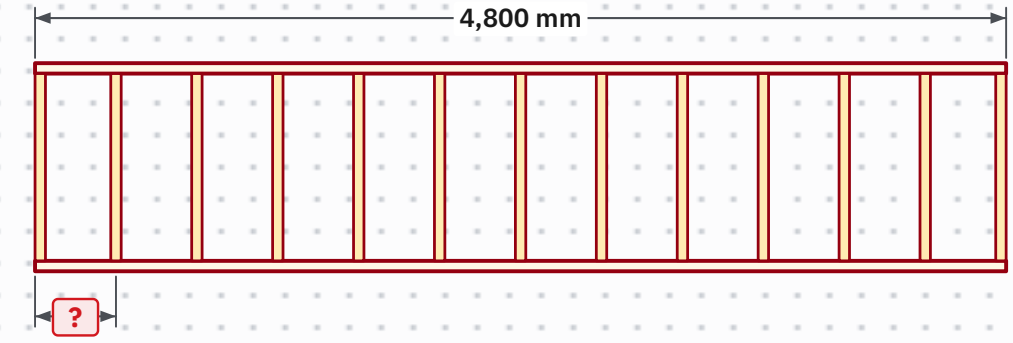


WORKING

Answer _____

2 A 4,800 mm wall has 13 studs, evenly spaced, with one at each end. Find the spacing.

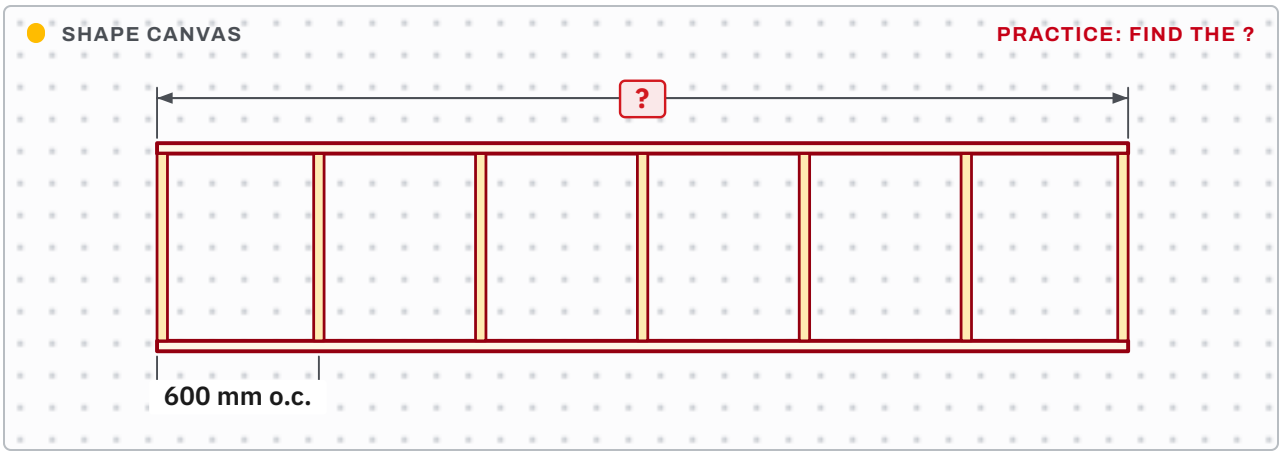
SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

3 Seven studs are set 600 mm o.c., with one at each end. How long is the wall?



WORKING

Answer _____

4 A floor is 4,880 mm long. The joists may be no more than 400 mm o.c., with a joist at each end. How many joists are needed, and how wide is the last space if you lay them out at 400 mm from one end?

WORKING

Answer _____

5 Use Table 1.7. An exterior wall supports a roof plus one floor. It is framed with 38 × 89 studs at 600 mm o.c. Does this meet the table? If not, how many studs does a 6,000 mm length of this wall need at the largest spacing the table allows, with a stud at each end?

WORKING

Answer _____

6 You need 10 cripple studs, each 610 mm long, cut from boards 2,440 mm long. Allow 3 mm of kerf for each piece. How many pieces come from one board, and how many boards do you need?

WORKING

Answer _____

7 Walls with a total length of 14,640 mm are sheathed with sheets 1,220 mm wide, in 2 rows. Ignore openings. Estimate first, then find the number of sheets.

WORKING

Answer _____

CHAPTER

2

Your Calculator: TI-30Xa and Sharp EL-531XT

A scientific calculator is a shop tool like a tape measure or a square. This book shows every calculator step on two calculators, the TI-30Xa and the Sharp EL-531XT. Both give the same answers, but they do not always take the keys in the same order. This chapter walks you around both faceplates: which keys do what, how each calculator wants its keys pressed, and how to read the display and check a result.

In this chapter you will

- Find the main keys on both calculators
- Turn on, clear and fix a mistyped number
- Use the 2nd and 2ndF keys
- Key in negatives, brackets, squares, roots, powers and π
- Use the fraction, percent and memory keys
- Set DEG and use the trig keys, then check the answer

Key words

- key sequence
- display
- second function
- D.A.L.
- memory



TI-30Xa



Sharp EL-531XT

2.1 Meet the two calculators

Both calculators have the same kinds of keys: number keys, the four operations (+ − × ÷), an equals key, and function keys for squares, roots, powers, π, fractions, memory and angles. The keys are in different places, so take a few minutes with the drawings below and your own calculator.

The **display** is the screen. The TI-30Xa display shows a number of up to 10 digits, with small indicators such as DEG and 2nd. The Sharp EL-531XT shows two lines: the calculation you keyed in on the top line and the answer on the bottom line.

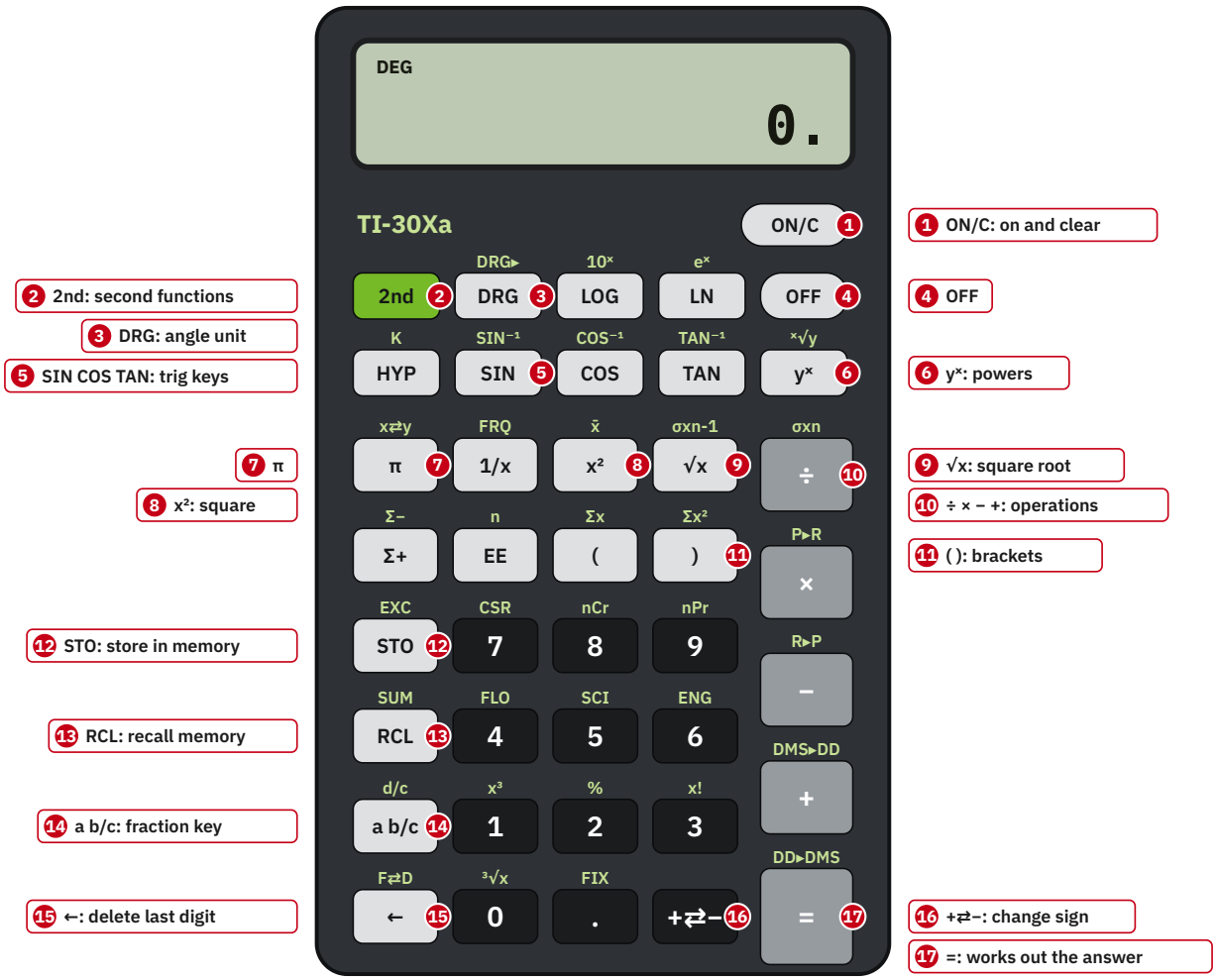


Figure 2.1 The TI-30Xa faceplate. The small labels above the keys are second functions, used with the 2nd key. The numbered notes name the keys used most in this book.

A **key sequence** is the list of keys you press, in order, to do a calculation. Every calculator box in this book shows the key sequence for both calculators and the display each one shows at the end. Table 2.1 lists the keys that look or work differently on the two calculators.

Job	TI-30Xa	Sharp EL-531XT
Turn on	ON/C	ON/C
Turn off	OFF	2ndF then ON/C (OFF is printed above it)
Second functions	2nd	2ndF
Change the sign of a number	+⇄− after the digits	+/− before the number
Square root	√x after the number	√ before the number, then =
Square	x² after the number	x² after the number, then =
Store in memory	STO then 1, 2 or 3	STO then a letter key (A to F, X, Y, M)
Fix a mistyped digit	←	DEL

Table 2.1 Keys that differ between the two calculators.

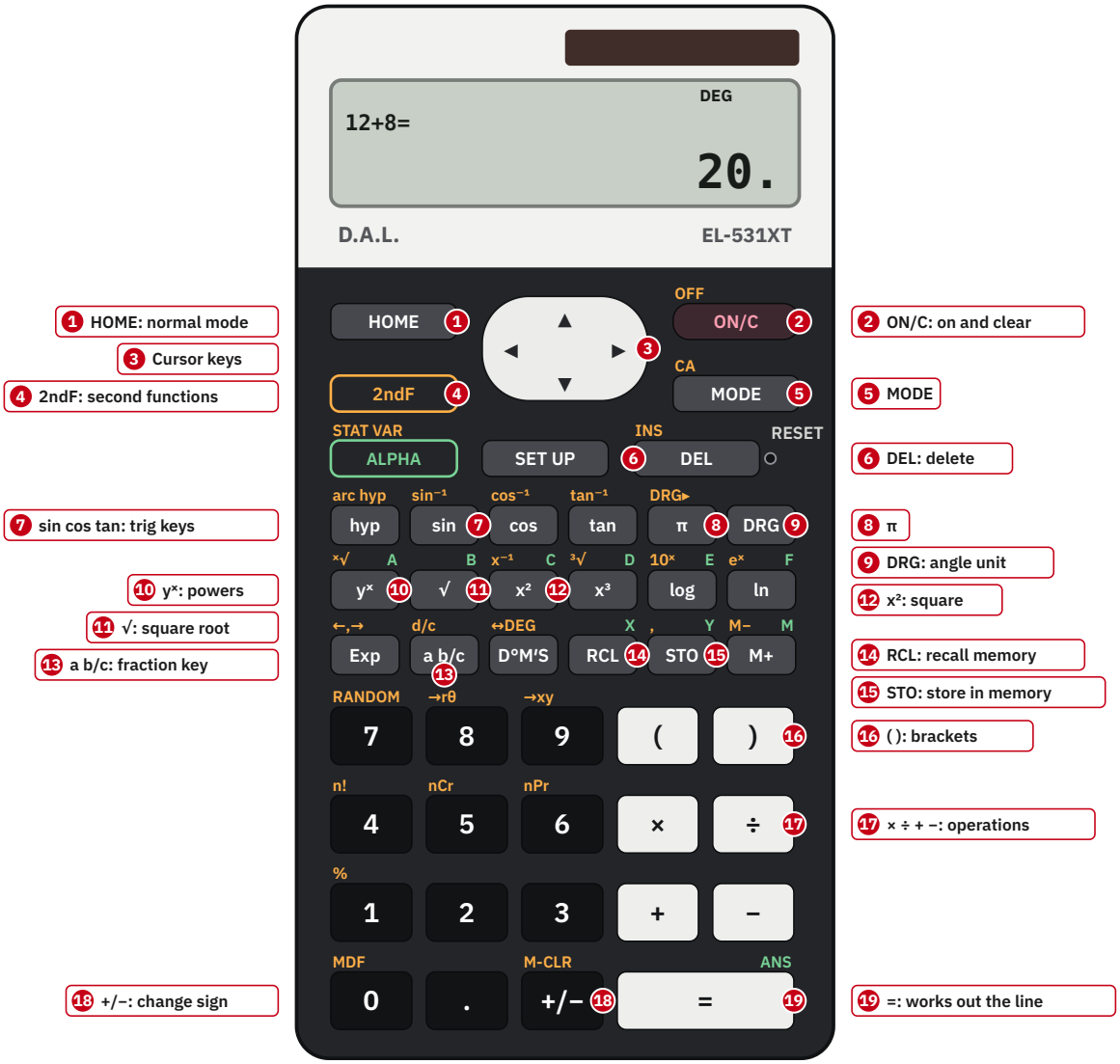


Figure 2.2 The Sharp EL-531XT faceplate. Orange labels above the keys are second functions, used with the 2ndF key. The letters at the upper right of some keys (green in this drawing) are memory names.

2.2 Turning on, clearing and fixing mistakes

Press ON/C to turn either calculator on. To turn the TI-30Xa off, press OFF. To turn the Sharp off, press 2ndF and then ON/C, because OFF is the second function printed in orange above the ON/C key. These steps are for the battery TI-30Xa, the model drawn in this book. The TI-30Xa Solar works differently: its guidebook says to expose the solar panel to light to turn it on and to cover the panel with the slide case to turn it off.

Start every calculation with a clear calculator. The calculator boxes in this book all start that way: press ON/C twice before the keys shown, and check that DEG is on the display.

Example 2.1 Fixing a wrong number

You meant to key 7 + 9 but pressed 8 instead of 9. Then you meant to key 125 × 4 but typed 1255. Fix each mistake without starting over.

SOLUTION

1. TI-30Xa: press ON/C once. One press clears only the number you were keying, so 7 + is still waiting. Key 9 and press =.
2. TI-30Xa: press ← to delete the last digit typed. 1255 becomes 125. Carry on with × 4 =.
3. Sharp: press DEL. When the cursor is at the right end of the line, DEL deletes the last character. Then key the right digit.

CALCULATOR

TI-30Xa

ON/C once

7 + 8 ON/C 9 =

Display 16.

Delete the last digit

1255 ← × 4 =

Display 500.

Sharp EL-531XT

ON/C once

7 + 9 =

Display 16.

Delete the last digit

125 × 4 =

Display 500.

The Sharp column shows each line keyed without the mistake. On the Sharp, use DEL to make the fix. ON/C on the Sharp clears the entry.

Answer **7 + 9 = 16 and 125 × 4 = 500.**

WHEN IN DOUBT, START AGAIN

On the TI-30Xa, pressing ON/C twice clears the display and every waiting operation. On the Sharp, ON/C clears the entry. If you are not sure what the calculator is holding, clear it and key the calculation again from the start.

2.3 Second functions: the 2nd and 2ndF keys

There are more jobs than keys, so most keys do two jobs. The job printed on the key is its first function. The job printed above the key is its **second function**. To use a second function, press 2nd on the TI-30Xa or 2ndF on the Sharp, then press the key.

On the TI-30Xa, 2nd shows on the display while it waits for the next key. On the Sharp, the second functions are printed in orange, and 2ndF shows on the display after you press 2ndF. Some functions are a second function on one calculator and a key of their own on the other. The cube, x^3, is an example.

Example 2.2 Cube and cube root

Find 5^3 (5 × 5 × 5) and the cube root of 27.

SOLUTION

1. On the TI-30Xa, x^3 and $\sqrt[3]{x}$ are second functions, so press 2nd before each one.
2. On the Sharp, x^3 has its own key. $\sqrt[3]{}$ is the second function printed above the x^3 key, so press 2ndF first.

CALCULATOR

TI-30Xa

5^3

[5] [2nd] [x^3]

Display: 125.

Cube root of 27

[27] [2nd] [$\sqrt[3]{x}$]

Display: 3.

Sharp EL-531XT

5^3

[5] [x^3] [=]

Display: 125.

Cube root of 27

[2ndF] [$\sqrt[3]{}$] [27] [=]

Display: 3.

Answer **5^3 = 125 and the cube root of 27 is 3.**

READ THE KEY BEFORE YOU PRESS IT

If the label you want is above the key, you need 2nd or 2ndF first.

2.4 Two ways to take keys

The two calculators work out the same answer but want some keys in a different order.

On the TI-30Xa, a function key such as x^2, $\sqrt{x}$ or SIN goes after the number, and it works at once. Key 144, press $\sqrt{x}$, and 12 shows straight away.

The Sharp has **D.A.L.** printed on its faceplate. For you, D.A.L. means this: press the keys in the same order the calculation is written. For √144 that is √ first, then 144, then =. The line is worked out when you press =. The top line of the display shows what you keyed, so you can read it back first.

Example 2.3 Square and square root

Find 12^2 and √144.

SOLUTION

1. 12^2 means 12 × 12. Both calculators take x^2 after the number. The Sharp needs = to finish.
2. √144 asks which number times itself makes 144. The TI-30Xa takes √x after the number. The Sharp takes √ first, as written.

CALCULATOR

TI-30Xa

12^2

[12] [x²]

Display 144.

√144

[144] [√x]

Display 12.

Sharp EL-531XT

12^2

[12] [x²] [=]

Display 144.

√144

[√] [144] [=]

Display 12.

Answer **12^2 = 144 and √144 = 12. Each answer checks the other.**

TRY IT 2.4

1. Write the TI-30Xa keys for √81.
2. Write the Sharp keys for √81.
3. Find 15^2.
4. Find √196.
5. Find 4^3.
6. Find the cube root of 64.

Answers are in the Answer Key.

2.5 Negative numbers

A negative number is less than zero, such as −5. Both calculators have a change-sign key that makes a number negative. It is not the minus key. The minus key subtracts one number from another.

On the TI-30Xa, key the digits first and then press +⇄−. On the Sharp, press +/− first and then the digits, the way the number is written.

Example 2.4 A calculation with a negative number

Find 42 × (−5) + 120.

SOLUTION

1. The −5 is a negative number, so it needs the change-sign key, not the minus key.
2. 42 × (−5) = −210, then −210 + 120 = −90. Both displays show the minus sign in front.

CALCULATOR

TI-30Xa

42 × 5 +⇄− + 120 =

Display -90.

Sharp EL-531XT

42 × +/− 5 + 120 =

Display -90.

Answer **−90**

CHANGE SIGN OR SUBTRACT?

Use the minus key between two numbers: 50 − 8. Use the change-sign key for a number that is negative on its own: 6 × (−4).

TRY IT 2.5

1. −8 + 15
2. 6 × (−4)
3. −20 − 12
4. (−36) ÷ 9

Answers are in the Answer Key.

Section 3.8 in Chapter 3 covers positive and negative numbers in full: the number line, adding and subtracting with signs, and the sign rules for × and ÷.

2.6 Brackets, powers and π

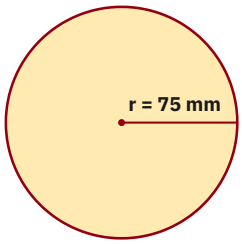
Brackets tell the calculator which part to work out first. Both calculators have (and) keys and take them in the order written: for (12 + 8) × 3, key (12 + 8) × 3 = and both show 60. Chapter 3 explains when you need them.

An **exponent** says how many of that number are multiplied together. $2^5 = 2 \times 2 \times 2 \times 2 \times 2$. Use the y^x key: the number, then y^x, then the exponent, then =. For 2^5, key 2 y^x 5 = on either calculator. Both show 32, and the Sharp shows 2^5= on its top line.

The π key enters the number π, about 3.14. Use it instead of typing 3.14, so the answer keeps all the digits.

Example 2.5 Using the π key

The distance around a circle is π × D, where D is the diameter (across the circle through its centre). The area is π × r², where r is the radius (half the diameter). A round duct has D = 150 mm and r = 75 mm. Find both, to two decimal places.



SOLUTION

1. Distance around: π × 150 = 471.2388..., which rounds to 471.24 mm.
2. Area: 75² = 5,625, then π × 5,625 = 17,671.458..., which rounds to 17,671.46 mm². Key 75 x² × π = on both calculators.

CALCULATOR

TI-30Xa

π × D

[π] [×] [150] [=]

Display 471.238898

π × r²

[75] [x²] [×] [π] [=]

Display 17671.45868

Sharp EL-531XT

π × D

[π] [×] [150] [=]

Display 471.238898

π × r²

[75] [x²] [×] [π] [=]

Display 17'671.45868

Answer **About 471.24 mm around and 17,671.46 mm² of area.**

TRY IT 2.6

1. (25 + 15) × 4
2. 3^4
3. 10^3 using y^x
4. π × 200, to two decimal places
5. π × 40², to two decimal places
6. (18 – 6) ÷ 4

Answers are in the Answer Key.

2.7 The fraction key

Both calculators have an a b/c key for fractions. To key a mixed number such as 3¼, key the whole number, a b/c, the top number, a b/c, the bottom number. The answer comes back as a fraction. Chapter 4 covers fractions in full.

Example 2.6 Fractions of an inch

Add 3¼ in and 2⅝ in. Then change ⅜ in to a decimal.

SOLUTION

1. Key 3 a b/c 1 a b/c 4 for 3¼ and 2 a b/c 5 a b/c 8 for 2⅝.
2. The TI-30Xa shows 5⌟7⌟8 and the Sharp shows 5 ⌜7 ⌜8. Both mean 5 and 7/8: the whole number, then the top number, then the bottom number.
3. To change a fraction answer to a decimal, press 2nd F⇄D on the TI-30Xa. On the Sharp, press a b/c after the answer. The second row changes ⅜ to a decimal.

CALCULATOR

TI-30Xa

3¼ + 2⅝

3 | a b/c | 1 | a b/c | 4 | + | 2 | a b/c | 5 | a b/c | 8 | =

Display: 5⌟7⌟8

⅜ as a decimal

3 | a b/c | 8 | 2nd | F⇄D

Display: 0.375

Sharp EL-531XT

3¼ + 2⅝

3 | a b/c | 1 | a b/c | 4 | + | 2 | a b/c | 5 | a b/c | 8 | =

Display: 5 ⌜7 ⌜8

⅜ as a decimal

3 | a b/c | 8 | = | a b/c

Display: 0.375

Answer $5\frac{7}{8}$ **in, and ⅜ in = 0.375 in.**

TRY IT 2.7

1. 1½ + 2¾ with the fraction key
2. Change ⅝ to a decimal.
3. Change 2⅜ to a decimal.
4. ¾ × 120

Answers are in the Answer Key.

Section 2.12 shows how to key a tape length in feet, inches and fractions of an inch.

2.8 Percent

The percent key (%) is a second function on both calculators: 2nd % on the TI-30Xa and 2ndF % on the Sharp, where % is printed above the 1 key. On the TI-30Xa you finish with =. On the Sharp, 2ndF % works out the line itself, so you do not press = after it.

Example 2.7 Percent of a number, and adding a percent

An order is for 250 bolts. Find 15% of 250. Then find 250 plus 15% more.

SOLUTION

1. 15% of 250: on the TI-30Xa key 250 × 15 2nd % =. On the Sharp key 250 × 15 2ndF %.
2. 250 plus 15%: on the TI-30Xa key 250 + 15 2nd % =. On the Sharp key 250 + 15 2ndF %. The calculator adds 15% of 250 to 250.

CALCULATOR

TI-30Xa

15% of 250

250 × 15 2nd % =

Display 37.5

250 + 15%

250 + 15 2nd % =

Display 287.5

Sharp EL-531XT

15% of 250

250 × 15 2ndF %

Display 37.5

250 + 15%

250 + 15 2ndF %

Display 287.5

Answer **15% of 250 is 37.5. Adding 15% gives 287.5, so order 288 bolts.**

PERCENT WITHOUT THE % KEY

15% means 15 out of 100. You can always key it as × 15 ÷ 100: 250 × 15 ÷ 100 = 37.5. Chapter 6 covers percent in full.

2.9 Memory: STO and RCL

A **memory** holds a number so you do not have to write it down and key it again. STO (store) puts a number in a memory. RCL (recall) brings it back.

The TI-30Xa has three memories, 1, 2 and 3. Press STO then 1 to store the display in memory 1. The Sharp names its memories with letters: A to F, X, Y and M. The letters are printed at the upper right above some keys. A is above y^x, so STO then y^x stores in A, and the box shows that key as A. On the Sharp, STO finishes the line you keyed and stores the answer.

Example 2.8 Storing a cut list total

Two pieces, 1,250 mm and 375 mm, are cut from a 6,000 mm bar. Ignore the saw cuts. Store the total of the pieces, clear the display, then find the offcut.

SOLUTION

1. 1,250 + 375 = 1,625. Store it in memory 1 (A on the Sharp).
2. Press ON/C. The memory keeps the number.
3. 6,000 − RCL 1 = 6,000 − 1,625 = 4,375.

CALCULATOR

TI-30Xa

1250 + 375 = STO 1 ON/C 6000 − RCL 1 =

Display 4375.

Sharp EL-531XT

1250 + 375 STO A ON/C 6000 − RCL A =

Display 4’375.

The TI-30Xa shows 4375. and the Sharp shows 4’375. The memory kept 1,625 after ON/C on both calculators.

Answer **4,375 mm**

OLD NUMBERS STAY IN MEMORY

Clearing with ON/C does not empty the memories, so a memory can still hold a number from an earlier job. Store the number you need before you recall it.

2.10 Angles: DEG and the trig keys

The trig keys SIN, COS and TAN work with angles. Chapter 13 explains what they mean. Angles can be measured in degrees, radians or grads, and the calculator must be set to the unit you are using. This book uses degrees.

Each press of DRG changes the angle unit in the order DEG, RAD, GRAD. The unit shows at the top of the display. Before any trig calculation, check that DEG is showing. If RAD or GRAD shows, press DRG until DEG comes back.



Figure 2.3 DEG at the top of each display means the calculator is set to degrees.

The trig keys follow the same rule as the other functions. On the TI-30Xa, key the angle and then SIN. On the Sharp, press sin, the angle and =. The inverse keys, SIN^{-1}, COS^{-1} and TAN^{-1}, are second functions. They take a number and give back an angle.

Example 2.9 Trig keys and their inverses

Find sin 30°. Find the angle whose tangent is 0.75, to two decimal places. Then find 600 × tan 35°, to two decimal places.

SOLUTION

1. sin 30° = 0.5.
2. The angle whose tangent is 0.75 uses TAN^{-1}: 2nd TAN^{-1} on the TI-30Xa, 2ndF $\tan^{-1}$ on the Sharp. The display shows 36.86989765, which rounds to 36.87°.
3. 600 × tan 35°: on the TI-30Xa, 35 TAN gives tan 35°, then × 600. On the Sharp, key tan 35 × 600 = as written. 420.1245229 rounds to 420.12.

CALCULATOR

TI-30Xa

sin 30°

[30] [SIN]

Display 0.5

$\tan^{-1}$ 0.75

[0.75] [2nd] [TAN^{-1}]

Display 36.86989765

600 × tan 35°

[35] [TAN] [×] [600] [=]

Display 420.1245229

Sharp EL-531XT

sin 30°

[sin] [30] [=]

Display 0.5

$\tan^{-1}$ 0.75

[2ndF] [$\tan^{-1}$] [0.75] [=]

Display 36.86989765

600 × tan 35°

[tan] [35] [×] [600] [=]

Display 420.1245229

Answer **sin 30° = 0.5, the angle is 36.87°, and 600 × tan 35° = 420.12.**

TRY IT 2.10

1. cos 60°
2. sin 45°, to four decimal places
3. tan 20°, to four decimal places
4. The angle whose sine is 0.5
5. The angle whose cosine is 0.8, to two decimal places
6. 250 × cos 40°, to two decimal places

Answers are in the Answer Key.

2.11 Checking a result

A calculator does exactly what you key, including your mistakes. Before you cut or order, check the answer in at least one of these ways.

1. Estimate first. Round the numbers and work it out in your head, as in Chapter 1. The calculator answer should be close.
2. Work it backwards. If $\sqrt{144} = 12$, then 12^2 should give 144. If 6,000 − 1,625 = 4,375, then 4,375 + 1,625 should give 6,000.
3. Read the Sharp top line. It shows the calculation you keyed, so you can spot a wrong number before you press =.
4. Key it again. If the second answer differs from the first, key it a third time slowly.

An answer with more than 10 digits does not fit on the display. The calculator then shows it with a small number at the right, called an exponent of 10. For 123,456,789 × 1,000, both calculators show 1.23456789 with a small 11, which means $1.23456789 \times 10^{11}$, or 123,456,789,000.

> **WRITE THE ANSWER WITH ITS UNIT**
>
> The display shows a number, not a size. Write the unit next to every answer you copy down: 4,375 mm, not 4375. A number without a unit is a guess on the shop floor.

2.12 Tape lengths in feet, inches and fractions

A carpenter's tape reads feet, inches and fractions of an inch, such as $8' \; 5\frac{3}{8}''$ (8 feet $5\frac{3}{8}$ inches). To work with a length like this on either calculator, change it to inches first. One foot is 12 inches, so the length in inches is feet × 12 + inches. The fraction key carries the fraction of an inch.

BEDMAS works the × 12 before the +, so no brackets are needed: key 8 × 12 + 5 a b/c 3 a b/c 8 =. Both calculators show the answer as a mixed number of inches, the same way they show any fraction answer (Section 2.7).

Example 2.10 Keying a tape length

A stud is cut to 8′ $5\frac{3}{8}$″. Find its length in inches. Then find it in millimetres, to the nearest millimetre. One inch is exactly 25.4 mm.

SOLUTION

1. Inches: $8 \times 12 + 5\frac{3}{8} = 96 + 5\frac{3}{8} = 101\frac{3}{8}$ in. The TI-30Xa shows 101⌟3⌟8 and the Sharp shows 101 ⌜3 ⌜8.
2. Millimetres: carry on from the answer with × 25.4 =. $101\frac{3}{8} \times 25.4 = 2{,}574.925$, which rounds to 2,575 mm.
3. On the Sharp, the second row puts the inches in brackets, because it keys the whole line in written order.

CALCULATOR

TI-30Xa

Feet and inches to inches

[8] [×] [12] [+] [5] [a b/c] [3] [a b/c] [8] [=]

Display: 101⌟3⌟8

Inches to mm

[8] [×] [12] [+] [5] [a b/c] [3] [a b/c] [8] [=] [×] [25.4] [=]

Display: 2574.925

Sharp EL-531XT

Feet and inches to inches

[8] [×] [12] [+] [5] [a b/c] [3] [a b/c] [8] [=]

Display: 101 ⌜3 ⌜8

Inches to mm

[(] [8] [×] [12] [+] [5] [a b/c] [3] [a b/c] [8] [)] [×] [25.4] [=]

Display: 2′574.925

Answer **$101\frac{3}{8}$ in, or about 2,575 mm.**

Example 2.11 Adding tape lengths

Two pieces of trim, $7' \; 4\frac{5}{8}''$ and $3' \; 9\frac{3}{4}''$, are butted end to end. Find the total length in feet and inches.

SOLUTION

1. Change both to inches in one line: $7 \times 12 + 4\frac{5}{8} + 3 \times 12 + 9\frac{3}{4} = 134\frac{3}{8}$ in. BEDMAS does both × 12 before the additions.
2. Whole feet: $134\frac{3}{8} \div 12$. The display shows 11 and a fraction. The whole number at the front, 11, is the number of whole feet.
3. Inches left over: $134\frac{3}{8} - 11 \times 12 = 134\frac{3}{8} - 132 = 2\frac{3}{8}$ in.

CALCULATOR

TI-30Xa

Total in inches

7 × 12 + 4 a b/c 5 a b/c 8 + 3 × 12 + 9 a b/c 3 a b/c 4 =

Display: 134⌟3⌟8

Whole feet

134 a b/c 3 a b/c 8 ÷ 12 =

Display: 11⌟19⌟96

Inches left over

134 a b/c 3 a b/c 8 − 11 × 12 =

Display: 2⌟3⌟8

Sharp EL-531XT

Total in inches

7 × 12 + 4 a b/c 5 a b/c 8 + 3 × 12 + 9 a b/c 3 a b/c 4 =

Display: 134⌜3⌜8

Whole feet

134 a b/c 3 a b/c 8 ÷ 12 =

Display: 11⌜19⌜96

Inches left over

134 a b/c 3 a b/c 8 − 11 × 12 =

Display: 2⌜3⌜8

Answer $11' \; 2\frac{3}{8}''$ $(134\frac{3}{8}$ in)

FEET AND INCHES ARE NOT DECIMALS

$8' \; 6''$ is not 8.6 ft. There are 12 inches in a foot, not 10. Key 8 × 12 + 6 = to get 102 in, and 102 ÷ 12 = 8.5 ft.

TRY IT 2.12

1. $6' \, 3\frac{1}{2}''$ in inches
2. $10' \, 7\frac{3}{4}''$ in inches
3. $2' \, 9\frac{1}{16}''$ in inches
4. $4' \, 2\frac{5}{8}'' + 5' \, 11\frac{1}{8}''$, in feet and inches
5. $12'$ minus $3' \, 8\frac{1}{4}''$, in feet and inches. Put the length you take away in brackets.
6. $92\frac{5}{8}''$ in millimetres, to the nearest millimetre

Answers are in the Answer Key.

WORKSHEET 2A

Key it in

Write the keys for both calculators, then the answer

Name _____

Date _____

For each problem, write the key sequence for the TI-30Xa and for the Sharp EL-531XT. Then work it out on your own calculator. Start each one with ON/C ON/C and DEG on the display.

1 2,450 + 1,375 − 980

WORKING

Answer _____

2 18^2

WORKING

Answer _____

3 √529

WORKING

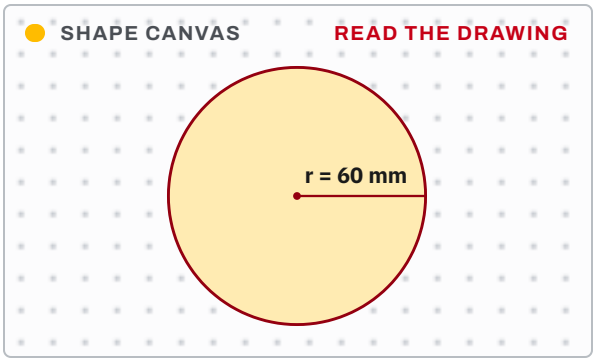
Answer _____

4 (−12) × 4 + 60

WORKING

Answer _____

5 The circle has the radius shown. Find D = 2 × r, then the distance around, π × D, to two decimal places.



WORKING

Answer _____

6 Add 2¾ in and 1⅝ in with the fraction key. Write the answer as each display shows it, then in the usual way.

WORKING

Answer _____

7 Find 20% of 340 kg.

WORKING

Answer _____

8 Find 800 × sin 25°, to two decimal places. Check that DEG is on the display first.

WORKING

Answer _____

WORKSHEET 2B

Know your keys, read the display

Name the lettered keys and read each display

Name _____

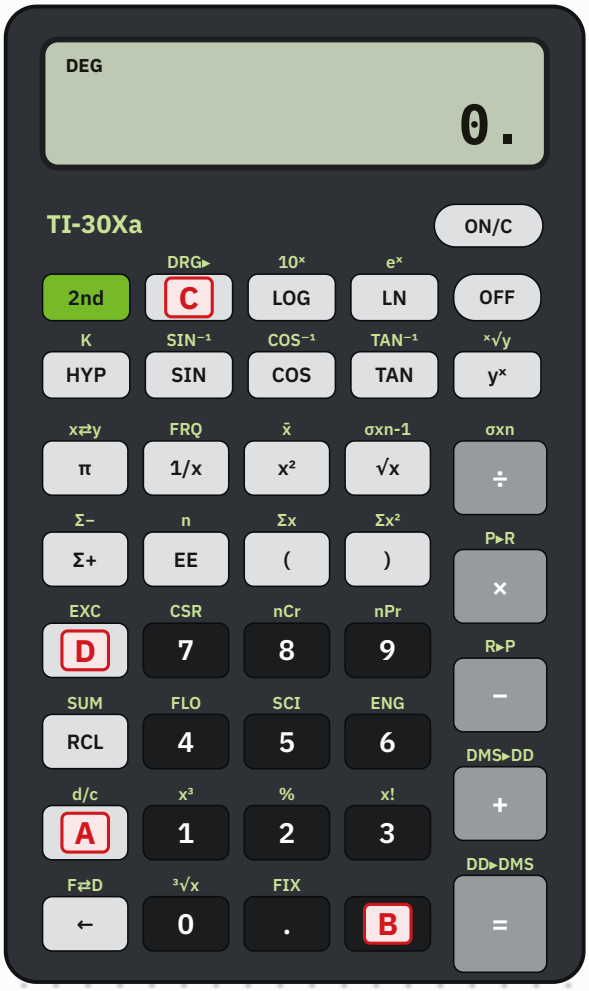
Date _____

The red letters mark keys on each faceplate. Name each key and say what it does. For the displays, write the number in the usual way.

1 Name keys A to H and say what each one does. A to D are on the TI-30Xa, E to H on the Sharp EL-531XT.

SHAPE CANVAS

NAME THE KEYS





WORKING

Answer _____

2 Write the number this display shows.

SHAPE CANVAS READ



TI-30Xa

WORKING Answer _____

3 What calculation was keyed? What is the answer?

SHAPE CANVAS READ



Sharp EL-531XT

WORKING Answer _____

4 This display shows RAD. What do you press before you use SIN?

SHAPE CANVAS READ



TI-30Xa

WORKING Answer _____

CHAPTER

3

Order of Operations

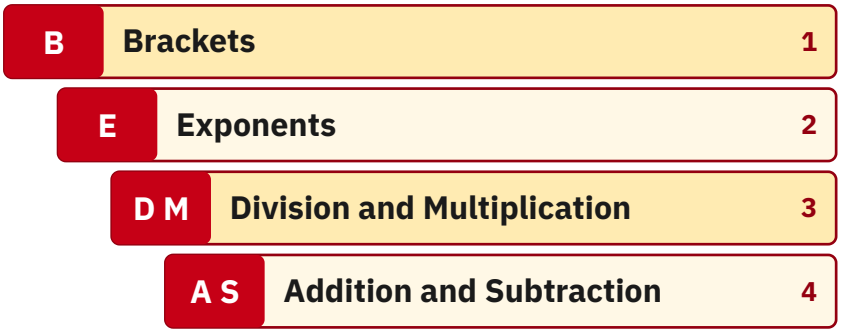
What is 3 + 4 × 5? Add first and you get 35. Multiply first and you get 23. Only one answer can be right, so everyone who works with numbers follows the same order, and so does your calculator. This chapter sets out that order, BEDMAS, and shows when the calculator needs brackets that are not written on the page.

In this chapter you will

- Use BEDMAS to work a calculation step by step
- Work out brackets, powers and roots in the right order
- Work left to right for × and ÷, and for + and −
- Treat a fraction bar as a grouping symbol
- Key brackets so the calculator follows the same order

Key words

- order of operations
- BEDMAS
- brackets
- exponent
- square
- square root



3.1 Why order matters

A calculation with more than one operation can be read in more than one way. The **order of operations** is the agreed order for working one out. When everyone uses the same order, a calculation written on a drawing or in a formula means the same thing to everyone who reads it.

Example 3.1 Bolts in a crate

A crate holds 3 loose bolts and 4 bags of 5 bolts. The count is written 3 + 4 × 5. How many bolts are in the crate?

SOLUTION

1. The 4 bags of 5 bolts are 4 × 5 = 20 bolts. That multiplication has to happen first.
2. Add the loose bolts: 3 + 20 = 23.
3. Adding first, 3 + 4 = 7 and 7 × 5 = 35, counts the 3 loose bolts as if they were in bags. That is wrong.

CALCULATOR

TI-30Xa

3 + 4 × 5 =

Display 23.

Sharp EL-531XT

3 + 4 × 5 =

Display 23.

Both calculators multiply before they add, so both show 23 even though the keys are pressed left to right.

Answer 23 bolts

3.2 BEDMAS

BEDMAS is a memory aid for the order of operations. Each letter stands for one kind of operation, from the first to be done to the last. The ladder on the first page of this chapter shows the four steps: start at the top step and work down.

Letter	Stands for	Example
B	Brackets ( ) and [ ]	(12 + 8) × 3: add 12 + 8 first
E	Exponents: squares, powers and roots	100 − 5^2: work out 5^2 first
D and M	Division and Multiplication, left to right	24 ÷ 4 × 3: divide first, because ÷ comes first
A and S	Addition and Subtraction, left to right	20 − 8 + 2: subtract first, because − comes first

Table 3.1 What each letter of BEDMAS means.

KEY IDEA

D and M are one step, not two. Neither one goes first. Work them in the order they appear, left to right. The same goes for A and S.

Example 3.2 Working through BEDMAS

Find 8 − 3 × 2 + 1.

SOLUTION

1. No brackets and no exponents.
2. Multiply: 3 × 2 = 6. The calculation is now 8 − 6 + 1.
3. Add and subtract left to right: 8 − 6 = 2, then 2 + 1 = 3.

CALCULATOR

TI-30Xa

8 − 3 × 2 + 1 =

Display 3.

Sharp EL-531XT

8 − 3 × 2 + 1 =

Display 3.

Answer 3

REWRITE THE LINE AFTER EACH STEP

Write the whole calculation again after each step, with the part you just worked out replaced by its answer. Each line gets shorter until one number is left.

TRY IT 3.2

1. 6 + 4 × 3
2. 20 − 12 ÷ 4
3. 5 × 6 − 4 × 3
4. 18 ÷ 3 + 2 × 7

Answers are in the Answer Key.

3.3 Brackets

Brackets group part of a calculation so it is worked out first. They come first in BEDMAS. Inside the brackets, use BEDMAS again.

Sometimes there are brackets inside brackets. Square brackets [ ] are often used for the outer pair so they are easy to tell apart. Work out the innermost brackets first, then work outward. On the calculator, every bracket is keyed with the (and) keys.

Example 3.3 Brackets change the answer

Find 12 + 8 × 3 and (12 + 8) × 3.

SOLUTION

1. 12 + 8 × 3: multiply first, 8 × 3 = 24, then 12 + 24 = 36.
2. (12 + 8) × 3: brackets first, 12 + 8 = 20, then 20 × 3 = 60.

CALCULATOR

TI-30Xa

No brackets

12 + 8 × 3 =

Display 36.

With brackets

(12 + 8) × 3 =

Display 60.

Sharp EL-531XT

No brackets

12 + 8 × 3 =

Display 36.

With brackets

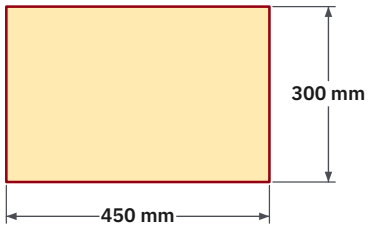
(12 + 8) × 3 =

Display 60.

Answer **36 and 60. The same numbers give different answers.**

Example 3.4 A frame and brackets inside brackets

A rectangular frame is 450 mm by 300 mm. The distance around it is 2 × (450 + 300). Find it. Then find 2 × [15 − (4 + 6)].



SOLUTION

1. Frame: brackets first, 450 + 300 = 750. Then 2 × 750 = 1,500 mm.
2. Inner brackets first: 4 + 6 = 10. The line is now 2 × [15 − 10].
3. Outer brackets: 15 − 10 = 5. Then 2 × 5 = 10.

On the calculator, square brackets are keyed with the same (and) keys: 2 × (15 − (4 + 6)) =. Close every bracket you open.

Answer **The frame is 1,500 mm around, and 2 × [15 − (4 + 6)] = 10.**

KEY THE × BEFORE A BRACKET

On paper, 3(4 + 5) means 3 × (4 + 5). The × is often left out next to a bracket. On the calculator, always key the × yourself: 3 × (4 + 5) = 27.

TRY IT 3.3

1. $(6 + 4) \times (3 - 2)$
2. $(25 - 5) \div 4$
3. $3 \times (7 + 8)$
4. $2 \times (1{,}200 + 800)$

Answers are in the Answer Key.

3.4 Exponents, squares and square roots

An **exponent** is the small raised number that says how many of that number are multiplied together. In 5^3 the exponent is 3: $5^3 = 5 \times 5 \times 5 = 125$.

The **square** of a number is the number multiplied by itself. The square of 5 is written 5^2 and read "5 squared": $5 \times 5 = 25$. The **square root** undoes a square. $\sqrt{25} = 5$ because $5 \times 5 = 25$.

Exponents and roots come right after brackets in BEDMAS. An exponent belongs only to the number or bracket it touches.

Example 3.5 What does the square belong to?

Find $100 - 5^2$, $(100 - 5)^2$, 5×3^2 and $(5 \times 3)^2$.

SOLUTION

1. $100 - 5^2$: the square belongs to the 5. $5^2 = 25$, then $100 - 25 = 75$.
2. $(100 - 5)^2$: brackets first, 95, then $95^2 = 9{,}025$.
3. 5×3^2: the square belongs to the 3. $3^2 = 9$, then $5 \times 9 = 45$.
4. $(5 \times 3)^2$: brackets first, 15, then $15^2 = 225$.

CALCULATOR

TI-30Xa

$100 - 5^2$

[100] [−] [5] [x²] [=]

Display: 75.

$(100 - 5)^2$

[(] [100] [−] [5] [)] [x²]

Display: 9025.

Sharp EL-531XT

$100 - 5^2$

[100] [−] [5] [x²] [=]

Display: 75.

$(100 - 5)^2$

[(] [100] [−] [5] [)] [x²] [=]

Display: 9′025.

Answer **75, 9,025, 45 and 225.**

A square root sign works like a bracket. The bar over the top covers everything under it, and all of that is worked out before the root is taken.

Example 3.6 A root over a sum

Find √(9 + 16) and √9 + √16.

SOLUTION

1. √(9 + 16): the root covers the sum. 9 + 16 = 25, then √25 = 5.
2. √9 + √16: two separate roots. √9 = 3 and √16 = 4, then 3 + 4 = 7.
3. On the Sharp, key √ and then the sum in brackets. On the TI-30Xa, key the sum in brackets, then √x after the closing bracket.

CALCULATOR √(9 + 16)

TI-30Xa

(9 + 16) √x

Display 5.

Sharp EL-531XT

√ (9 + 16) =

Display 5.

Answer **√(9 + 16) = 5 but √9 + √16 = 7. They are not the same.**

A **perfect square** is a whole number that is the square of a whole number. The table below lists the squares of 1 to 20. Read it from left to right to square a number: $13^2 = 169$. Read it from right to left to take a square root: 324 sits next to 18, so √324 = 18. A number that is not in a square column, such as 200, has a square root that is not a whole number. √200 is between 14 and 15, because 200 is between 196 and 225.

Number	Square	Number	Square	Number	Square	Number	Square
1	**1**	6	**36**	11	**121**	16	**256**
2	**4**	7	**49**	12	**144**	17	**289**
3	**9**	8	**64**	13	**169**	18	**324**
4	**16**	9	**81**	14	**196**	19	**361**
5	**25**	10	**100**	15	**225**	20	**400**

TRY IT 3.4

1. 4^2
2. 2×3^2
3. $(2 \times 3)^2$
4. $10 + 2^3$

Answers are in the Answer Key.

3.5 Left to right

Division and multiplication share one step of BEDMAS. Addition and subtraction share the last step. When a line has two operations from the same step, work them in the order you meet them, from left to right.

MULTIPLICATION DOES NOT ALWAYS COME FIRST

The M comes after the D in BEDMAS, but that does not mean divide before multiply. They share a step. 24 ÷ 4 × 3 = 18, not 24 ÷ 12 = 2. In the same way, 50 − 20 + 5 = 35, not 50 − 25 = 25.

A bar is 6,000 mm long. You cut off 1,250 mm, then 875 mm, and then weld a 400 mm piece back on. Ignoring saw cuts, its length is 6,000 − 1,250 − 875 + 400. Left to right, that is 4,750, then 3,875, then 4,275 mm.

Example 3.7 Left to right

Find 48 ÷ 6 ÷ 2, 24 ÷ 4 × 3 and 20 − 8 − 2.

SOLUTION

1. 48 ÷ 6 ÷ 2: left to right, 48 ÷ 6 = 8, then 8 ÷ 2 = 4. Working 6 ÷ 2 first would give 48 ÷ 3 = 16, which is wrong.
2. 24 ÷ 4 × 3: ÷ comes first, 24 ÷ 4 = 6, then 6 × 3 = 18.
3. 20 − 8 − 2: 20 − 8 = 12, then 12 − 2 = 10. Working 8 − 2 first would give 14, which is wrong.

CALCULATOR

TI-30Xa

48 ÷ 6 ÷ 2

48 ÷ 6 ÷ 2 =

Display 4.

24 ÷ 4 × 3

24 ÷ 4 × 3 =

Display 18.

Sharp EL-531XT

48 ÷ 6 ÷ 2

48 ÷ 6 ÷ 2 =

Display 4.

24 ÷ 4 × 3

24 ÷ 4 × 3 =

Display 18.

Answer **4, 18 and 10.**

TRY IT 3.5

1. 48 ÷ 4 × 2
2. 90 ÷ 3 ÷ 5
3. 40 − 16 ÷ 4 × 2
4. 30 − 10 + 5

Answers are in the Answer Key.

3.6 The fraction bar groups

A fraction bar means divide. It also groups: everything above the bar is worked out as one number, everything below the bar is worked out as one number, and then the top is divided by the bottom. It acts like a pair of brackets around the top and another pair around the bottom.

On a calculator there is no fraction bar for a calculation like this, so you must key the brackets yourself: (top) ÷ (bottom).

Example 3.8 Brackets for a fraction bar

Find $\frac{18 + 6}{15 - 8}$ to two decimal places.

SOLUTION

1. Top: 18 + 6 = 24. Bottom: 15 − 8 = 7.
2. Divide: 24 ÷ 7 = 3.428571429..., which rounds to 3.43.
3. Keyed without brackets, 18 + 6 ÷ 15 − 8 = gives 10.4, because the calculator divides 6 ÷ 15 first. That answer is wrong.

CALCULATOR

TI-30Xa

With brackets

(18 + 6) ÷ (15 − 8) =

Display 3.428571429

Without brackets (wrong)

18 + 6 ÷ 15 − 8 =

Display 10.4

Sharp EL-531XT

With brackets

(18 + 6) ÷ (15 − 8) =

Display 3.428571429

Without brackets (wrong)

18 + 6 ÷ 15 − 8 =

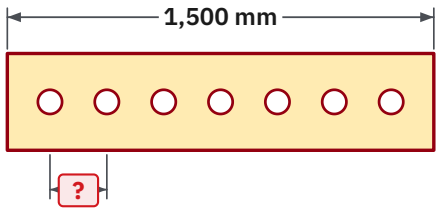
Display 10.4

This is the same calculation as an example in the Sharp manual, keyed with brackets.

Answer **3.43**

Example 3.9 Hole spacing

Seven holes are evenly spaced along a 1,500 mm plate, with the end holes 150 mm from each end. Seven holes make 6 spaces. The spacing is $\frac{1{,}500 - 2 \times 150}{6}$. Find it.



SOLUTION

1. Top first, using BEDMAS inside it: 2 × 150 = 300, then 1,500 − 300 = 1,200.
2. Divide by the bottom: 1,200 ÷ 6 = 200.
3. On either calculator, key (1500 − 2 × 150) ÷ 6 =.

Answer **200 mm centre to centre.**

TRY IT 3.6

1. $\frac{45 + 27}{14 - 6}$
2. $\frac{312 + 305 + 298 + 309}{4}$
3. $\frac{100}{2 + 3}$
4. $\frac{2{,}400 - 2 \times 200}{5}$

Answers are in the Answer Key.

3.7 Your calculator and BEDMAS

Both calculators follow BEDMAS. Key a calculation from left to right, as it is written, and the calculator holds back an addition until any multiplication after it is done. That is why 3 + 4 × 5 = gives 23 on both. The TI-30Xa guidebook and the Sharp manual both list this order, with anything in brackets worked out first.

The calculator only knows what you key. It cannot see a fraction bar, a root bar over a sum, or a square that belongs to a whole bracket. Before you key a formula from a drawing or a manual, find every grouping on paper, then key each one as brackets. On the Sharp, read the top line of the display before you press =.

Written	Key it as	Why
$\frac{18 + 6}{15 - 8}$	(18 + 6) ÷ (15 − 8) =	A fraction bar groups the top and the bottom
√(9 + 16)	TI-30Xa: (9 + 16) √x Sharp: √ (9 + 16) =	The root bar covers the whole sum
$(100 - 5)^2$	TI-30Xa: (100 − 5) x^2 Sharp: (100 − 5) x^2 =	The square belongs to the bracket
3(4 + 5)	3 × (4 + 5) =	Key the × that is not written

Table 3.2 When the calculator needs brackets.

3.8 Positive and negative numbers

An **integer** is a whole number, zero, or a whole number below zero. Numbers above zero are positive. Numbers below zero are negative and carry a minus sign in front: −5 is read "negative five". A positive number can be written with a plus sign, +5, or with no sign at all.

Negative numbers show up wherever a job measures from a zero point: temperatures below 0 °C, elevations below a benchmark, and cuts that come out under the target length. Section 2.5 shows how to key a negative number with the change-sign key on both calculators.

A **number line** puts the integers in order. Zero is in the middle, positive numbers go to the right and negative numbers go to the left. A number is greater than every number to its left. So 4 is greater than −6, and −3 is greater than −8, because −3 is closer to zero. On a thermometer, −8 °C is colder than −3 °C.

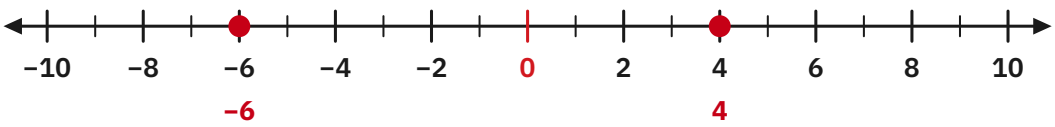


Figure 3.1 A number line from −10 to 10. −6 is to the left of 4, so −6 is less than 4.

ADDING AND SUBTRACTING WITH SIGNS

Adding a positive number moves right on the number line. Adding a negative number moves left, so it is the same as subtracting: 6 + (−9) = 6 − 9 = −3. Subtracting a negative number moves right, so it is the same as adding: 5 − (−3) = 5 + 3 = 8.

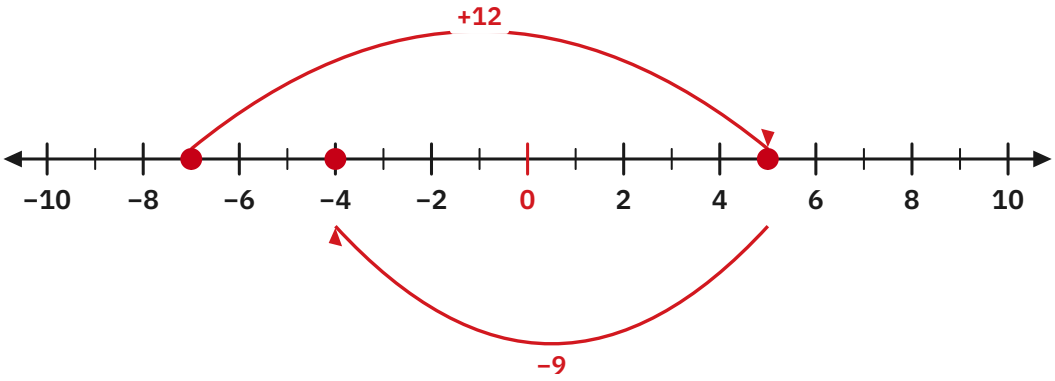


Figure 3.2 Start at −7 and move 12 to the right to reach 5. Then move 9 to the left to reach −4.

Example 3.10 Temperature on a job site

At 6:00 the temperature on a job site is −7 °C. By noon it has risen 12 degrees. By evening it has fallen 9 degrees from the noon reading. Find the noon and evening temperatures.

SOLUTION

1. Noon: −7 + 12. Start at −7 and move 12 to the right. The answer is 5 °C.
2. Evening: 5 − 9. Start at 5 and move 9 to the left. The answer is −4 °C.
3. Key the −7 with the change-sign key, not the minus key.

CALCULATOR

TI-30Xa

Noon

7 +⇄− + 12 =

Display 5.

Evening

7 +⇄− + 12 − 9 =

Display -4.

Sharp EL-531XT

Noon

+/− 7 + 12 =

Display 5.

Evening

+/− 7 + 12 − 9 =

Display -4.

Answer 5 °C at noon and −4 °C in the evening.

On a building section, heights are often given as elevations from a benchmark, a level that is called 0. Levels above it are positive and levels below it are negative. The distance between two levels is always the higher elevation minus the lower one.

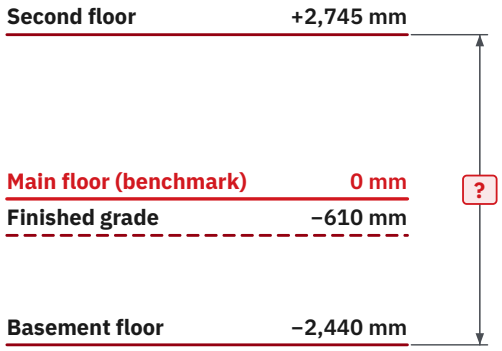


Figure 3.3 Elevations on a house section, measured from the main floor at 0 mm.

Example 3.11 Floor and grade elevations

In the drawing, the second floor is at +2,745 mm, finished grade is at −610 mm and the basement floor is at −2,440 mm. Find the height from the basement floor to the second floor, and the height from the basement floor up to finished grade.

SOLUTION

1. Higher minus lower: 2,745 − (−2,440). Subtracting a negative is the same as adding, so this is 2,745 + 2,440 = 5,185 mm.
2. Finished grade minus basement floor: −610 − (−2,440) = −610 + 2,440 = 1,830 mm.
3. Key each negative elevation with the change-sign key.

CALCULATOR

TI-30Xa

Basement floor to second floor

2745 | − | 2440 | +⇄− | =

Display: 5185.

Basement floor to grade

610 | +⇄− | − | 2440 | +⇄− | =

Display: 1830.

Sharp EL-531XT

Basement floor to second floor

2745 | − | +/− | 2440 | =

Display: 5′185.

Basement floor to grade

+/− | 610 | − | +/− | 2440 | =

Display: 1′830.

Answer **5,185 mm from the basement floor to the second floor, and 1,830 mm from the basement floor to finished grade.**

Example 3.12 Over and under the target length

Five blocks are cut to a target length of 368 mm. Each one is measured and written as over (+) or under (−) the target: +2, −3, +1, −4 and +1 mm. Find the total and the average difference from the target.

SOLUTION

1. Total: 2 + (−3) + 1 + (−4) + 1. Adding a negative is subtracting: 2 − 3 + 1 − 4 + 1 = −3 mm.
2. Average: the total divided by the number of blocks. The fraction bar groups the top, so key brackets: (−3) ÷ 5 = −0.6 mm.
3. The blocks run 0.6 mm under the target on average.

CALCULATOR Average difference

TI-30Xa

(2 + 3 +/− + 1 + 4 +/− + 1) ÷ 5 =

Display -0.6

Sharp EL-531XT

(2 + +/− 3 + 1 + +/− 4 + 1) ÷ 5 =

Display -0.6

Answer **A total of −3 mm, an average of −0.6 mm (0.6 mm under).**

Multiplying and dividing follow two short sign rules. The table shows each case, worked on the calculator.

Signs	Answer	Example
positive × positive	positive	4 × 3 = 12
negative × negative	positive	(−4) × (−3) = 12
positive × negative	negative	4 × (−3) = −12
negative × positive	negative	(−4) × 3 = −12
negative ÷ negative	positive	(−12) ÷ (−3) = 4
negative ÷ positive	negative	(−12) ÷ 3 = −4

Table 3.3 Sign rules for multiplying and dividing.

SIGN RULES

Same signs give a positive answer. Different signs give a negative answer. This holds for × and for ÷.

Example 3.13 A cooler and a cold front

A walk-in cooler is at 3 °C when its door is propped open in winter. The air inside drops 4 degrees each hour for 5 hours. Find the temperature after 5 hours. On another day, the outside temperature falls from 4 °C to −14 °C over 6 hours. Find the average change per hour.

SOLUTION

1. Five drops of 4 degrees are 5 × (−4) = −20. BEDMAS multiplies before it adds: 3 + 5 × (−4) = 3 + (−20) = −17 °C.
2. Change = new − old = −14 − 4 = −18 degrees. Per hour: (−18) ÷ 6 = −3. Different signs give a negative answer.
3. The fraction bar of the average needs brackets on the calculator: (−14 − 4) ÷ 6.

CALCULATOR

TI-30Xa

3 + 5 × (−4)

3 + 5 × 4 +⇄− =

Display -17.

(−14 − 4) ÷ 6

(14 +⇄− − 4) ÷ 6 =

Display -3.

Sharp EL-531XT

3 + 5 × (−4)

3 + 5 × +/− 4 =

Display -17.

(−14 − 4) ÷ 6

(+/− 14 − 4) ÷ 6 =

Display -3.

Answer **−17 °C after 5 hours, and an average change of −3 degrees per hour (3 degrees colder each hour).**

A SQUARE OF A NEGATIVE NUMBER

An exponent belongs only to what it touches. $(-3)^2 = (-3) \times (-3) = 9$, because the bracket puts the minus sign under the square. In -3^2 the square belongs to the 3 only, so $-3^2 = -(3 \times 3) = -9$. On the calculator, key $(-3)^2$ with brackets: TI-30Xa 3 +⇄− x^2, Sharp (+/− 3) x^2 =.

TRY IT 3.8

1. Which is colder, −12 °C or −9 °C?
2. −9 + 14
3. 6 − 11
4. −4 − 7
5. 3 − (−8)
6. (−6) × 7
7. (−48) ÷ (−6)
8. $(-5)^2$
9. A floor drain is at −35 mm from a benchmark and the floor around it is at −10 mm. How much higher is the floor than the drain?
10. A rod should be 1,200 mm. Four rods measure +1, −2, −2 and +3 mm from the target. Find the total difference.

Answers are in the Answer Key.

3.9 Framing and stairs: BEDMAS on the job

Carpenters write a few calculations again and again. Two of them, the stud count for a wall and the total rise and total run of a stair, are good BEDMAS practice.

Studs are set a fixed distance apart, measured from the centre of one stud to the centre of the next. This distance is the spacing on centre, written o.c. When a wall is laid out from one end, the number of spaces is the wall length divided by the spacing, and there is one more stud than there are spaces.

STUDS IN A WALL

studs = length ÷ spacing + 1

Divide first, then add 1. If length ÷ spacing is not a whole number, round it up before you add 1, so no space is wider than the spacing. This counts the regular studs only. Studs at corners and beside openings are counted on their own.

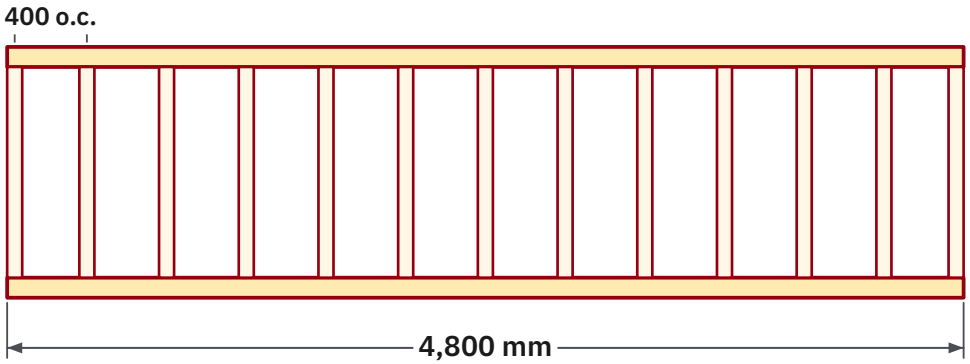


Figure 3.4 A 4,800 mm wall with studs at 400 mm on centre: 12 spaces and 13 studs.

The spacing comes from the plans, and it may not be more than the maximum in Table 9.23.10.1 of the BC Building Code, which Sentence 9.23.10.1.(1) points to (BCBC 2024, Division B, page 9-182). For an exterior wall of 38 × 89 studs that carries a roof, with or without attic storage, the table lists a maximum stud

spacing of 600 mm. For the same studs carrying a roof plus one floor, it lists 400 mm.

Example 3.14 Counting studs

A wall is 4,800 mm long, with studs at 400 mm o.c. How many studs does it need? Then find the studs for a 12 ft wall at 16 in o.c.

SOLUTION

1. BEDMAS divides before it adds: 4,800 ÷ 400 + 1 = 12 + 1 = 13 studs. No brackets are needed.
2. Brackets in the wrong place change the job: 4,800 ÷ (400 + 1) = 11.97, which is not a stud count.
3. Feet to inches first: 12 × 12 = 144 in. Then 144 ÷ 16 + 1 = 9 + 1 = 10 studs. × and ÷ are worked left to right, then the + comes last.

CALCULATOR

TI-30Xa

4,800 ÷ 400 + 1

4800 ÷ 400 + 1 =

Display 13.

Wrong brackets

4800 ÷ (400 + 1) =

Display 11.97007481

12 × 12 ÷ 16 + 1

12 × 12 ÷ 16 + 1 =

Display 10.

Sharp EL-531XT

4,800 ÷ 400 + 1

4800 ÷ 400 + 1 =

Display 13.

Wrong brackets

4800 ÷ (400 + 1) =

Display 11.97007481

12 × 12 ÷ 16 + 1

12 × 12 ÷ 16 + 1 =

Display 10.

Answer **13 studs for the 4,800 mm wall and 10 studs for the 12 ft wall.**

A stair has risers, the vertical part of each step, and treads, the part you step on. The unit rise is the height of one step, nosing to nosing. The unit run is the depth of one step, nosing to nosing. In the stairs in this book the upper floor is the top step, so a flight has one tread fewer than it has risers.

STAIR TOTALS

$$\text{total rise} = \text{number of risers} \times \text{unit rise}$$

$$\text{total run} = (\text{number of risers} - 1) \times \text{unit run}$$

The brackets matter: the subtraction is worked out before the multiplication.

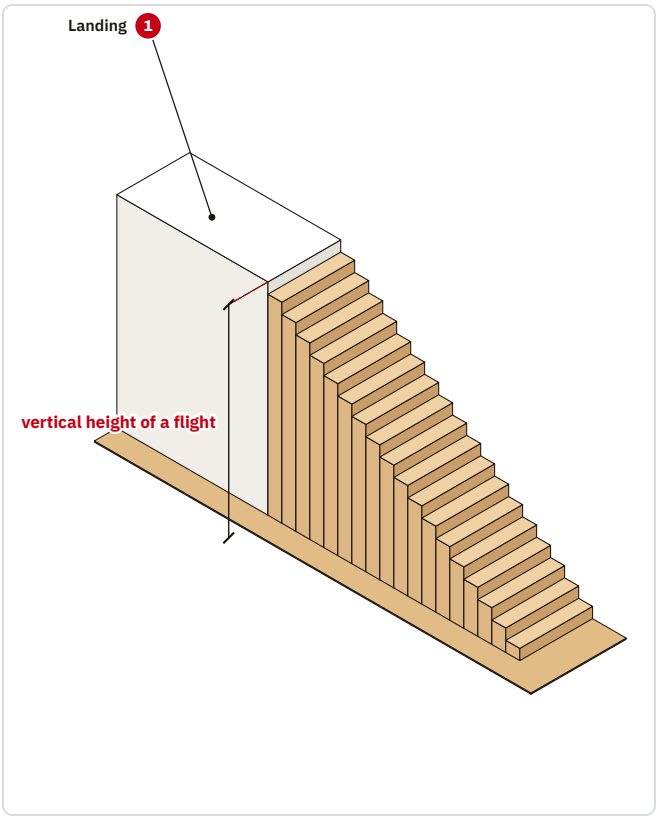


Figure 3.5 A flight of 19 risers of 190 mm. Its vertical height is 19 × 190 = 3,610 mm.
Source: BCBC 2024, Division B, Article 9.8.3.3. (page 9-30)

The BC Building Code limits the height of one flight. Article 9.8.3.3, Sentence (1), says: "The vertical height of any flight of stairs shall not exceed 3.7 m." (BCBC 2024, Division B, page 9-30). The flight in the figure is 3,610 mm, which is under 3.7 m (3,700 mm). For the unit rise of a private stair, Table 9.8.4.1 in Sentence 9.8.4.1.(1) gives a maximum of 200 mm and a minimum of 125 mm (page 9-30). For the unit run of rectangular treads in a private stair, Table 9.8.4.2 in Sentence 9.8.4.2.(1) gives a maximum of 355 mm and a minimum of 255 mm (page 9-31).

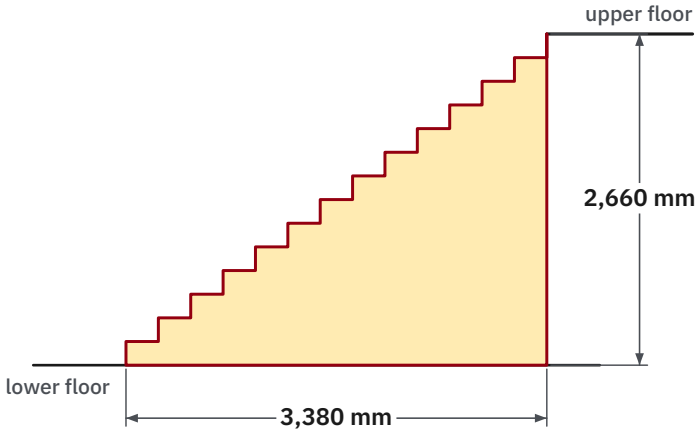


Figure 3.6 A straight flight in side view: 14 risers of 190 mm and 13 treads with a unit run of 260 mm.

Example 3.15 Total rise and total run

A private stair has 14 risers with a unit rise of 190 mm and treads with a unit run of 260 mm. Find the total rise and the total run. Is the total rise under the 3.7 m limit for one flight?

SOLUTION

1. Total rise: 14 × 190 = 2,660 mm. That is less than 3,700 mm, so one flight is within Article 9.8.3.3.
2. Total run: brackets first, 14 − 1 = 13 treads. Then 13 × 260 = 3,380 mm.
3. Without the brackets, 14 − 1 × 260 multiplies first and gives 14 − 260 = −246, which is not a length.

CALCULATOR

TI-30Xa

Total rise

14 × 190 =

Display 2660.

Total run

(14 − 1) × 260 =

Display 3380.

No brackets (wrong)

14 − 1 × 260 =

Display -246.

Sharp EL-531XT

Total rise

14 × 190 =

Display 2′660.

Total run

(14 − 1) × 260 =

Display 3′380.

No brackets (wrong)

14 − 1 × 260 =

Display -246.

Answer **Total rise 2,660 mm, which is under 3.7 m. Total run 3,380 mm.**

TRY IT 3.9

1. Studs for a 3,600 mm wall at 400 mm o.c.
2. Studs for a 6,000 mm wall at 600 mm o.c.
3. Studs for a 16 ft wall at 16 in o.c.
4. Studs for a 4,100 mm wall at 400 mm o.c.
5. The total rise of 16 risers of 175 mm
6. The total run of a flight with 16 risers and a unit run of 255 mm
7. A total rise of 2,850 mm is split into 15 equal risers. Find the unit rise. Is it within the private stair limits of Table 9.8.4.1?
8. One flight has 20 risers of 190 mm. Find its height. Is it within Article 9.8.3.3?

Answers are in the Answer Key.

WORKSHEET 3A

Order on the canvas

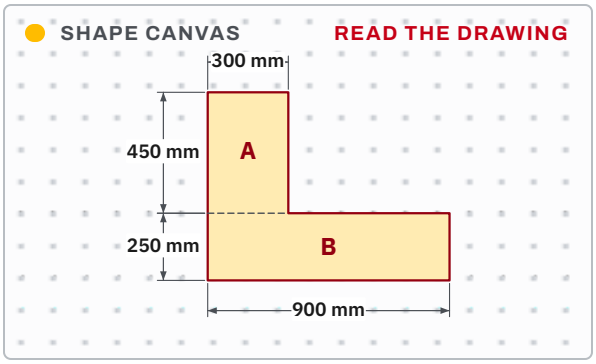
Write the calculation, then work it out with BEDMAS

Name _____

Date _____

Each drawing comes with a calculation. Work it out step by step on paper, rewriting the line after each step, then check it on your calculator.

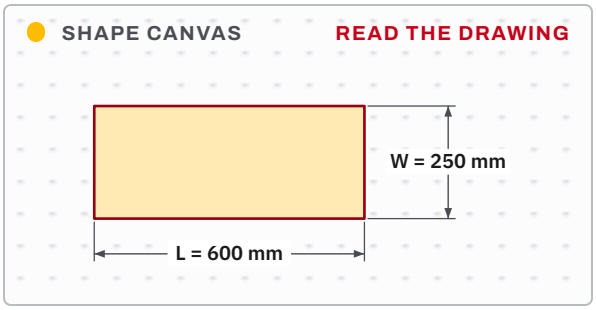
1 The plate is split into A and B. Its area in mm² is 300 × 450 + 900 × 250. Work it out.



WORKING

Answer _____

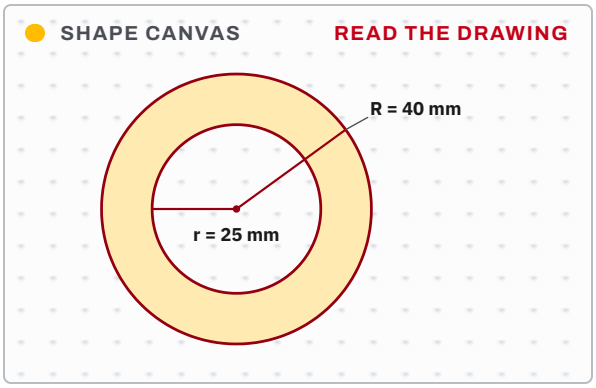
2 The distance around the frame is 2 × (L + W). Find it. If L doubles, what is the new distance around?



WORKING

Answer _____

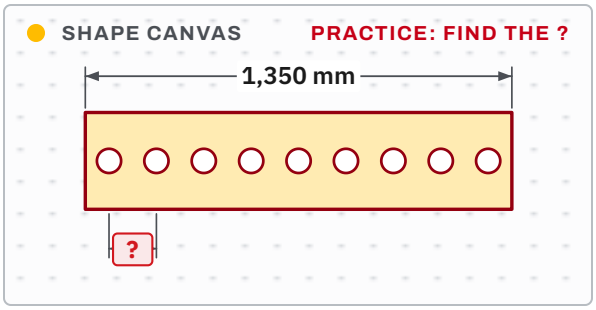
3 The ring has R = 40 mm and r = 25 mm. Its area is π × (R² − r²). Find it to two decimal places.



WORKING

Answer _____

4 Nine holes are evenly spaced on a 1,350 mm bar, with the end holes 75 mm from the ends. The spacing is (1,350 − 2 × 75) ÷ 8. Find it.



WORKING

Answer _____

WORKSHEET 3B

Brackets or not?

Work out, add brackets, and key it right

Name _____

Date _____

Use BEDMAS for each problem. Where you add brackets, check the new line gives the answer asked for.

1 Find $6 + 4 \times 3 - 2$.

WORKING

Answer _____

2 Find $9 + 36 \div 4 - 2$.

WORKING

Answer _____

3 Find 2×3^2 and $(2 \times 3)^2$. Why are they different?

WORKING

Answer _____

4 Find $40 - 16 \div 4 \times 2$.

WORKING

Answer _____

5 Add one pair of brackets to make this true: $12 + 8 \div 4 = 5$.

WORKING

Answer _____

6 Add one pair of brackets to make this true: $3 \times 5 - 2 + 4 = 13$.

WORKING

Answer _____

7 Find $\sqrt{(36 + 64)}$ and $\sqrt{36} + \sqrt{64}$.

WORKING

Answer _____

8 Write the keys for $\frac{45 + 27}{14 - 6}$ and find the answer.

WORKING

Answer _____

9 Three 1,500 mm pieces are cut from a 6,000 mm bar with three cuts. The kerf is 3 mm. The offcut is $6{,}000 - (3 \times 1{,}500 + 3 \times 3)$. Find it.

WORKING

Answer _____

10 In 5×3^2, change the 3 to a 4. What is the new answer? By how much did it grow?

WORKING

Answer _____

11 A crew splits 120 m and 60 m of cable equally among 3 jobs. A learner keys $120 + 60 \div 3 =$ and gets 140. What went wrong? Write the right keys and the right answer.

WORKING

Answer _____

CHAPTER

4

Fractions

Many inch sizes are fractions. A tape reads $2\frac{5}{16}$ in, a drawing calls for $\frac{3}{8}$ in plate, and a cut list adds up lengths like $36\frac{3}{8}$ in. To read, add and divide these sizes you need to work with fractions.

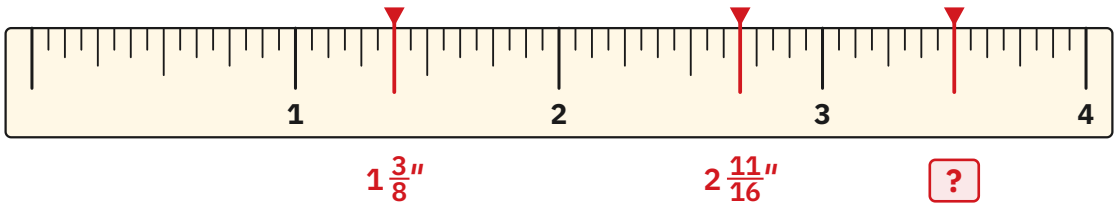
This chapter starts with what a fraction means and how to read an inch tape and a steel rule. It then covers equivalent fractions, lowest terms, mixed numbers and the four operations. Each method is shown by hand and then with the fraction key on the TI-30Xa and the Sharp EL-531XT.

In this chapter you will

- Read an inch tape to the nearest 1/16 in and a rule to 1/32 and 1/64 in
- Write equivalent fractions and put fractions in lowest terms
- Change between improper fractions and mixed numbers
- Compare fractions using a common denominator
- Add, subtract, multiply and divide fractions and mixed numbers
- Use the fraction key on both calculators

Key words

- fraction
- numerator and denominator
- proper and improper fraction
- mixed number
- equivalent fractions
- lowest terms
- common denominator
- reciprocal



4.1 What a fraction is

A **fraction** names part of a whole. The whole is split into equal parts, and the fraction tells you how many of those parts you have.

In $\frac{3}{8}$ the bottom number is the **denominator**. It tells how many equal parts the whole is split into: 8. The top number is the **numerator**. It tells how many of those parts you have: 3.

The line between the two numbers is the fraction bar. It means divide: $\frac{3}{8}$ is the same as $3 \div 8$. Chapter 5 uses this to change fractions into decimals.

A fraction whose numerator is smaller than its denominator is a **proper fraction**. It is worth less than one whole. $\frac{5}{8}$, $\frac{3}{4}$ and $\frac{15}{16}$ are proper fractions.

Example 4.1 Parts of a bar

A flat bar is marked into 8 equal lengths. Three lengths are cut off. What fraction of the bar is cut off, and what fraction is left?



SOLUTION

1. The bar is split into 8 equal parts, so the denominator is 8.
2. 3 parts are cut off, so $\frac{3}{8}$ of the bar is cut off.
3. $8 - 3 = 5$ parts are left, so $\frac{5}{8}$ of the bar is left.

Answer **$\frac{3}{8}$ is cut off and $\frac{5}{8}$ is left. Together they make $\frac{8}{8}$, the whole bar.**



Figure 4.1 One part of each bar is shaded. The bigger the denominator, the smaller each part: $\frac{1}{16}$ in is smaller than $\frac{1}{8}$ in.

TRY IT 4.1

1. A bar is split into 4 equal parts and 3 are shaded. What fraction is shaded?
2. In $\frac{7}{16}$, which number is the numerator?
3. In $\frac{7}{16}$, how many equal parts is the whole split into?
4. A box holds 12 couplings and 5 are used. What fraction is used? What fraction is left?
5. Which is smaller, $\frac{1}{4}$ in or $\frac{1}{8}$ in?
6. Is $\frac{9}{8}$ a proper fraction? Explain.

Answers are in the Answer Key.

4.2 Reading an inch tape and rule

Inch tapes and rules split each inch into equal parts. On the rule below, the longest lines mark whole inches. The next longest line marks the half inch. Shorter lines mark quarters, then eighths, and the shortest lines mark sixteenths. This book reads tapes to the nearest $\frac{1}{16}$ in, and steel rules to $\frac{1}{32}$ in or $\frac{1}{64}$ in.

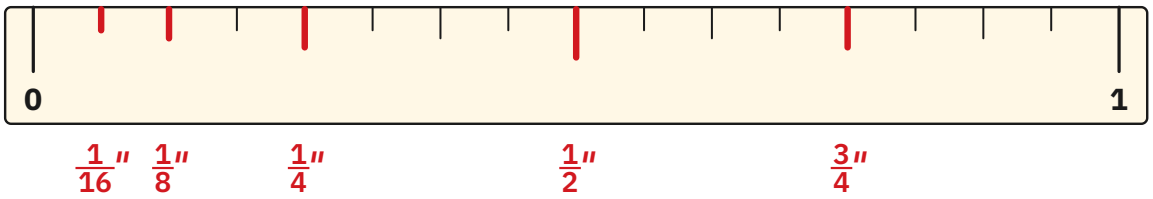


Figure 4.2 One inch split into sixteenths. $\frac{1}{2} = \frac{8}{16}$, $\frac{1}{4} = \frac{4}{16}$ and $\frac{1}{8} = \frac{2}{16}$. The length of each line shows which fraction it marks.

1. Find the last whole inch line before the mark. That gives the whole number.
2. Count the sixteenth spaces from that inch line to the mark.
3. Write the count over 16. That is the fraction part.
4. Put the fraction in lowest terms (Section 4.3 shows how).

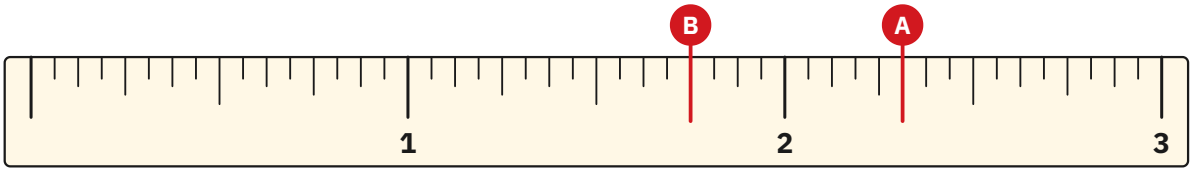


Figure 4.3 A rule marked in sixteenths. Marks A and B are read in Example 4.2.

Example 4.2 Reading marks A and B

Read marks A and B on the rule above.

SOLUTION

1. A: the last whole inch line before A is the 2. Count the spaces from the 2 line to A: 5 sixteenths. 5 is odd, so $\frac{5}{16}$ is in lowest terms. A reads $2\frac{5}{16}$ in.
2. B: the last whole inch line before B is the 1. Count 12 sixteenths past it. Divide the top and bottom of $\frac{12}{16}$ by 4 to get $\frac{3}{4}$. B reads $1\frac{3}{4}$ in.
3. Check: B sits on a quarter line, the line for $\frac{3}{4}$.

Answer **A is $2\frac{5}{16}$ in and B is $1\frac{3}{4}$ in.**

COUNT SPACES, NOT LINES

Start counting at the first space after the whole inch line. If you count the inch line itself as 1, every reading comes out $\frac{1}{16}$ in too long.

A rule marked in 32nds splits each inch into 32 equal parts, and a rule marked in 64ths splits each inch into 64. The lines are close together, so read them in two steps. First find the last eighth line before the mark, then count the small spaces past it.

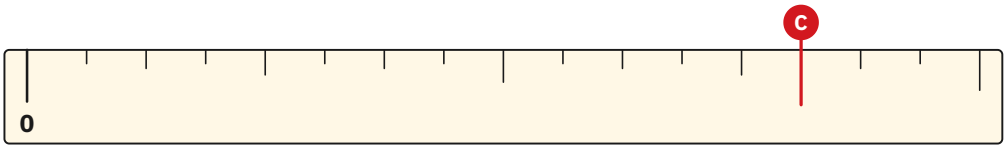


Figure 4.4 The first half inch of a rule marked in 32nds.

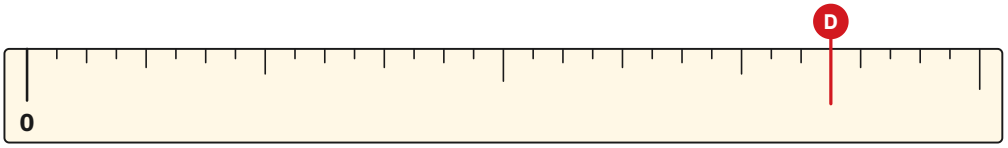


Figure 4.5 The first half inch of a rule marked in 64ths.

Example 4.3 Reading 32nds and 64ths

Read mark C on the rule marked in 32nds and mark D on the rule marked in 64ths.

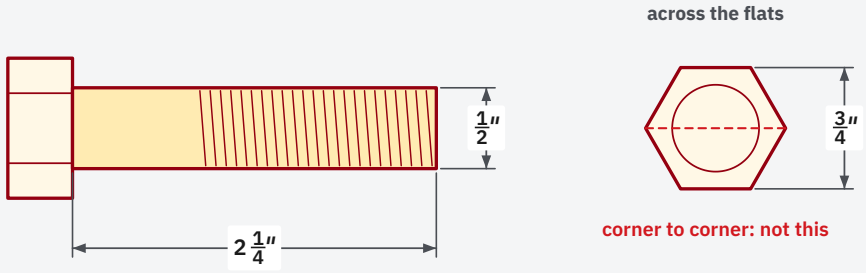
SOLUTION

1. C: the last eighth line before C is the $\frac{3}{8}$ line, and $\frac{3}{8} = \frac{12}{32}$. C is 1 small space past it: 12 + 1 = 13. C reads $\frac{13}{32}$ in.
2. D: the last eighth line before D is also the $\frac{3}{8}$ line, and $\frac{3}{8} = \frac{24}{64}$. D is 3 small spaces past it: 24 + 3 = 27. D reads $\frac{27}{64}$ in.
3. 13 and 27 are odd, so both readings are in lowest terms.

Answer **C is $\frac{13}{32}$ in and D is $\frac{27}{64}$ in.**

MEASURING HARDWARE

Bolt, drill and wrench sizes in inches are written as fractions, and they are read the same way as a tape. A drill bit marked $\frac{3}{8}$ cuts a hole $\frac{3}{8}$ in across. A bolt is sized by its diameter, measured across the threads, and its length is measured from under the head to the end. A wrench or socket fits the head across the flats, from one flat face to the opposite flat face, not corner to corner. The bolt in the drawing has these sizes as an example: diameter $\frac{1}{2}$ in, length $2\frac{1}{4}$ in, head $\frac{3}{4}$ in across the flats.



In a set marked in sixteenths, sizes step by $\frac{1}{16}$ in. To find the size after $\frac{1}{2}$ in, write it as $\frac{8}{16}$ and add one sixteenth: $\frac{9}{16}$ in. To compare a $\frac{5}{8}$ socket with a $\frac{9}{16}$ socket, write $\frac{5}{8}$ as $\frac{10}{16}$: the $\frac{5}{8}$ is bigger by $\frac{1}{16}$ in.

Read each mark on the rule below for Try it 4.2. Give each reading to the nearest $\frac{1}{16}$ in, in lowest terms.

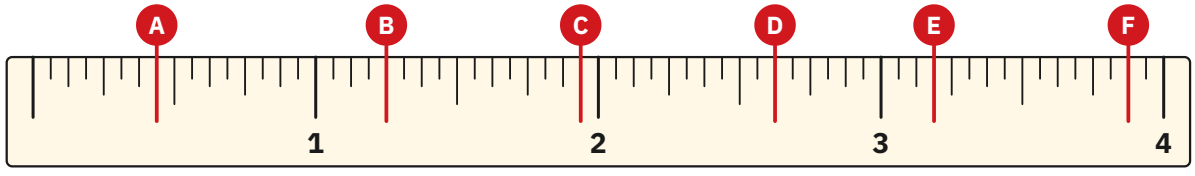


Figure 4.6 Marks A to F for Try it 4.2.

TRY IT 4.2

1. Mark A
2. Mark B
3. Mark C
4. Mark D
5. Mark E
6. Mark F

Answers are in the Answer Key.

4.3 Equivalent fractions and lowest terms

Equivalent fractions are written differently but have the same value. In the figure below, $\frac{1}{2}$, $\frac{2}{4}$, $\frac{4}{8}$ and $\frac{8}{16}$ of the bars are shaded. The four shaded parts are the same length.



Figure 4.7 $\frac{1}{2} = \frac{2}{4} = \frac{4}{8} = \frac{8}{16}$. The dashed line is at the half.

MAKING AN EQUIVALENT FRACTION

Multiply or divide the numerator and the denominator by the same number. The value of the fraction does not change. To write $\frac{3}{4}$ in sixteenths, multiply the top and bottom by 4: $\frac{3}{4} = \frac{12}{16}$.

A fraction is in **lowest terms** when no whole number bigger than 1 divides evenly into both the numerator and the denominator. To put a fraction in lowest terms, divide the top and the bottom by a number that goes into both, and repeat until no number bigger than 1 goes into both. This book writes tape readings in lowest terms: $1\frac{3}{4}$ in, not $1\frac{12}{16}$ in.

Example 4.4 Reducing a reading

Write $\frac{12}{16}$ and $\frac{24}{64}$ in lowest terms.

SOLUTION

1. $\frac{12}{16}$: both numbers are even. Divide both by 2 to get $\frac{6}{8}$, then by 2 again to get $\frac{3}{4}$. 3 is odd, so stop.
2. $\frac{24}{64}$: divide by 8 in one step. $24 \div 8 = 3$ and $64 \div 8 = 8$, so $\frac{24}{64} = \frac{3}{8}$.

CALCULATOR

TI-30Xa

[12] [a b/c] [16] [=]

Display: 3⌟4

Sharp EL-531XT

[12] [a b/c] [16] [=]

Display: 3⌜4

When you press =, the display shows the fraction in lowest terms. The TI-30Xa shows 3⌟4 and the Sharp shows 3⌜4. Read either one as $\frac{3}{4}$.

Answer $\frac{12}{16} = \frac{3}{4}$ **and** $\frac{24}{64} = \frac{3}{8}$

A QUICK TEST FOR TAPE FRACTIONS

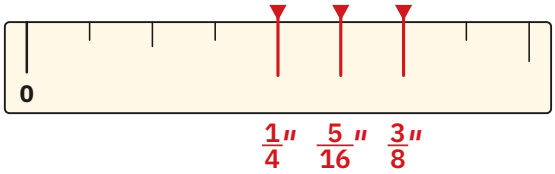
Tape and rule fractions have 2, 4, 8, 16, 32 or 64 as the denominator. If the numerator is odd, the fraction is already in lowest terms. If the numerator is even, divide the top and the bottom by 2 and test again.

Comparing fractions

When two fractions have the same denominator, the one with the bigger numerator is bigger: $\frac{9}{16}$ is bigger than $\frac{7}{16}$. When the denominators are different, rewrite the fractions with the same denominator first. A denominator that the fractions share is a **common denominator**. For tape fractions, 16 works for halves, quarters, eighths and sixteenths. Use 32 or 64 when a fraction has 32 or 64 on the bottom.

Example 4.5 Putting sizes in order

Three bolts have diameters of $\frac{3}{8}$ in, $\frac{5}{16}$ in and $\frac{1}{4}$ in. List them from smallest to largest.



SOLUTION

1. Change each to sixteenths: $\frac{3}{8} = \frac{6}{16}$, $\frac{5}{16}$ stays as it is, and $\frac{1}{4} = \frac{4}{16}$.
2. Put the numerators in order: 4, 5, 6.
3. The rule shows the same order from left to right.

Answer $\frac{1}{4}, \frac{5}{16}, \frac{3}{8}$

For fractions that are not tape fractions, use the smallest number that both denominators divide into evenly. For $\frac{2}{3}$ and $\frac{3}{4}$ that number is 12: $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$, so $\frac{3}{4}$ is bigger.

TRY IT 4.3

1. Write $\frac{10}{16}$ in lowest terms.
2. Write $\frac{14}{16}$ in lowest terms.
3. Write $\frac{20}{32}$ in lowest terms.
4. Write $\frac{48}{64}$ in lowest terms.
5. Write $\frac{5}{8}$ in sixteenths.
6. Write $\frac{3}{4}$ in 32nds.
7. Which is bigger, $\frac{3}{8}$ or $\frac{7}{16}$?
8. Which is bigger, $\frac{5}{32}$ or $\frac{1}{8}$?
9. Put in order from smallest: $\frac{3}{4}, \frac{5}{8}, \frac{11}{16}$.
10. Which is bigger, $\frac{2}{3}$ or $\frac{5}{8}$? Use 24 as the common denominator.

Answers are in the Answer Key.

4.4 Improper fractions and mixed numbers

An **improper fraction** has a numerator equal to or bigger than its denominator, so it is worth one whole or more. A **mixed number** is a whole number and a proper fraction written together. Tape readings such as $2\frac{3}{4}$ in are mixed numbers. Improper fractions make multiplying and dividing easier.

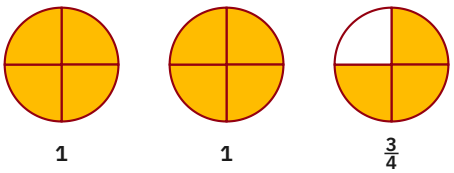


Figure 4.8 Eleven quarters make 2 whole circles and 3 quarters more: $\frac{11}{4} = 2\frac{3}{4}$.

Example 4.6 Improper to mixed

Change $\frac{43}{16}$ to a mixed number.

SOLUTION

1. Divide the numerator by the denominator: 43 ÷ 16 = 2 with a remainder. 2 × 16 = 32 and 43 − 32 = 11.
2. The whole number is 2. The remainder 11 goes over the same denominator: $\frac{11}{16}$.

Answer $\mathbf{\frac{43}{16} = 2\frac{11}{16}}$

Example 4.7 Mixed to improper

Change $3\frac{5}{8}$ to an improper fraction.

SOLUTION

1. Multiply the whole number by the denominator: 3 whole inches hold 3 × 8 = 24 eighths.
2. Add the numerator: 24 + 5 = 29. Write it over the same denominator.

Answer $\mathbf{3\frac{5}{8} = \frac{29}{8}}$

TRY IT 4.4

1. Change $\frac{7}{4}$ to a mixed number.
2. Change $\frac{29}{16}$ to a mixed number.
3. Change $\frac{21}{8}$ to a mixed number.
4. Change $2\frac{3}{8}$ to an improper fraction.
5. Change $5\frac{1}{2}$ to an improper fraction.
6. Change $1\frac{15}{16}$ to an improper fraction.

Answers are in the Answer Key.

4.5 The fraction key

The TI-30Xa and the Sharp EL-531XT each have a fraction key, marked a b/c. To key in a mixed number, press the whole number, the fraction key, the numerator, the fraction key again and then the denominator. For a fraction with no whole number, start with the numerator.

CALCULATOR Keying in $3\frac{5}{8}$

TI-30Xa

Key in $3\frac{5}{8}$ and press =

[3] [a b/c] [5] [a b/c] [8] [=]

Display: 3⌟5⌟8

Change it to an improper fraction

[3] [a b/c] [5] [a b/c] [8] [=] [2nd] [d/c]

Display: 29⌟8

Change it to a decimal

[3] [a b/c] [5] [a b/c] [8] [=] [2nd] [F⇄D]

Display: 3.625

Sharp EL-531XT

Key in $3\frac{5}{8}$ and press =

[3] [a b/c] [5] [a b/c] [8] [=]

Display: 3 ⌜5 ⌜8

Change it to an improper fraction

[3] [a b/c] [5] [a b/c] [8] [=] [2ndF] [d/c]

Display: 29 ⌜8

Change it to a decimal

[3] [a b/c] [5] [a b/c] [8] [=] [a b/c]

Display: 3.625

The TI-30Xa marks the parts of a fraction with ⌟ and the Sharp uses ⌜. Read 3⌟5⌟8 and 3 ⌜5 ⌜8 as $3\frac{5}{8}$, and 29⌟8 as $\frac{29}{8}$.

Job	TI-30Xa	Sharp EL-531XT
Key in a fraction	a b/c between the parts	a b/c between the parts
Change a result between a fraction and a decimal	2nd, then F⇄D	a b/c after the =
Change between a mixed number and an improper fraction	2nd, then d/c	2ndF, then d/c
Change a decimal such as 0.375 to a fraction	0.375, 2nd, F⇄D	0.375, =, a b/c

Table 4.1 Fraction keys on the two calculators. Both show 0.375 as $\frac{3}{8}$.

CALCULATOR A decimal as a fraction

TI-30Xa

[0.375] [2nd] [F⇄D]

Display: 3⌟8

Sharp EL-531XT

[0.375] [=] [a b/c]

Display: 3 ⌜8

Both displays read $\frac{3}{8}$.

A DECIMAL IN A FRACTION CALCULATION

On the Sharp, if any number in a fraction calculation is keyed as a decimal, the answer comes out as a decimal: its manual shows 1.25 + 2/5 = 1.65. Key every number as a fraction when you want a fraction answer. To switch a fraction answer to a decimal and back, press 2nd F⇄D on the TI-30Xa or a b/c on the Sharp.

TRY IT 4.5

1. Write the keys to enter $2\frac{7}{16}$.
2. The TI-30Xa shows 1⌟3⌟8. What number is that?
3. The Sharp shows 15 ⌐16. What number is that?
4. Use the fraction key to put $\frac{28}{32}$ in lowest terms.
5. Use d/c to change $4\frac{3}{4}$ to an improper fraction.
6. Use the calculator to change 0.625 to a fraction.

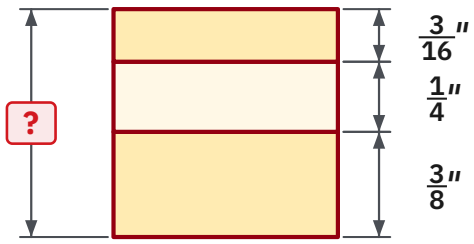
Answers are in the Answer Key.

4.6 Adding and subtracting fractions

To add or subtract fractions with the same denominator, add or subtract the numerators and keep the denominator: $\frac{3}{16} + \frac{5}{16} = \frac{8}{16} = \frac{1}{2}$. When the denominators are different, change to a common denominator first: $\frac{7}{8} - \frac{5}{16} = \frac{14}{16} - \frac{5}{16} = \frac{9}{16}$. Then put the answer in lowest terms.

Example 4.8 Stacking plates

Three plates are stacked. They are $\frac{3}{8}$ in, $\frac{1}{4}$ in and $\frac{3}{16}$ in thick. How thick is the stack?



SOLUTION

1. Change to sixteenths: $\frac{6}{16}$, $\frac{4}{16}$ and $\frac{3}{16}$.
2. Add the numerators: 6 + 4 + 3 = 13.
3. The total is $\frac{13}{16}$. 13 is odd, so it is in lowest terms.

CALCULATOR

TI-30Xa

3 | a b/c | 8 | + | 1 | a b/c | 4 | + | 3 | a b/c | 16 | =

Display: 13⌟16

Sharp EL-531XT

3 | a b/c | 8 | + | 1 | a b/c | 4 | + | 3 | a b/c | 16 | =

Display: 13 ⌐16

The TI-30Xa shows 13⌟16 and the Sharp shows 13 ⌐16. Both mean $\frac{13}{16}$.

Answer $\frac{13}{16}$ **in**

To add mixed numbers, add the whole numbers and add the fractions. If the fractions come to one whole or more, change that part to a mixed number and carry the whole number.

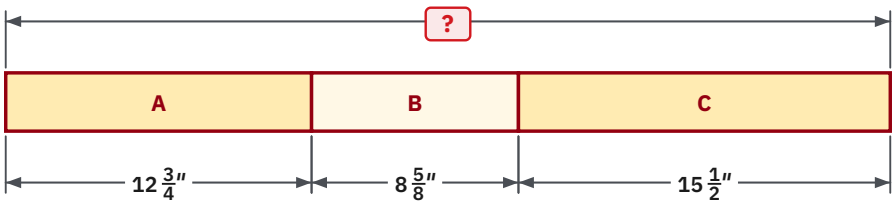


Figure 4.9 Three pieces on a cut list. The total length is hidden.

Example 4.9 Total of a cut list

Find the total length of the three pieces: $12\frac{3}{4}$ in, $8\frac{5}{8}$ in, $15\frac{1}{2}$ in.

SOLUTION

1. Add the whole numbers: 12 + 8 + 15 = 35.
2. Change the fractions to eighths and add them: $\frac{6}{8} + \frac{5}{8} + \frac{4}{8} = \frac{15}{8}$.
3. $\frac{15}{8} = 1\frac{7}{8}$. Carry the 1: 35 + 1 = 36. The total is $36\frac{7}{8}$ in.

CALCULATOR

TI-30Xa

12 | a b/c | 3 | a b/c | 4 | + | 8 | a b/c | 5 | a b/c | 8 | + | 15 | a b/c | 1 | a b/c | 2 | =

Display: 36⌟7⌟8

Sharp EL-531XT

12 | a b/c | 3 | a b/c | 4 | + | 8 | a b/c | 5 | a b/c | 8 | + | 15 | a b/c | 1 | a b/c | 2 | =

Display: 36 ⌐7 ⌐8

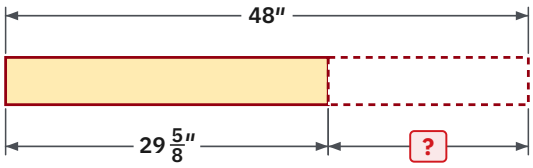
The TI-30Xa shows 36⌟7⌟8 and the Sharp shows 36 ⌐7 ⌐8. Both mean $36\frac{7}{8}$.

Answer $36\frac{7}{8}$ in

To subtract mixed numbers, subtract the fractions first and then the whole numbers. When the fraction on top is smaller than the fraction you take away, borrow 1 from the whole number. Write the borrowed 1 as a fraction with the same denominator, such as $\frac{8}{8}$ or $\frac{16}{16}$, and add it to the fraction. For example, $5\frac{1}{4} - 2\frac{7}{16} = 5\frac{4}{16} - 2\frac{7}{16} = 4\frac{20}{16} - 2\frac{7}{16} = 2\frac{13}{16}$.

Example 4.10 What is left of the bar?

A piece $29\frac{5}{8}$ in long is cut from a 48 in bar. Ignore the saw cut. How long is the offcut?



SOLUTION

1. There is no fraction in 48 to take $\frac{5}{8}$ from, so borrow 1: $48 = 47\ \frac{8}{8}$.
2. Fractions: $\frac{8}{8} - \frac{5}{8} = \frac{3}{8}$.
3. Whole numbers: $47 - 29 = 18$.

CALCULATOR

TI-30Xa

48 | − | 29 | a b/c | 5 | a b/c | 8 | =

Display: 18⌟3⌟8

Sharp EL-531XT

48 | − | 29 | a b/c | 5 | a b/c | 8 | =

Display: 18 ⌜3 ⌜8

The TI-30Xa shows 18⌟3⌟8 and the Sharp shows 18 ⌜3 ⌜8. Both mean $18\frac{3}{8}$.

Answer $\mathbf{18\frac{3}{8}}$ **in**

CHECK BY ADDING

Add your answer to the number you took away. You should get the number you started with: $18\frac{3}{8} + 29\frac{5}{8} = 48$.

TRY IT 4.6

1. $\frac{3}{8} + \frac{7}{16}$
2. $\frac{5}{8} + \frac{3}{4}$
3. $10\frac{7}{8} + 6\frac{3}{4}$
4. Pieces of $18\frac{3}{8}$ in, $24\frac{1}{2}$ in and $9\frac{15}{16}$ in. What is the total length?
5. $\frac{15}{16} - \frac{3}{8}$
6. $6 - 2\frac{3}{8}$
7. $12\frac{3}{16} - 5\frac{5}{8}$
8. A piece $22\frac{11}{16}$ in long is cut from a 36 in bar. Ignore the saw cut. How long is the offcut?

Answers are in the Answer Key.

4.7 Multiplying and dividing fractions

To multiply fractions, multiply the numerators, multiply the denominators and put the answer in lowest terms: $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12} = \frac{1}{2}$. A whole number can be written over 1: $6 = \frac{6}{1}$. Change mixed numbers to improper fractions before you multiply.

In word problems, the word "of" means multiply. $\frac{3}{4}$ of a 24 in bar is $\frac{3}{4} \times \frac{24}{1} = \frac{72}{4} = 18$ in.

SAW CUTS IN FRACTIONS

When a problem gives the **kerf** as a fraction of an inch, work with it like any other fraction. With a kerf of $\frac{1}{8}$ in, 4 cuts remove $4 \times \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$ in of stock. The problems in this book give the kerf whenever it counts.

Example 4.11 Pieces and saw cuts

Four pieces, each $11\frac{3}{4}$ in long, are cut from a 48 in bar. You make 4 cuts and the kerf is $\frac{1}{8}$ in. How long is the offcut?

SOLUTION

1. Pieces: $4 \times 11\frac{3}{4} = 4 \times \frac{47}{4} = \frac{188}{4} = 47$ in.
2. Saw cuts: $4 \times \frac{1}{8} = \frac{1}{2}$ in.
3. Offcut: $48 - 47 - \frac{1}{2} = \frac{1}{2}$ in.

CALCULATOR

TI-30Xa

48 | − | 4 | × | 11 | a b/c | 3 | a b/c | 4 | − | 4 | × | 1 | a b/c | 8 | =

Display: 1⌟2

Sharp EL-531XT

48 | − | 4 | × | 11 | a b/c | 3 | a b/c | 4 | − | 4 | × | 1 | a b/c | 8 | =

Display: 1⌜2

Answer $\frac{1}{2}$ **in**

The **reciprocal** of a fraction is the fraction turned upside down. The reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$, and the reciprocal of 5 (or $\frac{5}{1}$) is $\frac{1}{5}$. A number times its reciprocal is 1: $\frac{3}{4} \times \frac{4}{3} = \frac{12}{12} = 1$.

DIVIDING BY A FRACTION

To divide by a fraction, multiply by its reciprocal. Change mixed numbers to improper fractions first. Then turn the number you are dividing by upside down and multiply.

For example, a $30\frac{1}{2}$ in bar cut into 4 equal pieces, ignoring the saw cuts, gives $30\frac{1}{2} \div 4 = \frac{61}{2} \times \frac{1}{4} = \frac{61}{8}$ $= 7\frac{5}{8}$ in for each piece.

Example 4.12 How many pieces?

How many spacers $1\frac{3}{4}$ in long can be cut from a 24 in bar? Ignore the saw cuts. How long is the offcut?

SOLUTION

1. $24 \div 1\frac{3}{4} = \frac{24}{1} \times \frac{4}{7} = \frac{96}{7} = 13\frac{5}{7}$.
2. Only whole spacers count: 13 spacers.
3. 13 spacers use $13 \times 1\frac{3}{4} = 22\frac{3}{4}$ in. The offcut is $24 - 22\frac{3}{4} = 1\frac{1}{4}$ in.

CALCULATOR

TI-30Xa

Spacers

24 ÷ 1 a b/c 3 a b/c 4 =

Display 13⌟5⌟7

Offcut

24 − 13 × 1 a b/c 3 a b/c 4 =

Display 1⌟1⌟4

Sharp EL-531XT

Spacers

24 ÷ 1 a b/c 3 a b/c 4 =

Display 13 ⌜5 ⌜7

Offcut

24 − 13 × 1 a b/c 3 a b/c 4 =

Display 1 ⌜1 ⌜4

The first display reads $13\frac{5}{7}$. Only the whole number of spacers counts.

Answer **13 spacers, with an offcut of $1\frac{1}{4}$ in.**

LENGTHS THAT ARE NOT ON THE TAPE

Some answers have denominators that are not on a tape, such as thirds. To mark them, multiply the fraction part by 16 and round to the nearest whole number of sixteenths. A 25 in bar cut into 3 equal pieces gives $\frac{25}{3} = 8\frac{1}{3}$ in. $\frac{1}{3} \times 16 = 5\frac{1}{3}$, which rounds to 5, so mark each piece at $8\frac{5}{16}$ in. Chapter 5 does the same job with decimal inches.

TRY IT 4.7

1. $\frac{3}{8} \times 4$
2. $\frac{2}{3}$ of 18
3. $1\frac{1}{2} \times 2\frac{1}{4}$
4. 6 pieces at $2\frac{3}{8}$ in. What is the total length, ignoring saw cuts?
5. What is the reciprocal of $\frac{5}{8}$?
6. $\frac{7}{8} \div 2$
7. $10\frac{1}{2} \div 1\frac{3}{4}$
8. A $29\frac{1}{4}$ in bar is cut into 3 equal pieces. Ignore the saw cuts. How long is each piece?

Answers are in the Answer Key.

4.8 Carpentry: tape, cut lengths and kerf

Carpenters read the tape on long boards, so the whole number of inches is often large. The method from Section 4.2 does not change: find the last whole inch, then count the sixteenths past it. A cut list then adds and subtracts those readings, and every saw cut takes a kerf.

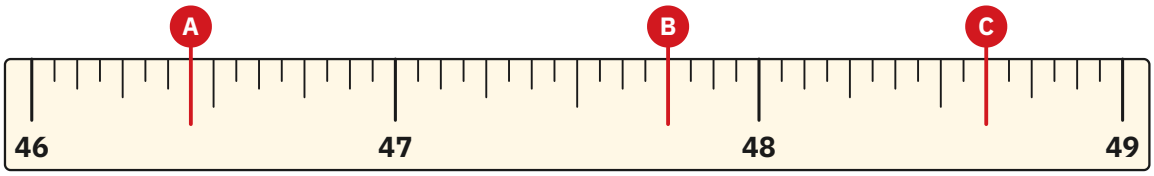


Figure 4.10 Part of a tape on a board, from 46 in to 49 in, marked in sixteenths.

Example 4.13 Reading the tape on a board

Read marks A, B and C in Figure 4.10.

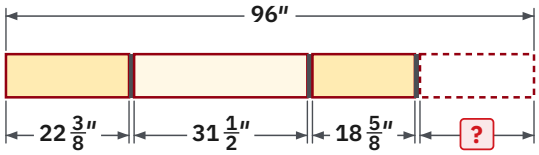
SOLUTION

1. A: past the 46 line by 7 sixteenths. A reads $46\frac{7}{16}$ in.
2. B: past the 47 line by 12 sixteenths, and $\frac{12}{16} = \frac{3}{4}$. B reads $47\frac{3}{4}$ in.
3. C: past the 48 line by 10 sixteenths, and $\frac{10}{16} = \frac{5}{8}$. C reads $48\frac{5}{8}$ in.

Answer **A = $46\frac{7}{16}$ in, B = $47\frac{3}{4}$ in, C = $48\frac{5}{8}$ in**

Example 4.14 A cut list from one board

Three pieces, $22\frac{3}{8}$ in, $31\frac{1}{2}$ in, $18\frac{5}{8}$ in, are cut from a 96 in board. You make 3 cuts and the kerf is $\frac{1}{8}$ in. How long is the offcut?



SOLUTION

1. Add the pieces: $22\frac{3}{8} + 31\frac{1}{2} + 18\frac{5}{8} = 72\frac{1}{2}$ in.
2. Saw cuts: $3 \times \frac{1}{8} = \frac{3}{8}$ in.
3. Offcut: $96 - 72\frac{1}{2} - \frac{3}{8} = 23\frac{1}{8}$ in.

CALCULATOR

TI-30Xa

96 | − | 22 | a b/c | 3 | a b/c | 8 | − | 31
a b/c | 1 | a b/c | 2 | − | 18 | a b/c | 5
a b/c | 8 | − | 3 | × | 1 | a b/c | 8 | =

Display: 23⌟1⌟8

Sharp EL-531XT

96 | − | 22 | a b/c | 3 | a b/c | 8 | − | 31
a b/c | 1 | a b/c | 2 | − | 18 | a b/c | 5
a b/c | 8 | − | 3 | × | 1 | a b/c | 8 | =

Display: 23 ⌜1 ⌜8

The TI-30Xa shows 23⌟1⌟8 and the Sharp shows 23 ⌜1 ⌜8. Both mean $23\frac{1}{8}$.

Answer $23\frac{1}{8}$ in

Example 4.15 How many blocks from a board?

Blocking pieces $14\frac{1}{2}$ in long are cut from a 96 in board. You make one cut for each block and the kerf is $\frac{1}{8}$ in. How many blocks do you get, and how long is the offcut?

SOLUTION

1. Each block uses its length plus one cut: $14\frac{1}{2} + \frac{1}{8} = 14\frac{5}{8}$ in.
2. $96 \div 14\frac{5}{8} = \frac{96}{1} \times \frac{8}{117} = 6\frac{22}{39}$. Only whole blocks count: 6 blocks.
3. $6 \times 14\frac{5}{8} = 87\frac{3}{4}$ in. The offcut is $96 - 87\frac{3}{4} = 8\frac{1}{4}$ in.

CALCULATOR

TI-30Xa

Blocks

96 ÷ 14 a b/c 5 a b/c 8 =

Display 6⌟22⌟39

Offcut

96 − 6 × 14 a b/c 5 a b/c 8 =

Display 8⌟1⌟4

Sharp EL-531XT

Blocks

96 ÷ 14 a b/c 5 a b/c 8 =

Display 6┌22┌39

Offcut

96 − 6 × 14 a b/c 5 a b/c 8 =

Display 8┌1┌4

Answer 6 blocks, with an offcut of $8\frac{1}{4}$ in.

Fractions in the building code

The BC Building Code limits holes and notches in framing as a fraction of the member's depth. Article 9.23.5.1, Sentence (1) says holes drilled in roof, floor or ceiling framing members shall be not larger than one-quarter the depth of the member and shall be located not less than 50 mm from the edges, unless the depth of the member is increased by the size of the hole (page 9-175). The code works in millimetres, so these answers are in millimetres.

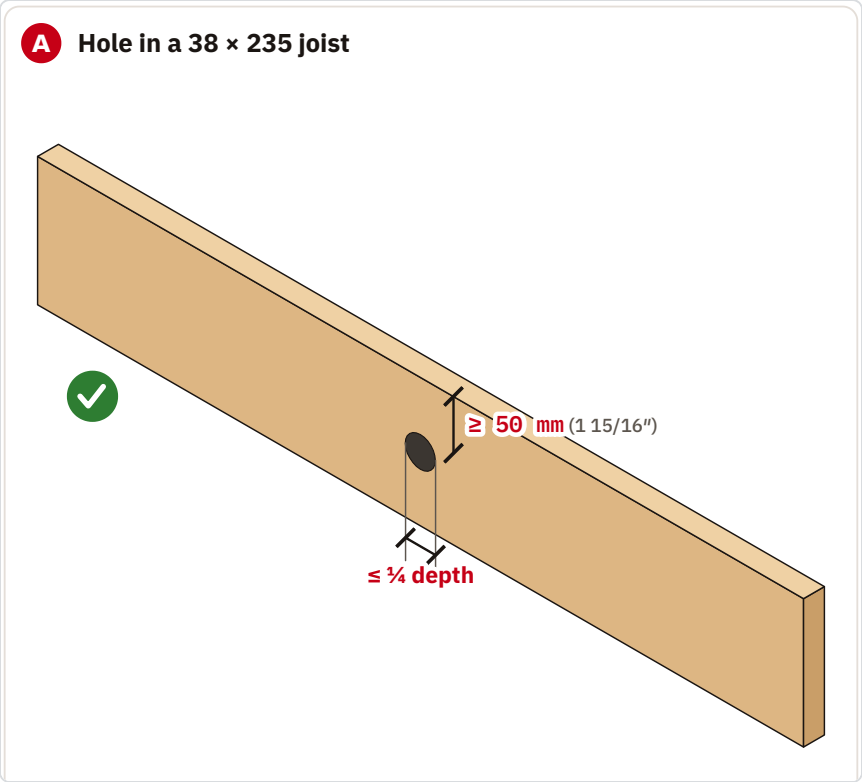


Figure 4.11 A hole in a joist: not more than one-quarter of the depth, and 50 mm or more from the edges.

Source: BCBC 2024, Division B, Article 9.23.5.1. (page 9-175)

Example 4.16 The largest hole in a joist

A floor joist is 38 mm by 235 mm, so its depth is 235 mm. What is the largest hole Article 9.23.5.1 allows in it?

SOLUTION

1. One-quarter of the depth: $\frac{1}{4} \times 235 = \frac{235}{4}$.
2. $\frac{235}{4} = 58\frac{3}{4}$ mm, which is 58.75 mm.

CALCULATOR

TI-30Xa	Sharp EL-531XT
235 ÷ 4 =	235 ÷ 4 =
Display 58.75	Display 58.75

Answer $58\frac{3}{4}$ **mm (58.75 mm), placed 50 mm or more from the top and bottom edges.**

Example 4.17 Notches and studs

Article 9.23.5.2, Sentence (1) allows a notch on the top of a floor, roof or ceiling member within half the joist depth from the edge of bearing, not deeper than one-third the joist depth (page 9-176). Article 9.23.5.3, Sentence (1) says the undamaged part of a loadbearing stud shall not be less than two-thirds the depth of the stud (page 9-176). Find (a) the deepest notch and the farthest it may sit from the bearing edge in a 286 mm deep joist, and (b) the deepest hole or notch in a loadbearing stud 89 mm deep.

SOLUTION

1. (a) Depth: $\frac{1}{3} \times 286 = \frac{286}{3} = 95\frac{1}{3}$ mm. Distance: $\frac{1}{2} \times 286 = 143$ mm.
2. (b) Two-thirds must stay: $\frac{2}{3} \times 89 = \frac{178}{3} = 59\frac{1}{3}$ mm.
3. The rest may be removed: $89 - 59\frac{1}{3} = 29\frac{2}{3}$ mm, which is $\frac{1}{3}$ of 89.

CALCULATOR

TI-30Xa

(a) notch

286 ÷ 3 =

Display 95.33333333

(b) removed

89 − 2 a b/c 3 × 89 =

Display 29⌟2⌟3

Sharp EL-531XT

(a) notch

286 ÷ 3 =

Display 95.33333333

(b) removed

89 − 2 a b/c 3 × 89 =

Display 29┌2┌3

Answer (a) $95\frac{1}{3}$ mm deep, within 143 mm of the bearing edge. (b) $29\frac{2}{3}$ mm.

TRY IT 4.8

1. A 96 in board has pieces of $30\frac{3}{8}$ in and $41\frac{11}{16}$ in cut off with 2 cuts. The kerf is $\frac{1}{8}$ in. How long is the offcut?
2. How many $22\frac{1}{2}$ in pieces come from a 120 in board with a $\frac{1}{8}$ in kerf for each piece?
3. Find the centre of a $92\frac{5}{8}$ in stud.
4. What is the largest hole Article 9.23.5.1 allows in a joist 184 mm deep?
5. Article 9.23.3.2, Sentence (1) says nails shall be long enough that not less than half their required length penetrates into the second member. A nail of required length 76 mm goes how far into the second member, at least?

Answers are in the Answer Key.

WORKSHEET 4A

Read the rule

Read each red pointer, then find the hidden lengths

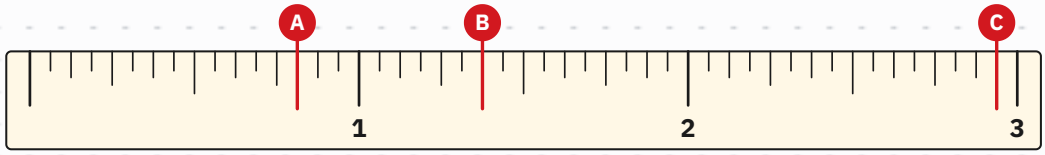
Name _____

Date _____

Give every reading in lowest terms. Ignore saw cuts unless the problem gives a kerf. Check your answers with the fraction key.

1 Read marks A, B and C to the nearest $\frac{1}{16}$ in.

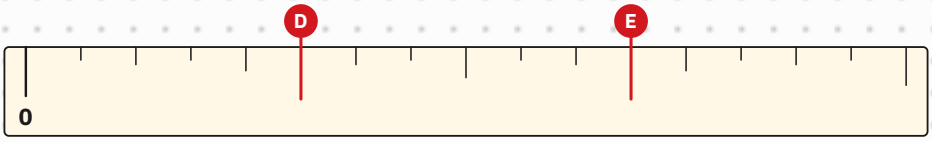
SHAPE CANVAS READ THE DRAWING



WORKING Answer _____

2 This half inch of a rule is marked in 32nds. Read marks D and E.

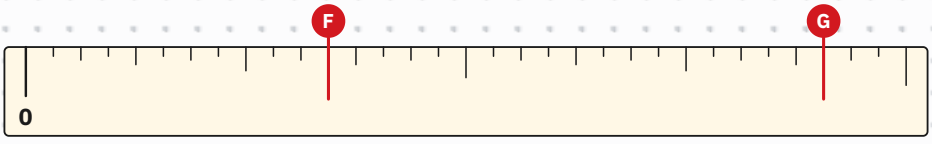
SHAPE CANVAS READ THE DRAWING



WORKING Answer _____

3 This half inch of a rule is marked in 64ths. Read marks F and G.

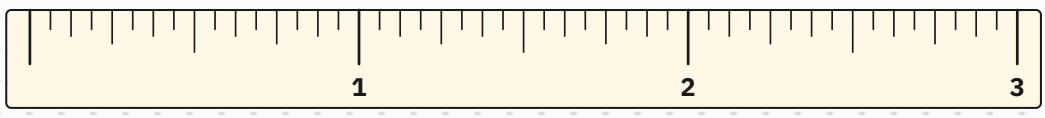
SHAPE CANVAS READ THE DRAWING



WORKING Answer _____

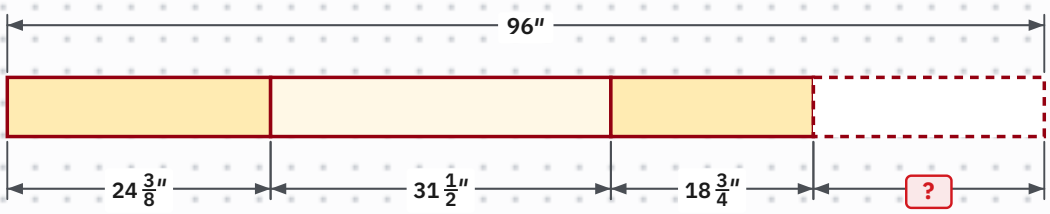
4 Draw a pointer on the rule at $1\frac{3}{8}$ in and another at $2\frac{11}{16}$ in. Label them.

SHAPE CANVAS READ THE DRAWING



5 Three pieces are cut from a 96 in bar. How long is the offcut?

SHAPE CANVAS — PRACTICE: FIND THE ?

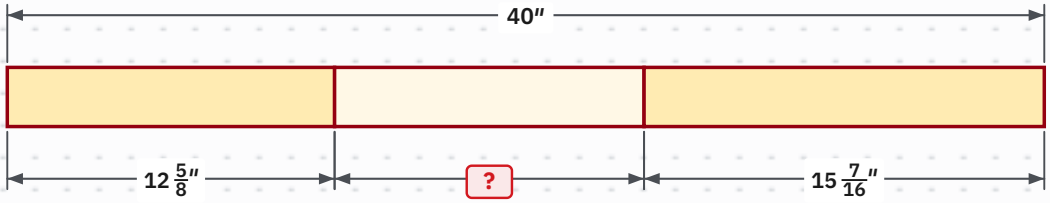


WORKING

Answer _____

6 The overall length is 40 in. Find the middle piece.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

7 A frame needs 2 pieces at $36\frac{3}{8}$ in and 2 pieces at $24\frac{5}{8}$ in, cut from a 144 in bar. You make 4 cuts and the kerf is $\frac{1}{8}$ in. How long is the offcut?

WORKING

Answer _____

8 Change one value: 8 pieces are each $5\frac{3}{8}$ in long. What is their total length? If each piece is made $\frac{1}{16}$ in longer, how much more stock is needed?

WORKING

Answer _____

9 A 72 in bar is cut into 5 equal pieces. Ignore the saw cuts. How long is each piece, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

WORKSHEET 4B

Break it down: parts in inches

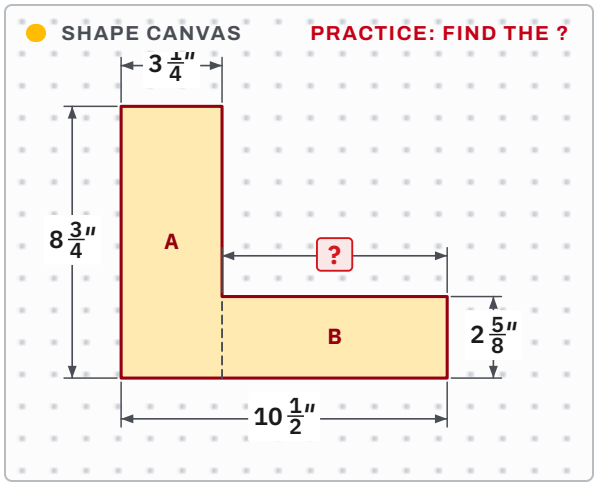
Split each part into pieces you can work with

Name _____

Date _____

Each drawing is to scale. Find every red ?. Show the fractions you add or take away in the working box.

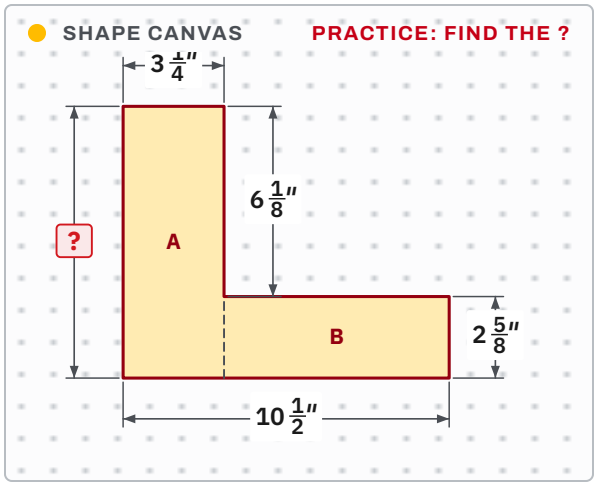
1 Find the hidden width.



WORKING

Answer _____

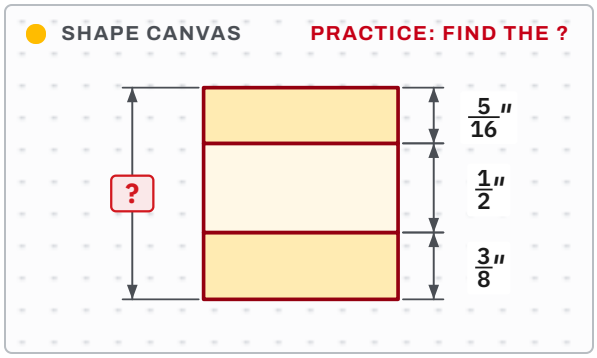
2 Find the overall height.



WORKING

Answer _____

3 Three plates are stacked. How thick is the stack?



WORKING

Answer _____

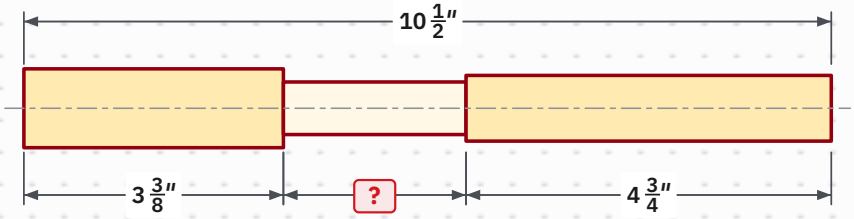
4 You cut $2\frac{1}{4}$ in pieces from a 30 in bar. You make one cut for each piece and the kerf is $\frac{1}{8}$ in. How many pieces do you get, and how long is the offcut?

WORKING

Answer _____

5 The shaft is $10\frac{1}{2}$ in long overall. Find the length of the middle step.

SHAPE CANVAS — PRACTICE: FIND THE ?

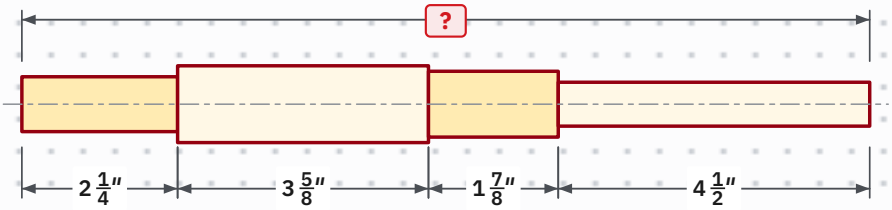


WORKING

Answer _____

6 Find the overall length of the shaft.

SHAPE CANVAS — PRACTICE: FIND THE ?

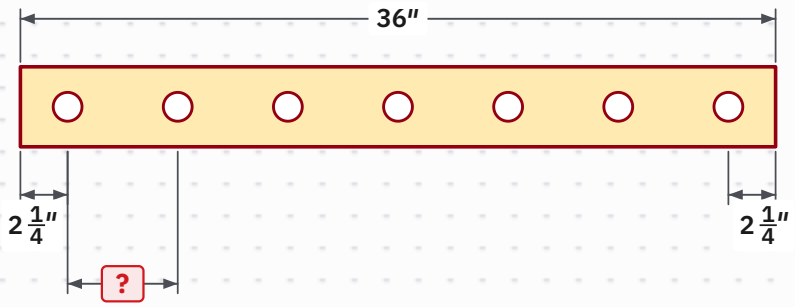


WORKING

Answer _____

7 Seven holes are evenly spaced along a 36 in plate. The end holes are $2\frac{1}{4}$ in from the ends. Find the spacing.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

WORKSHEET 4C

Carpentry: tape, cuts and code fractions

Read the tape on a board, work out cut lists and find code limits

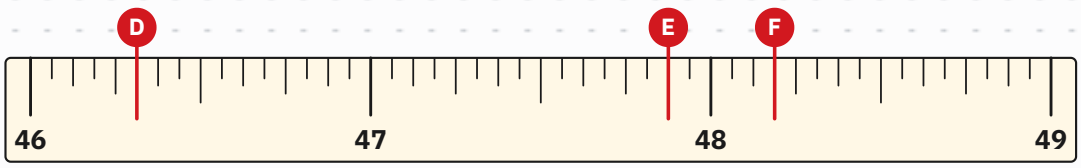
Name _____

Date _____

Give tape readings in lowest terms. The kerf is $\frac{1}{8}$ in for every cut. Code limits come from the Articles in Section 4.8.

1 Read marks D, E and F on the tape.

SHAPE CANVAS — READ THE DRAWING

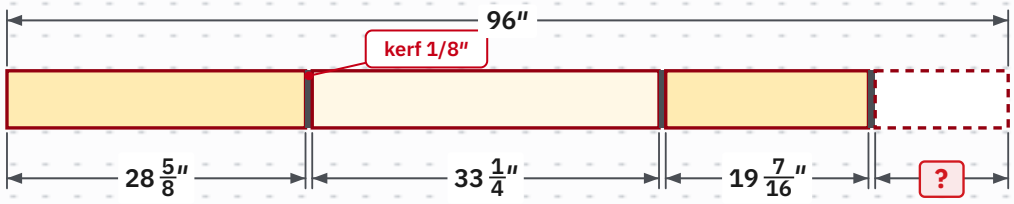


WORKING

Answer _____

2 Three pieces are cut from a 96 in board with 3 cuts. How long is the offcut?

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

3 How many $14\frac{1}{2}$ in blocks come from a 120 in board, with one cut for each block? How long is the offcut?

WORKING

Answer _____

4 Find the centre of a board $78\frac{3}{8}$ in long.

WORKING

Answer _____

5 Article 9.23.5.1: what is the largest hole allowed in a joist 286 mm deep?

WORKING

Answer _____

6 Article 9.23.5.2: how deep may a notch be in a joist 184 mm deep, and how far from the bearing edge may it sit?

WORKING

Answer _____

7 Article 9.23.5.3: a loadbearing stud is 140 mm deep. How much of its depth must stay undamaged, and what is the deepest notch?

WORKING

Answer _____

8 Article 9.23.3.2: a nail has a required length of 82 mm. How far must it go into the second member, at least?

WORKING

Answer _____

CHAPTER

5

Decimals

A metric drawing might give a size of 76.2 mm, flat bar might cost $3.99 a metre, and a calculator answer might come out as 2.83 in. Each of these is a decimal.

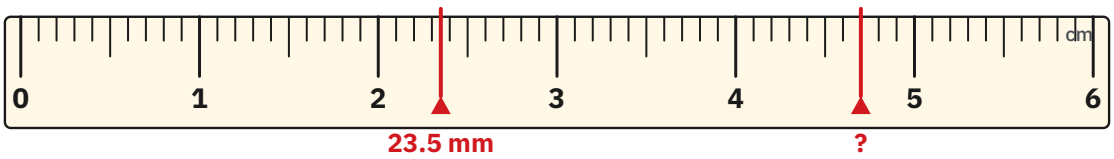
This chapter covers place value after the decimal point, reading a metric rule, rounding, and adding, subtracting, multiplying and dividing decimals. It then links decimals to the fractions in Chapter 4, so you can change a decimal inch to the nearest sixteenth and mark it on a tape.

In this chapter you will

- Read and write decimals using place value
- Read a metric rule to the nearest 0.5 mm
- Compare and round decimals
- Add, subtract, multiply and divide decimals
- Change fractions to decimals and decimals to fractions
- Change a decimal inch to the nearest 1/16 or 1/32 in
- Work with money to the nearest cent

Key words

- decimal
- decimal point
- decimal place
- decimal inch



5.1 Decimals and place value

A **decimal** is a number written with a **decimal point**. The point separates the whole number part from the part that is less than one. Each place after the point is worth one tenth of the place to its left: tenths, then hundredths, then thousandths.

Number	Tens	Ones	.	Tenths	Hundredths	Thousandths
25.4	2	5	.	4		
76.2	7	6	.	2		
3.175		3	.	1	7	5
0.375		0	.	3	7	5

Table 5.1 Place value on both sides of the decimal point.

Each digit after the point fills one **decimal place**. 2.375 has three decimal places. A decimal is a fraction with 10, 100 or 1,000 on the bottom: $0.7 = \frac{7}{10}$, $0.25 = \frac{25}{100}$ and $0.375 = \frac{375}{1{,}000}$. Write a 0 in front of the point when there is no whole number, as in 0.375. It makes the point easy to see.

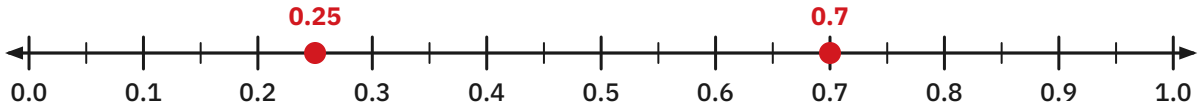


Figure 5.1 Tenths from 0 to 1. 0.25 sits halfway between 0.2 and 0.3, and 0.7 is seven tenths.

Example 5.1 Reading a decimal size

A drawing gives a hole position as 12.07 mm. Read the number in words and say what the 7 is worth.

SOLUTION

1. The whole number part is 12.
2. The 0 is in the tenths place and the 7 is in the hundredths place.
3. Read it as twelve and seven hundredths. The 7 is worth $\frac{7}{100}$, or 0.07.

Answer **Twelve and seven hundredths millimetres. The 7 is worth 0.07 mm.**

Reading a metric rule

On the metric rules in this book, each small line is 1 mm, the longer lines mark every 5 mm, and the numbered lines mark every 10 mm (1 cm). This book reads metric rules to the nearest 0.5 mm. A mark halfway between two millimetre lines reads as .5.

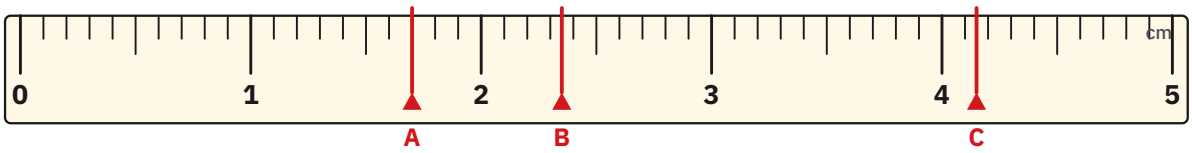


Figure 5.2 A metric rule. Each small space is 1 mm.

Example 5.2 Reading the metric rule

Read marks A, B and C on the rule above.

SOLUTION

1. A: past the 1 cm line (10 mm) by 7 small spaces: 10 + 7 = 17 mm.
2. B: past the 20 mm line by 3 spaces and then half a space: 23.5 mm.
3. C: past the 40 mm line by 1 space and half a space: 41.5 mm.

Answer **A = 17 mm, B = 23.5 mm, C = 41.5 mm**

TRY IT 5.1

1. Write 3.06 in words.
2. What is the 5 worth in 0.375?
3. How many decimal places does 9.525 have?
4. Write $\frac{9}{10}$ as a decimal.
5. Write $\frac{45}{100}$ as a decimal.
6. Write 0.125 as a fraction over 1,000.

Answers are in the Answer Key.

5.2 Comparing and rounding decimals

To compare decimals, line up the decimal points and compare the digits from the left. Add zeros on the right so both numbers have the same number of decimal places. Zeros on the right of a decimal do not change its value: 0.38 = 0.380. To compare a 0.375 in plate with a 0.38 in plate, write 0.375 and 0.380. The tenths are the same, and 8 hundredths is more than 7 hundredths, so the 0.38 in plate is thicker, by 0.005 in.

MORE DIGITS IS NOT BIGGER

0.375 has more digits than 0.38, but it is the smaller number. Compare place by place from the left, not by counting digits.

Decimals round the same way as whole numbers. Find the place you are rounding to, look at the digit to its right, and round up if that digit is 5 or more. Then drop every digit after the rounding place.

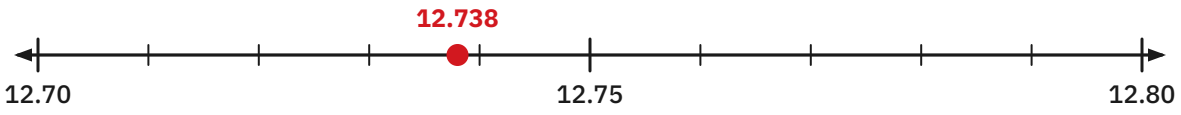


Figure 5.3 12.738 is closer to 12.74 than to 12.73, and closer to 12.7 than to 12.8.

Example 5.3 Rounding a size

Round 12.738 mm to one decimal place and to two decimal places.

SOLUTION

1. One place: the tenths digit is 7. The digit to its right is 3, so the 7 stays: 12.7 mm.
2. Two places: the hundredths digit is 3. The digit to its right is 8, so the 3 goes up to 4: 12.74 mm.

Answer **12.7 mm to one place, 12.74 mm to two places.**

Example 5.4 Rounding to the nearest 0.5 mm

A calculation gives 23.38 mm. Round it to the nearest 0.5 mm so you can mark it with a metric rule.

SOLUTION

1. Double the number: 23.38 × 2 = 46.76.
2. Round to the nearest whole number: 47.
3. Halve it: 47 ÷ 2 = 23.5.

Answer **23.5 mm**

TRY IT 5.2

1. Which is bigger, 0.6 or 0.58?
2. Put in order from smallest: 0.5, 0.125, 0.25.
3. Round 6.35 to one decimal place.
4. Round 9.525 to two decimal places.
5. Round 3.175 mm to the nearest 0.5 mm.
6. Round 18.74 mm to the nearest whole millimetre.

Answers are in the Answer Key.

5.3 Adding and subtracting decimals

To add or subtract decimals, line up the decimal points, one under the other. Fill empty places with zeros, then add or subtract as with whole numbers. The point in the answer goes straight below the points above it.

Example 5.5 Total of a metric cut list

Find the total length of four pieces: 152.4 mm, 76.2 mm, 38.1 mm, 12.7 mm.

SOLUTION

1. Line up the points: each number has one decimal place.
2. Tenths: 4 + 2 + 1 + 7 = 14. Write 4, carry 1.
3. Whole numbers: 152 + 76 + 38 + 12 + 1 (carried) = 279.

CALCULATOR

TI-30Xa

152.4 + 76.2 + 38.1 + 12.7 =

Display 279.4

Sharp EL-531XT

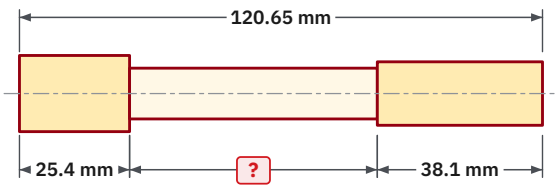
152.4 + 76.2 + 38.1 + 12.7 =

Display 279.4

Answer **279.4 mm**

Example 5.6 Finding a hidden length

A stepped pin is 120.65 mm long overall. Two of its steps are 25.4 mm and 38.1 mm long. How long is the third step?



SOLUTION

1. Add the known steps: 25.4 + 38.1 = 63.5.
2. Subtract from the overall length: write 63.5 as 63.50 and line up the points.
3. 120.65 − 63.50 = 57.15.

CALCULATOR

TI-30Xa

120.65 − 25.4 − 38.1 =

Display 57.15

Sharp EL-531XT

120.65 − 25.4 − 38.1 =

Display 57.15

Answer **57.15 mm**

Decimal inches add the same way. Pieces of 2.375 in, 1.25 in and 0.625 in come to 2.375 + 1.250 + 0.625 = 4.25 in.

TRY IT 5.3

1. 4.5 + 0.75
2. 12.7 + 6.35 + 3.175
3. 10 − 3.25
4. 76.2 − 19.05
5. A 600 mm bar has pieces of 152.4 mm and 228.6 mm cut off. Ignore the saw cuts. How much is left?
6. Shims of 0.5 mm, 0.25 mm and 0.1 mm are stacked. How thick is the stack?

Answers are in the Answer Key.

5.4 Multiplying decimals

To multiply decimals, multiply as if there were no points. Then count the decimal places in the numbers you multiplied and give the answer that many decimal places. For 1.5 × 0.25: 15 × 25 = 375. 1.5 has one place and 0.25 has two, so the answer has three: 0.375.

Multiplying by 10, 100 or 1,000 moves the point 1, 2 or 3 places to the right: 2.375 × 1,000 = 2,375.

Example 5.7 Twelve equal pieces

You need 12 pieces, each 152.4 mm long. What total length do they use? Ignore the saw cuts.

SOLUTION

1. Estimate first: 12 × 150 = 1,800.
2. Multiply without the point: 12 × 1,524 = 18,288.
3. 152.4 has one decimal place, so the answer has one: 1,828.8. It is close to the estimate.

CALCULATOR

TI-30Xa

12 × 152.4 =

Display 1828.8

Sharp EL-531XT

12 × 152.4 =

Display 1’828.8

The TI-30Xa shows 1828.8. The Sharp shows 1’828.8, with a small mark between the thousands and the hundreds.

Answer 1,828.8 mm

Example 5.8 Inches to millimetres

A drawing calls for a $\frac{3}{8}$ in plate. How thick is that in millimetres? Round to the nearest 0.5 mm.

SOLUTION

1. There are exactly 25.4 mm in 1 in. Chapter 7 covers unit conversion.
2. $\frac{3}{8}$ in = 0.375 in. 0.375 × 25.4 = 9.525 mm.
3. To the nearest 0.5 mm: 9.525 × 2 = 19.05, which rounds to 19, and 19 ÷ 2 = 9.5.

CALCULATOR

TI-30Xa

3 ÷ 8 × 25.4 =

Display 9.525

Sharp EL-531XT

3 ÷ 8 × 25.4 =

Display 9.525

Answer **9.525 mm, or 9.5 mm to the nearest 0.5 mm.**

TRY IT 5.4

1. 0.5 × 0.5
2. 3 × 2.375
3. 1.2 × 0.15
4. 4.75 × 100
5. 8 pieces at 76.2 mm. Total length?
6. Change $\frac{5}{8}$ in (0.625 in) to millimetres.

Answers are in the Answer Key.

5.5 Dividing decimals

When you divide by a whole number, divide as usual and keep the point in the same place in the answer. When you divide by a decimal, move the point to the right in both numbers until the number you divide by is a whole number. Dividing by 10, 100 or 1,000 moves the point 1, 2 or 3 places to the left: 254 ÷ 100 = 2.54.

When a division does not come out even, keep going past the last digit: write zeros after the point and bring them down. Section 5.8 shows this by hand, and how to round the answer to hundredths or thousandths.

Example 5.9 Dividing by a decimal

How many 0.25 m lengths can be cut from a 7.5 m bar? Ignore the saw cuts.

SOLUTION

1. 0.25 has two decimal places, so move the point two places to the right in both numbers: 7.5 ÷ 0.25 becomes 750 ÷ 25 = 30.

Answer **30 lengths**

Example 5.10 Pieces with saw cuts

How many 152.4 mm pieces can you cut from a 2,438.4 mm bar? You make one cut for each piece and the kerf is 3.2 mm. How long is the offcut?

SOLUTION

1. Each piece uses its length plus one saw cut: 152.4 + 3.2 = 155.6 mm.
2. 2,438.4 ÷ 155.6 = 15.67. Only whole pieces count: 15 pieces.
3. 15 pieces use 15 × 155.6 = 2,334 mm. The offcut is 2,438.4 − 2,334 = 104.4 mm.

CALCULATOR

TI-30Xa

Pieces

2438.4 ÷ 155.6 =

Display 15.67095116

Offcut

2438.4 − 15 × 155.6 =

Display 104.4

Sharp EL-531XT

Pieces

2438.4 ÷ 155.6 =

Display 15.67095116

Offcut

2438.4 − 15 × 155.6 =

Display 104.4

Answer **15 pieces, with an offcut of 104.4 mm.**

TRY IT 5.5

1. 9.6 ÷ 4
2. 1.5 ÷ 0.3
3. 25.4 ÷ 0.5
4. 635 ÷ 100
5. A 914.4 mm bar is cut into 6 equal pieces. Ignore the saw cuts. How long is each piece?
6. 100 mm split into 3 equal parts. How long is each, to two decimal places?

Answers are in the Answer Key.

5.6 Fractions and decimals

The fraction bar means divide, so to change a fraction to a decimal, divide the numerator by the denominator: $\frac{3}{8} = 3 \div 8 = 0.375$. An inch size written this way is a **decimal inch**: 0.375 in is $\frac{3}{8}$ in. For a mixed number, change only the fraction part and keep the whole number: $2\frac{5}{8}$ in = 2.625 in.

CALCULATOR Fraction to decimal

TI-30Xa

$\frac{3}{8}$ by dividing

[3] [÷] [8] [=]

Display 0.375

$2\frac{5}{8}$ with the fraction key

[2] [a b/c] [5] [a b/c] [8] [2nd] [F⇄D]

Display 2.625

Sharp EL-531XT

$\frac{3}{8}$ by dividing

[3] [÷] [8] [=]

Display 0.375

$2\frac{5}{8}$ with the fraction key

[2] [a b/c] [5] [a b/c] [8] [=] [a b/c]

Display 2.625

Fraction	Decimal inch	Millimetres
$\frac{1}{8}$ in	0.125 in	3.175 mm
$\frac{1}{4}$ in	0.25 in	6.35 mm
$\frac{3}{8}$ in	0.375 in	9.525 mm
$\frac{1}{2}$ in	0.5 in	12.7 mm
$\frac{5}{8}$ in	0.625 in	15.875 mm
$\frac{3}{4}$ in	0.75 in	19.05 mm
$\frac{7}{8}$ in	0.875 in	22.225 mm
1 in	1 in	25.4 mm

Table 5.2 Eighths of an inch as decimal inches and millimetres. All of these are exact.

To change a decimal with three places or fewer to a fraction, write it over 10, 100 or 1,000 and put it in **lowest terms**: $0.375 = \frac{375}{1{,}000} = \frac{3}{8}$ (divide both by 125). The fraction key does this for you (2nd F⇄D on the TI-30Xa, a b/c on the Sharp), as Chapter 4 showed.

Some fractions never end as a decimal. $\frac{2}{3} = 0.666\ldots$ and $\frac{5}{12} = 0.4166\ldots$ go on forever. Section 5.8 divides them by hand and rounds them to three decimal places.

Decimal inches to the nearest sixteenth

A tape is marked in sixteenths, but a calculator gives decimal inches such as 2.83 in. To mark the length, change the decimal part to the nearest $\frac{1}{16}$ in. For finer work on a steel rule, use the nearest $\frac{1}{32}$ in. Always say which one you used.

1. Keep the whole number of inches.
2. Multiply the decimal part by 16 (or by 32).
3. Round to the nearest whole number. That is the number of sixteenths (or 32nds).

4 Write it over 16 (or 32) and put it in lowest terms.

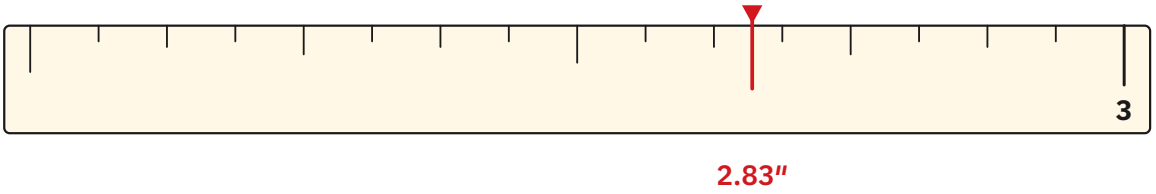


Figure 5.4 The half inch from $2\frac{1}{2}$ in to 3 in, marked in 32nds. 2.83 in lies between the $2\frac{26}{32}$ line and the $2\frac{27}{32}$ line, a little closer to $2\frac{27}{32}$.

Example 5.11 Marking 2.83 in on a tape

A calculation gives 2.83 in. Change it to the nearest $\frac{1}{16}$ in, and to the nearest $\frac{1}{32}$ in.

SOLUTION

1. Keep the 2 inches. The decimal part is 0.83.
2. Nearest $\frac{1}{16}$: 0.83 × 16 = 13.28, which rounds to 13. That gives $\frac{13}{16}$, so $2\frac{13}{16}$ in.
3. Nearest $\frac{1}{32}$: 0.83 × 32 = 26.56, which rounds to 27. That gives $\frac{27}{32}$, so $2\frac{27}{32}$ in.

CALCULATOR

TI-30Xa

Sixteenths

0.83 × 16 =

Display 13.28

32nds

0.83 × 32 =

Display 26.56

Sharp EL-531XT

Sixteenths

0.83 × 16 =

Display 13.28

32nds

0.83 × 32 =

Display 26.56

Answer $2\frac{13}{16}$ **in to the nearest** $\frac{1}{16}$ **in, or** $2\frac{27}{32}$ **in to the nearest** $\frac{1}{32}$ **in.**

Example 5.12 Millimetres to the nearest sixteenth

A metric part is 50 mm long. What is that to the nearest $\frac{1}{16}$ in?

SOLUTION

1. Change to inches by dividing by 25.4, and multiply by 16 in the same line to count sixteenths: 50 ÷ 25.4 × 16 = 31.496 (to three places).
2. Round to the nearest whole number: 31.496 is less than 31.5, so 31 sixteenths.
3. $\frac{31}{16} = 1\frac{15}{16}$.

CALCULATOR

TI-30Xa

50 ÷ 25.4 × 16 =

Display: 31.49606299

Sharp EL-531XT

50 ÷ 25.4 × 16 =

Display: 31.49606299

The display shows 31.49606299. That is just under 31.5, so it rounds down to 31 sixteenths.

Answer $1\frac{15}{16}$ **in to the nearest** $\frac{1}{16}$ **in.**

THE FRACTION KEY DOES NOT ROUND TO THE TAPE

The fraction key gives the exact fraction, not the nearest sixteenth. On the TI-30Xa, 0.83 2nd F⇄D shows 83⌟100. On the Sharp, 0.83 = a b/c shows 83 ┌100. Both mean $\frac{83}{100}$, not a tape reading. Use the multiply-by-16 method instead.

CALCULATOR The fraction key on 0.83

TI-30Xa

0.83 2nd F⇄D

Display: 83⌟100

Sharp EL-531XT

0.83 = a b/c

Display: 83 ┌100

TRY IT 5.6

1. Change $\frac{3}{4}$ to a decimal.
2. Change $1\frac{7}{8}$ in to a decimal inch.
3. Change 0.625 to a fraction in lowest terms.
4. Change 3.4 in to the nearest $\frac{1}{16}$ in.
5. Change 0.59 in to the nearest $\frac{1}{32}$ in.
6. Change 100 mm to the nearest $\frac{1}{16}$ in.

Answers are in the Answer Key.

5.7 Money

Money is written with two decimal places: dollars, then cents. $3.99 is 3 dollars and 99 cents. Round money answers to the nearest cent, which is two decimal places. The calculator does not show zeros at the end of a decimal: 4 × 12.85 shows 51.4, which means $51.40. Always write money with two decimal places.

Flat bar at $3.99 per metre: 7.5 m cost 7.5 × 3.99 = $29.925. The third decimal digit is 5, so round up to the nearest cent: $29.93.

Example 5.13 Adding up an order

An order has 4 boxes of bolts at $12.85 a box and 2 cans of cutting oil at $8.49 a can. What is the total?

SOLUTION

1. Bolts: 4 × 12.85 = 51.40.
2. Oil: 2 × 8.49 = 16.98.
3. Total: 51.40 + 16.98 = 68.38.

CALCULATOR

TI-30Xa

4 × 12.85 + 2 × 8.49 =

Display 68.38

Sharp EL-531XT

4 × 12.85 + 2 × 8.49 =

Display 68.38

The calculator multiplies before it adds, so no brackets are needed. Chapter 3 explains why. The display shows 68.38, which is $68.38.

Answer **$68.38**

TRY IT 5.7

1. Round $14.736 to the nearest cent.
2. 3 m of pipe at $6.25 per metre. Cost?
3. 12 washers at $0.18 each. Cost?
4. 2.4 m of angle at $5.49 per metre. Cost, to the nearest cent?
5. You pay for a $37.65 order with $50. How much change?
6. Five workers split a $68.40 lunch bill equally. How much each?

Answers are in the Answer Key.

5.8 Dividing to thousandths and rounding

Many divisions do not come out even. To carry on, divide as usual and write the point in the answer straight above the point in the number you are dividing. When you run out of digits, write a 0 at the end of the number and bring it down. Keep going until you have one more decimal place than you need, then round.

1. Divide the whole number part as usual.
2. Write the point in the answer above the point in the number.
3. Write a 0 after the last digit, bring it down and divide again. Repeat.
4. Stop one place past the place you round to. Then round.

Example 5.14 Dividing past the last digit

A 9.25 m length of trim is cut into 4 equal pieces. Ignore the saw cuts. How long is each piece? Round to two and to three decimal places.

SOLUTION

1. Start with 9. 4 goes into 9 twice: 2 × 4 = 8, and 9 − 8 = 1.
2. Write the point in the answer. Bring down the 2 to make 12. 4 goes into 12 3 times: 3 × 4 = 12, and 12 − 12 = 0.
3. Bring down the 5 to make 5. 4 goes into 5 once: 1 × 4 = 4, and 5 − 4 = 1.
4. Write a 0 at the end of the number and bring it down to make 10. 4 goes into 10 twice: 2 × 4 = 8, and 10 − 8 = 2.
5. Write a 0 at the end of the number and bring it down to make 20. 4 goes into 20 5 times: 5 × 4 = 20, and 20 − 20 = 0.
6. The answer is 2.3125 m. It ends here, because the remainder is 0.
7. To two decimal places: the digit after the hundredths is 2, so 2.3125 rounds to 2.31. To three places: the digit after the thousandths is 5, so it rounds up to 2.313.

CALCULATOR

TI-30Xa

9.25 ÷ 4 =

Display 2.3125

Sharp EL-531XT

9.25 ÷ 4 =

Display 2.3125

Answer **2.31 m to two decimal places, 2.313 m to three. Three decimal places of a metre is the nearest millimetre: 2,313 mm.**

Example 5.15 A division that never ends

A 2 kg bag of grout is shared equally among 7 forms. How much goes in each form? Round to three decimal places.

SOLUTION

1. Start with 2. 7 goes into 2 zero times. Write 0.
2. Write the point in the answer. Write a 0 at the end of the number and bring it down to make 20. 7 goes into 20 twice: 2 × 7 = 14, and 20 − 14 = 6.
3. Write a 0 at the end of the number and bring it down to make 60. 7 goes into 60 8 times: 8 × 7 = 56, and 60 − 56 = 4.
4. Write a 0 at the end of the number and bring it down to make 40. 7 goes into 40 5 times: 5 × 7 = 35, and 40 − 35 = 5.
5. Write a 0 at the end of the number and bring it down to make 50. 7 goes into 50 7 times: 7 × 7 = 49, and 50 − 49 = 1.
6. The digits keep coming, so stop at four places: 0.2857.
7. The fourth digit is 7, which is 5 or more, so the third digit goes up: 0.286.

CALCULATOR

TI-30Xa

2 ÷ 7 =

Display 0.285714286

Sharp EL-531XT

2 ÷ 7 =

Display 0.285714285

The display shows 0.285714286. Round it the same way.

Answer **0.286 kg in each form, to three decimal places.**

Example 5.16 A fraction that does not end

Change $\frac{5}{12}$ to a decimal by hand. Round to three decimal places.

SOLUTION

1. Start with 5. 12 goes into 5 zero times. Write 0.
2. Write the point in the answer. Write a 0 at the end of the number and bring it down to make 50. 12 goes into 50 4 times: 4 × 12 = 48, and 50 − 48 = 2.
3. Write a 0 at the end of the number and bring it down to make 20. 12 goes into 20 once: 1 × 12 = 12, and 20 − 12 = 8.
4. Write a 0 at the end of the number and bring it down to make 80. 12 goes into 80 6 times: 6 × 12 = 72, and 80 − 72 = 8.
5. Write a 0 at the end of the number and bring it down to make 80. 12 goes into 80 6 times: 6 × 12 = 72, and 80 − 72 = 8.
6. The 6 repeats forever: 0.4166...
7. To three places, look at the fourth digit, 6. Round up: 0.417.

CALCULATOR

TI-30Xa

5 ÷ 12 =

Display `0.416666667`

Sharp EL-531XT

5 ÷ 12 =

Display `0.416666666`

Answer $\frac{5}{12} \approx 0.417$. **The ≈ sign means "is about".**

CARRY ONE DIGIT PAST THE PLACE

To round to thousandths you need the digit after the thousandths. Divide to four decimal places, round to three, and write only the three.

TRY IT 5.8

1. 7 ÷ 8 by hand
2. 10 ÷ 3, to three decimal places
3. 1 ÷ 6, to two decimal places
4. 22 ÷ 7, to three decimal places
5. 4.5 ÷ 7, to three decimal places
6. Change $\frac{7}{9}$ to a decimal, to three decimal places.

Answers are in the Answer Key.

5.9 Carpentry: decimal feet and pricing

A builder's plans or a calculator may give lengths in decimal feet, such as 12.75 ft. A tape reads feet, inches and sixteenths. A foot has 12 inches, so to change the decimal part of a foot to inches, multiply it by 12. To go the other way, divide the inches by 12.

Example 5.17 Decimal feet to a tape reading

A rafter calculation gives 8.6 ft. What do you mark on the tape, to the nearest $\frac{1}{16}$ in?

SOLUTION

1. Keep the 8 feet. The decimal part is 0.6 ft.
2. 0.6 × 12 = 7.2 in. Keep the 7 inches.
3. Change the 0.2 in to sixteenths: 0.2 × 16 = 3.2, which rounds to 3.

CALCULATOR

TI-30Xa

Inches

0.6 × 12 =

Display 7.2

Sixteenths

0.2 × 16 =

Display 3.2

Sharp EL-531XT

Inches

0.6 × 12 =

Display 7.2

Sixteenths

0.2 × 16 =

Display 3.2

Answer 8 ft $7\frac{3}{16}$ in

Example 5.18 Feet and inches to decimal feet

A wall is 14 ft 5 in long. Write it in decimal feet, to three decimal places.

SOLUTION

1. 5 in is $\frac{5}{12}$ ft. Divide: 5 ÷ 12 = 0.4166…
2. Round to three places: 0.417.
3. Add the whole feet: 14 + 0.417 = 14.417.

CALCULATOR

TI-30Xa	Sharp EL-531XT
14 + 5 ÷ 12 =	14 + 5 ÷ 12 =
Display 14.41666667	Display 14.41666667

Answer **14.417 ft**

Building code values are metric decimals. Article 9.23.3.1, Sentence (2) says nails used for the framing and sheathing nailing tables shall have a diameter not less than that stated in Table 9.23.3.1. The table gives the minimum diameter for each minimum nail length (BCBC 2024, Division B, Article 9.23.3.1. (page 9-166)).

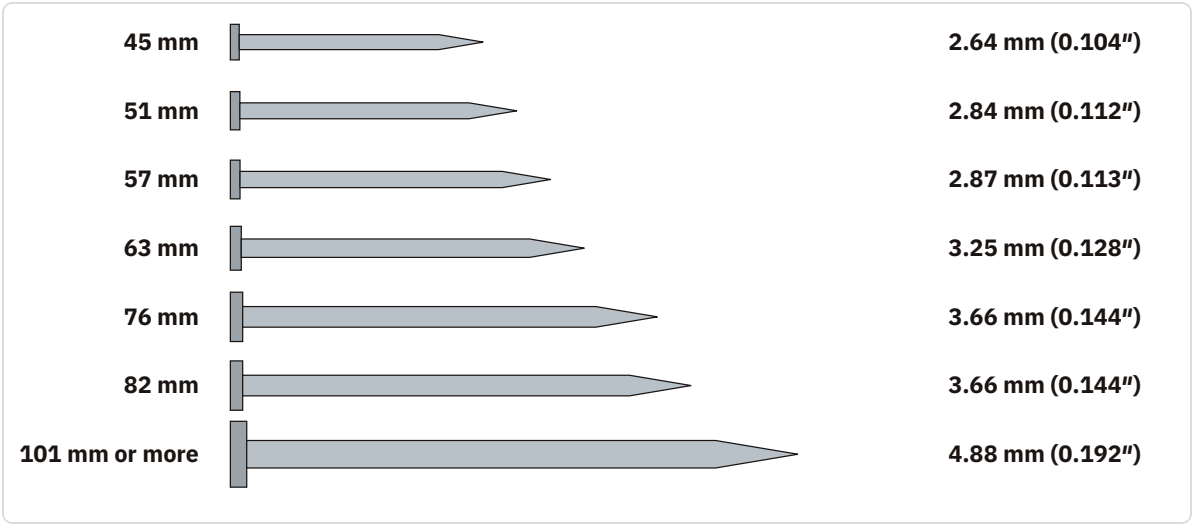


Figure 5.5 Table 9.23.3.1 drawn to scale. Left: the minimum length of each nail. Right: its minimum diameter in millimetres, with decimal inches in brackets.

Source: BCBC 2024, Division B, Article 9.23.3.1. (page 9-166)

Minimum length of nails, mm	Minimum diameter of nails, mm
45	2.64
51	2.84
57	2.87
63	3.25
76	3.66
82	3.66
101 or greater	4.88

Table 5.3 Table 9.23.3.1, Diameter of Nails. BCBC 2024, Division B, Article 9.23.3.1. (page 9-166).

Example 5.19 Comparing nail diameters

Use Table 5.3. By how much is the minimum diameter of a 76 mm nail larger than that of a 63 mm nail? Which two lengths have the closest diameters?

SOLUTION

1. 3.66 − 3.25 = 0.41 mm.
2. Line up the points and compare each diameter with the next: 2.84 and 2.87 differ by only 0.03 mm. The 76 mm and 82 mm nails have the same diameter, 3.66 mm.

Answer **0.41 mm larger. The 51 mm and 57 mm nails have the closest different diameters, 0.03 mm apart.**

Example 5.20 Pricing lumber and sheets

An order has 24 pieces of 38 × 89 lumber, each 2.44 m long, at $2.79 per metre, and 12 sheets of sheathing at $54.99 each. The prices are from the supplier's quote. Find the total before tax.

SOLUTION

1. Metres of lumber: 24 × 2.44 = 58.56 m.
2. Lumber: 58.56 × 2.79 = $163.3824, which rounds to $163.38.
3. Sheets: 12 × 54.99 = $659.88.
4. Total: 163.38 + 659.88 = $823.26.

CALCULATOR

TI-30Xa

Lumber

24 × 2.44 × 2.79 =

Display 163.3824

Sheets

12 × 54.99 =

Display 659.88

Sharp EL-531XT

Lumber

24 × 2.44 × 2.79 =

Display 163.3824

Sheets

12 × 54.99 =

Display 659.88

Answer **$823.26 before tax.**

TRY IT 5.9

1. Change 10.25 ft to feet and inches.
2. Change 6 ft 9 in to decimal feet.
3. Change 3.4 ft to feet and inches, to the nearest $\frac{1}{16}$ in.
4. Change 9 ft 7 in to decimal feet, to three decimal places.
5. In Table 5.3, what is the minimum diameter of a 57 mm nail?
6. 16 studs at $4.87 each. Cost?

Answers are in the Answer Key.

WORKSHEET 5A

Decimal parts on the canvas

Metric parts with decimal sizes. Find every red ?

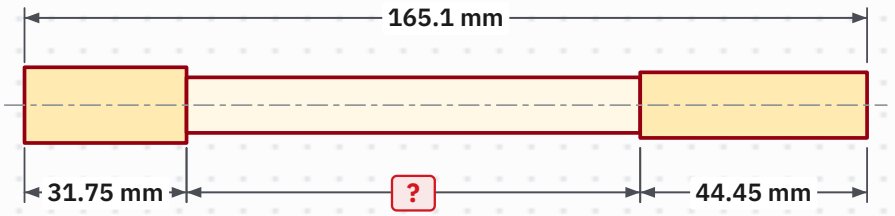
Name _____

Date _____

Line up the decimal points when you add or subtract. Estimate first, then check with your calculator.

1 The pin is 165.1 mm long overall. Find the length of the middle step.

SHAPE CANVAS — PRACTICE: FIND THE ?

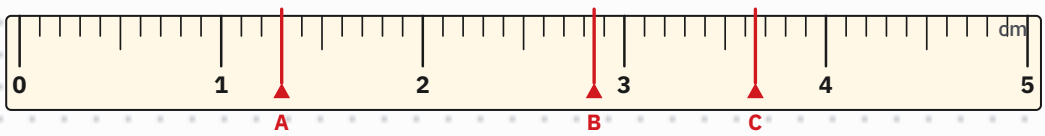


WORKING

Answer _____

2 Read marks A, B and C to the nearest 0.5 mm.

SHAPE CANVAS — READ THE DRAWING

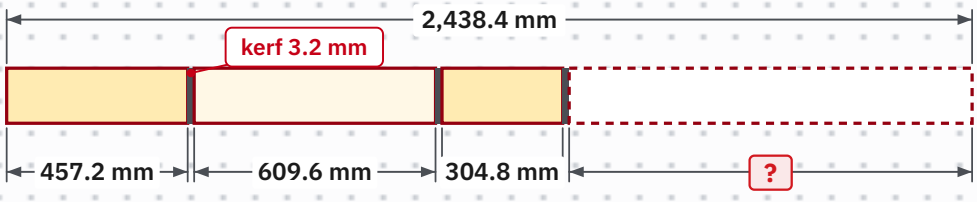


WORKING

Answer _____

3 Three pieces are cut from a 2,438.4 mm bar. You make 3 cuts and the kerf is 3.2 mm. How long is the offcut?

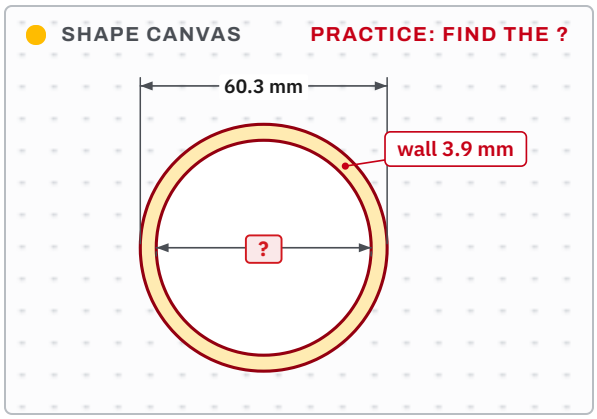
SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

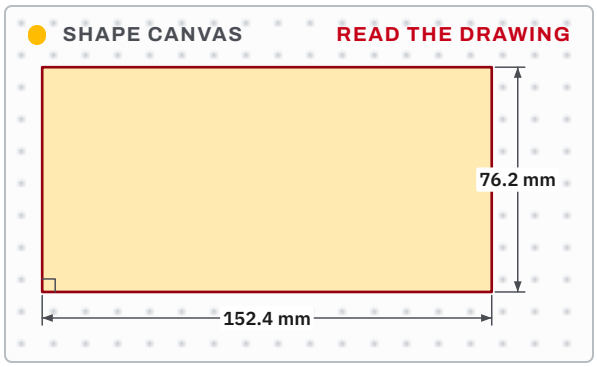
4 A pipe has an outside diameter of 60.3 mm and a wall 3.9 mm thick. Find the inside diameter.



WORKING

Answer _____

5 Find the distance around the plate by adding its four sides.



WORKING

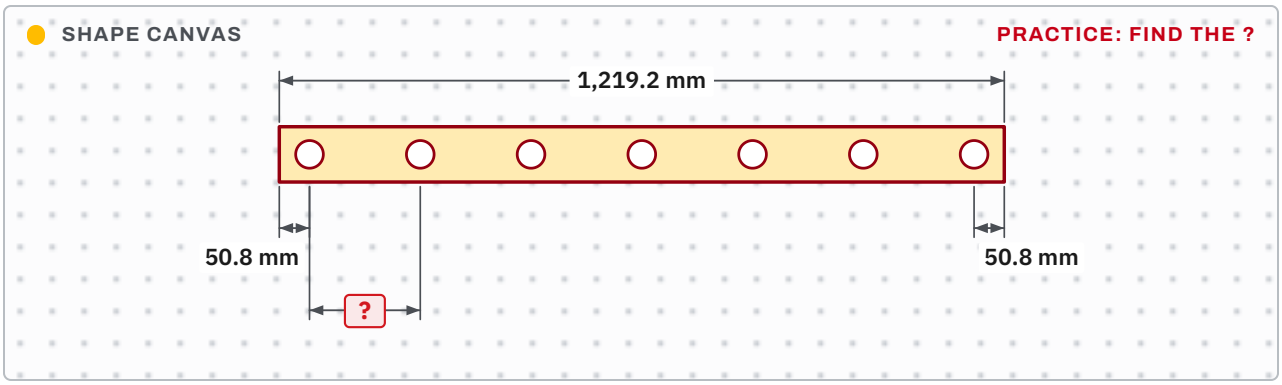
Answer _____

6 Shims of 0.8 mm, 0.5 mm, 0.25 mm, 1.5 mm are stacked. How thick is the stack? Is it more or less than 3 mm?

WORKING

Answer _____

7 Seven holes are evenly spaced along a 1,219.2 mm plate. The end holes are 50.8 mm from the ends. Find the spacing to two decimal places.



WORKING

Answer _____

WORKSHEET 5B

Decimal inches and money

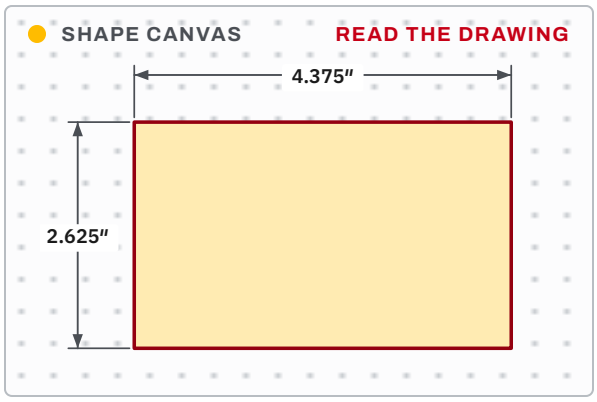
Change decimal inches to tape fractions and work out costs

Name _____

Date _____

For each tape reading, say whether you rounded to the nearest 1/16 in or the nearest 1/32 in. Round money to the nearest cent.

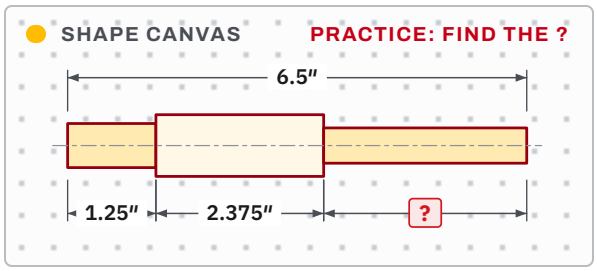
1 Write each size on the plate as a fraction in lowest terms.



WORKING

Answer _____

2 The pin is 6.5 in long overall. Find the missing step as a decimal inch, then as a fraction.



WORKING

Answer _____

3 Change 1.71 in to the nearest $\frac{1}{16}$ in and to the nearest $\frac{1}{32}$ in.

WORKING

Answer _____

4 A metric part is 75 mm long. What is that to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

5 Change $\frac{5}{16}$ in to millimetres, to two decimal places.

WORKING

Answer _____

6 Three 6 m lengths of flat bar at $4.15 per metre and 12 bolts at $0.38 each. What is the total cost?

WORKING

Answer _____

7 Change one value: a cut list has 6 pieces at 2.375 in. Find the total length as a decimal inch and as a fraction. If each piece is cut 0.125 in shorter, what is the new total?

WORKING

Answer _____

WORKSHEET 5C

Carpentry: decimal feet and pricing

Feet, inches and decimal feet, lumber prices and nail sizes

Name _____

Date _____

A foot has 12 inches. Round tape readings to the nearest 1/16 in and money to the nearest cent. Prices are given in each problem. Code values come from Table 5.3.

1 Change 11.5 ft to feet and inches.

WORKING

Answer _____

2 Change 7 ft 8 in to decimal feet, to three decimal places.

WORKING

Answer _____

3 A plan gives 5.3 ft. What is that in feet and inches, to the nearest $\frac{1}{16}$ in?

WORKING

Answer _____

4 Divide by hand: 15 ft of trim is cut into 7 equal pieces. Ignore the saw cuts. How long is each piece, to three decimal places?

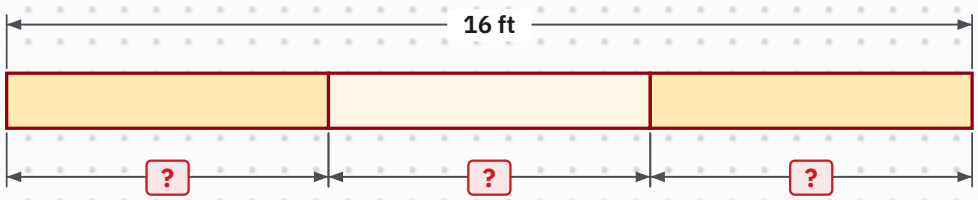
WORKING

Answer _____

5 A 16 ft board is cut into 3 equal pieces. Ignore the saw cuts. Find each piece in decimal feet to three decimal places, then in feet and inches.

SHAPE CANVAS

PRACTICE: FIND THE ?



WORKING

Answer _____

6 An order has 30 studs at $4.87 each and 8 sheets at $62.45 each. What is the total before tax?

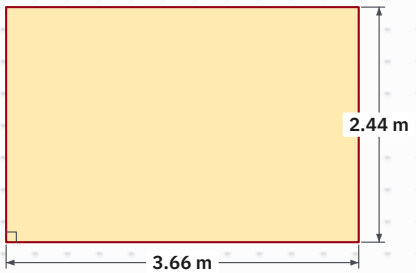
WORKING

Answer _____

7 Baseboard runs around a room with walls of 3.66 m, 2.44 m, 3.66 m and 2.44 m. Ignore the door. Baseboard costs $3.15 per metre. How many metres, and what does it cost?

SHAPE CANVAS

READ THE DRAWING



WORKING

Answer _____

8 Use Table 5.3. Write the minimum diameters of the 45 mm, 63 mm and 101 mm nails from smallest to largest. How much bigger is the largest than the smallest?

WORKING

Answer _____

CHAPTER

6

Percent, Ratio and Proportion

Shop math is full of comparisons. A waste allowance adds 10% to an order. A coating is mixed 4 parts to 1. A pump moves so many litres per minute. A drawing shows a 2,400 mm wall as a 120 mm line.

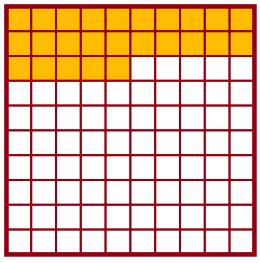
This chapter shows how percents, ratios, rates and proportions work, how they connect to the fractions and decimals you already know, and how to read a scale drawing. Percent is shown on both calculators, because the Sharp EL-531XT and the TI-30Xa use the % key in different ways.

In this chapter you will

- Change between percents, decimals and fractions
- Find a percent of a number, and find what percent one number is of another
- Add a waste allowance and work out increases and decreases
- Write and simplify ratios, and share an amount in a ratio
- Work with rates and solve proportions
- Read and make scale drawings in metric and imperial

Key words

- percent
- ratio
- rate
- proportion
- scale
- gross pay
- net pay
- unit price
- inverse proportion



25 of 100 squares
= 25%
= 0.25
= $\frac{1}{4}$

6.1 What percent means

Percent means “out of 100”. The sign % stands for percent. 25% means 25 out of every 100. If a square is split into 100 equal parts and 25 of them are shaded, 25% of the square is shaded.

A percent is one more way to write a part of a whole. You already know two others: a **fraction** and a **decimal**. 25% = $\frac{25}{100}$ = 0.25. All three describe the same amount.

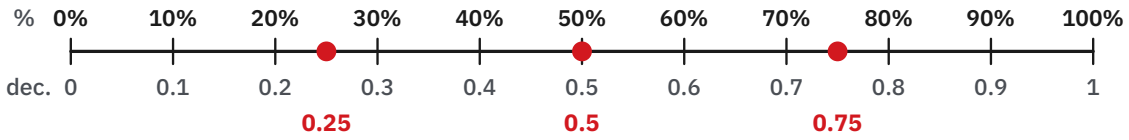


Figure 6.1 Percent and decimal on the same line. 100% is the whole, which is 1.

MOVING BETWEEN PERCENT AND DECIMAL

Percent to decimal: divide by 100, which moves the decimal point two places left. 35% = 0.35.

Decimal to percent: multiply by 100, which moves the decimal point two places right. 0.08 = 8%.

Example 6.1 Percent to decimal and fraction

Write 35%, 6.5% and 150% as decimals. Write 35% as a fraction in lowest terms.

SOLUTION

1. 35% = 35 ÷ 100 = 0.35.
2. 6.5% = 6.5 ÷ 100 = 0.065. The point moves two places left, so a zero fills the empty place.
3. 150% = 150 ÷ 100 = 1.5. A percent over 100 is more than one whole.
4. 35% = $\frac{35}{100}$. Divide the top and bottom by 5: $\frac{35}{100} = \frac{7}{20}$.

Answer **0.35, 0.065 and 1.5. 35% = $\frac{7}{20}$.**

Example 6.2 Fraction to percent

A crew has finished $\frac{3}{8}$ of a pipe run. What percent is finished?

SOLUTION

1. Change the fraction to a decimal: 3 ÷ 8 = 0.375.
2. Multiply by 100 to change the decimal to a percent: 0.375 × 100 = 37.5.

CALCULATOR

TI-30Xa

3 ÷ 8 × 100 =

Display 37.5

Sharp EL-531XT

3 ÷ 8 × 100 =

Display 37.5

Answer **37.5% is finished.**

Percent	1%	5%	10%	12.5%	20%	25%	50%	75%	100%	150%
Decimal	0.01	0.05	0.1	0.125	0.2	0.25	0.5	0.75	1	1.5
Fraction	$\frac{1}{100}$	$\frac{1}{20}$	$\frac{1}{10}$	$\frac{1}{8}$	$\frac{1}{5}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1	$1\frac{1}{2}$

Table 6.1 Percents you will meet often, with their decimals and fractions in lowest terms.

MIXED NUMBERS AND PERCENTS OVER 100%

One whole is 100%, so every whole number in a mixed number adds 100%. To change a mixed number to a percent, change it to a decimal first, then multiply by 100. $2\frac{3}{4}$ = 2 + 3 ÷ 4 = 2.75, and 2.75 × 100 = 275%. Check: 2 wholes are 200% and $\frac{3}{4}$ is 75%, so the total is 275%.

Going the other way, 275% = 2.75 = $2\frac{3}{4}$. A percent over 100% always gives a number bigger than 1.

TRY IT 6.1

1. Write 42% as a decimal.
2. Write 7.5% as a decimal.
3. Write 0.06 as a percent.
4. Write 1.25 as a percent.
5. Write 40% as a fraction in lowest terms.
6. Write $\frac{5}{8}$ as a percent.

Answers are in the Answer Key.

6.2 Finding a percent of a number

In a percent question, the word “of” means multiply. To find 15% of 240, change 15% to a decimal and multiply: 0.15 × 240.

Both calculators also have a % key. They use it in different ways. On the TI-30Xa the % key is a second function, and you finish with = (TI-30Xa guidebook, p. 4). On the Sharp EL-531XT, 2ndF % works the calculation out by itself, so there is no = at the end (Sharp EL-531XT Calculation Examples [4]).

Example 6.3 Brackets for a second job

A pallet holds 240 brackets. 15% of them are set aside for a second job. How many brackets is that?

SOLUTION

1. Change the percent to a decimal: 15% = 0.15.
2. Multiply: 0.15 × 240 = 36.

CALCULATOR

TI-30Xa

240 × 15 2nd % =

Display 36.

Sharp EL-531XT

240 × 15 2ndF %

Display 36.

With the % key, the TI-30Xa keys end with =, and the Sharp keys end with 2ndF %. The decimal method, 0.15 × 240 =, gives 36 on both calculators too.

Answer **36 brackets.**

THE SHARP % KEY ENDS THE CALCULATION

On the Sharp EL-531XT, do not press = after 2ndF %. The answer is already on the display. On the TI-30Xa, press = after % to finish.

A QUICK 10% CHECK

10% of a number is the number divided by 10: move the decimal point one place left. 10% of 240 is 24, so 15% must be about one and a half times that, 36. Use this to check your calculator answer.

TRY IT 6.2

1. 20% of 450
2. 8% of 1,250
3. 75% of 64
4. 2.5% of 3,000
5. A box of 500 bolts. 4% are rejected. How many bolts are rejected?
6. 10% of 6,096 mm

Answers are in the Answer Key.

6.3 Finding the percent and the whole

Sometimes you know the part and the whole and you want the percent. Divide the part by the whole, then multiply by 100.

PERCENT OF A WHOLE

Percent = Part ÷ Whole × 100

The part and the whole must be in the same unit.

Example 6.4 Welds that failed inspection

An inspector checks 240 welds and 18 of them fail. What percent fail?

SOLUTION

1. The part is 18 and the whole is 240.
2. 18 ÷ 240 = 0.075.
3. 0.075 × 100 = 7.5.

CALCULATOR

TI-30Xa

18 ÷ 240 2nd % =

Display 7.5

Sharp EL-531XT

18 ÷ 240 2ndF %

Display 7.5

Here the % key does the × 100. Keying 18 ÷ 240 × 100 = gives the same 7.5.

Answer **7.5% of the welds fail.**

When you know the part and the percent, you can find the whole. Change the percent to a decimal and divide the part by it. In Example 6.3, 36 brackets were 15% of the pallet. If you did not know the pallet size, you could find it: 36 ÷ 0.15 = 240 brackets.

THREE QUESTIONS, ONE RELATIONSHIP

Part = decimal × Whole. Find the part by multiplying. Find the decimal by dividing the part by the whole. Find the whole by dividing the part by the decimal.

TRY IT 6.3

1. 45 is what percent of 180?
2. 12 is what percent of 400?
3. 9 of 60 parts are scrap. What percent is scrap?
4. 30 is 25% of what number?
5. 18 is 6% of what number?
6. A 750 mm piece is cut from a 3,000 mm bar. What percent of the bar is the piece?

Answers are in the Answer Key.

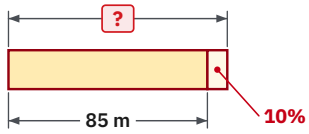
6.4 Waste allowance, increase and decrease

Some material is always lost to cutting, fitting and mistakes. A waste allowance is extra material added to an order to cover that loss. A 10% waste allowance means you order the amount you need plus 10% of it.

Adding a percent is a percent increase. Taking a percent away is a percent decrease. You can do either in two steps (find the percent, then add or subtract it) or in one step by multiplying. To add 10%, multiply by 1.10, which is 110%. To take off 30%, multiply by 0.70, which is 70%.

Example 6.5 Ordering pipe with a waste allowance

A job needs 85 m of pipe. Add a 10% waste allowance. The pipe is sold in 6 m lengths. How many lengths do you order?



SOLUTION

1. 10% of 85 = 0.1 × 85 = 8.5 m.
2. Add it on: 85 + 8.5 = 93.5 m.
3. Lengths: 93.5 ÷ 6 = 15.5833…
4. A part length still has to be bought, so round up to 16.

CALCULATOR

TI-30Xa

85 | + | 10 | 2nd | % | =

Display 93.5

Sharp EL-531XT

85 | + | 10 | 2ndF | %

Display 93.5

Pressing + and then % adds 10% of 85 to 85 on both calculators (TI-30Xa guidebook, p. 4, and Sharp EL-531XT Calculation Examples [4]). In one step without the % key: 85 × 1.1 = 93.5.

Answer **93.5 m of pipe, which is 16 lengths.**

Example 6.6 A sale price

A tool is priced at $400.00. It is on sale at 30% off. What is the sale price?

SOLUTION

1. 30% of $400.00 = 0.3 × 400 = $120.00.
2. Subtract: $400.00 − $120.00 = $280.00. In one step: 0.7 × 400 = 280.

CALCULATOR

TI-30Xa

400 | − | 30 | 2nd | % | =

Display: 280.

Sharp EL-531XT

400 | − | 30 | 2ndF | %

Display: 280.

Pressing − and then % takes 30% of 400 away from 400 on both calculators (TI-30Xa guidebook, p. 4, and Sharp EL-531XT Calculation Examples [4]).

Answer **The sale price is $280.00.**

To find a percent change, find the amount of the change, divide it by the original amount (the amount before the change), and multiply by 100.

PERCENT CHANGE

$$\text{Percent change} = (\text{New} - \text{Original}) \div \text{Original} \times 100$$

A positive answer is an increase. A negative answer is a decrease.

Example 6.7 A pay raise

An apprentice's wage goes from $24.00 to $25.20 an hour. What is the percent increase?

SOLUTION

1. Change: $25.20 − $24.00 = $1.20.
2. Divide by the original wage: 1.20 ÷ 24.00 = 0.05.
3. Multiply by 100: 5%.

CALCULATOR

TI-30Xa

(25.2 − 24) ÷ 24 × 100 =

Display 5.

Sharp EL-531XT

(25.2 − 24) ÷ 24 × 100 =

Display 5.

The brackets make both calculators find the change before dividing.

Answer **A 5% increase.**

UP 10% AND DOWN 10% IS NOT A ROUND TRIP

Start with 200. Up 10% gives 200 × 1.1 = 220. Down 10% from there gives 220 × 0.9 = 198, not 200. Each percent is taken of a different starting amount.

TRY IT 6.4

1. Add a 15% waste allowance to 60 m of cable.
2. You need 24 sheets. Add a 10% allowance and round up to whole sheets.
3. A $250.00 price is cut by 20%. New price?
4. A 1,200 mm piece grows by 5%. New length?
5. A price rises from $80.00 to $92.00. Percent increase?
6. A tank level drops from 1,500 L to 1,200 L. Percent decrease?

Answers are in the Answer Key.

6.5 Ratio

A **ratio** compares two amounts. A coating mixed 4 parts base to 1 part hardener has a ratio of base to hardener of 4 to 1. You can write it three ways: 4 : 1, 4 to 1 or $\frac{4}{1}$.

Order matters. The ratio of hardener to base is 1 : 4, not 4 : 1. A ratio is simplified the same way as a fraction: divide both numbers by the same number until they share no common factor. Before you compare two amounts, put them in the same unit.

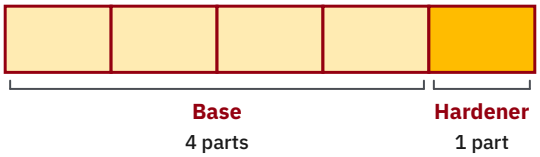
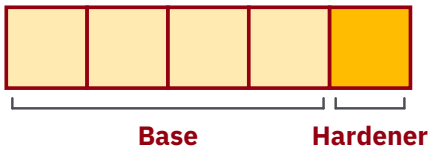


Figure 6.2 A mix of 4 : 1. Every part is the same size.

For example, to compare a 750 mm piece with a 2.5 m piece, first write 2.5 m as 2,500 mm. The ratio 750 : 2,500 simplifies to 3 : 10 when you divide both numbers by 250.

Example 6.8 Sharing an amount in a ratio

You need 750 mL of mixed coating. The label gives a mix of 4 parts base to 1 part hardener. How much of each do you measure?



SOLUTION

1. Add the parts: 4 + 1 = 5 parts.
2. One part: 750 ÷ 5 = 150 mL.
3. Base: 4 × 150 = 600 mL. Hardener: 1 × 150 = 150 mL.
4. Check: 600 + 150 = 750 mL.

CALCULATOR

TI-30Xa

750 ÷ 5 × 4 =

Display: 600.

Sharp EL-531XT

750 ÷ 5 × 4 =

Display: 600.

Answer **600 mL of base and 150 mL of hardener.**

Carpenters write a roof slope as a ratio too, such as 4 in 12. Section 6.11 compares roof slopes with the limits in the BC Building Code.

TRY IT 6.5

1. Write 12 : 18 in lowest terms.
2. Write 48 teeth to 16 teeth in lowest terms.
3. Write 400 mm to 1.2 m in lowest terms.
4. Share 900 mL in the ratio 2 : 1.
5. Share 2 L in the ratio 3 : 1. Give the answers in mL.
6. A crew has 3 journeypersons and 5 apprentices. Ratio of apprentices to journeypersons?

Answers are in the Answer Key.

6.6 Rate

A **rate** compares two amounts in different units: litres per minute, dollars per metre, millimetres per minute. A unit rate is the amount for one unit, such as 150 L in one minute. Find it by dividing.

Example 6.9 Pump rate

A pump moves 1,200 L in 8 minutes. What is its rate? At that rate, how long does it take to fill a 2,700 L tank?

SOLUTION

1. Rate: 1,200 ÷ 8 = 150 L per minute.
2. Time: 2,700 ÷ 150 = 18 minutes.

CALCULATOR

TI-30Xa

Rate

1200 ÷ 8 =

Display 150.

Time

2700 ÷ 150 =

Display 18.

Sharp EL-531XT

Rate

1200 ÷ 8 =

Display 150.

Time

2700 ÷ 150 =

Display 18.

Answer **150 L per minute. The tank takes 18 minutes.**

TRY IT 6.6

1. 450 mm of weld in 3 minutes. Rate in mm per minute?
2. 96 fittings in 8 hours. Fittings per hour?
3. A pump moves 75 L per minute. How long for 1,800 L?
4. Supplier A: 12 m of cable for $86.40. Supplier B: 20 m for $138.00. Which costs less per metre?
5. At $7.25 per metre, what do 18 m cost?
6. A drill bit cuts 120 holes in 40 minutes. Holes per minute?

Answers are in the Answer Key.

6.7 Proportion

A **proportion** says that two ratios are equal. $\frac{1}{4} = \frac{5}{20}$ is a proportion, because both ratios simplify to 1 : 4. In a proportion the cross products are equal: 1 × 20 = 4 × 5 = 20.

When one of the four numbers is missing, set up the two ratios with the same units in the same places, then solve for the missing number.

SOLVING A PROPORTION

$$a \div b = c \div x$$

$$x = b \times c \div a$$

Multiply the two numbers that sit across from each other, then divide by the third.

Example 6.10 Welding time for a larger order

A welder finishes 8 brackets in 2.5 hours. At the same rate, how long do 20 brackets take?

SOLUTION

1. Write the proportion with brackets on top and hours below: $\frac{8}{2.5} = \frac{20}{x}$.
2. Cross multiply: $8 \times x = 2.5 \times 20 = 50$.
3. Divide: $x = 50 \div 8 = 6.25$ hours.
4. 6.25 hours is 6 hours and $0.25 \times 60 = 15$ minutes.

CALCULATOR

TI-30Xa

2.5 × 20 ÷ 8 =

Display 6.25

Sharp EL-531XT

2.5 × 20 ÷ 8 =

Display 6.25

Answer **6.25 hours, which is 6 hours 15 minutes.**

KEEP THE UNITS LINED UP

If brackets are on top in the first ratio, brackets go on top in the second ratio too. Mixing them up gives an answer that is far too big or far too small. Check it with common sense: more brackets must take more time.

In every proportion in this section, both amounts grow together: more brackets take more hours. Some amounts work the other way. More workers on a job take fewer hours. That is an inverse proportion, and Section 6.10 shows how to solve it.

TRY IT 6.7

1. Is $\frac{3}{4} = \frac{15}{20}$ a proportion?
2. Solve $\frac{2}{5} = \frac{x}{45}$.
3. Solve $\frac{6}{x} = \frac{9}{24}$.
4. 5 brackets use 3 m of flat bar. How much bar do 35 brackets use?
5. A crew lays 18 m of pipe in 4 hours. How long for 45 m?
6. A mix is 3 : 1. You have 1.8 L of the first part. How much of the second part?

Answers are in the Answer Key.

6.8 Scale drawings

A drawing of a large part is smaller than the part. The **scale** is the ratio of a length on the drawing to the real length. A scale of 1 : 20 means 1 mm on the drawing stands for 20 mm on the real part.

Drawing to real: multiply by the second number. Real to drawing: divide by it. A larger second number gives a smaller drawing. A scale such as 2 : 1 makes the drawing larger than the part, which is useful for small details.

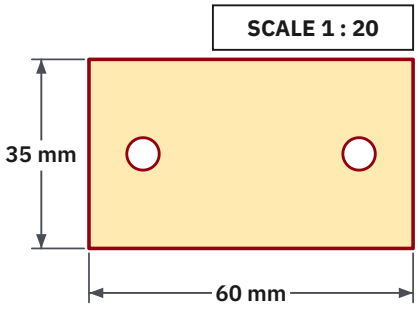
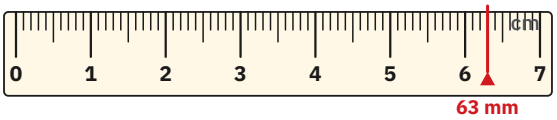


Figure 6.3 A plate drawn at 1 : 20. The sizes shown are the lengths measured on the paper.

The plate above measures 60 mm by 35 mm on the paper, so the real plate is 60 × 20 = 1,200 mm long and 35 × 20 = 700 mm wide.

Example 6.11 Measuring a drawing

A line on a drawing at 1 : 50 measures 63 mm with a rule, as shown. How long is the real part?



SOLUTION

1. Drawing to real, so multiply by 50.
2. 63 × 50 = 3,150 mm.

CALCULATOR

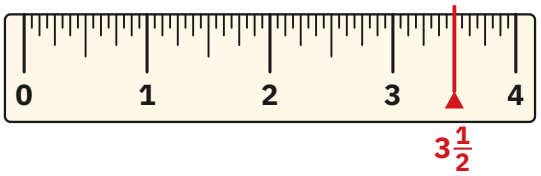
TI-30Xa	Sharp EL-531XT
63 × 50 =	63 × 50 =
Display 3150.	Display 3′150.

Answer **3,150 mm, which is 3.15 m.**

A scale can also be written as two lengths. A drawing that states its scale as $\frac{1}{4}$″ = 1′-0″ uses $\frac{1}{4}$ inch on the drawing for 1 foot on the real job. Since 1 ft = 12 in, that ratio is $\frac{1}{4}$: 12, which is 1 : 48.

Example 6.12 An imperial scale

A drawing is marked $\frac{1}{4}$″ = 1′-0″. A line measures $3\frac{1}{2}$ in, as shown. How long is the real length?



SOLUTION

1. Count how many quarter inches are in the line:
   3.5 ÷ 0.25 = 14.
2. Each quarter inch is 1 ft, so the real length is 14 ft.

CALCULATOR

TI-30Xa	Sharp EL-531XT
3.5 ÷ 0.25 =	3.5 ÷ 0.25 =
Display 14.	Display 14.

Answer **14 ft**

Floor plans for houses are scale drawings too. Section 6.11 reads a metric plan and an imperial plan.

TRY IT 6.8

1. At 1 : 10, a line measures 48 mm. Real length?
2. At 1 : 25, a line measures 72 mm. Real length?
3. A 3,000 mm beam is drawn at 1 : 50. Drawing length?
4. A 12 mm part is drawn at 5 : 1. Drawing length?
5. A 1,500 mm part is drawn 75 mm long. What is the scale?
6. At $\frac{1}{4}$″ = 1′-0″, a line measures $2\frac{3}{4}$ in. Real length?

Answers are in the Answer Key.

6.9 Money on the job

Percents, rates and rounding come together when you look at a pay stub or a till receipt. **Gross pay** is hours worked times the hourly rate. **Net pay** is what is left after deductions are taken off. Money is rounded to the cent, two decimal places.

Overtime is paid at a higher rate. The rate is set by the job, so it is given in each problem here. A rate of 1.5 times means each overtime hour pays one and a half regular hours.

Example 6.13 A week's pay with overtime

An apprentice earns $25.40 an hour. On this job, hours over 40 in a week are paid at 1.5 times the hourly rate. This week the apprentice works 45 hours. What is the gross pay?

SOLUTION

1. Regular pay: 40 × $25.40 = $1,016.00.
2. Overtime rate: 1.5 × $25.40 = $38.10 an hour.
3. Overtime pay: 5 × $38.10 = $190.50.
4. Gross pay: $1,016.00 + $190.50 = $1,206.50.

CALCULATOR

TI-30Xa

40 × 25.4 + 5 × 25.4 × 1.5 =

Display 1206.5

Sharp EL-531XT

40 × 25.4 + 5 × 25.4 × 1.5 =

Display 1′206.5

BEDMAS does both multiplications before the addition, on both calculators.

Answer **Gross pay is $1,206.50.**

Most things you buy in British Columbia carry two sales taxes. The federal goods and services tax (GST) is collected by the Canada Revenue Agency. The provincial sales tax (PST) is collected by the Province of British Columbia. The B.C. government says that, generally, the rate of PST is 7% on the purchase or lease price of goods and services, with some exceptions. Its guide for businesses says PST is charged on the price before the GST is added, so each tax is found on the price before tax.

Tax	Rate in B.C.	Source	Read
GST	5%	Canada Revenue Agency, "Charge and collect the GST/HST", table "Which rate to charge", British Columbia row. canada.ca/en/revenue-agency/services/tax/businesses/topics/gst-hst-businesses/charge-collect-which-rate.html	October 19, 2026
PST	7% (general rate)	Province of British Columbia, "B.C. provincial sales tax (PST)", last updated September 21, 2026, and "Small business guide to PST". www2.gov.bc.ca/gov/content/taxes/sales-taxes/pst	October 19, 2026

Table 6.2 Sales tax rates used in this book. Rates can change. Check the current rates on these pages before you rely on them.

Example 6.14 GST and PST on a purchase

A drill costs $249.00 before tax. Find the GST, the PST and the total.

SOLUTION

1. GST: 5% of $249.00 = 0.05 × 249 = $12.45.
2. PST: 7% of $249.00 = 0.07 × 249 = $17.43.
3. Total: $249.00 + $12.45 + $17.43 = $278.88.

CALCULATOR

TI-30Xa

GST

249 × 5 2nd % =

Display 12.45

PST

249 × 7 2nd % =

Display 17.43

Sharp EL-531XT

GST

249 × 5 2ndF %

Display 12.45

PST

249 × 7 2ndF %

Display 17.43

Answer **GST $12.45, PST $17.43, total $278.88.**

Example 6.15 Discount first, then tax

Safety boots are listed at $189.95. They are on sale at 20% off. What do you pay with GST and PST?

SOLUTION

1. Take the discount off first: $189.95 − 20% = 0.8 × 189.95 = $151.96.
2. GST on the sale price: 0.05 × 151.96 = 7.598, which rounds to $7.60.
3. PST on the sale price: 0.07 × 151.96 = 10.6372, which rounds to $10.64.
4. Total: $151.96 + $7.60 + $10.64 = $170.20.

CALCULATOR

TI-30Xa

Sale price

189.95 | − | 20 | 2nd | % | =

Display 151.96

GST

151.96 | × | 0.05 | =

Display 7.598

PST

151.96 | × | 0.07 | =

Display 10.6372

Sharp EL-531XT

Sale price

189.95 | − | 20 | 2ndF | %

Display 151.96

GST

151.96 | × | 0.05 | =

Display 7.598

PST

151.96 | × | 0.07 | =

Display 10.6372

Answer **You pay $170.20.**

The **unit price** is the cost of one unit: one litre, one metre, one screw. Divide the price by the number of units. It lets you compare packages of different sizes.

Example 6.16 Which bag is the better buy?

Cable ties come in a bag of 100 for $6.49 and a bag of 500 for $27.95. Find the cost of one tie in each bag, in cents.

SOLUTION

1. Bag of 100: $6.49 ÷ 100 = $0.0649, which is 6.49¢ a tie.
2. Bag of 500: $27.95 ÷ 500 = $0.0559, which is 5.59¢ a tie.
3. 5.59¢ is less than 6.49¢.

CALCULATOR

TI-30Xa

Bag of 100

6.49 ÷ 100 × 100 =

Display 6.49

Bag of 500

27.95 ÷ 500 × 100 =

Display 5.59

Sharp EL-531XT

Bag of 100

6.49 ÷ 100 × 100 =

Display 6.49

Bag of 500

27.95 ÷ 500 × 100 =

Display 5.59

Answer **The bag of 500 is the better buy, at 5.59¢ a tie.**

A QUICK TAX CHECK

GST and PST together are 12% of the price before tax. 10% is easy to find in your head, so the total tax is a little more than 10%. On $249.00, 10% is $24.90, and the two taxes together are $29.88.

TRY IT 6.9

1. A helper works 36 hours at $22.75 an hour. Gross pay?
2. At $28.00 an hour, with overtime paid at 1.5 times, what is one overtime hour worth?
3. A $80.00 item. Find the GST, the PST and the total.
4. A $150.00 tool is 10% off. Find the sale price, then the total with GST and PST.
5. Gross pay $1,150.00, deductions $262.40. Net pay?
6. A 4 L can costs $31.96. What is the price per litre?

Answers are in the Answer Key.

6.10 Inverse proportion

In Section 6.7 both amounts grew together. Twice as many brackets took twice as long. In an **inverse proportion**, one amount goes up as the other goes down. Twice as many workers take half as long. The two amounts multiply to the same number every time.

To solve one, multiply the two numbers you know that belong together. That product is the total job. Then divide it by the new number.

INVERSE PROPORTION

$$a \times b = c \times x$$

$$x = a \times b \div c$$

a and b belong together. c is the new amount and x is the one you are finding.

Example 6.17 More hands, less time

3 pipefitters can install a run of pipe in 8 hours. Working at the same rate, how long would 4 pipefitters take?

SOLUTION

1. The whole job is 3 × 8 = 24 worker-hours.
2. Share it among 4 workers: 24 ÷ 4 = 6 hours.
3. Check with common sense: more workers, so less time.

CALCULATOR

TI-30Xa

3 × 8 ÷ 4 =

Display 6.

Sharp EL-531XT

3 × 8 ÷ 4 =

Display 6.

Answer **6 hours.**

Example 6.18 A faster pump

A tank fills in 45 minutes when a pump moves 120 L per minute. How long does it take at 150 L per minute?

SOLUTION

1. The tank holds 120 × 45 = 5,400 L.
2. At the new rate: 5,400 ÷ 150 = 36 minutes.

CALCULATOR

TI-30Xa

120 × 45 ÷ 150 =

Display 36.

Sharp EL-531XT

120 × 45 ÷ 150 =

Display 36.

Answer **36 minutes.**

DIRECT OR INVERSE?

Ask what happens when one amount doubles. If the other amount doubles too, it is a direct proportion: set two ratios equal, as in Section 6.7. If the other amount halves, it is an inverse proportion: multiply, then divide.

TRY IT 6.10

1. 2 welders finish a job in 12 hours. How long for 3 welders?
2. 5 workers take 6 days. How long for 10 workers?
3. A tank drains in 40 minutes at 90 L per minute. How long at 60 L per minute?
4. A crew of 4 takes 15 hours. How many workers are needed to finish in 10 hours?
5. Direct or inverse? The number of brackets and the length of flat bar they use.
6. Direct or inverse? The speed of a pump and the time to fill a tank.

Answers are in the Answer Key.

6.11 Carpentry: roof slope, waste and scale

Carpenters write the slope of a roof as a rise in a run: 4 in 12 means the roof rises 4 in for every 12 in it runs across. That is the ratio 4 : 12, which simplifies to 1 : 3. The BC Building Code writes the same idea as “1 in 3”: 1 unit of rise in 3 units of run.

Table 9.26.3.1, Roofing Types and Slope Limits, forms part of Sentence 9.26.3.1.(1) of the BC Building Code 2024 (Division B, page 9-241). It sets the slopes on which each kind of roofing may be applied. For asphalt shingles it gives a minimum slope of 1 in 3 in normal application and 1 in 6 in low slope application.

Sentences (2) and (3) of the Article allow lower slopes for some built-up roofs and metal roof systems.

TABLE 9.26.3.1. ROOFING TYPES AND SLOPE LIMITS
Forming Part of Sentence 9.26.3.1.(1)

Type of Roofing	Minimum Slope	Maximum Slope
Asphalt Shingles		
Low slope application	**1 in 6**	**no limit**
Normal application	**1 in 3**	**no limit**

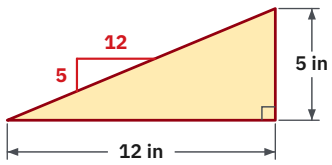
Figure 6.4 The top of Table 9.26.3.1 on the ACCESS plate for Article 9.26.3.1. A minimum slope of 1 in 3 means at least 1 unit of rise for every 3 units of run.
Source: BCBC 2024, Division B, Article 9.26.3.1. (page 9-241)

Type of roofing	Minimum slope in the code	As a rise in 12
Asphalt shingles, low slope application	1 in 6	2 in 12
Asphalt shingles, normal application	1 in 3	4 in 12
Cedar shakes	1 in 3	4 in 12
Wood shingles	1 in 4	3 in 12
Clay tile	1 in 2	6 in 12
Slate shingles	1 in 2	6 in 12

Table 6.3 Some minimum slopes from Table 9.26.3.1 (BCBC 2024, Division B, Article 9.26.3.1. (page 9-241)). None of these rows has a maximum slope. The last column is worked out here: 12 ÷ the second number.

Example 6.19 Does the roof meet the code?

A garage roof is framed at 5 in 12. Write the slope as "1 in ?". Is it steep enough for asphalt shingles in normal application?



SOLUTION

1. 5 in 12 is the ratio 5 : 12. Divide both numbers by 5 to make the rise 1: 1 : 2.4.
2. So the slope is 1 in 2.4. The code minimum for asphalt shingles in normal application is 1 in 3.
3. A smaller second number means less run for each unit of rise, so a steeper roof. 2.4 is less than 3, so 1 in 2.4 is steeper than 1 in 3.

CALCULATOR

TI-30Xa

12 ÷ 5 =

Display 2.4

Sharp EL-531XT

12 ÷ 5 =

Display 2.4

Answer **1 in 2.4. Yes, it is steeper than the 1 in 3 minimum in Table 9.26.3.1.**

Example 6.20 Sheathing with a waste allowance

A roof takes 38 sheets of sheathing. The estimator adds a 12% waste allowance for cuts. How many whole sheets do you order?

SOLUTION

1. 12% of 38 = 0.12 × 38 = 4.56 sheets.
2. Add it on: 38 + 4.56 = 42.56.
3. A part sheet still has to be bought, so round up to 43.

CALCULATOR

TI-30Xa

38 + 12 2nd % =

Display 42.56

Sharp EL-531XT

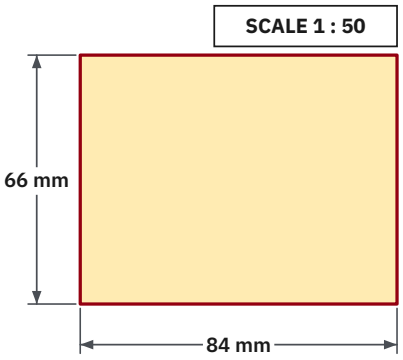
38 + 12 2ndF %

Display 42.56

Answer **43 sheets.**

Example 6.21 Reading a floor plan

On a floor plan drawn at 1 : 50, a room measures 84 mm by 66 mm, as shown. What are the real sizes? What is the floor area in m²?



SOLUTION

1. Drawing to real: multiply by 50. 84 × 50 = 4,200 mm and 66 × 50 = 3,300 mm.
2. In metres: 4.2 m by 3.3 m.
3. Area: 4.2 × 3.3 = 13.86 m².

CALCULATOR

TI-30Xa

4.2 × 3.3 =

Display 13.86

Sharp EL-531XT

4.2 × 3.3 =

Display 13.86

Answer **4,200 mm by 3,300 mm, and 13.86 m².**

SCALE THE LENGTHS, NOT THE AREA

Multiply each length by the scale number, then find the area. The area of the drawing times 50 is not the real area. Each side is 50 times longer, so the area is 50 × 50 = 2,500 times larger.

TRY IT 6.11

1. Write 6 in 12 as “1 in ?”.
2. Write a slope of 1 in 4 as a rise in 12.
3. Is 3 in 12 steep enough for asphalt shingles in normal application (1 in 3)?
4. A roof at 4 in 12 runs 4,200 mm. How much does it rise?
5. A floor needs 28 m² of plywood. Add a 15% waste allowance.
6. On a plan at 1 : 50, a wall measures 76 mm. Real length in metres?

Answers are in the Answer Key.

WORKSHEET 6A

Percent on the canvas

Find each red ?

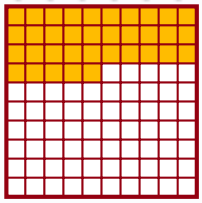
Name _____

Date _____

Show the decimal you multiply or divide by. Check each answer with a 10% estimate.

1 Write the shaded part as a percent, a decimal and a fraction in lowest terms.

SHAPE CANVAS READ THE DRAWING

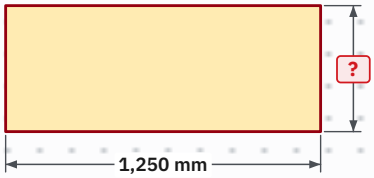


WORKING

Answer _____

2 The width is 40% of the length. Find the width.

SHAPE CANVAS PRACTICE: FIND THE ?

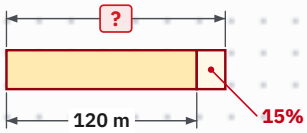


WORKING

Answer _____

3 A run needs 120 m of conduit. Add a 15% waste allowance. How much do you order?

SHAPE CANVAS PRACTICE: FIND THE ?

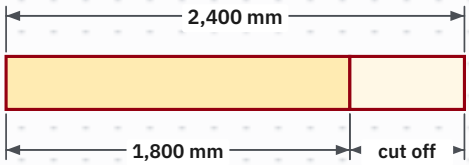


WORKING

Answer _____

4 A 2,400 mm bar is cut down to 1,800 mm. What is the percent decrease?

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

5 Change one value: a 750 mm piece is made 20% longer. Then the new piece is made 20% shorter. Find both lengths. Is it back to 750 mm?

WORKING

Answer _____

WORKSHEET 6B

Ratio, proportion and scale

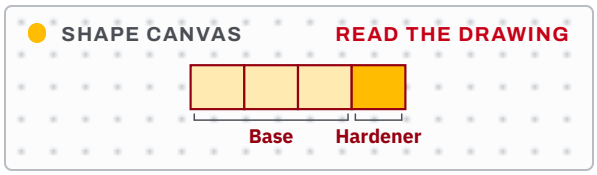
Mixes, similar shapes and scale drawings

Name _____

Date _____

Write each ratio or proportion before you calculate. Keep the same units in the same places.

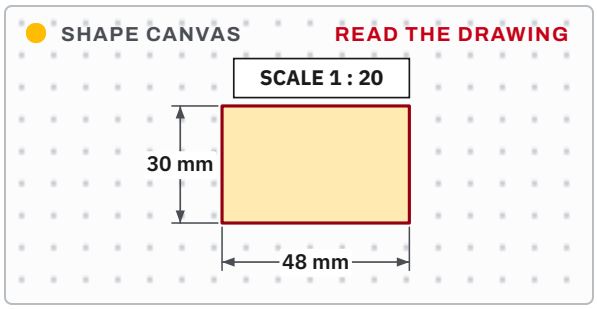
1 A coating is mixed as shown. How many mL of base and of hardener make 2.4 L of mix?



WORKING

Answer _____

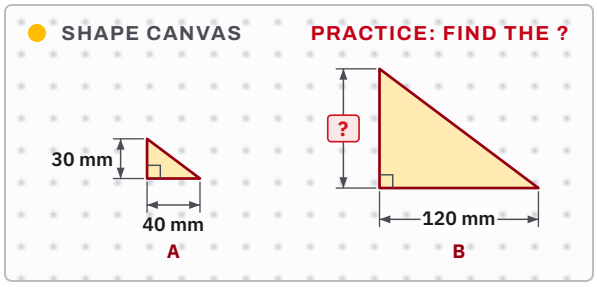
2 The sizes shown are measured on a 1 : 20 drawing. Find the real length and width.



WORKING

Answer _____

3 Triangle B is triangle A made larger, so their sides are in proportion. Find the height of B.



WORKING

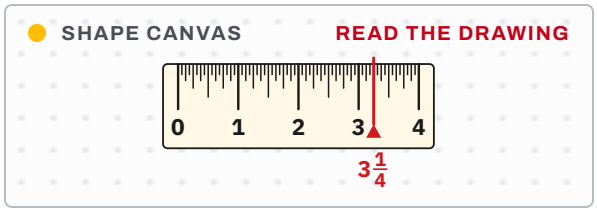
Answer _____

4 Change one value: the plate in Problem 2 is redrawn at 1 : 10. How long is its 960 mm side on the new drawing? What happens to every line?

WORKING

Answer _____

5 At $\frac{1}{4}$″ = 1′-0″, find the real length of this line.



WORKING

Answer _____

6 A pump fills 1,500 L in 12 minutes. Find its rate, and the time to fill 2,250 L.

WORKING

Answer _____

WORKSHEET 6C

Carpentry: slope, waste and scale

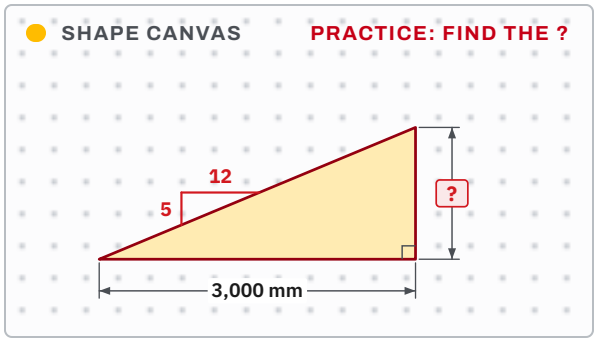
Roofs, floor plans and material orders

Name _____

Date _____

The red slope mark on each roof gives the rise for 12 of run. Write each ratio before you calculate. Waste allowances are given in each problem. Section 6.11 has the BC Building Code slope limits.

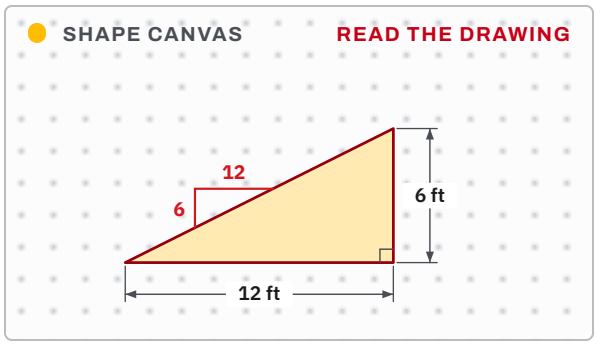
1 The roof is framed at 5 in 12. Find the rise.



WORKING

Answer _____

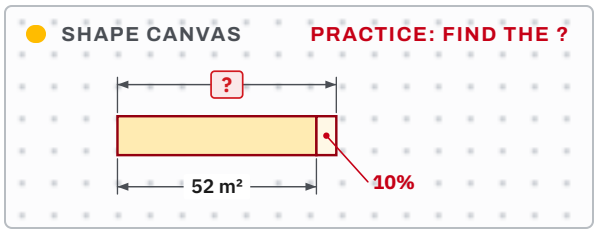
2 Write this roof slope as "1 in ?". Is it steep enough for asphalt shingles in normal application (1 in 3, Table 9.26.3.1)?



WORKING

Answer _____

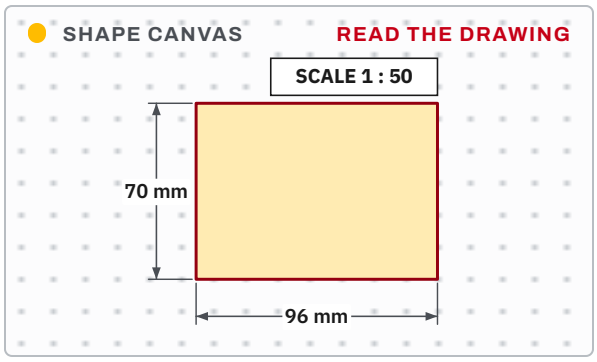
3 A floor is 52 m². Add a 10% waste allowance. How much flooring do you order?



WORKING

Answer _____

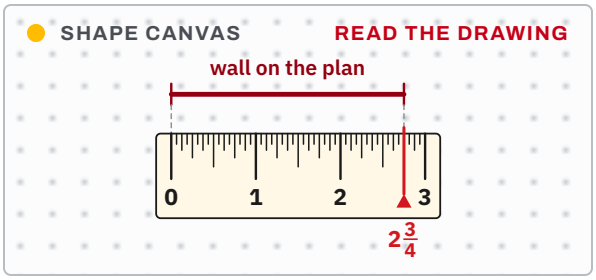
4 The sizes are measured on a floor plan drawn at 1 : 50. Find the real sizes and the floor area in m².



WORKING

Answer _____

5 A plan is drawn at $\frac{1}{4}$″ = 1′-0″. Find the real length of the wall line, measured from 0 to the red mark.



WORKING

Answer _____

6 Change one value: a roof at 4 in 12 runs 3,600 mm. Find the rise. Then find it at 8 in 12. What happens to the rise when the first number of the slope doubles?

WORKING

Answer _____

7 A house takes 64 sheets of subfloor and 112 sheets of wall sheathing. The estimator adds a 10% waste allowance to each. How many whole sheets of each do you order?

WORKING

Answer _____

8 A roof at 4 in 12 runs 15 ft. A second roof at 6 in 12 runs 9 ft. Which roof rises more, and by how much?

WORKING

Answer _____

9 Two framers can frame a set of walls in 15 hours. How long would 3 framers take, at the same rate? How many framers are needed to finish in 6 hours?

WORKING

Answer _____

CHAPTER

7

Measurement and Unit Conversion

A metric drawing, an inch tape, a pump rated in US gallons and a heater set in °F can all turn up on the same job. Before you can add, compare or cut, every number has to be in the same unit.

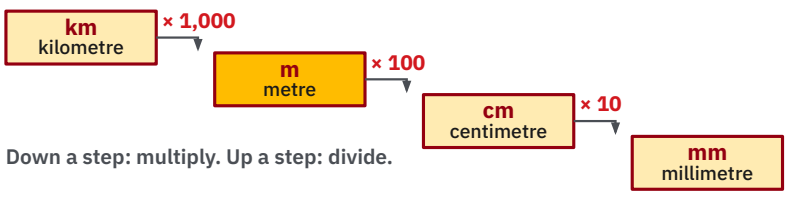
This chapter covers the metric system and its prefixes, imperial units and feet-and-inches, and how to convert between them with exact conversion factors. It finishes with units of area and volume, litres and gallons, and temperature. Every factor in this chapter is exact or is taken from a published source, and the source is given with each table.

In this chapter you will

- Use SI units and the prefixes kilo, centi and milli
- Work with inches, feet and yards, and with feet and inches
- Convert between metric and imperial with exact factors
- Convert units of area and volume
- Convert between litres, US gallons and imperial gallons
- Convert temperatures between °C and °F

Key words

- unit
- metric system
- SI
- prefix
- imperial system
- conversion factor



7.1 Units and the metric system

A **unit** is a fixed amount used to measure something: a metre, an inch, a litre. A measurement is always a number and a unit. “2,400” on its own means nothing on the job. 2,400 mm and 2,400 in are very different lengths.

The **metric system** is built on powers of ten. Its modern form is the **SI**, the International System of Units. The SI base units include the metre (m) for length, the kilogram (kg) for mass and the second (s) for time (NIST SP 1038, section 4.1.1). The litre (L) is accepted for use with the SI for volume (NIST SP 1038, section 4.1.2 and the note to the volume table in section 5.1.5).

A **prefix** in front of a unit multiplies it by a power of ten. Kilo means 1,000, so 1 km = 1,000 m. Milli means $\frac{1}{1{,}000}$, so 1 mm = $\frac{1}{1{,}000}$ m.

Prefix	Symbol	Multiplies the unit by	Example
kilo	k	1,000	1 km = 1,000 m
hecto	h	100	1 hL = 100 L
deca	da	10	1 dam = 10 m
deci	d	0.1	1 dm = 0.1 m
centi	c	0.01	1 cm = 0.01 m
milli	m	0.001	1 mm = 0.001 m
micro	μ	0.000 001	1 μm = 0.000 001 m

Table 7.1 SI prefixes. Source: Weights and Measures Act (Canada), Schedule I, Part V, and NIST SP 1038, Table 2. NIST notes that deci, deka (deca) and hecto are rarely used.

Each step on the ladder in the chapter opener is a power of ten, so a metric conversion only moves the decimal point. Going to a smaller unit, the number gets bigger, so multiply. Going to a bigger unit, the number gets smaller, so divide.

Example 7.1 Metric conversions

Convert: (a) 2.45 m to mm, (b) 375 mm to m, (c) 1,250 mL to L, (d) 0.85 t to kg.

SOLUTION

1. (a) 1 m = 1,000 mm. Smaller unit, so multiply: 2.45 × 1,000 = 2,450 mm. The point moves 3 places right.
2. (b) Bigger unit, so divide: 375 ÷ 1,000 = 0.375 m. The point moves 3 places left.
3. (c) 1 L = 1,000 mL. 1,250 ÷ 1,000 = 1.25 L.
4. (d) 1 t (tonne) = 1,000 kg (Weights and Measures Act, Schedule I, Part IV). 0.85 × 1,000 = 850 kg.

Answer **2,450 mm, 0.375 m, 1.25 L and 850 kg.**

Very large and very small numbers, such as 1,000,000 mm² in a square metre, can also be written with a power of ten. Section 7.7 shows this scientific notation and the EE key on both calculators.

TRY IT 7.1

1. 3.6 m to mm
2. 850 mm to m
3. 42 cm to mm
4. 6,096 mm to m
5. 2.5 L to mL
6. 1,800 kg to t

Answers are in the Answer Key.

7.2 Imperial units, feet and inches

The **imperial system** uses the inch, foot, yard, pound and gallon. Canadian law defines the yard as 0.9144 m, the foot as $\frac{1}{3}$ yard and the inch as $\frac{1}{36}$ yard (Weights and Measures Act, Schedule II). The US defines the same inch, foot and yard (NIST SP 1038), so a US tape and a Canadian tape agree. Table 7.2 lists the length units.

Lengths on the job are often written in feet and inches, such as 7′ $9\frac{1}{2}$″ (7 feet $9\frac{1}{2}$ inches). To work with them on a calculator, change them to inches first. Multiply the feet by 12 and add the inches.

For example, 7′ $9\frac{1}{2}$″ is 7 × 12 + 9.5 = 93.5 in.

Example 7.2 Inches to feet and inches

A cut list gives a length of $142\frac{3}{4}$ in. Write it in feet and inches.

SOLUTION

1. Divide by 12: 142.75 ÷ 12 = 11.8958… The whole number part is 11 ft.
2. Inches left over: 142.75 − 11 × 12 = 10.75 in.
3. 10.75 in is $10\frac{3}{4}$ in.

CALCULATOR

TI-30Xa

142.75 ÷ 12 =

Display 11.89583333

Sharp EL-531XT

142.75 ÷ 12 =

Display 11.89583333

Only the whole number part, 11, is feet. Find the inches left with 142.75 − 11 × 12 =, which shows 10.75.

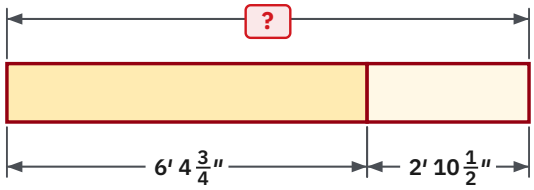
Answer **11′ $10\frac{3}{4}$″**

Unit	Symbol	Equals
inch	in or ″	$\frac{1}{36}$ yard
foot	ft or ′	12 in = $\frac{1}{3}$ yard
yard	yd	3 ft = 36 in
mile	mi	1,760 yd = 5,280 ft

Table 7.2 Imperial length units. Source: Weights and Measures Act (Canada), Schedule II, Measurement of Length.

Example 7.3 Adding feet and inches

Two pieces of $6' \, 4\frac{3}{4}''$ and $2' \, 10\frac{1}{2}''$ are laid end to end. Find the total length.



SOLUTION

1. Add the feet: 6 + 2 = 8 ft. Add the inches: $4\frac{3}{4} + 10\frac{1}{2} = 15\frac{1}{4}$ in.
2. $15\frac{1}{4}$ in is more than a foot: $15\frac{1}{4} - 12 = 3\frac{1}{4}$ in, and carry 1 ft.
3. Total: 8 + 1 = 9 ft and $3\frac{1}{4}$ in.
4. Check in inches: 76.75 + 34.5 = 111.25 in = $9' \, 3\frac{1}{4}''$.

CALCULATOR

TI-30Xa

76.75 | + | 34.5 | =

Display 111.25

Sharp EL-531XT

76.75 | + | 34.5 | =

Display 111.25

Answer $9' \, 3\frac{1}{4}''$

Some plans and site layouts give lengths in decimal feet, such as 12.35 ft. Section 7.8 changes feet, inches and sixteenths to decimal feet and back.

TRY IT 7.2

1. Write $5' \, 7''$ in inches.
2. Write $10' \, 2\frac{1}{4}''$ in inches.
3. Write 86 in in feet and inches.
4. Write $125\frac{1}{2}$ in in feet and inches.
5. How many feet are in 4 yd?
6. Add $3' \, 8\frac{1}{2}''$ and $4' \, 7\frac{3}{4}''$.

Answers are in the Answer Key.

7.3 Converting with exact factors

A **conversion factor** is the number you multiply or divide by to change units. The inch is defined as exactly 25.4 mm, so 25.4 is the factor between inches and millimetres. It is exact, not rounded, and it gives the same answer on every job.

INCHES AND MILLIMETRES

$$\text{mm} = \text{in} \times 25.4$$

$$\text{in} = \text{mm} \div 25.4$$

1 in = 25.4 mm exactly (NIST SP 1038, section 5.1.3). It also follows from the Weights and Measures Act: 0.9144 m ÷ 36 = 0.0254 m.

From	To	Multiply by	Exact?
inch (in)	millimetre (mm)	25.4	exact
inch (in)	centimetre (cm)	2.54	exact
foot (ft)	metre (m)	0.3048	exact
yard (yd)	metre (m)	0.9144	exact
mile (mi)	kilometre (km)	1.609 344	exact
pound (lb)	kilogram (kg)	0.453 592 37	exact

Table 7.3 Length and mass factors. Source: NIST SP 1038, sections 5.1.3 and 5.2, which prints exact factors in bold. The yard and the pound are also defined this way in the Weights and Measures Act (Canada), Schedule II. To go the other way, divide. Long decimals are printed in groups of three digits, as in the source. Key them without the spaces.

Example 7.4 A fraction of an inch in millimetres

A drawing calls for $\frac{3}{8}$ in plate. What is that in millimetres?

SOLUTION

1. Inches to millimetres: multiply by 25.4.
2. $\frac{3}{8} \times 25.4 = 0.375 \times 25.4 = 9.525$ mm.

CALCULATOR

TI-30Xa

[3] [a b/c] [8] [×] [25.4] [=]

Display: 9.525

Sharp EL-531XT

[3] [a b/c] [8] [×] [25.4] [=]

Display: 9.525

The fraction key keys $\frac{3}{8}$ directly. Keying 3 ÷ 8 × 25.4 = gives the same result.

Answer **9.525 mm**

Example 7.5 Millimetres to the nearest sixteenth

A metric drawing shows 1,480 mm. You have an inch tape. What do you mark, to the nearest $\frac{1}{16}$ in?

SOLUTION

1. Dividing by 25.4 changes millimetres to inches. Multiplying by 16 then counts sixteenths of an inch, so do both in one line.
2. 1,480 ÷ 25.4 × 16 = 932.28… sixteenths, which rounds to 932 sixteenths.
3. 932 ÷ 16 = 58.25 in, which is $58\frac{4}{16} = 58\frac{1}{4}$ in.

CALCULATOR

TI-30Xa

Sixteenths

[1480] [÷] [25.4] [×] [16] [=]

Display: 932.2834646

Inches

[932] [÷] [16] [=]

Display: 58.25

Sharp EL-531XT

Sixteenths

[1480] [÷] [25.4] [×] [16] [=]

Display: 932.2834646

Inches

[932] [÷] [16] [=]

Display: 58.25

Answer **Mark $58\frac{1}{4}$″ (that is 4′ $10\frac{1}{4}$″).**

Example 7.6 Feet and inches to millimetres

A pipe run is 7′ 9 $\frac{1}{2}$″ long. How long is it in millimetres?

SOLUTION

1. Change to inches first, as in Section 7.2: 7 × 12 + 9.5 = 93.5 in.
2. Multiply by 25.4: 93.5 × 25.4 = 2,374.9 mm.

CALCULATOR

TI-30Xa

(7 × 12 + 9.5) × 25.4 =

Display 2374.9

Sharp EL-531XT

(7 × 12 + 9.5) × 25.4 =

Display 2′374.9

The brackets make both calculators finish the inches before multiplying by 25.4.

Answer **2,374.9 mm, which is about 2.37 m.**

Example 7.7 Pounds to kilograms

A bag of grout is marked 50 lb. What is its mass in kilograms, to two decimal places?

SOLUTION

1. Pounds to kilograms: multiply by 0.453 592 37.
2. 50 × 0.453 592 37 = 22.6796185 kg.
3. Round to two decimal places: 22.68 kg.

CALCULATOR

TI-30Xa

50 × 0.45359237 =

Display 22.6796185

Sharp EL-531XT

50 × 0.45359237 =

Display 22.6796185

Answer **22.68 kg**

MULTIPLY OR DIVIDE?

A millimetre is much smaller than an inch, so a length in millimetres is always the bigger number. If you convert $\frac{3}{8}$ in and get 0.015 mm, you divided when you should have multiplied. Check every answer this way.

TRY IT 7.3

1. 12 in to mm
2. $\frac{5}{16}$ in to mm
3. 20 ft to m
4. 762 mm to inches
5. 2,000 mm to inches, to the nearest $\frac{1}{16}$ in
6. 110 lb to kg, to two decimal places

Answers are in the Answer Key.

7.4 Units of area and volume

Area is measured in square units and volume in cubic units. A **square unit** is a square 1 unit on each side. A **cubic unit** is a cube 1 unit on each side. Because a square has two sides that are measured, the length factor is used twice for area. For volume it is used three times.

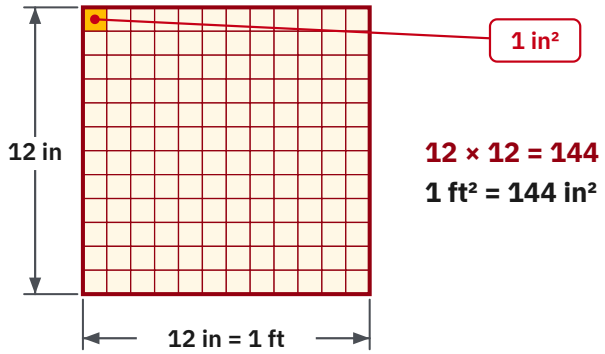


Figure 7.1 One square foot is a square 12 in on each side. It holds 12 rows of 12 square inches.

Area	Volume
1 m² = 1,000 × 1,000 = 1,000,000 mm²	1 m³ = 1,000 L
1 ft² = 12 × 12 = 144 in²	1 L = 1,000,000 mm³ = 1,000 cm³
1 in² = 645.16 mm² (exact)	1 ft³ = 12 × 12 × 12 = 1,728 in³
1 ft² = 0.092 903 04 m² (exact)	1 in³ = 16.387 064 cm³ (exact)

Table 7.4 Area and volume factors. Sources: NIST SP 1038, sections 5.1.4 and 5.1.5, for the square inch, square foot and cubic inch. The Weights and Measures Act (Canada), Schedule I, Part IV, for the litre ($\frac{1}{1,000}$ m³), and Schedule II for 1 in² = $\frac{1}{144}$ ft² and 1 in³ = $\frac{1}{1,728}$ ft³.

Example 7.8 A plate in square metres

A plate is 2,400 mm by 1,200 mm. Find its area in mm² and in m².

SOLUTION

1. Area = length × width = 2,400 × 1,200 = 2,880,000 mm².
2. 1 m² = 1,000,000 mm², so divide: 2,880,000 ÷ 1,000,000 = 2.88 m².
3. Check with metres first: 2.4 m × 1.2 m = 2.88 m².

CALCULATOR

TI-30Xa

2400 × 1200 ÷ 1000000 =

Display 2.88

Sharp EL-531XT

2400 × 1200 ÷ 1000000 =

Display 2.88

Answer **2,880,000 mm² = 2.88 m²**

Example 7.9 A sheet in square feet and square metres

A sheet is 4 ft by 8 ft. Find its area in ft², then in m² to two decimal places.

SOLUTION

1. Area = 4 × 8 = 32 ft².
2. Multiply by 0.092 903 04: 32 × 0.092 903 04 = 2.97289728 m².
3. Round: 2.97 m².

CALCULATOR

TI-30Xa

32 × 0.09290304 =

Display 2.97289728

Sharp EL-531XT

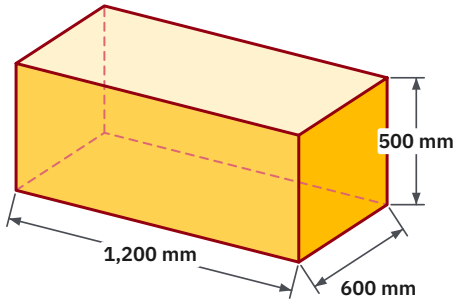
32 × 0.09290304 =

Display 2.97289728

Answer **32 ft² = 2.97 m²**

Example 7.10 Litres in a tank

A rectangular tank measures 1,200 mm × 600 mm × 500 mm inside. How many litres does it hold?



SOLUTION

1. Volume = 1,200 × 600 × 500 = 360,000,000 mm³.
2. 1 L = 1,000,000 mm³, so divide:
   360,000,000 ÷ 1,000,000 = 360 L.

CALCULATOR

TI-30Xa

1200 × 600 × 500 ÷ 1000000 =

Display 360.

Sharp EL-531XT

1200 × 600 × 500 ÷ 1000000 =

Display 360.

Answer 360 L

DO NOT USE 1,000 FOR SQUARE MILLIMETRES

1 m = 1,000 mm, but 1 m² = 1,000,000 mm². A square metre is 1,000 mm long and 1,000 mm wide, so it holds 1,000 × 1,000 square millimetres.

TRY IT 7.4

1. 3 m² to mm²
2. 450,000 mm² to m²
3. 5 ft² to in²
4. 10 in² to mm²
5. A tank 1,000 mm × 800 mm × 600 mm. Litres?
6. 2 ft³ to in³

Answers are in the Answer Key.

7.5 Litres and gallons

There are two gallons, and they are not the same size. The US gallon is 231 in³ (NIST SP 1038, section 3.2). Since 1 in³ = 16.387 064 cm³ exactly, the US gallon is 231 × 16.387 064 = 3,785.411 784 cm³, which is exactly 3.785 411 784 L. The imperial gallon is defined in the Weights and Measures Act (Canada), Schedule II, as $\frac{454\,609}{100\,000\,000}$ m³, which is exactly 4.546 09 L.

Gallon	Litres (exact)	Source
US gallon (US gal)	3.785 411 784	NIST SP 1038: 231 in³, and 1 in³ = 16.387 064 cm³
Imperial gallon (imp gal)	4.546 09	Weights and Measures Act (Canada), Schedule II

Table 7.5 The two gallons. The volume table in NIST SP 1038 (section 5.1.5) rounds the US gallon to 3.785 412 L.

Example 7.11 US gallons to litres

A drum holds 55 US gal. How many litres is that, to two decimal places?

SOLUTION

1. Multiply by 3.785 411 784: 55 × 3.785 411 784 = 208.197648… L.
2. Round: 208.20 L.

CALCULATOR

TI-30Xa

55 × 3.785411784 =

Display 208.1976481

Sharp EL-531XT

55 × 3.785411784 =

Display 208.1976481

Answer **208.20 L**

Example 7.12 Imperial gallons, litres and US gallons

An older tank is marked 40 imp gal. How many litres does it hold? How many US gallons? Round to two decimal places.

SOLUTION

1. Imperial gallons to litres: 40 × 4.546 09 = 181.8436 L.
2. Litres to US gallons: divide by 3.785 411 784. 181.8436 ÷ 3.785 411 784 = 48.038…
3. Round: 181.84 L and 48.04 US gal.

CALCULATOR

TI-30Xa

Litres

40 × 4.54609 =

Display 181.8436

US gallons

40 × 4.54609 ÷ 3.785411784 =

Display 48.03799702

Sharp EL-531XT

Litres

40 × 4.54609 =

Display 181.8436

US gallons

40 × 4.54609 ÷ 3.785411784 =

Display 48.03799702

Answer **181.84 L, which is 48.04 US gal. The imperial gallon is the larger gallon.**

TRY IT 7.5

1. 10 US gal to L, to two decimal places
2. 10 imp gal to L, to two decimal places
3. The 360 L tank in Example 7.10, in US gal to two decimal places
4. 1 imp gal in US gal, to three decimal places
5. 1 ft^3 in US gal, to two decimal places (1 ft^3 = 1,728 in^3)
6. 500 L to imp gal, to two decimal places

Answers are in the Answer Key.

7.6 Temperature

Temperature is measured in degrees Celsius (°C) in the metric system and degrees Fahrenheit (°F) in the imperial system. The two scales are linked by an exact formula. NIST SP 1038 (section 5.3, note 33) gives °C = (°F − 32) ÷ 1.8, and turned around it gives °F = 1.8 × °C + 32. Both are exact definitions, so round only the final answer.

CELSIUS AND FAHRENHEIT

$$°F = 1.8 \times °C + 32$$

$$°C = (°F - 32) \div 1.8$$

Exact. Follow BEDMAS: in the first formula multiply before you add, and in the second subtract inside the brackets before you divide.

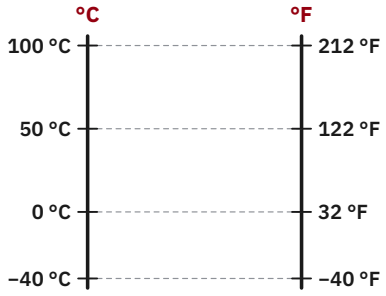


Figure 7.2 The two scales side by side. Each °F mark comes from °F = 1.8 × °C + 32. At −40 the two scales read the same number.

Example 7.13 Celsius to Fahrenheit

A water heater is set to 60 °C. What is that in °F?

SOLUTION

1. °F = 1.8 × 60 + 32. Multiply first: 1.8 × 60 = 108.
2. Then add: 108 + 32 = 140.

CALCULATOR

TI-30Xa

1.8 × 60 + 32 =

Display 140.

Sharp EL-531XT

1.8 × 60 + 32 =

Display 140.

Answer **140 °F**

Example 7.14 Fahrenheit to Celsius

An oven for drying electrodes is set to 350 °F. What is that in °C, to one decimal place?

SOLUTION

1. °C = (350 − 32) ÷ 1.8. Brackets first: 350 − 32 = 318.
2. Divide: 318 ÷ 1.8 = 176.6667..., which rounds to 176.7.

CALCULATOR

TI-30Xa

(350 − 32) ÷ 1.8 =

Display 176.6666667

Sharp EL-531XT

(350 − 32) ÷ 1.8 =

Display 176.6666667

Without the brackets, both calculators divide 32 by 1.8 first and give a wrong answer.

Answer **176.7 °C**

A CHANGE IN TEMPERATURE HAS NO 32

A rise or drop is converted with the 1.8 alone: a change of 1 °C is a change of 1.8 °F (NIST SP 1038, note 33). A part that warms up by 25 °C warms up by 25 × 1.8 = 45 °F.

Worksheet 7C at the end of the chapter is a full page of temperature practice.

TRY IT 7.6

1. 20 °C to °F
2. −10 °C to °F
3. 212 °F to °C
4. 50 °F to °C
5. 400 °F to °C, to one decimal place
6. A temperature drops by 15 °C. How many °F is that drop?

Answers are in the Answer Key.

7.7 Powers of ten and scientific notation

The metric prefixes in Section 7.1 are powers of ten. 1,000 is 10 × 10 × 10, which is written 10^3: ten to the power 3. 1,000,000 is 10^6. A negative power means divide: 10^{-3} is 1 ÷ 1,000 = 0.001, the milli in millimetre.

Scientific notation writes a number as a number from 1 up to 10, times a power of ten. 6,400,000 is 6.4×10^6. 0.0254 is 2.54×10^{-2}. The power tells you how many places the decimal point moves: right for a positive power, left for a negative one.

Example 7.15 Writing numbers in scientific notation

Write (a) 2,880,000 mm² and (b) 0.000 35 m in scientific notation. (c) Write 3.6×10^4 as an ordinary number.

SOLUTION

1. (a) Put the point after the first digit: 2.88. The point moved 6 places left, so the power is 6. 2,880,000 mm² = 2.88×10^6 mm².
2. (b) The first digit that is not zero is 3. The point moves 4 places right to make 3.5, so the power is −4. 0.000 35 m = 3.5×10^{-4} m.
3. (c) A power of 4 moves the point 4 places right: 3.6×10^4 = 36,000.

Answer **2.88×10^6 mm², 3.5×10^{-4} m and 36,000.**

Both calculators take a power of ten with one key. On the TI-30Xa it is EE. On the Sharp EL-531XT it is Exp. Key the number, then EE or Exp, then the power. A calculator also switches to scientific notation by itself when an answer has too many digits for the display.

Example 7.16 Scientific notation on the calculator

A tank holds 3.6×10^8 mm³. There are 1×10^6 mm³ in a litre. How many litres does it hold?

SOLUTION

1. Divide the volume by the size of one litre: $3.6 \times 10^8 \div (1 \times 10^6)$.
2. Dividing powers of ten takes the powers away: $8 - 6 = 2$, so the answer is $3.6 \times 10^2 = 360$.

CALCULATOR

TI-30Xa

3.6 | EE | 8 | ÷ | 1 | EE | 6 | =

Display: 360.

Sharp EL-531XT

3.6 | Exp | 8 | ÷ | 1 | Exp | 6 | =

Display: 360.

EE (TI-30Xa) and Exp (Sharp EL-531XT) key the power of ten that belongs to the number in front of them, so no brackets are needed.

Answer **360 L. This is the tank of Example 7.10: 1,200 × 600 × 500 = 360,000,000 mm³.**

TRY IT 7.7

1. Write 45,000 in scientific notation.
2. Write 0.006 in scientific notation.
3. Write 7.2×10^5 as an ordinary number.
4. Write 1.5×10^{-3} m in millimetres.
5. How many mm² are in 1 m², as a power of ten?
6. Work out $4.8 \times 10^7 \div (2 \times 10^4)$.

Answers are in the Answer Key.

7.8 Decimal feet

Site plans and some layout tools give lengths in **decimal feet**: feet and hundredths of a foot, such as 12.35 ft. A hundredth of a foot is $0.01 \times 12 = 0.12$ in, a little less than $\frac{1}{8}$ in. Your tape reads feet, inches and sixteenths, so you need to change between the two.

CHANGING BETWEEN THE TWO

To decimal feet: change the inches and sixteenths to inches, divide by 12, and add the whole feet.

From decimal feet: keep the whole feet. Multiply the decimal part by 12 to get inches. Keep the whole inches, and multiply what is left by 16 to count sixteenths.

Example 7.17 Feet, inches and sixteenths to decimal feet

Write $10' \, 4\frac{3}{16}''$ in decimal feet, to two decimal places.

SOLUTION

1. Inches: 4 + 3 ÷ 16 inches. Divide by 12 to change them to feet.
2. Add the whole feet: 10 + (4 + 3 ÷ 16) ÷ 12 = 10.349… ft.
3. Round to two decimal places: 10.35 ft.

CALCULATOR

TI-30Xa

10 + (4 + 3 ÷ 16) ÷ 12 =

Display 10.34895833

Sharp EL-531XT

10 + (4 + 3 ÷ 16) ÷ 12 =

Display 10.34895833

The brackets make both calculators add the inches before dividing by 12.

Answer **10.35 ft**

Example 7.18 Decimal feet to feet, inches and sixteenths

A plan gives a length of 12.35 ft. What do you mark on the tape, to the nearest $\frac{1}{16}$ in?

SOLUTION

1. Whole feet: 12 ft. The decimal part is 0.35 ft.
2. Inches: 0.35 × 12 = 4.2 in. Keep the 4 whole inches.
3. Sixteenths: 0.2 × 16 = 3.2, which rounds to 3 sixteenths.
4. Put it together: $12' \; 4\frac{3}{16}''$.

CALCULATOR

TI-30Xa

Inches

0.35 × 12 =

Display 4.2

Sixteenths

0.2 × 16 =

Display 3.2

Sharp EL-531XT

Inches

0.35 × 12 =

Display 4.2

Sixteenths

0.2 × 16 =

Display 3.2

Answer $12' \; 4\frac{3}{16}''$

TRY IT 7.8

1. Write 6′ 6″ in decimal feet.
2. Write 8′ 3″ in decimal feet.
3. Write $4' \; 7\frac{1}{2}''$ in decimal feet, to two decimal places.
4. Write 9.75 ft in feet and inches.
5. Write 15.4 ft in feet and inches, to the nearest $\frac{1}{16}$ in.
6. Write 22.18 ft in feet and inches, to the nearest $\frac{1}{16}$ in.

Answers are in the Answer Key.

7.9 Carpentry: code values, board feet and floor area

The BC Building Code states its limits in millimetres. On a job laid out in inches you convert them, but the code value itself never changes. The rise limit is 200 mm, not the inch number you worked out. When you round a converted limit to a tape reading, round so that you stay inside the limit: round a maximum down and a minimum up.

Table 9.8.4.1 forms part of Sentence 9.8.4.1.(1) (BCBC 2024, Division B, Article 9.8.4.1. (page 9-30)). For private stairs, which serve single dwelling units, it gives a maximum rise of 200 mm and a minimum rise of 125 mm. Table 9.8.4.2, in Sentence 9.8.4.2.(1) (BCBC 2024, Division B, Article 9.8.4.2. (page 9-31)), gives private stairs a minimum run of 255 mm and a maximum run of 355 mm.

TABLE 9.8.4.1. RISE, MM

Stair type	Max.	Min.
Private (1)	**200 (7⅞″)**	**125 (4 15/16″)**
Public (2)	**180 (7 1/16″)**	**125 (4 15/16″)**

Figure 7.3 Table 9.8.4.1 on the ACCESS plate for Article 9.8.4.1. The code value is the millimetre number. The inch values in brackets are conversions to the nearest sixteenth.

Source: BCBC 2024, Division B, Article 9.8.4.1. (page 9-30)

Example 7.19 A code maximum in inches

The maximum rise for private stairs is 200 mm (Table 9.8.4.1). Convert it to sixteenths of an inch. What is the largest tape reading that stays within the limit?

SOLUTION

1. 200 ÷ 25.4 × 16 = 125.98... sixteenths.
2. The nearest sixteenth is 126 sixteenths, which is $7\frac{14}{16} = 7\frac{7}{8}$ in. But 126 ÷ 16 × 25.4 = 200.03 mm, a little over 200 mm.
3. For a maximum, round down: 125 sixteenths = $7\frac{13}{16}$ in, which is 198.44 mm.
4. The code value is still 200 mm.

CALCULATOR

TI-30Xa

Sixteenths

200 ÷ 25.4 × 16 =

Display 125.984252

Check 125 sixteenths

125 ÷ 16 × 25.4 =

Display 198.4375

Sharp EL-531XT

Sixteenths

200 ÷ 25.4 × 16 =

Display 125.984252

Check 125 sixteenths

125 ÷ 16 × 25.4 =

Display 198.4375

Answer **$7\frac{13}{16}$ in. The limit stays 200 mm.**

Example 7.20 A code minimum in inches

The minimum run for private stairs is 255 mm (Table 9.8.4.2). What is the smallest tape reading, to a sixteenth, that meets it?

SOLUTION

1. 255 ÷ 25.4 × 16 = 160.63... sixteenths.
2. For a minimum, round up: 161 sixteenths.
3. 161 ÷ 16 = 10 and 1 sixteenth left: $10\frac{1}{16}$ in. Check: 161 ÷ 16 × 25.4 = 255.59 mm, which is at least 255 mm.

CALCULATOR

TI-30Xa

Sixteenths

255 ÷ 25.4 × 16 =

Display 160.6299213

Check 161 sixteenths

161 ÷ 16 × 25.4 =

Display 255.5875

Sharp EL-531XT

Sixteenths

255 ÷ 25.4 × 16 =

Display 160.6299213

Check 161 sixteenths

161 ÷ 16 × 25.4 =

Display 255.5875

Answer $10\frac{1}{16}$ in. The limit stays 255 mm.

Lumber is often counted in **board feet**. NIST SP 1038 (section 5.1.5) gives 1 board foot = 0.002 359 737 m³. Divided by 1 in³ = 16.387 064 cm³, that is 144 in³: a board 12 in by 12 in by 1 in. For a board with its thickness and width in inches and its length in feet, the formula below follows from 144 in³. Use the sizes the problem or the order gives.

BOARD FEET

$$\text{Board feet} = \text{thickness (in)} \times \text{width (in)} \times \text{length (ft)} \div 12$$

Thickness × width × (length × 12) ÷ 144 is the same as thickness × width × length ÷ 12.

Example 7.21 Board feet in a lumber order

An order has 25 pieces sized 2 in × 8 in × 12 ft. How many board feet is that? What is the volume in m³, to three decimal places?

SOLUTION

1. One piece: 2 × 8 × 12 ÷ 12 = 16 board feet.
2. 25 pieces: 25 × 16 = 400 board feet.
3. In m³: 400 × 0.002 359 737 = 0.943895..., which rounds to 0.944 m³.

CALCULATOR

TI-30Xa

Board feet

25 × 2 × 8 × 12 ÷ 12 =

Display 400.

Cubic metres

400 × 0.002359737 =

Display 0.9438948

Sharp EL-531XT

Board feet

25 × 2 × 8 × 12 ÷ 12 =

Display 400.

Cubic metres

400 × 0.002359737 =

Display 0.9438948

Answer 400 board feet, about 0.944 m³.

Example 7.22 Floor area in square feet

A room measures 4.2 m by 3.6 m. The flooring is priced by the square foot. Find the area in m² and in ft², to two decimal places.

SOLUTION

1. Area: 4.2 × 3.6 = 15.12 m².
2. 1 ft² = 0.092 903 04 m², so divide: 15.12 ÷ 0.092 903 04 = 162.7503...
3. Round: 162.75 ft².

CALCULATOR

TI-30Xa

4.2 × 3.6 ÷ 0.09290304 =

Display 162.7503255

Sharp EL-531XT

4.2 × 3.6 ÷ 0.09290304 =

Display 162.7503255

Answer 15.12 m² = 162.75 ft²

TRY IT 7.9

1. The maximum rise for public stairs is 180 mm (Table 9.8.4.1). Convert it to inches, rounded down to a sixteenth.
2. Guards inside a dwelling unit must be at least 900 mm high (Sentence 9.8.8.3.(2), page 9-40). Convert to inches, rounded up to a sixteenth.
3. How many board feet are in one piece 2 in × 6 in × 10 ft?
4. How many board feet are in 12 pieces 2 in × 4 in × 8 ft? Give two decimal places.
5. A deck is 3.6 m × 4.8 m. Area in ft², to two decimal places?
6. A sheet is 4 ft × 8 ft. Area in m², to two decimal places?

Answers are in the Answer Key.

WORKSHEET 7A

Convert on the canvas

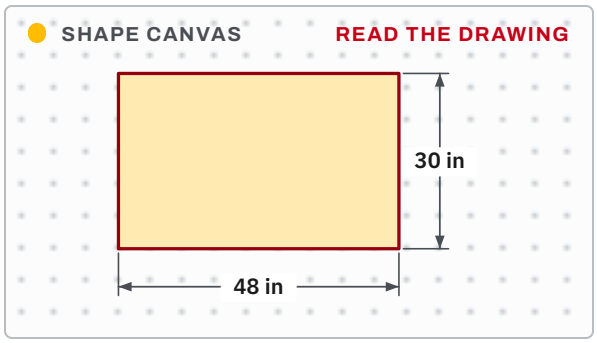
Lengths and masses in both systems

Name _____

Date _____

Write the factor you use on every line. Use 1 in = 25.4 mm and 1 lb = 0.453 592 37 kg.

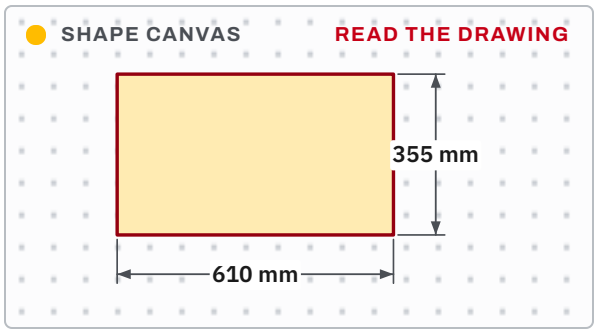
1 The sheet is sized in inches. Write both sizes in millimetres.



WORKING

Answer _____

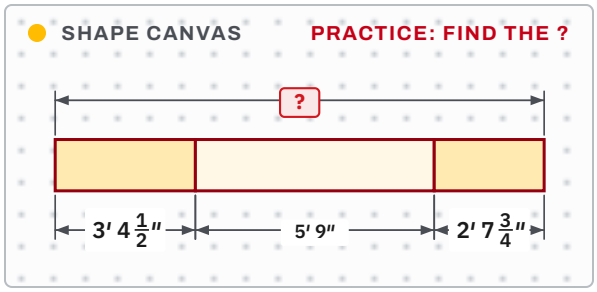
2 The plate is sized in millimetres. Write both sizes in inches, to the nearest sixteenth of an inch.



WORKING

Answer _____

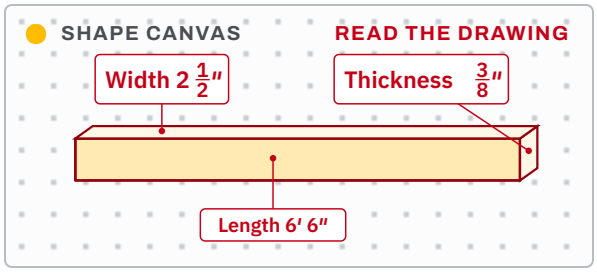
3 Find the total length in feet and inches, then in millimetres.



WORKING

Answer _____

4 Convert each size on the flat bar to millimetres.



WORKING

Answer _____

5 Three bags of 50 lb are loaded on a cart. What is their total mass in kilograms, to two decimal places?

WORKING

Answer _____

6 A drawing gives 2,400 mm. Write it in metres, in centimetres, and in feet and inches to the nearest sixteenth of an inch.

WORKING

Answer _____

WORKSHEET 7B

Area, volume, capacity and temperature

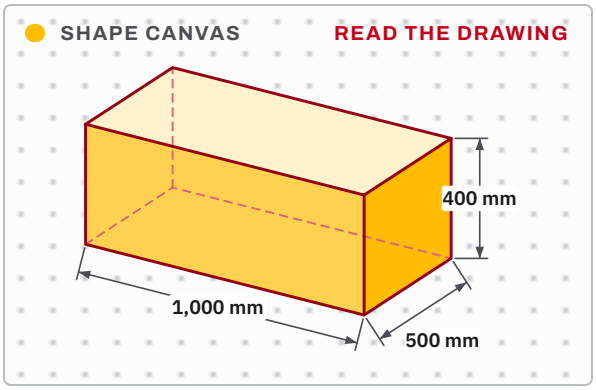
Square and cubic units, gallons and degrees

Name _____

Date _____

Use 1 L = 1,000,000 mm³, 1 US gal = 231 in³ and 1 ft² = 0.092 903 04 m².

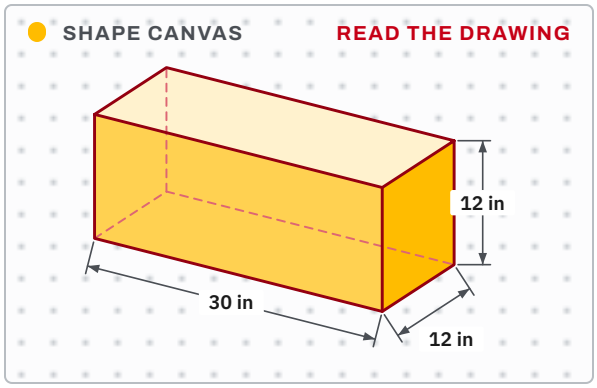
1 How many litres does the tank hold?



WORKING

Answer _____

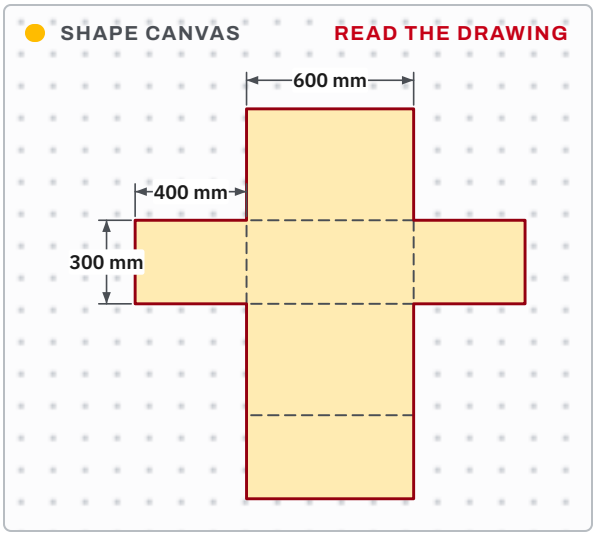
2 The tank is measured in inches. How many US gallons does it hold, to two decimal places?



WORKING

Answer _____

3 Unfold it: the net of a closed box. Add the six rectangles. Give the area in mm² and m².



WORKING

Answer _____

4 A procedure gives a preheat of 150 °C. What is that in °F? A part cools by 40 °C. How many °F is that drop?

WORKING

Answer _____

5 Change one value: a cube is 100 mm on each side. How many litres does it hold? How many if each side is 200 mm?

WORKING

Answer _____

WORKSHEET 7C

Temperature practice

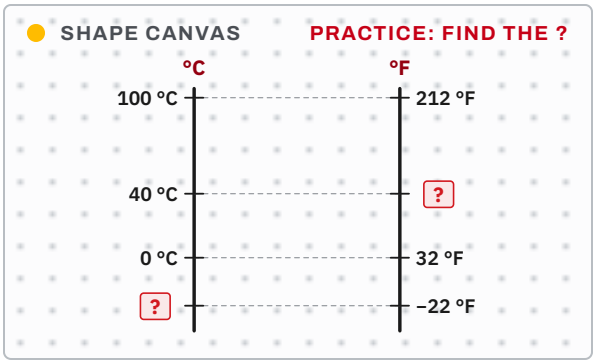
Celsius and Fahrenheit, readings and changes

Name _____

Date _____

Use °F = 1.8 × °C + 32 and °C = (°F − 32) ÷ 1.8 for a reading. Use 1.8 alone for a change in temperature. Round only the final answer.

1 Find each red ? on the two scales.



WORKING

Answer _____

2 A shop is heated to 22 °C. What is that in °F?

WORKING

Answer _____

3 A curing oven is set to 425 °F. What is that in °C, to one decimal place?

WORKING

Answer _____

4 A winter night drops to −25 °C. What is that in °F?

WORKING

Answer _____

5 A US weather report gives 5 °F. What is that in °C?

WORKING

Answer _____

6 Water in a loop warms up by 18 °C. How many °F is that rise?

WORKING

Answer _____

7 A part cools by 45 °F. How many °C is that drop?

WORKING

Answer _____

8 Which is hotter, 38 °C or 100 °F? By how many °F?

WORKING

Answer _____

9 A product label gives a working range of 10 °C to 32 °C. Write the range in °F.

WORKING

Answer _____

10 At what temperature do the two scales read the same number? Check it with both formulas.

WORKING

Answer _____

11 Change one value: a heater is set to 50 °C. Find the °F reading. It is turned up by 10 °C. Find the new reading. By how much did the °F reading go up?

WORKING

Answer _____

12 Write 0 °F in °C, to one decimal place.

WORKING

Answer _____

13 Write 60 °F in °C, to one decimal place.

WORKING

Answer _____

14 A recipe oven is set to 180 °C. What is that in °F?

WORKING

Answer _____

15 Write 37 °C in °F.

WORKING

Answer _____

16 A thermostat is turned up from 68 °F to 72 °F. How many °C is that change, to one decimal place?

WORKING

Answer _____

17 A pipe warms from 15 °C to 60 °C. Find the rise in °C and in °F.

WORKING

Answer _____

WORKSHEET 7D

Carpentry: code values, board feet and area

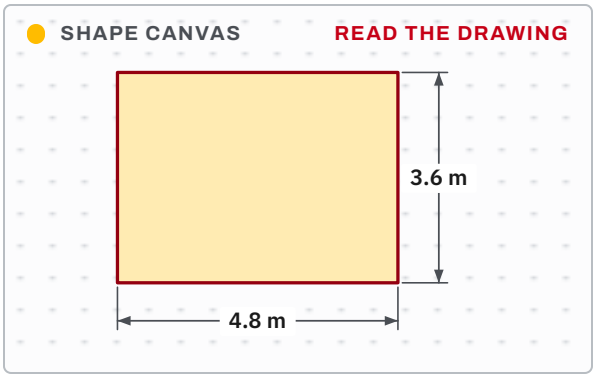
Convert without changing the code value

Name _____

Date _____

Code values are quoted from the BC Building Code 2024, Division B. Round a converted maximum down and a converted minimum up, to the nearest $\frac{1}{16}$ in. Use 1 ft² = 0.092 903 04 m² and board feet = thickness (in) × width (in) × length (ft) ÷ 12.

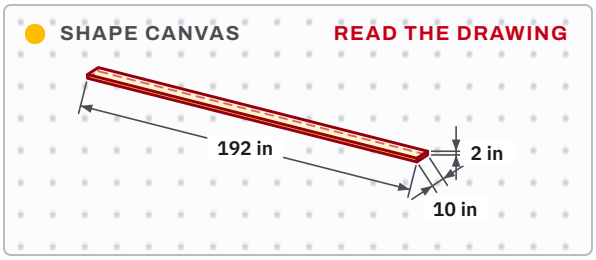
1 A room is drawn with metric sizes. Find its area in m², then in ft² to two decimal places.



WORKING

Answer _____

2 Find the board feet in one piece, to two decimal places. Then find the board feet in 30 pieces.



WORKING

Answer _____

3 The maximum rise for public stairs is 180 mm (Table 9.8.4.1, page 9-31). Give the largest tape reading, to a sixteenth, that stays within it.

WORKING

Answer _____

4 Stairs serving a single dwelling unit need a clear height of at least 1 950 mm (Sentence 9.8.2.2.(3), page 9-30). Write it in feet and inches, rounded up to a sixteenth.

WORKING

Answer _____

5 Required handrails must be 865 mm to 1 070 mm high (Sentence 9.8.7.4.(2), page 9-36). Write the range in inches, rounded to a sixteenth so that it stays inside the range.

WORKING

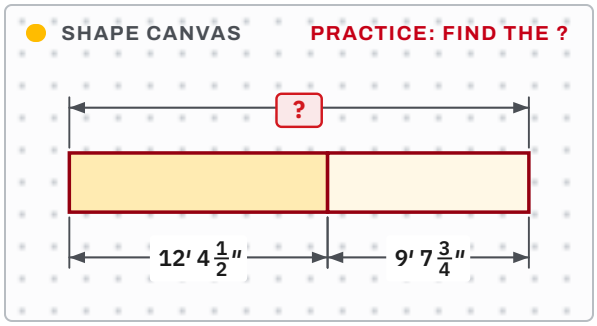
Answer _____

6 A layout gives a wall as 14.65 ft. Write it in feet, inches and sixteenths.

WORKING

Answer _____

7 Find the total length of the rim joist pieces in feet and inches, in decimal feet to two places, and in millimetres.



WORKING

Answer _____

8 A sheet of sheathing is 4 ft × 8 ft. A roof needs 46 sheets. What is the area covered in ft² and in m², to two decimal places?

WORKING

Answer _____

9 The minimum rise for private stairs is 125 mm (Table 9.8.4.1, page 9-31). Give the smallest tape reading, to a sixteenth, that meets it.

WORKING

Answer _____

10 The maximum run for private stairs is 355 mm (Table 9.8.4.2, page 9-31). Give the largest tape reading, to a sixteenth, that stays within it.

WORKING

Answer _____

11 A lumber order is 40 pieces 2 in × 6 in × 12 ft and 20 pieces 2 in × 10 in × 14 ft. How many board feet in all, to the nearest whole board foot?

WORKING

Answer _____

12 Write 9′ $5\frac{1}{4}$″ in decimal feet, to two decimal places.

WORKING

Answer _____

CHAPTER

8

Formulas and Transposing

A formula is a short rule that tells you how to work something out. Shop drawings, code books and manuals are full of them: the area of a plate, the distance around a pipe, the volume of a tank.

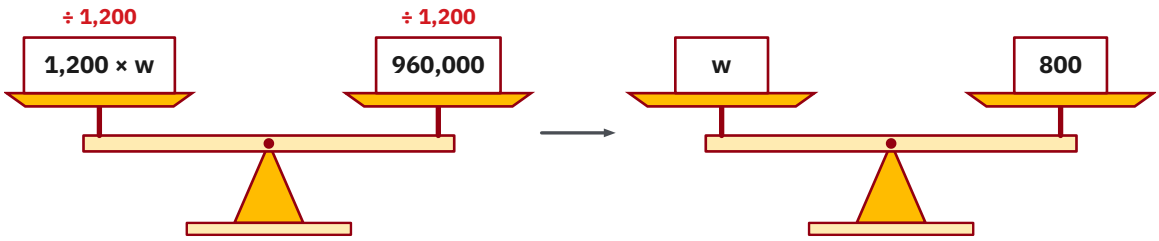
This chapter shows you how to read a formula, put your own numbers into it, and turn it around to find a different value. Turning a formula around is called transposing. It lets you start from the answer you need, such as an area, and find the size that gives it. Every result is checked by putting it back into the formula.

In this chapter you will

- Read a formula and name its subject, variables and constants
- Substitute values into a formula and work it out with BEDMAS
- Use inverse operations to keep an equation balanced
- Transpose one-step and two-step formulas
- Transpose formulas with squares and square roots, such as $A = \pi \times r^2$
- Check every answer by substituting it back

Key words

- formula
- variable
- constant
- subject of a formula
- substitute
- inverse operation
- transpose



8.1 Reading a formula

A **formula** is a rule written with letters and symbols. The formula for the area of a rectangle is $A = l \times w$. It says: the area A equals the length l times the width w.

A letter that stands for a quantity that can change is a **variable**. A value that never changes is a **constant**. In $C = \pi \times D$, C and D are variables and π is a constant.

The letter on its own on the left of the equals sign is the **subject** of the formula. The formula tells you how to work out the subject from the other values.

Formula	What it works out	What the letters mean
$A = l \times w$	Area of a rectangle	A area, l length, w width
$P = 2 \times (l + w)$	Perimeter of a rectangle	P perimeter (distance around)
$C = \pi \times D$	Distance around a circle	C circumference, D diameter
$A = \pi \times r^2$	Area of a circle	r radius (half the diameter)
$V = l \times w \times h$	Volume of a box shape	V volume, h height
$t_F = 1.8 \times t_C + 32$	Temperature in °F from °C	t_F degrees Fahrenheit, t_C degrees Celsius

Table 8.1 Formulas used in this chapter. Chapters 9, 10 and 11 explain where the shape formulas come from.

Some books leave out the × sign: lw means $l \times w$ and 2l means $2 \times l$. A small raised 2 means squared: r^2 means $r \times r$. This book writes the × sign every time.

TRY IT 8.1

1. What is the subject of $V = l \times w \times h$?
2. Name the constant in $A = \pi \times r^2$.
3. Name the variables in $C = \pi \times D$.
4. Rewrite 2lw with × signs.
5. What does s^2 mean?
6. Name the two constants in $t_F = 1.8 \times t_C + 32$.

Answers are in the Answer Key.

8.2 Substituting values

To **substitute** means to put a number in place of a letter. Write the formula, write it again with the numbers in place of the letters, then work it out using **BEDMAS** from Chapter 3.

1. Write the formula.
2. Write it again with each letter replaced by its value. Keep every bracket.
3. Work it out in BEDMAS order: brackets, exponents, division and multiplication, then addition and subtraction.
4. Write the answer with its unit.

Example 8.1 Area and perimeter of a plate

A plate is 1,200 mm long and 800 mm wide. Use A = l × w to find its area and P = 2 × (l + w) to find the distance around it.

SOLUTION

1. Area: A = 1,200 × 800 = 960,000.
2. Perimeter: P = 2 × (1,200 + 800). Brackets first, 1,200 + 800 = 2,000. Then 2 × 2,000 = 4,000.

CALCULATOR

TI-30Xa

Area

[1200] [×] [800] [=]

Display 960000.

Perimeter

[2] [×] [(] [1200] [+] [800] [)] [=]

Display 4000.

Sharp EL-531XT

Area

[1200] [×] [800] [=]

Display 960’000.

Perimeter

[2] [×] [(] [1200] [+] [800] [)] [=]

Display 4’000.

Key the brackets. Without them the calculator works 2 × 1,200 first and then adds 800.

Answer **A = 960,000 mm² (square millimetres) and P = 4,000 mm.**

Example 8.2 Formulas with π and a square

A disc has a diameter of 150 mm, so its radius is 75 mm. Use C = π × D to find the distance around it and A = π × r² to find its area. Round to two decimal places.

SOLUTION

1. Distance around: C = π × 150 = 471.24. Use the π key. It holds many more digits than 3.14.
2. Area: A = π × 75². The exponent comes first, so square only the 75: 75² = 5,625. Then π × 5,625 = 17,671.46.

CALCULATOR

TI-30Xa

Distance around

π × 150 =

Display 471.238898

Area

π × 75 x² =

Display 17671.45868

Sharp EL-531XT

Distance around

π × 150 =

Display 471.238898

Area

π × 75 x² =

Display 17'671.45868

The x² key squares only the number just before it, so the calculator follows BEDMAS for you.

Answer **C = 471.24 mm and A = 17,671.46 mm².**

Example 8.3 Converting a temperature

The formula $t_F = 1.8 \times t_C + 32$ changes a Celsius temperature to Fahrenheit. Change 20 °C to °F.

SOLUTION

1. $t_F = 1.8 \times 20 + 32$
2. Multiply first: 1.8 × 20 = 36. Then add: 36 + 32 = 68.

Answer **68 °F**

Electrical work uses formulas too. The two below link voltage, current, resistance and power. Substitute into them the same way: write the formula, put the numbers in place of the letters, then work it out.

ELECTRICAL FORMULAS

$$V = I \times R$$

$$P = V \times I$$

V voltage in volts (V), I current in amperes (A), R resistance in ohms (Ω), P power in watts (W). V = I × R is called Ohm's law. Section 8.9 works both formulas and turns them around.

USE THE SAME UNITS

Every length in a formula must be in the same unit. A plate 1.2 m long and 800 mm wide must be changed to 1,200 mm by 800 mm, or 1.2 m by 0.8 m, before you multiply. Chapter 7 shows how.

TRY IT 8.2

1. $A = l \times w$ with $l = 2.4$ m and $w = 1.2$ m.
2. $P = 2 \times (l + w)$ with $l = 600$ mm and $w = 450$ mm.
3. $C = \pi \times D$ with $D = 300$ mm. Round to two decimal places.
4. $A = \pi \times r^2$ with $r = 20$ cm. Round to two decimal places.
5. $V = l \times w \times h$ with $l = 2$ m, $w = 1.5$ m and $h = 0.5$ m.
6. $t_F = 1.8 \times t_C + 32$ with $t_C = 35$ °C.

Answers are in the Answer Key.

8.3 Keeping an equation balanced

An equals sign says that both sides have the same value. Think of a balance scale that is level. If you take something off one pan, you must take the same off the other pan, or the scale tips. The same rule keeps an equation true: whatever you do to one side, do to the other side.

To find an unknown value you undo what has been done to it. Each operation has an **inverse operation** that undoes it.

If the letter has been	Undo it by	Example
added to (+ 250)	subtracting (– 250)	x + 250 – 250 = x
subtracted from (– 75)	adding (+ 75)	x – 75 + 75 = x
multiplied (× 4)	dividing (÷ 4)	4 × x ÷ 4 = x
divided (÷ 3)	multiplying (× 3)	x ÷ 3 × 3 = x
squared (x^2)	taking the square root (√)	$\sqrt{(x^2)} = x$

Table 8.2 Operations and their inverses.

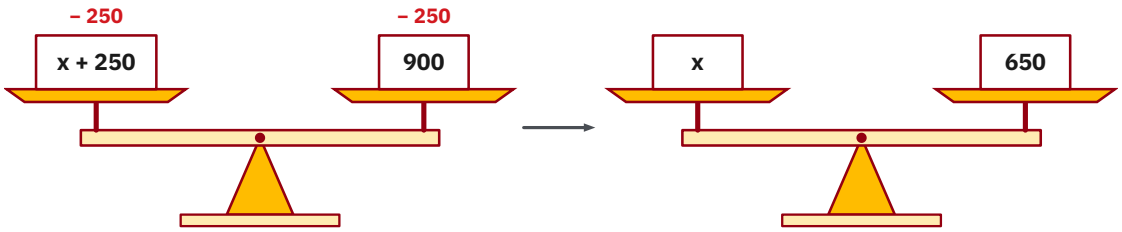
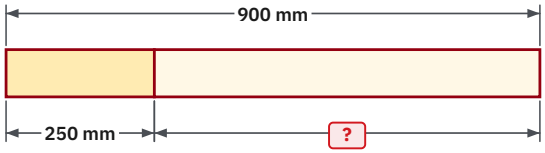


Figure 8.1 Take 250 off both pans and the scale stays level. The unknown x is left on its own.

Example 8.4 Undoing an operation

A 900 mm bar is cut into a 250 mm piece and a second piece, so x + 250 = 900. Solve it. Then solve 4 × x = 1,000 and x ÷ 3 = 125.



SOLUTION

1. 250 was added to x. Subtract 250 from both sides: x = 900 − 250 = 650. Check: 650 + 250 = 900.
2. x was multiplied by 4. Divide both sides by 4: x = 1,000 ÷ 4 = 250. Check: 4 × 250 = 1,000.
3. x was divided by 3. Multiply both sides by 3: x = 125 × 3 = 375. Check: 375 ÷ 3 = 125.

Answer **The second piece is 650 mm. Then x = 250 and x = 375.**

KEY IDEA

Do the same thing to both sides. Pick the inverse operation that leaves the letter on its own.

TRY IT 8.3

1. Solve x + 38 = 120.
2. Solve x − 45 = 155.
3. Solve 6 × x = 1,500.
4. Solve x ÷ 4 = 35.
5. Solve x + 1,275 = 6,000.
6. Solve 12 × x = 3,000.

Answers are in the Answer Key.

8.4 Transposing one-step formulas

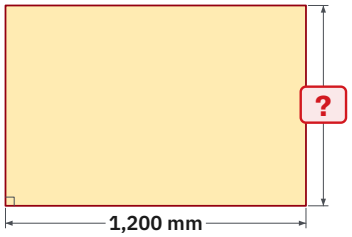
To **transpose** a formula means to rearrange it so that a different letter becomes the subject. You use the balance rule from the last section. The letters stay letters until the end, then you substitute the numbers.

1. Find the letter you want on its own.
2. Ask what has been done to it.
3. Undo that with the inverse operation, on both sides.
4. Write the new formula with the letter you want on the left.
5. Substitute the numbers, work it out, then check.

In the Shape Canvas, change one size and the numbers that depend on it update. Transposing lets you work the other way, from a value you need, such as an area, back to the size that gives it.

Example 8.5 Given the area, find the width

A rectangular plate must have an area of 960,000 mm². It is 1,200 mm long. Transpose A = l × w to find the width.



SOLUTION

1. w has been multiplied by l. Divide both sides by l.
2. A ÷ l = l × w ÷ l, so w = A ÷ l.
3. w = 960,000 ÷ 1,200 = 800

CALCULATOR

TI-30Xa

960000 ÷ 1200 =

Display 800.

Sharp EL-531XT

960000 ÷ 1200 =

Display 800.

Answer **w = 800 mm. Check: 1,200 × 800 = 960,000.**

Example 8.6 Given the distance around, find the diameter

A tape wrapped once around a round tank reads 1,000 mm. Transpose C = π × D and find the diameter to two decimal places.

SOLUTION

1. D has been multiplied by π. Divide both sides by π, so D = C ÷ π.
2. D = 1,000 ÷ π = 318.31

CALCULATOR

TI-30Xa

Diameter

1000 ÷ π =

Display 318.3098862

Check with the rounded answer

318.31 × π =

Display 1000.000358

Sharp EL-531XT

Diameter

1000 ÷ π =

Display 318.3098862

Check with the rounded answer

318.31 × π =

Display 1’000.000358

Answer **D = 318.31 mm. The check gives 1,000.00 mm to two decimal places, so the answer is right.**

Example 8.7 Find the height of a tank

A tank with a rectangular base 1.2 m long and 0.8 m wide must hold 0.576 m³. Transpose V = l × w × h to find the height.

SOLUTION

1. h has been multiplied by l and by w. Divide both sides by l × w, so h = V ÷ (l × w).
2. h = 0.576 ÷ (1.2 × 0.8) = 0.576 ÷ 0.96 = 0.6

CALCULATOR

TI-30Xa

0.576 ÷ (1.2 × 0.8) =

Display 0.6

Sharp EL-531XT

0.576 ÷ (1.2 × 0.8) =

Display 0.6

The brackets matter. Keyed as 0.576 ÷ 1.2 × 0.8, both calculators divide by 1.2 and then multiply by 0.8, which gives 0.384.

Answer **0.6 m. Check: 1.2 × 0.8 × 0.6 = 0.576.**

TRY IT 8.4

1. Make l the subject of A = l × w.
2. A = 2.88 m² and w = 1.2 m. Find l.
3. A sheet has an area of 1,152 in² and is 24 in wide. Find its length.
4. C = 942.48 mm. Find D to the nearest millimetre.
5. D = 2 × r. D = 168 mm. Find r.
6. V = 1.5 m³, l = 2 m and w = 1.5 m. Find h.
7. A square has P = 4 × s. P = 2,400 mm. Find s.

Answers are in the Answer Key.

Ohm's law turns around the same way: I = V ÷ R and R = V ÷ I. Section 8.9 has worked examples.

8.5 Transposing two-step formulas

In a two-step formula, two things have been done to the letter you want. Undo them in the reverse order. The last thing done is the first thing you undo, just as you take off your boots before your socks.

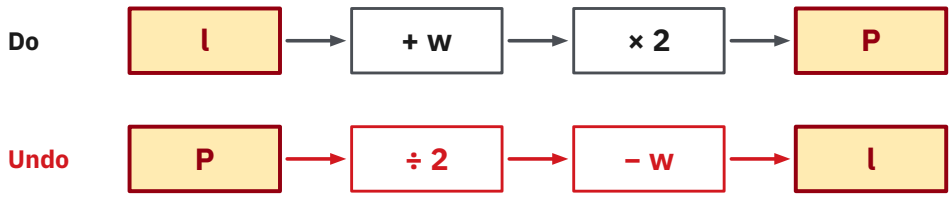


Figure 8.2 In P = 2 × (l + w), l has w added and then is multiplied by 2. To get l back, divide by 2 and then subtract w.

Example 8.8 Find the length from the perimeter

A rectangular frame uses 5,000 mm of flat bar all the way around. It is 900 mm wide. Transpose P = 2 × (l + w) to find the length.

SOLUTION

1. Undo the × 2: divide both sides by 2. P ÷ 2 = l + w
2. Undo the + w: subtract w from both sides. l = P ÷ 2 − w
3. l = 5,000 ÷ 2 − 900 = 2,500 − 900 = 1,600

CALCULATOR

TI-30Xa

5000 ÷ 2 − 900 =

Display 1600.

Sharp EL-531XT

5000 ÷ 2 − 900 =

Display 1’600.

Answer **1,600 mm. Check: 2 × (1,600 + 900) = 5,000.**

Example 8.9 Fahrenheit to Celsius

Transpose $t_F = 1.8 \times t_C + 32$ to make t_C the subject. Then change 350 °F to °C, to two decimal places.

SOLUTION

1. t_C was multiplied by 1.8 and then 32 was added. Undo the + 32 first: $t_F - 32 = 1.8 \times t_C$
2. Then undo the × 1.8: $t_C = (t_F - 32) \div 1.8$
3. $t_C = (350 - 32) \div 1.8 = 318 \div 1.8 = 176.67$

CALCULATOR

TI-30Xa

(350 − 32) ÷ 1.8 =

Display 176.6666667

Sharp EL-531XT

(350 − 32) ÷ 1.8 =

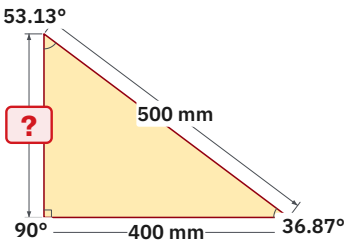
Display 176.6666667

Keyed without brackets, 350 − 32 ÷ 1.8 gives 332.22, because the calculator divides first.

Answer **176.67 °C. NIST publishes the same conversion: $t_C = (t_F - 32) / 1.8$.**

Example 8.10 Height of a gusset

A triangular gusset has an area given by A = b × h ÷ 2. Its base is 400 mm and its area must be 60,000 mm². Find the height.



SOLUTION

1. Undo the ÷ 2: 2 × A = b × h
2. Undo the × b: h = 2 × A ÷ b
3. h = 2 × 60,000 ÷ 400 = 300

CALCULATOR

TI-30Xa

2 × 60000 ÷ 400 =

Display 300.

Sharp EL-531XT

2 × 60000 ÷ 400 =

Display 300.

Answer **300 mm. Check: 400 × 300 ÷ 2 = 60,000.**

TRY IT 8.5

1. P = 2 × (l + w). P = 3,600 mm and l = 1,100 mm. Find w.
2. Make w the subject of P = 2 × (l + w).
3. Change 212 °F to °C.
4. Change −40 °F to °C.
5. An invoice uses T = 60 + 85 × h, with T in dollars and h in hours. T = $740. Find h.
6. A = b × h ÷ 2. A = 6,000 mm² and b = 120 mm. Find h.

Answers are in the Answer Key.

8.6 Formulas with squares and square roots

Squaring and taking the **square root** undo each other. If a letter has been squared, take the square root of both sides to get it back.

Example 8.11 Side of a square plate

A square plate must have an area of 62,500 mm². Transpose $A = s^2$ and find the side.

SOLUTION

1. s has been squared. Take the square root of both sides, so $s = \sqrt{A}$.
2. $s = \sqrt{62{,}500} = 250$

CALCULATOR

TI-30Xa	Sharp EL-531XT
62500 √x	√ 62500 =
Display 250.	Display 250.

Answer **250 mm. Check: $250^2 = 250 \times 250 = 62{,}500$.**

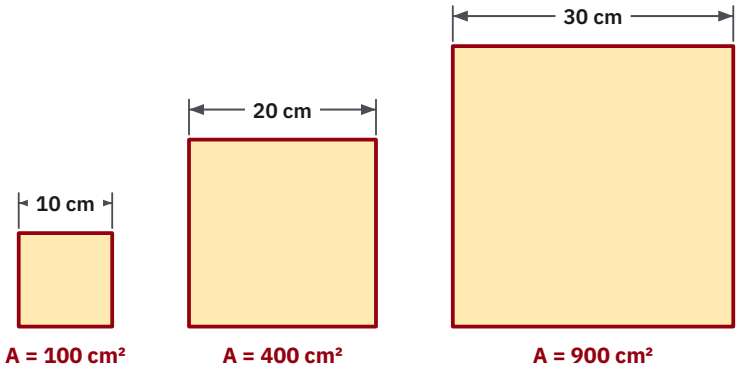


Figure 8.3 Change the side of a square and its area changes with it. Four times the area goes with twice the side. Nine times the area goes with three times the side.

In $A = \pi \times r^2$, two things are done to r: it is squared, then multiplied by π. Undo them in reverse order. Divide by π first, then take the square root.

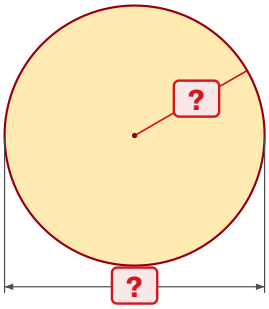
RADIUS FROM THE AREA OF A CIRCLE

$$r = \sqrt{(A \div \pi)} \qquad D = 2 \times r$$

Divide by π first, then take the square root. Double r to get the diameter.

Example 8.12 Given the area, find the radius

A round plate must have an area of 50,000 mm². Find its radius and its diameter, to two decimal places.



SOLUTION

1. Undo the × π: A ÷ π = r²
2. Undo the square: r = √(A ÷ π)
3. r = √(50,000 ÷ π) = 126.16
4. D = 2 × r = 2 × 126.1566… = 252.31 (use the unrounded r)

CALCULATOR

TI-30Xa

Radius

50000 ÷ π = √x

Display 126.1566261

Diameter

50000 ÷ π = √x × 2 =

Display 252.3132522

Sharp EL-531XT

Radius

√ (50000 ÷ π) =

Display 126.1566261

Diameter

√ (50000 ÷ π) × 2 =

Display 252.3132522

The calculator keeps every digit of r, so the diameter comes from the unrounded r.

Answer **r = 126.16 mm and D = 252.31 mm. Check: π × 126.16² = 50,002.7, very close to 50,000.**

Example 8.13 An imperial cover plate

A round cover plate must have an area of 1,000 in². Find its diameter to the nearest 1/16 in.

SOLUTION

1. r = √(1,000 ÷ π) = 17.841 in, so D = 2 × r = 35.682 in.
2. Change the decimal part to sixteenths: 0.682 × 16 = 10.91, which rounds to 11.

CALCULATOR

TI-30Xa

1000 ÷ π = √x × 2 =

Display 35.68248232

Sharp EL-531XT

√ (1000 ÷ π) × 2 =

Display 35.68248232

Answer **D = $35\frac{11}{16}$″**

DIVIDE BY Π BEFORE THE SQUARE ROOT

In r = √(A ÷ π) the division is inside the root. Taking √50,000 first and then dividing by π gives 71.18, which is wrong. On the TI-30Xa press = before √x. On the Sharp put the division in brackets after √.

TRY IT 8.6

1. A square has an area of 1,600 cm². Find its side.
2. A square has an area of 0.36 m². Find its side.
3. A circle has an area of 31,416 mm². Find r to the nearest millimetre.
4. A circle has an area of 2,000 mm². Find D to two decimal places.
5. A round tank has V = π × r² × h, r = 0.5 m and V = 1.5 m³. Find h to two decimal places.
6. The area of a square is made 9 times as big. What happens to its side?

Answers are in the Answer Key.

8.7 Checking by substituting back

A transposed formula can look right and still be wrong. The check takes a few seconds: put your answer back into the formula you started with, together with the values you were given. If the formula works out to the value you were given, your answer is right. If you rounded the answer, the check comes out close rather than exact.

Example 8.14 Catching a mistake

A learner transposes V = l × w × h and writes h = V × l ÷ w. With V = 0.45 m³, l = 1.5 m and w = 0.6 m this gives h = 1.125 m. Check the answer and correct it.

SOLUTION

1. Substitute back: 1.5 × 0.6 × 1.125 = 1.0125. That is not 0.45, so the answer is wrong.
2. The mistake: l multiplies h, so it must divide. h = V ÷ (l × w) = 0.45 ÷ (1.5 × 0.6) = 0.5
3. Substitute back: 1.5 × 0.6 × 0.5 = 0.45. It checks.

CALCULATOR

TI-30Xa

0.45 ÷ (1.5 × 0.6) =

Display 0.5

Sharp EL-531XT

0.45 ÷ (1.5 × 0.6) =

Display 0.5

Answer h = 0.5 m

Also ask whether the size is reasonable before you cut. A tank 1.5 m long and 0.6 m wide that holds less than half a cubic metre cannot be more than a metre tall.

KEY IDEA

Transpose with letters, substitute the numbers, then check by putting the answer back into the original formula.

8.8 From words to formulas and equations

On the job, math usually arrives as words: "the offcut is 315 mm shorter than the bar", "the bill is a $75 call-out plus $95 an hour". To work with it, turn the words into symbols.

An **expression** is a group of numbers, letters and operation signs with no equals sign, such as x + 250 or 3 × l. An **equation** says that two quantities are equal, so it has an equals sign: x + 250 = 900. A formula is an equation that works as a rule for any values you put in.

Operation	Key words	Example	In symbols
+ add	plus, more than, longer than, added to, increased by, total, sum, altogether	250 mm longer than the bar	b + 250
− subtract	minus, less than, shorter than, take away, decreased by, difference, left over	315 mm shorter than the bar	b − 315
× multiply	times, product, double, twice, triple	twice the width	2 × w
÷ divide	divided by, shared equally, cut into equal pieces, half, quarter	a bar cut into 4 equal pieces	b ÷ 4
= equals	is, equals, gives, comes to, totals, makes	the total is 900 mm	… = 900

Table 8.3 Key words and the operations they usually mean. Read the whole sentence: a key word is a clue, not a rule.

LESS THAN TURNS THE ORDER AROUND

"75 less than x" is x − 75, not 75 − x. Start from the amount that something is taken from. In the same way, "315 mm shorter than the bar" is b − 315.

To turn a story into an equation and solve it, work through these steps. Example 8.15 starts with the first part: writing expressions.

1. Read the story to the end. Find what you are asked for.
2. Choose a letter for the unknown and say what it stands for, with its unit.
3. Turn the key words into symbols and write the equation. The equals sign goes where the story says is, totals or comes to.
4. Solve with inverse operations, as in Sections 8.3 to 8.5.
5. Check the answer in the story, not only in the equation.
6. Write the answer as a sentence with its unit.

Example 8.15 Writing expressions

Let L be the length of a bar in millimetres. Write an expression for each: (a) 150 mm longer than the bar, (b) the bar less a 3 mm saw cut, (c) twice the bar, (d) the bar cut into 5 equal pieces, (e) 4 bars and a 600 mm offcut.

SOLUTION

1. (a) "longer than" adds: L + 150.
2. (b) "less" takes 3 away from the bar: L − 3.
3. (c) "twice" multiplies by 2: 2 × L.
4. (d) "cut into equal pieces" divides: L ÷ 5.
5. (e) 4 bars is 4 × L. Then add the offcut: 4 × L + 600.

Answer **L + 150, L − 3, 2 × L, L ÷ 5 and 4 × L + 600.**

Example 8.16 One-step equations from a story

(a) A piece is cut from a 2,440 mm board. The piece left over measures 1,685 mm. Ignore the saw cut. How long is the piece that was cut off? (b) Four equal shelves use 3,360 mm of board. How long is each shelf?

SOLUTION

1. (a) Let x be the piece cut off, in mm. The two pieces make the board: x + 1,685 = 2,440. Subtract 1,685 from both sides: x = 2,440 − 1,685 = 755. Check in the story: 755 + 1,685 = 2,440 mm.
2. (b) Let s be one shelf, in mm. Four shelves: 4 × s = 3,360. Divide both sides by 4: s = 3,360 ÷ 4 = 840. Check: 4 × 840 = 3,360 mm.

CALCULATOR

TI-30Xa

Piece cut off

2440 − 1685 =

Display 755.

One shelf

3360 ÷ 4 =

Display 840.

Sharp EL-531XT

Piece cut off

2440 − 1685 =

Display 755.

One shelf

3360 ÷ 4 =

Display 840.

Answer **The piece cut off is 755 mm. Each shelf is 840 mm.**

Example 8.17 A two-step equation from a story

A fence has a 1,200 mm gate and n boards, each 140 mm wide, set edge to edge. The whole fence is 3,580 mm long. How many boards are there?

SOLUTION

1. Let n be the number of boards. The boards give 140 × n, and the gate adds 1,200: 140 × n + 1,200 = 3,580.
2. Undo in reverse order. Undo the + 1,200 first: 140 × n = 3,580 − 1,200 = 2,380.
3. Then undo the × 140: n = 2,380 ÷ 140 = 17.
4. Check in the story: 140 × 17 + 1,200 = 2,380 + 1,200 = 3,580 mm.

CALCULATOR

TI-30Xa

(3580 − 1200) ÷ 140 =

Display 17.

Sharp EL-531XT

(3580 − 1200) ÷ 140 =

Display 17.

The brackets make the calculator subtract before it divides, the same order as the solution.

Answer **17 boards.**

Example 8.18 A bill written as an equation

A mobile welder charges a $75 call-out fee plus $95 an hour. The bill comes to $502.50. How many hours did the job take?

SOLUTION

1. Let h be the hours. "$95 an hour" means $95 for each hour, so the hours cost 95 × h. Add the call-out fee: 75 + 95 × h = 502.50.
2. Undo the + 75: 95 × h = 502.50 − 75 = 427.50.
3. Undo the × 95: h = 427.50 ÷ 95 = 4.5.
4. Check: 75 + 95 × 4.5 = 75 + 427.50 = $502.50.

CALCULATOR

TI-30Xa

(502.5 − 75) ÷ 95 =

Display 4.5

Sharp EL-531XT

(502.5 − 75) ÷ 95 =

Display 4.5

Answer **4.5 hours.**

KEY IDEA

Name the unknown, turn the key words into symbols, and put the equals sign where the story says is or totals. Solve, then check the answer in the story.

TRY IT 8.8

1. Write an expression for a length 75 mm shorter than a bar of length b.
2. Write an expression for the cost of n bolts at $0.85 each plus $12 delivery.
3. Write an expression for a 6,000 mm length cut into k equal pieces.
4. A pipe run needs 2,750 mm of pipe. 1,980 mm is already in place. Write and solve an equation for the length still needed.
5. Six equal studs use 14,640 mm of lumber. Ignore saw cuts. Write and solve an equation for the length of one stud.
6. Three times a length, plus 250 mm, is 1,600 mm. Find the length.
7. A plumber charges $90 to come out plus $110 an hour. The bill is $640. How many hours were worked?
8. Half a bar, less a 3 mm saw cut, is 1,497 mm. How long is the bar?

Answers are in the Answer Key.

8.9 Electrical formulas

Section 8.2 gave two electrical formulas: Ohm's law, $V = I \times R$, and the power formula, $P = V \times I$. Here they are used as math only: substitute, transpose and check, the same skills as the rest of this chapter.

The formulas follow from how the units are defined. NIST (the US National Institute of Standards and Technology) gives 1 Ω = 1 V/A, one ohm is one volt per ampere, and 1 V = 1 W/A, one volt is one watt per ampere.

Example 8.19 Ohm's law and power

A heating coil on a test bench has a resistance of 12 Ω and carries a current of 1.5 A. Use $V = I \times R$ to find the voltage across it. Then use $P = V \times I$ to find the power.

SOLUTION

1. $V = I \times R = 1.5 \times 12 = 18$ V.
2. $P = V \times I = 18 \times 1.5 = 27$ W. Use the voltage you just found.

CALCULATOR

TI-30Xa

Voltage

1.5 × 12 =

Display 18.

Power

18 × 1.5 =

Display 27.

Sharp EL-531XT

Voltage

1.5 × 12 =

Display 18.

Power

18 × 1.5 =

Display 27.

Answer **V = 18 V and P = 27 W.**

Example 8.20 Ohm's law turned around

Make I the subject of V = I × R and find the current when V = 24 V and R = 16 Ω. Then make I the subject of P = V × I and find the current when P = 60 W and V = 12 V.

SOLUTION

1. In V = I × R, I has been multiplied by R. Divide both sides by R: I = V ÷ R.
2. I = 24 ÷ 16 = 1.5 A.
3. In P = V × I, I has been multiplied by V. Divide both sides by V: I = P ÷ V = 60 ÷ 12 = 5 A.

CALCULATOR

TI-30Xa

I = V ÷ R

24 ÷ 16 =

Display 1.5

I = P ÷ V

60 ÷ 12 =

Display 5.

Sharp EL-531XT

I = V ÷ R

24 ÷ 16 =

Display 1.5

I = P ÷ V

60 ÷ 12 =

Display 5.

Answer **I = 1.5 A, then I = 5 A. Check: 1.5 × 16 = 24 and 12 × 5 = 60.**

TRY IT 8.9

1. $V = I \times R$ with $I = 0.5$ A and $R = 48\ \Omega$. Find V.
2. $P = V \times I$ with $V = 24$ V and $I = 2.5$ A. Find P.
3. Make R the subject of $V = I \times R$. Find R when $V = 36$ V and $I = 3$ A.
4. Find I when $V = 9$ V and $R = 4.5\ \Omega$.
5. Make V the subject of $P = V \times I$. Find V when $P = 90$ W and $I = 7.5$ A.
6. Putting $V = I \times R$ into $P = V \times I$ gives $P = I^2 \times R$. Find P when $I = 2$ A and $R = 6\ \Omega$.

Answers are in the Answer Key.

8.10 Stair formulas and the BC Building Code

A carpenter laying out a stair starts from one measurement, the total rise: the height from the finished floor at the bottom to the finished floor at the top. Each step up is a riser. The height of one step is the unit rise. The run of one step is the unit run, and the horizontal length of the whole flight is the total run.

The BC Building Code (BCBC 2024) sets the limits. It says the rise is "measured as the vertical nosing-to-nosing distance" (Division B, Sentence 9.8.4.1.(1), page 9-30), and it defines run as "the horizontal distance between two adjacent tread nosings on a stair" (Division A, Sentence 1.4.1.2.(1), page 1-23). A flight is "a series of steps between landings" (Division A, page 1-18).

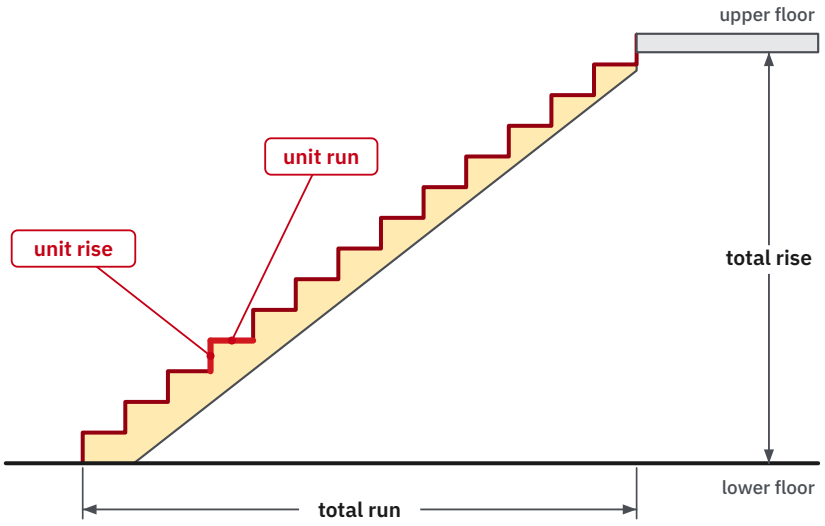


Figure 8.4 A straight flight of 14 risers from one floor to the next. The upper floor is the top step, so the flight has 13 treads. Chapter 12 finds the length of the stringer from the total rise and the total run with the Pythagorean theorem.

STAIR FORMULAS

number of risers = total rise ÷ maximum rise, rounded up to a whole number

unit rise = total rise ÷ number of risers

number of treads = number of risers − 1

total run = number of treads × unit run

Work in millimetres. Round the number of risers up, never down: rounding down makes each riser taller than the maximum.

Stair type	Rise, max.	Rise, min.	Run, max.	Run, min.
Private	200 mm	125 mm	355 mm	255 mm
Public	180 mm	125 mm	No limit	280 mm

Table 8.4 Rise and run for rectangular treads. BCBC 2024, Division B, Table 9.8.4.1 (Sentence 9.8.4.1.(1), page 9-30) and Table 9.8.4.2 (Sentence 9.8.4.2.(1), page 9-31).

The notes to both tables say that private stairs "are exterior and interior stairs that serve" single dwelling units, houses with a secondary suite including their common spaces, and garages that serve those houses. Public stairs "are all stairs not described as service stairs or private stairs." Article 9.8.3.3 adds that "the vertical height of any flight of stairs shall not exceed 3.7 m" (page 9-30).

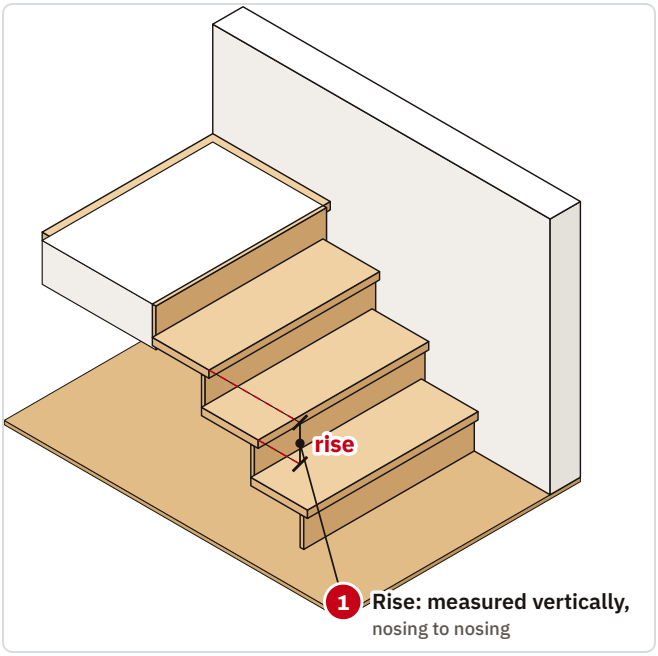


Figure 8.5 Part of plate 9.8.4.1. The rise of each step is measured vertically from one nosing to the next. The limits are in Table 8.4.

Source: BCBC 2024, Division B, Article 9.8.4.1. (page 9-30)

Example 8.21 Risers for a house stair

A private stair in a house has a total rise of 2,690 mm. Find the number of risers and the unit rise, to two decimal places. Check the unit rise against Table 9.8.4.1.

SOLUTION

1. Number of risers: 2,690 ÷ 200 = 13.45. Round up to 14.
2. Unit rise: 2,690 ÷ 14 = 192.14 mm.
3. Check: 192.14 mm is not more than 200 mm and not less than 125 mm, so it meets the table.
4. Why not 13 risers? 2,690 ÷ 13 = 206.92 mm, taller than the 200 mm maximum.

CALCULATOR

TI-30Xa

Risers

2690 ÷ 200 =

Display 13.45

Unit rise

2690 ÷ 14 =

Display 192.1428571

With 13 risers

2690 ÷ 13 =

Display 206.9230769

Sharp EL-531XT

Risers

2690 ÷ 200 =

Display 13.45

Unit rise

2690 ÷ 14 =

Display 192.1428571

With 13 risers

2690 ÷ 13 =

Display 206.9230769

Answer **14 risers of 192.14 mm.**

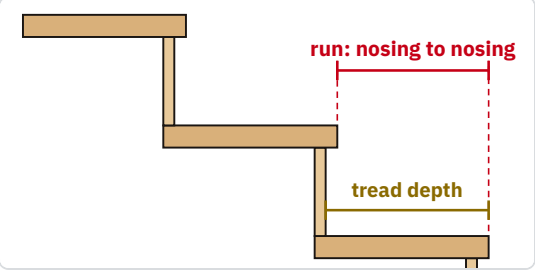


Figure 8.6 Part of plate 9.8.4.2. The run is measured horizontally from nosing to nosing. Sentence 9.8.4.2.(2) says the depth of a rectangular tread "shall be not less than its run and not more than its run plus 25 mm."

Source: BCBC 2024, Division B, Article 9.8.4.2. (page 9-31)

Example 8.22 Total run

The stair in Example 8.21 has 14 risers. The drawing gives a unit run of 260 mm. Check the unit run against Table 9.8.4.2, then find the total run.

SOLUTION

1. 260 mm is not less than 255 mm and not more than 355 mm, so it meets the table for a private stair.
2. Number of treads: 14 − 1 = 13.
3. Total run: 13 × 260 = 3,380 mm.

CALCULATOR

TI-30Xa

13 × 260 =

Display 3380.

Sharp EL-531XT

13 × 260 =

Display 3′380.

Answer **The total run is 3,380 mm.**

Example 8.23 A public stair

A public stair has a total rise of 3,150 mm. Check the height of the flight against Article 9.8.3.3. Find the number of risers and the unit rise. The drawing gives a unit run of 300 mm. Find the total run.

SOLUTION

1. 3,150 mm is less than 3.7 m = 3,700 mm, so one flight is allowed.
2. Risers: 3,150 ÷ 180 = 17.5. Round up to 18.
3. Unit rise: 3,150 ÷ 18 = 175 mm, between 125 mm and 180 mm.
4. The run of 300 mm is not less than 280 mm. Treads: 18 − 1 = 17. Total run: 17 × 300 = 5,100 mm.

CALCULATOR

TI-30Xa

Risers

3150 ÷ 180 =

Display 17.5

Unit rise

3150 ÷ 18 =

Display 175.

Total run

17 × 300 =

Display 5100.

Sharp EL-531XT

Risers

3150 ÷ 180 =

Display 17.5

Unit rise

3150 ÷ 18 =

Display 175.

Total run

17 × 300 =

Display 5’100.

Answer **18 risers of 175 mm and a total run of 5,100 mm.**

Every riser in a flight must be the same height, within a small tolerance. Sentence 9.8.4.4.(1) allows a maximum tolerance of "5 mm between adjacent treads or landings" and "10 mm between the tallest and shortest risers in a flight". Sentence 9.8.4.4.(3) sets the same 5 mm and 10 mm for the run of rectangular treads (page 9-32).

Example 8.24 Checking a built flight

The risers of a built flight measure, from the bottom: 188, 191, 190, 196, 189 and 191 mm. Do they meet Sentence 9.8.4.4.(1)?

SOLUTION

1. Adjacent risers: 191 − 188 = 3, 191 − 190 = 1, 196 − 190 = 6, 196 − 189 = 7, 191 − 189 = 2.
2. Two differences, 6 mm and 7 mm, are more than 5 mm.
3. Tallest and shortest: 196 − 188 = 8 mm, which is within 10 mm.

Answer **No. The flight meets the 10 mm rule but fails the 5 mm rule between adjacent risers, at the fourth riser.**

The stair formulas transpose like any other formula. From unit rise = total rise ÷ number of risers you get total rise = number of risers × unit rise, and from total run = number of treads × unit run you get unit run = total run ÷ number of treads.

TRY IT 8.10

1. A private stair has a total rise of 2,450 mm. Find the number of risers and the unit rise to two decimal places.
2. The stair in question 1 has a unit run of 280 mm. Find the total run.
3. A public stair has a total rise of 2,880 mm. Find the number of risers and the unit rise.
4. A stair has 15 risers of 186 mm. Find the total rise.
5. A private stair has 12 treads and a total run of 3,060 mm. Find the unit run. Does it meet Table 9.8.4.2?
6. Two adjacent risers measure 184 mm and 191 mm. Do they meet Sentence 9.8.4.4.(1)?

Answers are in the Answer Key.

WORKSHEET 8A

Given the area, find the side

Use the value you know to find the ?

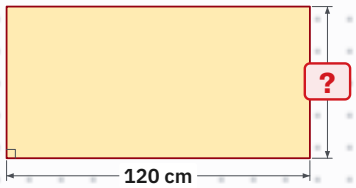
Name _____

Date _____

Transpose the formula first, then substitute. Check each answer by putting it back into the formula you started with. Round to two decimal places where the answer does not come out exactly.

1 The rectangle has an area of 7,200 cm² and a length of 120 cm. Use A = l × w to find the width.

SHAPE CANVAS PRACTICE: FIND THE ?

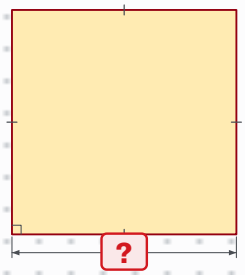


WORKING

Answer _____

2 The square has an area of 2,025 cm². Use A = s² to find the side.

SHAPE CANVAS PRACTICE: FIND THE ?

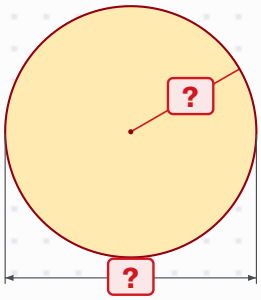


WORKING

Answer _____

3 The circle has an area of 5,000 mm². Find the radius and the diameter.

SHAPE CANVAS PRACTICE: FIND THE ?

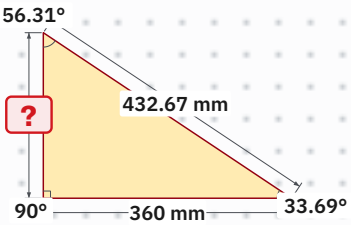


WORKING

Answer _____

4 The gusset has an area of 43,200 mm² and a base of 360 mm. Use A = b × h ÷ 2 to find the height.

SHAPE CANVAS PRACTICE: FIND THE ?

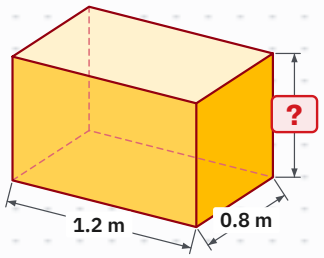


WORKING

Answer _____

5 The box holds 0.72 m³. Use V = l × w × h to find the height.

SHAPE CANVAS PRACTICE: FIND THE ?

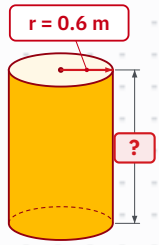


WORKING

Answer _____

6 The round tank holds 2 m³. Use V = π × r² × h to find the height.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

7 Change one value: the square in problem 2 is given an area 4 times as big, 8,100 cm². Find the new side. What happened to the side?

WORKING

Answer _____

8 Change one value: the rectangle in problem 1 keeps its area of 7,200 cm², but its length is doubled to 240 cm. Find the new width. What happened to the width?

WORKING

Answer _____

9 Find the mistake. A learner keys 0.72 ÷ 1.2 × 0.8 = to find the height of a box 1.2 m by 0.8 m that holds 0.72 m³, and gets 0.48 m. What went wrong? Find the correct height and check it.

WORKING

Answer _____

WORKSHEET 8B

Rearrange, solve and check

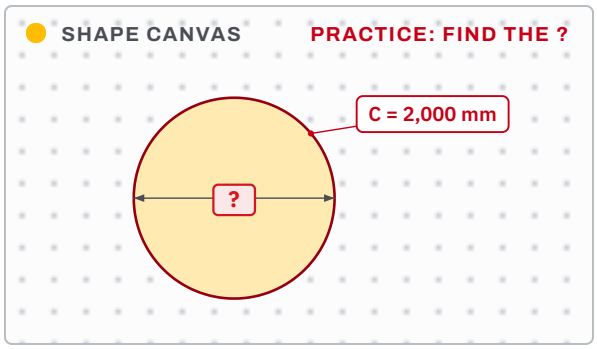
Transpose with letters first, then put in the numbers

Name _____

Date _____

Show the transposed formula, the substitution and the check. Round to two decimal places unless the problem says otherwise.

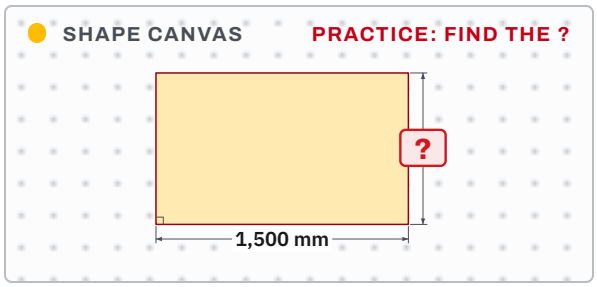
1 Make D the subject of C = π × D. Find the diameter of a pipe whose circumference is 2,000 mm.



WORKING

Answer _____

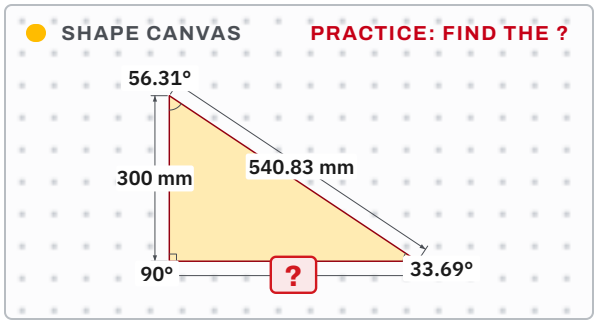
2 A frame uses 4,800 mm of bar all the way around and is 1,500 mm long. Use P = 2 × (l + w) to find the width.



WORKING

Answer _____

3 The gusset has an area of 67,500 mm² and a height of 300 mm. Make b the subject of A = b × h ÷ 2 and find the base.



WORKING

Answer _____

4 Make t_C the subject of $t_F = 1.8 \times t_C + 32$. Then change 450 °F to °C.

WORKING

Answer _____

5 An invoice uses T = 60 + 85 × h, with T in dollars and h in hours. The total is $527.50. How many hours were worked?

WORKING

Answer _____

CHAPTER

9

Perimeter and Circumference

How much flat bar goes around a guard? How long is the band that wraps a tank? How big a sheet rolls into a round duct? Each question asks for a distance around the outside of a shape.

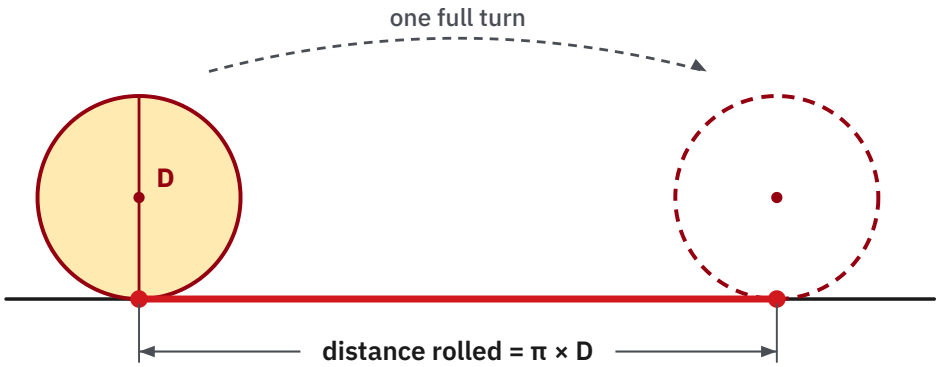
This chapter works out the distance around rectangles, triangles and other straight-sided shapes, and then around circles. The distance around a circle is always π × D, the number π times the diameter. The same formula gives the stretch-out, the length of flat sheet that rolls into a pipe or a round duct.

In this chapter you will

- Find the perimeter of straight-sided shapes
- Find a missing side from a perimeter
- Name the parts of a circle
- Use C = π × D, and find D from C
- Find the stretch-out of a pipe or duct
- Find arcs and composite perimeters

Key words

- perimeter
- radius
- diameter
- circumference
- pi (π)
- stretch-out
- arc



9.1 Perimeter of straight-sided shapes

The **perimeter** is the distance all the way around the outside of a flat shape. Picture walking along every edge until you are back where you started. Add up every edge you walked.

PERIMETER

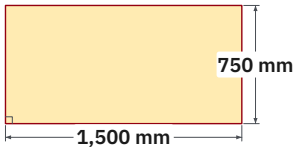
Rectangle: $P = 2 \times (l + w)$ **Square: $P = 4 \times s$**

Triangle: $P = a + b + c$ **Regular polygon: $P = n \times s$**

l length, w width, s side, a, b and c the three sides, n the number of equal sides

Example 9.1 A rectangular guard frame

A guard frame is 1,500 mm long and 750 mm wide. How much flat bar goes around it?



SOLUTION

1. Two lengths and two widths: $P = 2 \times (l + w)$.
2. $P = 2 \times (1{,}500 + 750) = 2 \times 2{,}250 = 4{,}500$

CALCULATOR

TI-30Xa

2 × (1500 + 750) =

Display 4500.

Sharp EL-531XT

2 × (1500 + 750) =

Display 4′500.

Answer **4,500 mm, which is 4.5 m.**

Example 9.2 A frame in fractions of an inch

A picture-style frame is $24\frac{3}{8}$ in by $18\frac{1}{2}$ in. Find the perimeter.

SOLUTION

1. Add a length and a width: $24\frac{3}{8} + 18\frac{1}{2} = 24\frac{3}{8} + 18\frac{4}{8} = 42\frac{7}{8}$ in.
2. Double it: $2 \times 42\frac{7}{8} = 84\frac{14}{8} = 85\frac{6}{8} = 85\frac{3}{4}$ in.

CALCULATOR

TI-30Xa

2 × (24 a b/c 3 a b/c 8 + 18 a b/c 1 a b/c 2) =

Display: 85⌟3⌟4

Sharp EL-531XT

2 × (24 a b/c 3 a b/c 8 + 18 a b/c 1 a b/c 2) =

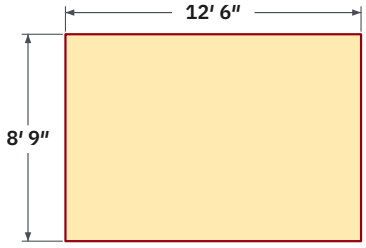
Display: 85 ⌜3 ⌜4

Both calculators show the answer as a mixed number: 85 and 3/4.

Answer $85\frac{3}{4}$ **in, which is 7′ 1**$\frac{3}{4}$**″.**

Example 9.3 Feet and inches

A concrete pad is 12 ft 6 in long and 8 ft 9 in wide. How much edge form goes around it? Give the answer in feet and inches.



SOLUTION

1. Change to inches. 12 ft 6 in = 12 × 12 + 6 = 150 in. 8 ft 9 in = 8 × 12 + 9 = 105 in.
2. P = 2 × (150 + 105) = 510 in.
3. 510 ÷ 12 = 42.5 ft, which is 42 ft and 0.5 × 12 = 6 in.

CALCULATOR

TI-30Xa

2 × (150 + 105) =

Display: 510.

Sharp EL-531XT

2 × (150 + 105) =

Display: 510.

Answer **42 ft 6 in**

When you know the perimeter and one side, transpose the formula as in Chapter 8. For a rectangle, w = P ÷ 2 − l. A rectangle with a perimeter of 3,000 mm and a length of 900 mm has a width of 3,000 ÷ 2 − 900 = 600 mm.

TRY IT 9.1

1. A square plate has sides of 850 mm. Find the perimeter.
2. A triangle has sides of 1.2 m, 0.9 m and 1.5 m. Find the perimeter.
3. A regular hexagon has sides of 125 mm. Find the perimeter.
4. A regular octagon has sides of 80 mm. Find the perimeter.
5. A rectangle is 3 ft 4 in by 2 ft 8 in. Find the perimeter in feet.
6. A rectangle has a perimeter of 2,600 mm and a length of 800 mm. Find the width.
7. A frame is $36\frac{1}{4}$ in by $24\frac{1}{2}$ in. Find the perimeter.

Answers are in the Answer Key.

9.2 The parts of a circle and π

The **radius** (r) is the distance from the centre of a circle to its edge. The **diameter** (D) is the distance across the circle through the centre, so it is two radii: D = 2 × r and r = D ÷ 2. The **circumference** (C) is the distance around the circle. It is the perimeter of a circle.

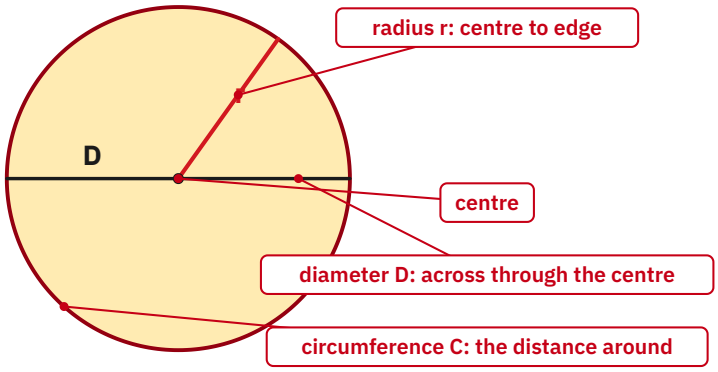


Figure 9.1 The parts of a circle.

Wrap a string once around any circle and lay it along the diameter. It fits a little more than 3 times, every time, for a washer or a tank. That number is **pi**, written π. It is about 3.14159, and its digits never end.

CALCULATOR The π key

TI-30Xa

π

Display 3.141592654

Sharp EL-531XT

π =

Display 3.141592654

Both calculators show π to 10 digits. Use the key every time instead of typing 3.14.

3.14 IS NOT Π

π × 600 = 1,884.96, but 3.14 × 600 = 1,884. The short value is almost 1 mm short on a 600 mm circle, and the error grows with the size of the job.

TRY IT 9.2

1. A circle has a radius of 45 mm. Find the diameter.
2. A pipe has a diameter of 168.3 mm. Find the radius.
3. A tank has a diameter of 2 ft. Find the radius in inches.
4. Which is longer, the diameter of a circle or its circumference? About how many times longer?

Answers are in the Answer Key.

9.3 Circumference

CIRCUMFERENCE

$$C = \pi \times D$$

If you know the radius, double it first to get D.

Example 9.4 From a diameter and from a radius

Find the circumference of a disc 450 mm in diameter, and of a ring with a radius of 125 mm. Round to two decimal places.

SOLUTION

1. Disc: C = π × 450 = 1,413.72.
2. Ring: double the radius first, D = 2 × 125 = 250. Then C = π × 250 = 785.40.

CALCULATOR

TI-30Xa

Disc

π × 450 =

Display 1413.716694

Ring

π × 250 =

Display 785.3981634

Sharp EL-531XT

Disc

π × 450 =

Display 1’413.716694

Ring

π × 250 =

Display 785.3981634

Answer **1,413.72 mm and 785.40 mm.**

Example 9.5 A band around a tank

A round tank is 48 in in diameter. How long is a band that goes once around it? Give the answer to the nearest 1/16 in.

SOLUTION

1. $C = \pi \times 48 = 150.796$ in.
2. Change the decimal part to sixteenths: $0.796 \times 16 = 12.74$, which rounds to 13.
3. $150\frac{13}{16}$ in is 12 ft $6\frac{13}{16}$ in, because $150 = 12 \times 12 + 6$.

CALCULATOR

TI-30Xa

[π] [×] [48] [=]

Display 150.7964474

Sharp EL-531XT

[π] [×] [48] [=]

Display 150.7964474

Answer **$150\frac{13}{16}$″, which is 12′ $6\frac{13}{16}$″.**

Example 9.6 Turns of a wheel

A wheel 600 mm in diameter rolls along a floor. In one turn it rolls one circumference. How far does it roll in 25 turns? How many turns does it make in 50 m?

SOLUTION

1. One turn: $\pi \times 600$ mm. 25 turns: $25 \times \pi \times 600 = 47{,}123.89$ mm, about 47.12 m.
2. Turns in 50 m = 50,000 mm: $50{,}000 \div (\pi \times 600) = 26.53$.

CALCULATOR

TI-30Xa

Distance

[25] [×] [π] [×] [600] [=]

Display 47123.8898

Turns

[50000] [÷] [(] [π] [×] [600] [)] [=]

Display 26.52582385

Sharp EL-531XT

Distance

[25] [×] [π] [×] [600] [=]

Display 47′123.8898

Turns

[50000] [÷] [(] [π] [×] [600] [)] [=]

Display 26.52582385

Answer **47,123.89 mm (47.12 m) and 26.53 turns.**

KEY IDEA

Double the diameter and the circumference doubles. Halve it and the circumference halves. C and D always keep the same ratio, π.

TRY IT 9.3

1. D = 200 mm. Find C to two decimal places.
2. r = 60 mm. Find C to two decimal places.
3. D = 10 in. Find C to the nearest 1/16 in.
4. A tape wrapped once around a pipe reads 2,500 mm. Find D to two decimal places.
5. C = 1,256.64 mm. Find D to the nearest millimetre.
6. A 500 mm wheel turns 10 times. How far does it roll, to two decimal places?
7. A circle with C = 1,000 mm has its diameter doubled. What is the new C?

Answers are in the Answer Key.

9.4 Stretch-out of a pipe or duct

Open out the side of a cylinder and it lies flat as a rectangle. One side of the rectangle is the length of the cylinder. The other side wrapped once around the cylinder, so it is the circumference. This length is the **stretch-out**. To make a round duct, you cut a sheet as long as the stretch-out and roll it.

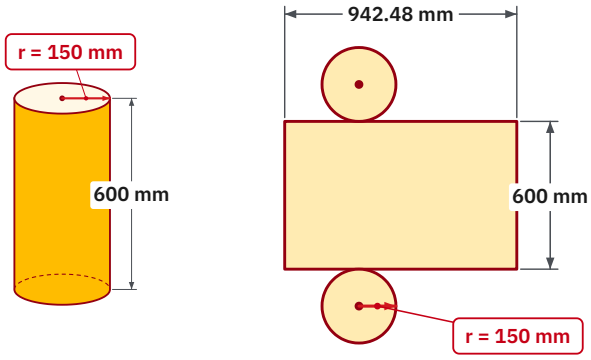


Figure 9.2 Unfold it. The cylinder on the left opens out to the net on the right. The stretch-out is π × D = π × 300 = 942.48 mm.

MEASURE AROUND WHEN YOU CAN

A tape wrapped once around a part reads the stretch-out directly. You need π only when you start from the diameter.

Example 9.7 Sheet for a round duct

A round duct is 300 mm in diameter and 1,200 mm long. The drawing adds 25 mm to the stretch-out for a lap seam. What size sheet is cut? Give the sheet length to the nearest millimetre.

SOLUTION

1. Stretch-out: π × 300 = 942.48 mm.
2. Add the seam allowance from the drawing: 942.48 + 25 = 967.48 mm.

CALCULATOR

TI-30Xa

π × 300 + 25 =

Display 967.4777961

Sharp EL-531XT

π × 300 + 25 =

Display 967.4777961

Answer **967 mm by 1,200 mm.**

Example 9.8 A jacket around a pipe

A sheet-metal jacket wraps around the outside of a pipe. The outside diameter is $6\frac{5}{8}$ in. Find the stretch-out to the nearest 1/16 in.

SOLUTION

1. The jacket wraps the outside surface, so use the outside diameter.
2. π × 6.625 = 20.813 in.
3. Sixteenths: 0.813 × 16 = 13.01, which rounds to 13.

CALCULATOR

TI-30Xa

π × 6 a b/c 5 a b/c 8 =

Display 20.81305133

Sharp EL-531XT

π × 6 a b/c 5 a b/c 8 =

Display 20.81305133

Answer $20\frac{13}{16}$″

TRY IT 9.4

1. A duct is 250 mm in diameter. Find the stretch-out to two decimal places.
2. A duct is 400 mm in diameter and the drawing adds 30 mm for the seam. Find the cut length to two decimal places.
3. A pipe has an outside diameter of $4\frac{1}{2}$ in. Find the stretch-out of a jacket to the nearest 1/16 in.
4. A stretch-out is 942.48 mm. What diameter does it roll into, to the nearest millimetre?

Answers are in the Answer Key.

9.5 Arcs

An **arc** is a part of the circumference. A 90° arc is a quarter of the circle, because a full turn is 360°. Find the fraction of the circle, then take that fraction of π × D.

ARC LENGTH

$$\text{arc} = \text{angle} \div 360 \times \pi \times D$$

Half circle: π × D ÷ 2. Quarter circle: π × D ÷ 4.

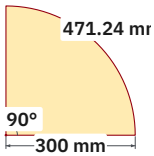


Figure 9.3 A 90° arc of a circle with a 300 mm radius. D = 600 mm, so the arc is 90 ÷ 360 × π × 600 = 471.24 mm.

Example 9.9 A curved handrail

A curved handrail follows a 120° arc of a circle 2,400 mm in diameter. How long is the rail? Round to two decimal places.

SOLUTION

1. 120° is 120 ÷ 360 = one third of the circle.
2. arc = 120 ÷ 360 × π × 2,400 = 2,513.27

CALCULATOR

TI-30Xa

120 ÷ 360 × π × 2400 =

Display 2513.274123

Sharp EL-531XT

120 ÷ 360 × π × 2400 =

Display 2’513.274123

Answer 2,513.27 mm

TRY IT 9.5

1. A half circle has D = 500 mm. Find the curved edge.
2. A quarter circle has r = 200 mm. Find the curved edge.
3. A 60° arc has D = 900 mm. Find its length.
4. A 270° arc has D = 100 mm. Find its length.

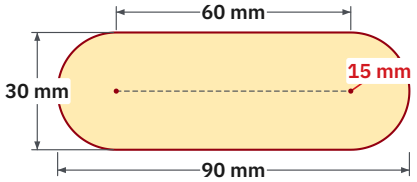
Answers are in the Answer Key.

9.6 Composite perimeters

A composite shape is made of simple parts. Break it down: split it into straight edges and arcs, find any missing lengths, then add only the edges on the outside. The dashed lines used to split a shape are not edges.

Example 9.10 A slot

A slot has straight sides 60 mm long and round ends 30 mm across. Find the distance around the slot.



SOLUTION

1. Two straight sides: 2 × 60 = 120 mm.
2. Two half circles make one whole circle with D = 30 mm: π × 30 = 94.25 mm.
3. Add: 120 + 94.25 = 214.25.

CALCULATOR

TI-30Xa

2 × 60 + π × 30 =

Display 214.2477796

Sharp EL-531XT

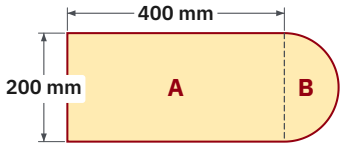
2 × 60 + π × 30 =

Display 214.2477796

Answer 214.25 mm

Example 9.11 A plate with a round end

A plate is a 400 mm by 200 mm rectangle with a half circle on one end. Find its perimeter.



SOLUTION

1. Part A gives three outside edges: 400 + 400 + 200 = 1,000 mm. Its right side is the dashed line, inside the plate.
2. Part B is a half circle with D = 200 mm: π × 200 ÷ 2 = 314.16 mm.
3. Add: 1,000 + 314.16 = 1,314.16.

Answer **1,314.16 mm**

Example 9.12 A belt around two pulleys

A belt runs around two pulleys, each 200 mm in diameter. The centres are 800 mm apart. How long is the belt?



SOLUTION

1. The two straight runs are each as long as the centre distance: 2 × 800 = 1,600 mm.
2. The belt wraps half of each pulley. Two halves make one circle: π × 200 = 628.32 mm.
3. Add: 1,600 + 628.32 = 2,228.32.

Answer **2,228.32 mm**

TRY IT 9.6

1. A slot has straight sides of 50 mm and ends 20 mm across. Find the perimeter.
2. A 600 mm by 400 mm plate has a 50 mm radius on each corner. Find the perimeter.
3. Two pulleys 150 mm in diameter have centres 600 mm apart. Find the belt length.
4. A quarter circle has r = 300 mm. Find the perimeter, both straight edges and the arc.

Answers are in the Answer Key.

9.7 Carpentry: plates, rim joists, fascia and trim

A carpenter measures around the outside of a building again and again: for the wall plates, for the rim joists of a floor, for the fascia along the roof edge, and for baseboard and trim inside. Each one is a perimeter.

In a wood-frame wall the studs stand on a bottom plate and carry top plates. The BC Building Code says "a bottom wall plate shall be provided in all cases" (BCBC 2024, Division B, Sentence 9.23.11.2.(1), page 9-184), and "at least 2 top plates shall be provided in loadbearing walls", except where Sentences (2) to (4)

allow one or none (Sentence 9.23.11.3.(1), page 9-185). A loadbearing wall with a double top plate therefore needs three plates along its length.

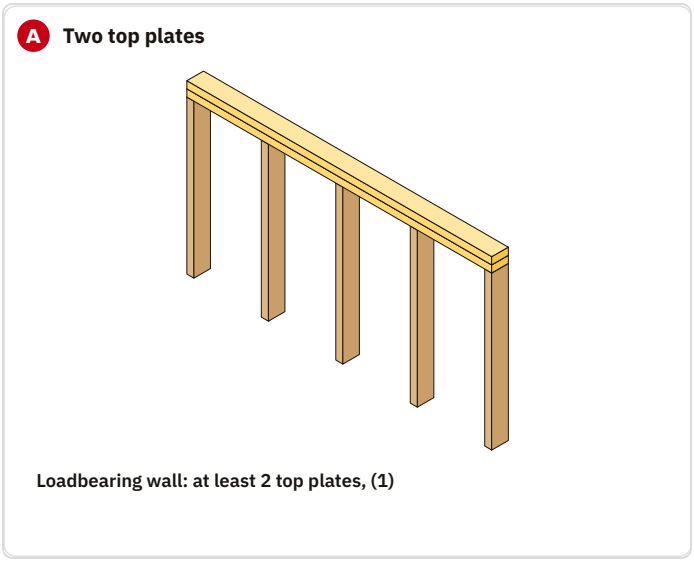


Figure 9.4 Panel A of plate 9.23.11.3: a loadbearing wall with two top plates over the studs.
Source: BCBC 2024, Division B, Article 9.23.11.3. (page 9-185)

Example 9.13 Wall plates for a house

The outside walls of a house are 12,200 mm by 8,500 mm. All four walls are loadbearing, with one bottom plate and two top plates. The estimate uses the outside perimeter and ignores openings. How much plate is needed? The plate comes in 4,880 mm lengths. How many lengths are ordered?

SOLUTION

1. Perimeter: P = 2 × (12,200 + 8,500) = 41,400 mm.
2. Three plates all the way around: 3 × 41,400 = 124,200 mm.
3. Lengths: 124,200 ÷ 4,880 = 25.45. Round up to 26, because 25 lengths are too short.

CALCULATOR

TI-30Xa

Plate

3 × 2 × (12200 + 8500) =

Display 124200.

Lengths

124200 ÷ 4880 =

Display 25.45081967

Sharp EL-531XT

Plate

3 × 2 × (12200 + 8500) =

Display 124’200.

Lengths

124200 ÷ 4880 =

Display 25.45081967

Answer **124,200 mm (124.2 m) of plate, so 26 lengths.**

The code defines a rim joist as "the outermost member in floor framing, other than blocking, be it parallel, perpendicular or on an angle to the floor joists" (Division A, Sentence 1.4.1.2.(1), page 1-23). The rim joists run all the way around the floor, so their total length is the perimeter of the floor framing.

Example 9.14 Rim joists

A floor frame is 9,760 mm by 7,320 mm. Find the total length of rim joist. The rim board comes in 4,880 mm lengths. How many lengths are needed, ignoring joints?

SOLUTION

1. P = 2 × (9,760 + 7,320) = 2 × 17,080 = 34,160 mm.
2. 34,160 ÷ 4,880 = 7 lengths exactly.

CALCULATOR

TI-30Xa

Rim joist

2 × (9760 + 7320) =

Display 34160.

Lengths

34160 ÷ 4880 =

Display 7.

Sharp EL-531XT

Rim joist

2 × (9760 + 7320) =

Display 34’160.

Lengths

34160 ÷ 4880 =

Display 7.

Answer **34,160 mm of rim joist, 7 lengths.**

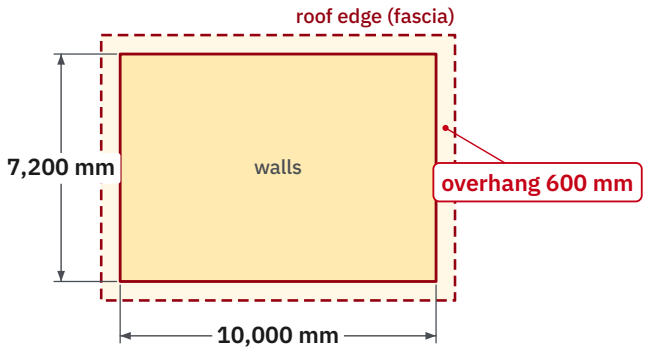


Figure 9.5 A roof plan. The roof edge sits 600 mm outside the walls on every side, so it is 1,200 mm longer and 1,200 mm wider than the walls.

Example 9.15 Fascia around a roof

A building has walls 10,000 mm by 7,200 mm. Its hip roof overhangs 600 mm on every side, and fascia runs along all four roof edges. How long is the fascia?

SOLUTION

1. The overhang adds 600 mm at each end: length 10,000 + 2 × 600 = 11,200 mm, width 7,200 + 2 × 600 = 8,400 mm.
2. P = 2 × (11,200 + 8,400) = 39,200 mm.
3. Compare the walls: 2 × (10,000 + 7,200) = 34,400 mm. The fascia is longer by 4,800 mm, which is 8 × 600.

CALCULATOR

TI-30Xa

2 × (10000 + 2 × 600 + 7200 + 2 × 600) =

Display 39200.

Sharp EL-531XT

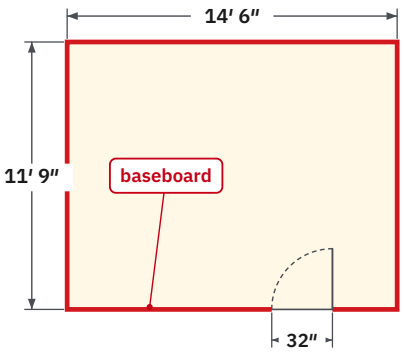
2 × (10000 + 2 × 600 + 7200 + 2 × 600) =

Display 39′200.

Answer **39,200 mm, or 39.2 m.**

Example 9.16 Baseboard in feet and inches

A room is 14 ft 6 in by 11 ft 9 in. Baseboard runs around the walls except across one door opening 32 in wide. How much baseboard is needed? Give feet and inches.



SOLUTION

1. Change to inches: 14 ft 6 in = 174 in, 11 ft 9 in = 141 in.
2. P = 2 × (174 + 141) = 630 in.
3. Take off the door: 630 − 32 = 598 in.
4. 598 ÷ 12 = 49 ft and 10 in left over, because 49 × 12 = 588.

CALCULATOR

TI-30Xa

2 × (174 + 141) − 32 =

Display 598.

Sharp EL-531XT

2 × (174 + 141) − 32 =

Display 598.

Answer **49′ 10″ of baseboard.**

TRY IT 9.7

1. A loadbearing wall 6,100 mm long has one bottom plate and two top plates. How much plate is that?
2. A floor frame is 8,540 mm by 6,100 mm. Find the total length of rim joist.
3. Walls are 9,000 mm by 6,000 mm and the roof overhangs 450 mm on every side. Find the length of fascia around all four edges.
4. A room is 3,600 mm by 3,000 mm with one door opening 810 mm wide. How much baseboard is needed?
5. A room is 12 ft by 10 ft 6 in with two door openings, each 30 in wide. How much baseboard is needed, in feet?
6. Fascia of 39,200 mm is cut from boards 4,880 mm long. Ignoring joints, how many boards are bought?

Answers are in the Answer Key.

WORKSHEET 9A

Around the outside

Find the perimeter, or find the ? from the perimeter

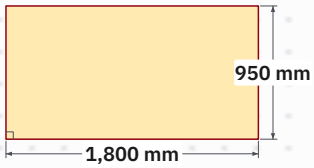
Name _____

Date _____

Add every outside edge. If a perimeter is given, transpose to find the ?

1 Find the perimeter of the plate.

SHAPE CANVAS READ THE DRAWING

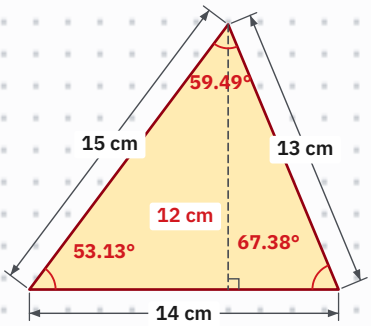


WORKING

Answer _____

2 Find the perimeter. The dashed line is the height of the triangle. It is not an edge.

SHAPE CANVAS READ THE DRAWING

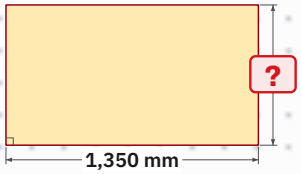


WORKING

Answer _____

3 The perimeter is 4,200 mm. Find the width.

SHAPE CANVAS PRACTICE: FIND THE ?

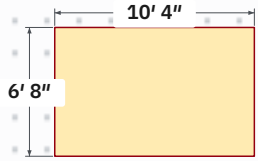


WORKING

Answer _____

4 Find the perimeter in feet and inches.

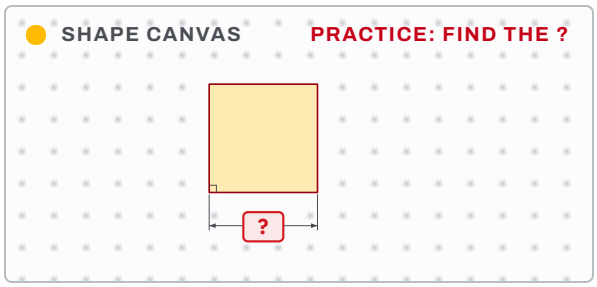
SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

5 The perimeter of the square is 2,000 mm. Find the side.



WORKING

Answer _____

6 Change one value: both sides of the plate in problem 1 are doubled. Find the new perimeter. What happened to it?

WORKING

Answer _____

WORKSHEET 9B

Circles, arcs and stretch-outs

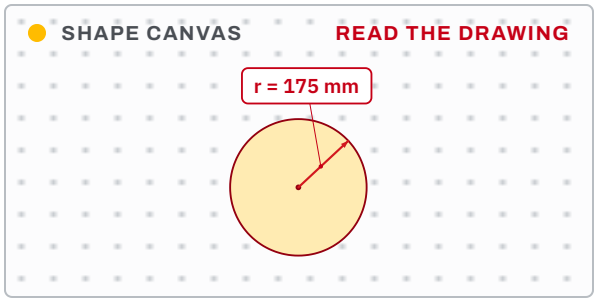
Distance around a circle is always π × D

Name _____

Date _____

Round to two decimal places unless a problem says otherwise. If a drawing gives r, double it to get D.

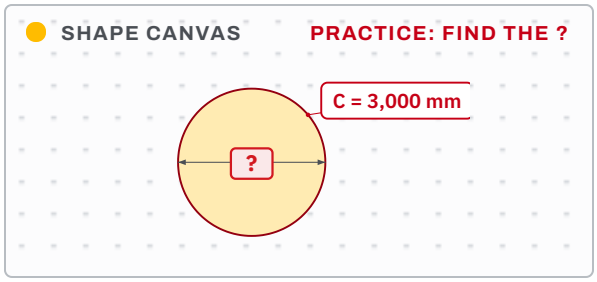
1 Find the circumference.



WORKING

Answer _____

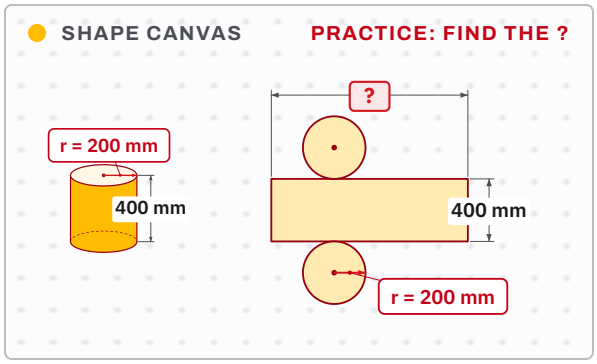
2 A tape around the tank reads 3,000 mm. Find the diameter.



WORKING

Answer _____

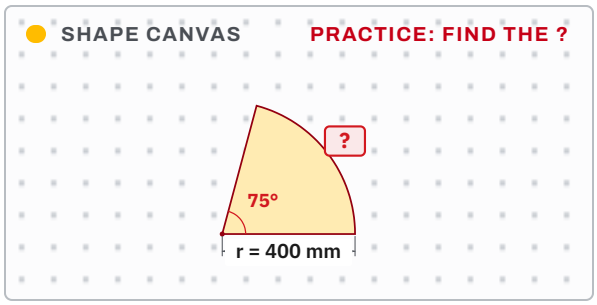
3 Unfold it. Find the stretch-out, the length marked ?, of the cylinder.



WORKING

Answer _____

4 Find the length of the arc.



WORKING

Answer _____

5 A wheel is 700 mm in diameter. How many turns does it make in 10 m?

WORKING

Answer _____

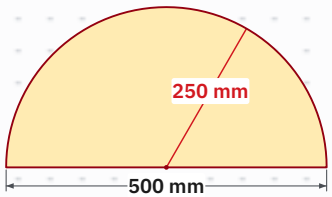
6 A jacket wraps a pipe with an outside diameter of $8\frac{5}{8}$ in. Find the stretch-out to the nearest 1/16 in.

WORKING

Answer _____

7 Find the curved edge and the whole perimeter of the half circle.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

WORKSHEET 9C

Break it down: composite shapes

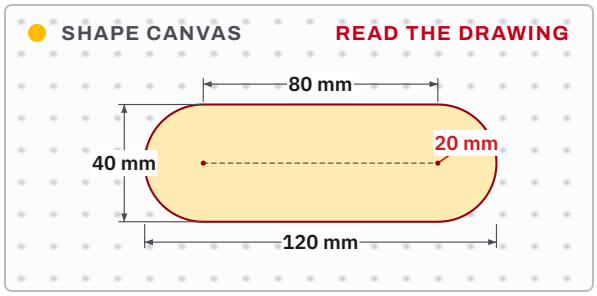
Split each shape into straight edges and arcs

Name _____

Date _____

Add only the outside edges. Dashed lines are not edges. Round to two decimal places.

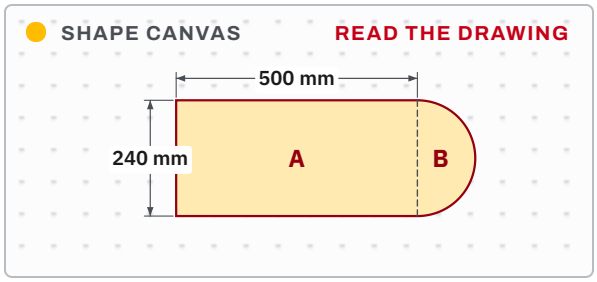
1 Find the perimeter of the slot.



WORKING

Answer _____

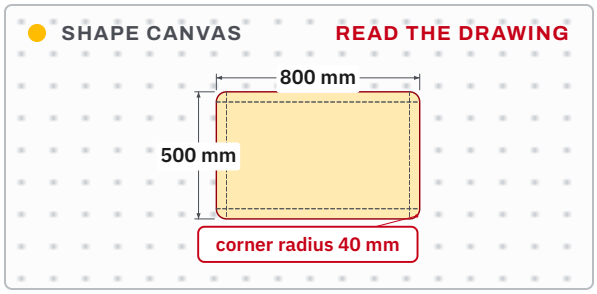
2 Find the perimeter of the plate. Part B is a half circle.



WORKING

Answer _____

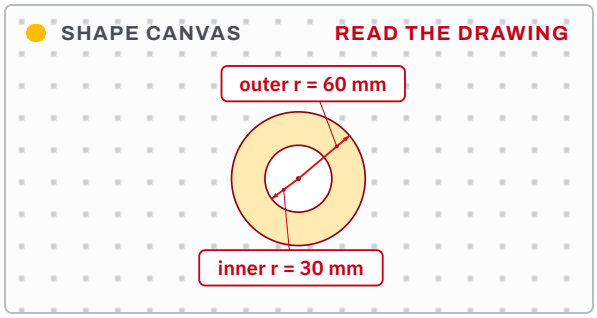
3 Each corner is a quarter circle. Find the perimeter.



WORKING

Answer _____

4 Find the total length of both edges of the washer, inside and outside.



WORKING

Answer _____

5 Find the length of the belt.

SHAPE CANVAS READ THE DRAWING

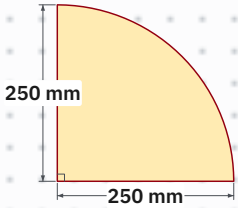


WORKING

Answer _____

6 Find the perimeter of the quarter circle.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

CHAPTER

10

Area

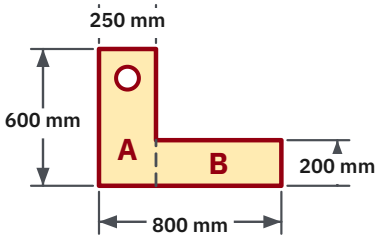
Area tells you how much flat surface a part covers. You use it to order plate and sheet, to work out how much coating a floor needs, and to lay out the sheet metal for a duct or a tank. This chapter builds the area formulas one shape at a time, then breaks composite parts into pieces and unfolds solids into flat nets.

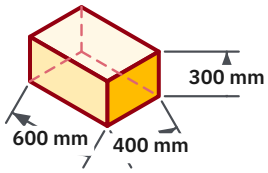
In this chapter you will

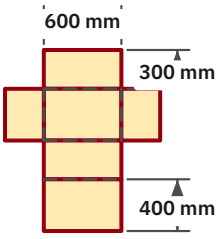
- Explain what area is and use square units
- Find the area of rectangles, triangles, parallelograms and trapezoids
- Find the area of circles and rings with $\pi \times r^2$
- Change mm^2 to m^2 and in^2 to ft^2
- Break a composite shape into basic shapes
- Unfold a solid into a net and find its surface area

Key words

- area
- square unit
- base and height
- composite shape
- surface area
- net







10.1 Area and square units

Area is the amount of flat surface inside a shape. A length is measured in one direction. Area covers two directions at once, so it is measured with squares.

A **square unit** is a square that is 1 unit long on each side. A square 1 cm on each side has an area of 1 square centimetre, written 1 cm². The small 2 shows that two lengths were multiplied: centimetres times centimetres.

The square units used in this book are mm², cm² and m² (square millimetres, square centimetres and square metres), and in² and ft² (square inches and square feet). A length in mm and an area in mm² measure different things, so always write the small 2 on an area.

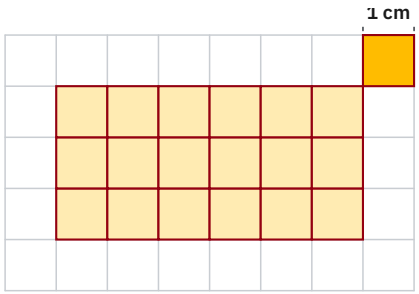


Figure 10.1 A plate drawn on a 1 cm grid covers 3 rows of 6 squares, so its area is 18 cm². The gold square is 1 cm².

TRY IT 10.1

1. A plate on a 1 cm grid covers 4 rows of 7 squares. What is its area?
2. How do you say 35 m² out loud?
3. Which is an area: 12 ft, 12 ft² or 12 ft³?
4. A floor is covered by 30 tiles. Each tile is exactly 1 ft². What is the area of the floor?

Answers are in the Answer Key.

10.2 Rectangles and squares

A rectangle that is 6 squares long and 3 squares wide holds 3 rows of 6 squares. Multiplying the length by the width counts them all at once, so you do not need a grid.

AREA OF A RECTANGLE AND A SQUARE

$$A = l \times w$$

$$A = s \times s = s^2$$

A = area, l = length, w = width, s = side of a square. Both lengths must be in the same unit.

Square millimetres make big numbers, and a large area is easier to read in square metres. One metre is 1,000 mm, so a square metre is 1,000 mm by 1,000 mm. In the same way, a square foot is 12 in by 12 in.

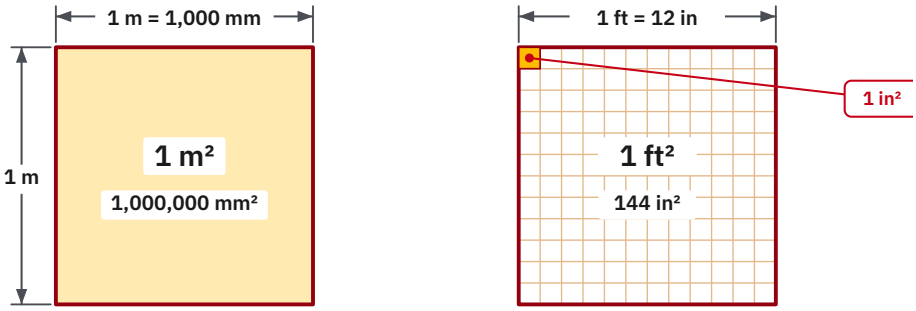
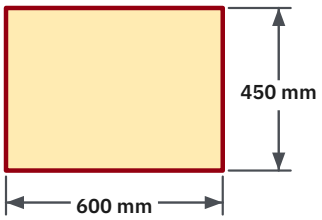


Figure 10.2 1 m² = 1,000 mm × 1,000 mm = 1,000,000 mm². 1 ft² = 12 in × 12 in = 144 in².

Example 10.1 Area of a base plate

A machine base plate is 600 mm long and 450 mm wide. Find its area in mm² and in m².



SOLUTION

1. A = l × w = 600 × 450 = 270,000 mm².
2. To change mm² to m², divide by 1,000,000: 270,000 ÷ 1,000,000 = 0.27 m².
3. Check: in metres the plate is 0.6 m by 0.45 m, and 0.6 × 0.45 = 0.27 m².

CALCULATOR

TI-30Xa

Area in mm²

600 × 450 =

Display 270000.

mm² to m²

270000 ÷ 1000000 =

Display 0.27

Sharp EL-531XT

Area in mm²

600 × 450 =

Display 270'000.

mm² to m²

270000 ÷ 1000000 =

Display 0.27

Answer **270,000 mm² = 0.27 m²**

DIVIDE BY 1,000,000, NOT BY 1,000

Changing mm to m divides a length by 1,000. An area has two lengths multiplied, so the factor is used twice: 1,000 × 1,000 = 1,000,000. Dividing 270,000 mm² by 1,000 gives 270, far too big for this plate in m².

The same idea works in imperial units. A sheet 4 ft by 8 ft has an area of 4 × 8 = 32 ft². In inches it is 48 in by 96 in, and 48 × 96 = 4,608 in². The check 32 × 144 = 4,608 shows the two answers agree.

Example 10.2 Coating a shop floor

A shop floor is 6 m by 4.5 m. The coating label says one litre covers 5 m². The coating is sold in 1 L tins. How many tins do you buy?

SOLUTION

1. Area: 6 × 4.5 = 27 m².
2. Litres: 27 ÷ 5 = 5.4 L.
3. A part tin still has to be bought, so round up.

CALCULATOR

TI-30Xa	Sharp EL-531XT
6 × 4.5 ÷ 5 =	6 × 4.5 ÷ 5 =
Display 5.4	Display 5.4

Answer **6 tins**

TRY IT 10.2

1. A plate is 750 mm by 400 mm. Find its area in mm².
2. A square cover plate has 250 mm sides. Find its area.
3. A table top is 3 ft by 5 ft. Find its area in ft² and in in².
4. A floor is 8 m by 5 m. One litre of coating covers 6 m². How many 1 L tins?

Answers are in the Answer Key.

10.3 Triangles

Every triangle is half of a rectangle or of a parallelogram. To use that idea you need two measurements, the base and the height.

The **base** is the side you measure from, usually the bottom side. The **height** is the distance from the base to the opposite corner, measured at right angles to the base. The height is not a sloping side.

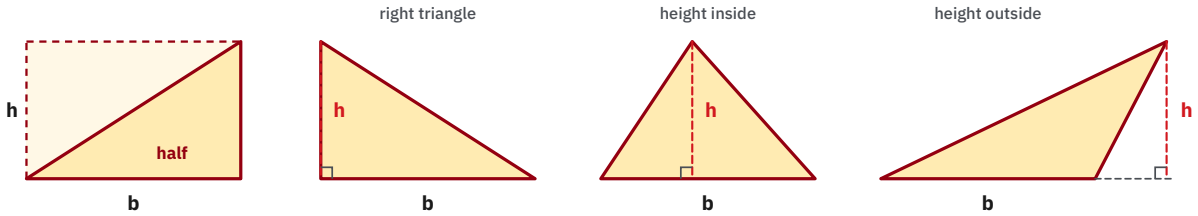


Figure 10.3 Left: a rectangle cut on its diagonal makes two equal triangles. Right: three triangles with the same base b and the same height h have the same area.

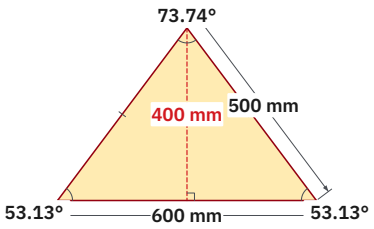
AREA OF A TRIANGLE

$$A = b \times h \div 2$$

b = base, h = height at right angles to the base.

Example 10.3 A triangular end panel

The end panel of a trough is a triangle with a base of 600 mm and a height of 400 mm. Its two sloping sides are each 500 mm long. Find the area in mm² and in m².



SOLUTION

1. Use the base and the height. The 500 mm sloping sides are not needed.
2. A = 600 × 400 ÷ 2 = 120,000 mm².
3. In m²: 120,000 ÷ 1,000,000 = 0.12 m².

CALCULATOR

TI-30Xa

600 × 400 ÷ 2 =

Display 120000.

Sharp EL-531XT

600 × 400 ÷ 2 =

Display 120’000.

Answer **120,000 mm² = 0.12 m²**

THE SLOPING SIDE IS NOT THE HEIGHT

Using the 500 mm side in the example above would give 600 × 500 ÷ 2 = 150,000 mm², which is too big. The height always meets the base at a right angle. A drawing often shows it as a dashed line with a small square at its foot.

TRY IT 10.3

1. A right-angled gusset: base 300 mm, height 200 mm. Area?
2. Base 18 in, height 10.5 in. Area?
3. Base 1.2 m, height 0.9 m. Area?
4. Base 300 mm, sloping sides 250 mm, height 200 mm. Area?

Answers are in the Answer Key.

10.4 Parallelograms and trapezoids

A parallelogram has two pairs of parallel sides. Cut the triangle off one end and move it to the other end: the parallelogram becomes a rectangle with the same base and the same height. So its area is base times height: **A = b × h**, where h is the height at right angles to the base, not the slanted side.

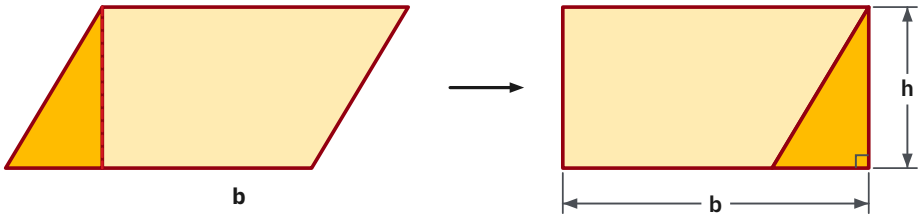


Figure 10.4 The gold triangle moves from the left end to the right end. The parallelogram and the rectangle have the same area.

A trapezoid has one pair of parallel sides, called a and b. Two copies of the same trapezoid, one turned around, fit together to make a parallelogram with base a + b. One trapezoid is half of it.

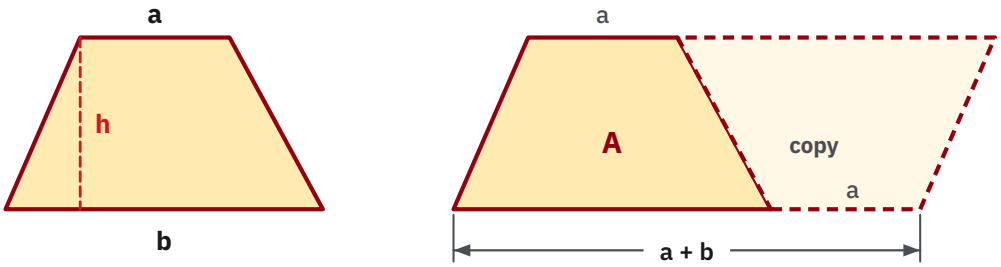


Figure 10.5 A trapezoid and a turned copy make a parallelogram with base a + b and the same height.

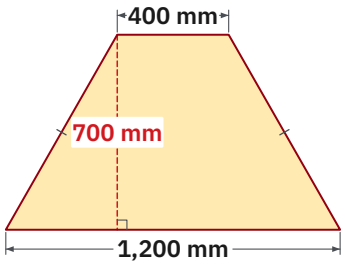
AREA OF A TRAPEZOID

$$A = (a + b) \div 2 \times h$$

a and b = the two parallel sides, h = the height between them. (a + b) ÷ 2 is the average width.

Example 10.4 A hopper side panel

A hopper side panel is a trapezoid. The parallel sides are 400 mm and 1,200 mm, and the height between them is 700 mm. Find the area in mm² and m².



SOLUTION

1. $A = (a + b) \div 2 \times h$.
2. Brackets first: 400 + 1,200 = 1,600.
3. Average width: 1,600 ÷ 2 = 800.
4. Times the height: 800 × 700 = 560,000 mm² = 0.56 m².

CALCULATOR

TI-30Xa

(400 + 1200) ÷ 2 × 700 =

Display 560000.

Sharp EL-531XT

(400 + 1200) ÷ 2 × 700 =

Display 560’000.

The brackets make both calculators add the parallel sides before dividing.

Answer 560,000 mm² = 0.56 m²

The quadrilateral family

A **quadrilateral** is any flat shape with four straight sides. The shapes in this chapter belong to one family. A square is a rectangle with four equal sides, and it is also a rhombus with four right angles. Rectangles and rhombuses are special parallelograms. The table lists what each member has, and the area formula to use.

Shape		Sides	Angles	Diagonals	Area
	**Square**	4 equal sides, opposite sides parallel	4 right angles	Equal length, cross at right angles, cut each other in half	$A = s \times s$
	**Rectangle**	Opposite sides equal and parallel	4 right angles	Equal length, cut each other in half	$A = l \times w$
	**Parallelogram**	Opposite sides equal and parallel	Opposite angles equal	Cut each other in half	$A = b \times h$
	**Rhombus**	4 equal sides, opposite sides parallel	Opposite angles equal	Cross at right angles, cut each other in half	$A = b \times h$ or $A = d_1 \times d_2 \div 2$
	**Trapezoid**	One pair of parallel sides, a and b	No fixed rule	No fixed rule	$A = (a + b) \div 2 \times h$
	**Kite**	Two pairs of equal sides that meet at a corner	One pair of opposite angles equal	Cross at right angles, one cuts the other in half	$A = d_1 \times d_2 \div 2$

The quadrilateral family. Tick marks show sides of equal length. h is always measured at right angles to the base, and d_1 and d_2 are the diagonals.

A rhombus and a kite can also be found from their two **diagonals**, the lines joining opposite corners. The diagonals cross at right angles, so the shape is exactly half of the rectangle that is d_1 long and d_2 wide: **$A = d_1 \times d_2 \div 2$**. A kite-shaped gusset with diagonals of 600 mm and 400 mm has an area of $600 \times 400 \div 2 = 120{,}000$ mm².

TRY IT 10.4

1. Parallelogram: base 500 mm, slanted side 340 mm, height 300 mm. Area?
2. Parallelogram: base 30 in, slanted side 15 in, height 12 in. Area?
3. Trapezoid: parallel sides 300 mm and 500 mm, height 200 mm. Area?
4. Trapezoid: parallel sides 14 in and 22 in, height 9 in. Area?

Answers are in the Answer Key.

10.5 Circles and rings

The area of a circle uses the **radius**, the distance from the centre to the edge. The radius is half the **diameter**.

Picture the circle inside a square. The square is 2 r on each side, so it holds 4 small squares that are r by r. The circle covers a bit more than 3 of those small squares: exactly π of them, and π is about 3.14.

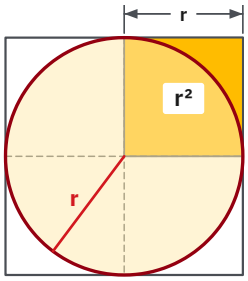


Figure 10.6 The gold square is r × r = r². The circle covers π of these squares.

AREA OF A CIRCLE

$$A = \pi \times r^2$$

r = radius. Square the radius first (exponents come before multiplying in BEDMAS), then multiply by π. Use the π key for the full value.

Example 10.5 A round blank

A round blank is cut with a radius of 150 mm. Find its area to two decimal places.

SOLUTION

1. A = π × r².
2. Square the radius: 150² = 150 × 150 = 22,500.
3. Multiply by π: π × 22,500 = 70,685.83.

CALCULATOR

TI-30Xa	Sharp EL-531XT
π × 150 x² =	π × 150 x² =
Display 70685.83471	Display 70'685.83471

Both calculators square 150 before multiplying by π, because they follow BEDMAS.

Answer 70,685.83 mm²

HALVE THE DIAMETER FIRST

A round cover for an opening 18 in across has r = 9 in, so its area is π × 9² = 254.47 in². Putting the diameter into the formula gives π × 18² = 1,017.88 in², four times too big. When a drawing gives Ø (diameter), divide by 2 before you square.

A half circle has half the area of the full circle, and a quarter circle has a quarter. A ring, such as a washer, a flange or the end of a pipe, is a big circle with a small circle taken out. Find the area of the outside circle (radius R) and subtract the area of the hole (radius r).

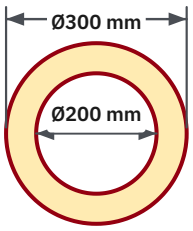
AREA OF A RING

$$A = \pi \times R^2 - \pi \times r^2 = \pi \times (R^2 - r^2)$$

R = outside radius, r = inside radius. Square each radius before subtracting.

Example 10.6 A flange ring

A flat flange ring has an outside diameter of 300 mm and an inside diameter of 200 mm. Find its area to two decimal places.



SOLUTION

1. Halve the diameters: R = 150 mm and r = 100 mm.
2. Square each one: $150^2 = 22{,}500$ and $100^2 = 10{,}000$.
3. Subtract: 22,500 − 10,000 = 12,500.
4. Multiply by π: π × 12,500 = 39,269.91.

CALCULATOR

TI-30Xa

π × (150 x^2 − 100 x^2) =

Display 39269.90817

Sharp EL-531XT

π × (150 x^2 − 100 x^2) =

Display 39′269.90817

Answer 39,269.91 mm²

SQUARE EACH RADIUS, THEN SUBTRACT

Subtracting the radii first, 150 − 100 = 50, and then squaring gives π × 50² = 7,853.98 mm². That is the area of a small circle, not the ring.

TRY IT 10.5

1. Circle, radius 80 mm. Area to two decimal places?
2. Half circle, radius 200 mm. Area to two decimal places?
3. Washer: outside diameter 40 mm, inside diameter 22 mm. Area to two decimal places?
4. Tube end: outside diameter 60 mm, wall 5 mm thick. Area of metal to two decimal places?

Answers are in the Answer Key.

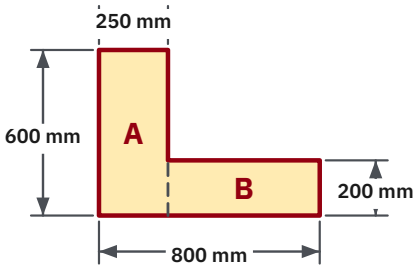
10.6 Composite shapes

A **composite shape** is made of two or more basic shapes: an L-shaped bracket, a plate with a hole, a lug with a round end. The Shape Canvas calls the method **break it down**.

1. Split the shape with dashed lines into basic shapes. Label them A, B, C.
2. Find any missing dimensions from the overall dimensions.
3. Find the area of each piece.
4. Add the pieces. Subtract any hole or cut-out.

Example 10.7 An L-shaped plate

Find the area of the L-shaped plate.



SOLUTION

1. Split on the dashed line into A (the upright) and B (the foot).
2. A: 250 × 600 = 150,000 mm².
3. B is 800 − 250 = 550 mm long: 550 × 200 = 110,000 mm².
4. Add: 150,000 + 110,000 = 260,000 mm².

CALCULATOR

TI-30Xa

250 × 600 + 550 × 200 =

Display 260000.

Sharp EL-531XT

250 × 600 + 550 × 200 =

Display 260’000.

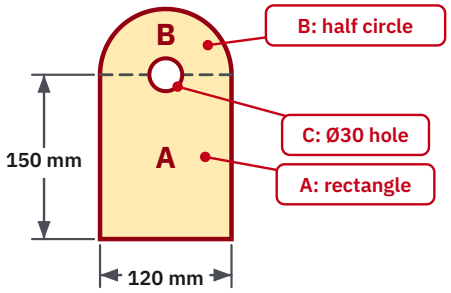
Answer **260,000 mm²**

TWO WAYS, ONE ANSWER

The L is also an 800 × 600 rectangle with a 550 × 400 corner missing.
800 × 600 − 550 × 400 = 260,000 mm², the same answer. When two methods agree, you can trust the answer.

Example 10.8 A lifting lug

A lifting lug is a 120 mm by 150 mm rectangle with a half circle on top. A 30 mm diameter hole is centred where the half circle meets the rectangle. Find the area of the lug to two decimal places.



SOLUTION

1. A: 120 × 150 = 18,000 mm².
2. B: the half circle is 120 mm across, so r = 60 mm. π × 60² ÷ 2 = 5,654.87 mm².
3. C: r = 15 mm. π × 15² = 706.86 mm².
4. A + B − C = 22,948.01 mm².

CALCULATOR

TI-30Xa

120 × 150 + π × 60 x² ÷ 2 − π × 15 x² =

Display 22948.00843

Sharp EL-531XT

120 × 150 + π × 60 x² ÷ 2 − π × 15 x² =

Display 22'948.00843

Answer **22,948.01 mm²**

TRY IT 10.6

1. A 500 mm by 300 mm rectangle with a 200 mm by 100 mm corner cut away. Area?
2. A 400 mm by 300 mm plate with a 160 mm diameter hole in the middle. Area to two decimal places?
3. A 300 mm by 200 mm plate with four 20 mm diameter holes. Area to two decimal places?
4. A 700 mm by 400 mm plate with a 200 mm by 150 mm notch cut out of one edge. Area?

Answers are in the Answer Key.

10.7 Surface area and nets

A solid, such as a box or a tank, is covered by faces. Its **surface area** is the total area of all its outside faces. Surface area tells you how much sheet a tank needs, or how much area there is to paint.

A **net** is a flat pattern that folds or rolls up to make the solid. The Shape Canvas calls this **unfold it**. Unfold the solid, find the area of each piece of the net, and add them.

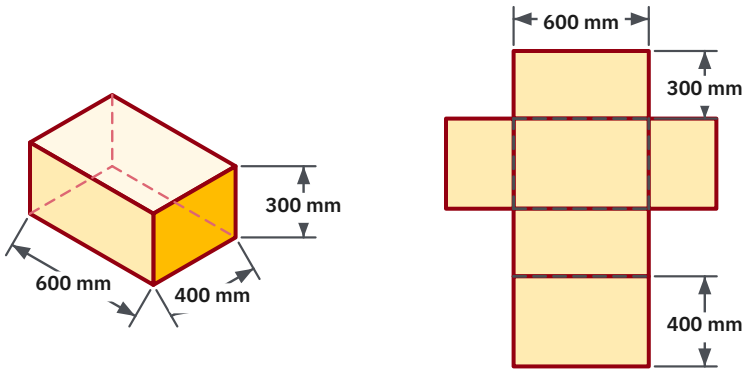


Figure 10.7 A closed box and its net. Opposite faces are the same size, so the net has three pairs of rectangles. Dashed lines are folds.

Example 10.9 Surface area of a closed box

A closed box is 600 mm long, 400 mm wide and 300 mm high. Find its surface area in mm² and m².

SOLUTION

1. Top and bottom: 2 × 600 × 400 = 480,000 mm².
2. Front and back: 2 × 600 × 300 = 360,000 mm².
3. Two ends: 2 × 400 × 300 = 240,000 mm².
4. Add the three pairs: 480,000 + 360,000 + 240,000 = 1,080,000 mm².
5. In m²: 1,080,000 ÷ 1,000,000 = 1.08 m².

CALCULATOR

TI-30Xa

2 × (600 × 400 + 600 × 300 + 400 × 300) =

Display 1080000.

Sharp EL-531XT

2 × (600 × 400 + 600 × 300 + 400 × 300) =

Display 1'080'000.

Answer **1,080,000 mm² = 1.08 m²**

A cylinder unrolls into two circles and one rectangle. The rectangle wraps around the circle, so its length is the distance around the circle, π × D. This length is the **stretch-out**. The width of the rectangle is the height of the cylinder.

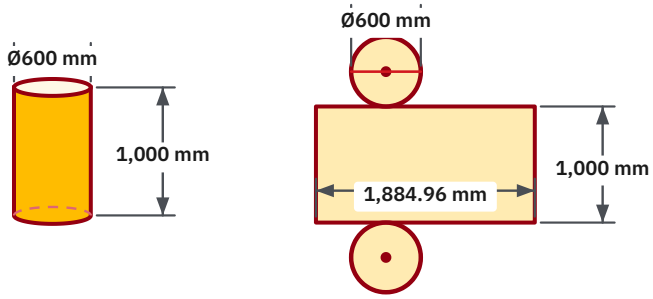


Figure 10.8 A closed cylinder and its net. The circles are 600 mm across, and the stretch-out is π × 600 = 1,884.96 mm.

SURFACE AREA OF A CLOSED CYLINDER

$$\text{Two ends: } 2 \times (\pi \times r^2)$$

$$\text{Side: } \pi \times D \times h$$

$$\text{SA} = \text{two ends} + \text{side}$$

r = radius, D = diameter, h = height. The side is the stretch-out π × D times the height.

Example 10.10 Sheet for a cylinder tank

A closed cylindrical tank has a radius of 300 mm and a height of 1,000 mm. Find its surface area to two decimal places in mm², then in m².

SOLUTION

1. Two ends: π × 300² = 282,743.34 mm² each, so 565,486.68 mm² for both.
2. Side: D = 2 × 300 = 600 mm, and π × 600 × 1,000 = 1,884,955.59 mm².
3. Total: 565,486.68 + 1,884,955.59 = 2,450,442.27 mm².
4. In m²: 2,450,442.27 ÷ 1,000,000 = 2.45 m² to two decimal places.

CALCULATOR

TI-30Xa

Two ends

π × 300 x^2 × 2 =

Display 565486.6776

Side

π × 600 × 1000 =

Display 1884955.592

Sharp EL-531XT

Two ends

π × 300 x^2 × 2 =

Display 565’486.6776

Side

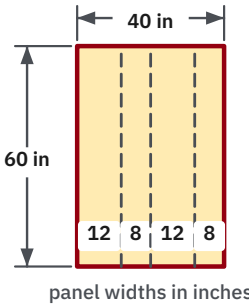
π × 600 × 1000 =

Display 1’884’955.592

Answer 2,450,442.27 mm², about 2.45 m²

Example 10.11 Sheet for a duct

A rectangular duct is 12 in by 8 in and 60 in long, open at both ends. Its net is one long rectangle: the four sides side by side. Ignore seams. Find the area of sheet in in² and in ft² to two decimal places.



panel widths in inches

SOLUTION

1. The width of the net is the **perimeter** of the duct: 12 + 8 + 12 + 8 = 40 in.
2. Area: 40 × 60 = 2,400 in².
3. In ft²: 2,400 ÷ 144 = 16.67 ft².

CALCULATOR

TI-30Xa

2 × (12 + 8) × 60 ÷ 144 =

Display 16.66666667

Sharp EL-531XT

2 × (12 + 8) × 60 ÷ 144 =

Display 16.66666667

Answer **2,400 in² = 16.67 ft²**

TRY IT 10.7

1. Closed box 500 mm by 400 mm by 200 mm. Surface area in mm²?
2. Open-top box 400 mm by 300 mm by 250 mm (no lid). Surface area in mm²?
3. Duct 10 in by 6 in, 72 in long, open ends. Sheet area in ft² to two decimal places?
4. Closed cylinder, radius 100 mm, height 400 mm. Surface area to two decimal places?

Answers are in the Answer Key.

10.8 Materials takeoff

A **takeoff** is the list of materials a job needs, worked out from the drawings. Area is the first step for drywall, sheathing, flooring, tile, paint and roofing. Sheets, boxes and cans are sold in whole units, so the count is always rounded up. A result of 3.1 sheets means you buy 4.

Sheet sizes, the area a box or can covers, the waste allowance and the price come from the job: the drawings, the product label and the supplier. In this section they are given in each problem.

1. Find the whole area of the wall, floor, ceiling or roof.
2. Subtract the openings (doors and windows) that the material does not cover.

3. If a waste allowance is given, add it: area × (1 + waste % ÷ 100).
4. Divide by the area that one unit covers.
5. Round up to the next whole unit, then multiply by the price for the cost.

MATERIALS TAKEOFF

Units = area ÷ coverage of one unit, rounded up

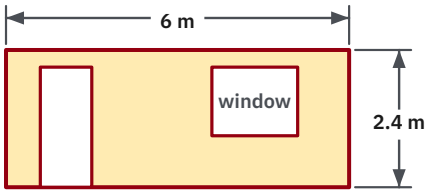
Area with waste = area × (1 + waste % ÷ 100)

Cost = units × price of one unit

Coverage is the area one sheet, box or can covers. A 10% waste allowance multiplies the area by 1.10. Two coats of paint double the area.

Example 10.12 Drywall for a wall with openings

A wall is 6 m long and 2.4 m high. It has a door 0.9 m by 2.1 m and a window 1.5 m by 1.2 m. Drywall comes in 1.2 m by 2.4 m sheets at $18.50 each. How many sheets do you buy, and what do they cost?



door 0.9 m × 2.1 m, window 1.5 m × 1.2 m

SOLUTION

1. Wall: 6 × 2.4 = 14.4 m².
2. Openings: 0.9 × 2.1 = 1.89 m² and 1.5 × 1.2 = 1.8 m².
3. Net area: 14.4 − 1.89 − 1.8 = 10.71 m².
4. One sheet covers 1.2 × 2.4 = 2.88 m². 10.71 ÷ 2.88 = 3.72, so buy 4 sheets.
5. Cost: 4 × $18.50 = $74.00.

CALCULATOR

TI-30Xa

Sheets

(6 × 2.4 − 0.9 × 2.1 − 1.5 × 1.2) ÷ 2.88 =

Display 3.71875

Cost

4 × 18.50 =

Display 74.

Sharp EL-531XT

Sheets

(6 × 2.4 − 0.9 × 2.1 − 1.5 × 1.2) ÷ 2.88 =

Display 3.71875

Cost

4 × 18.50 =

Display 74.

Answer **4 sheets, $74.00**

The thickness of the board is a code question, not a math question. BCBC 2024 Sentence 9.29.5.3.(1) says the spacing of supports for single-layer gypsum board must conform to Table 9.29.5.3. (page 9-275). For walls, the table allows 12.7 mm board on studs up to 600 mm o.c., whether the board runs parallel or perpendicular to the studs. 9.5 mm board on walls is listed only perpendicular to the framing, with supports up to 400 mm o.c.

A **Table 9.29.5.3., Spacing of Supports for Gypsum Board**

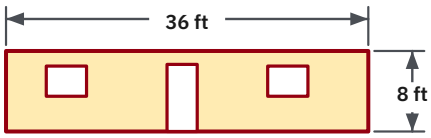
Thickness, mm	Orientation of Board to Framing	Maximum Spacing of Supports, mm o.c.		
		Walls	Ceilings	
			Painted Finish	Water-Based Texture Finish
Gypsum board conforming to Sentence 9.29.5.2.(1) (except Sections 9 and 12 of ASTM C1396/C1396M)				
9.5	parallel	-	-	-
	perpendicular	400	400	-
12.7	parallel	600	400	-
	perpendicular	600	600	400
15.9	parallel	600	400	-
	perpendicular	600	600	600
Gypsum ceiling board conforming to Clause 9.29.5.2.(1)(b) (only Section 12 of ASTM C1396/C1396M)				
12.7	parallel	600	400	-
	perpendicular	600	600	600

Figure 10.9 Read across a row: the board thickness, the way the board runs to the framing, then the largest spacing of supports for walls and ceilings.

Source: BCBC 2024, Division B, Article 9.29.5.3. (page 9-275)

Example 10.13 Wall sheathing in feet

An exterior wall is 36 ft long and 8 ft high. The plan subtracts two windows, each 4 ft by 3 ft, and a door 3 ft by 6 ft 8 in. Sheathing comes in 4 ft by 8 ft sheets at $24.95. How many sheets, and what is the cost?



windows 4 ft × 3 ft, door 3 ft × 6 ft 8 in

SOLUTION

1. Wall: 36 × 8 = 288 ft².
2. Windows: 2 × 4 × 3 = 24 ft². Door: 6 ft 8 in = 80 in, and 80 ÷ 12 = 6⅔ ft, so 3 × 6⅔ = 20 ft².
3. Net area: 288 − 24 − 20 = 244 ft².
4. One sheet covers 4 × 8 = 32 ft². 244 ÷ 32 = 7.625, so buy 8 sheets.
5. Cost: 8 × $24.95 = $199.60.

CALCULATOR

TI-30Xa

Net area, ft²

36 × 8 − 2 × 4 × 3 − 3 × 80 ÷ 12 =

Display 244.

Sheets

244 ÷ 32 =

Display 7.625

Cost

8 × 24.95 =

Display 199.6

Sharp EL-531XT

Net area, ft²

36 × 8 − 2 × 4 × 3 − 3 × 80 ÷ 12 =

Display 244.

Sheets

244 ÷ 32 =

Display 7.625

Cost

8 × 24.95 =

Display 199.6

Answer **8 sheets, $199.60**

Example 10.14 Floor tile with a waste allowance

A kitchen floor is 3.6 m by 2.7 m. The tiles are 300 mm by 300 mm, 12 to a box, at $42.00 a box. The plan allows 10% for cuts and breakage. How many boxes, and what is the cost?

SOLUTION

1. Floor: 3.6 × 2.7 = 9.72 m².
2. Add 10%: 9.72 × 1.10 = 10.692 m².
3. One box covers 12 × 0.3 × 0.3 = 1.08 m².
4. 10.692 ÷ 1.08 = 9.9, so buy 10 boxes.
5. Cost: 10 × $42.00 = $420.00.

CALCULATOR

TI-30Xa

3.6 × 2.7 × 1.10 ÷ 1.08 =

Display 9.9

Sharp EL-531XT

3.6 × 2.7 × 1.10 ÷ 1.08 =

Display 9.9

Answer **10 boxes, $420.00**

ADD THE WASTE BEFORE YOU ROUND

Work out the area with waste first, then divide and round up once at the end. Rounding the area or the box count early can leave you a box short.

Example 10.15 Paint for a room, two coats

A room is 4.2 m by 3.6 m and the walls are 2.4 m high. Leave out a door 0.9 m by 2.1 m and a window 1.2 m by 1 m. The walls get two coats. The label says one 3.78 L can covers 35 m². Cans are $54.99. How many cans, and what is the cost?

SOLUTION

1. The walls unfold into one long rectangle. Its length is the perimeter of the room: $2 \times (4.2 + 3.6) = 15.6$ m.
2. Wall area: $15.6 \times 2.4 = 37.44$ m².
3. Less the openings: $37.44 - 1.89 - 1.2 = 34.35$ m².
4. Two coats: $34.35 \times 2 = 68.7$ m². $68.7 \div 35 = 1.96$, so buy 2 cans.
5. Cost: $2 \times \$54.99 = \$109.98$.

CALCULATOR

TI-30Xa

(2 × (4.2 + 3.6) × 2.4 − 0.9 × 2.1 − 1.2 × 1) × 2 ÷ 35 =

Display 1.962857143

Sharp EL-531XT

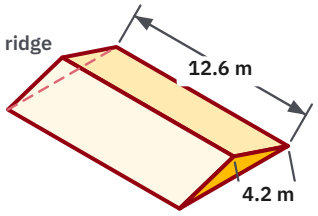
(2 × (4.2 + 3.6) × 2.4 − 0.9 × 2.1 − 1.2 × 1) × 2 ÷ 35 =

Display 1.962857143

Answer 2 cans, $109.98

Example 10.16 Sheathing a gable roof

A gable roof has two equal sides. Each side measures 12.6 m along the ridge and 4.2 m down the slope from the ridge to the eave. Roof sheathing comes in 1.22 m by 2.44 m sheets at $32.75. Allow 5% waste. How many sheets, and what is the cost?



SOLUTION

1. Use the slope length, not the level width of the building: the sheathing lies on the slope.
2. Two sides: $2 \times 12.6 \times 4.2 = 105.84$ m².
3. Add 5%: $105.84 \times 1.05 = 111.132$ m².
4. One sheet covers $1.22 \times 2.44 = 2.9768$ m². $111.132 \div 2.9768 = 37.33$, so buy 38 sheets.
5. Cost: $38 \times \$32.75 = \$1{,}244.50$.
6. The thickness comes from the code. For roofs not used as a walking deck, Sentence 9.23.16.7.(2) points to Table 9.23.16.7.-A (page 9-226). With rafters at 600 mm o.c., plywood needs at least 9.5 mm with the edges supported, or 12.5 mm with the edges unsupported.

CALCULATOR

TI-30Xa

Sheets

2 × 12.6 × 4.2 × 1.05 ÷ (1.22 × 2.44) =

Display 37.33270626

Cost

38 × 32.75 =

Display 1244.5

Sharp EL-531XT

Sheets

2 × 12.6 × 4.2 × 1.05 ÷ (1.22 × 2.44) =

Display 37.33270626

Cost

38 × 32.75 =

Display 1’244.5

Answer **38 sheets, $1,244.50**

TRY IT 10.8

1. A wall 4.8 m by 2.4 m has one door 0.9 m by 2.1 m. How many 1.2 m by 2.4 m drywall sheets?
2. A subfloor is 7.2 m by 4.8 m. Sheets are 1.22 m by 2.44 m. Allow 5% waste. How many sheets?
3. Laminate flooring for a room 5.4 m by 3.9 m, 8% waste. One box covers 2.2 m² and costs $38.50. Boxes and cost?
4. A ceiling 5 m by 4 m gets two coats. One can covers 35 m². How many cans?
5. A wall 20 ft by 8 ft has a window 5 ft by 4 ft. How many 4 ft by 8 ft sheets?

Answers are in the Answer Key.

WORKSHEET 10A

Find the area

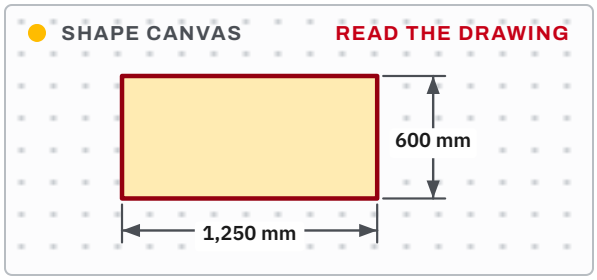
Practice mode: one drawing hides a dimension behind a red ?

Name _____

Date _____

Show the formula, the numbers you put into it and the answer with its square unit.

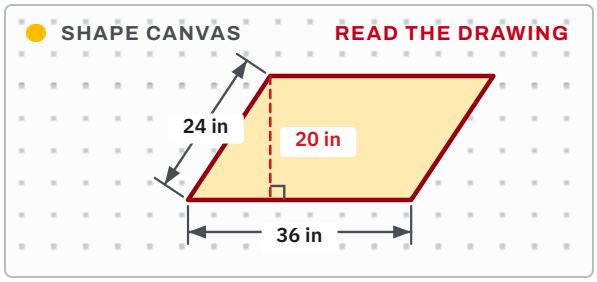
1 Find the area in mm² and in m².



WORKING

Answer _____

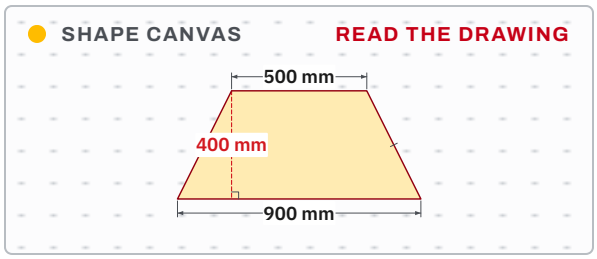
2 Find the area in in² and in ft².



WORKING

Answer _____

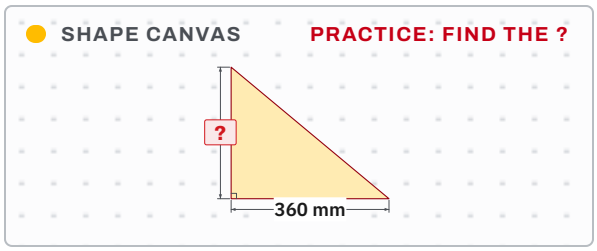
3 Find the area of the trapezoid.



WORKING

Answer _____

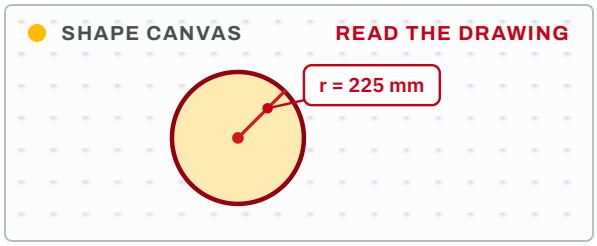
4 The triangle has an area of 54,000 mm². Find the hidden height.



WORKING

Answer _____

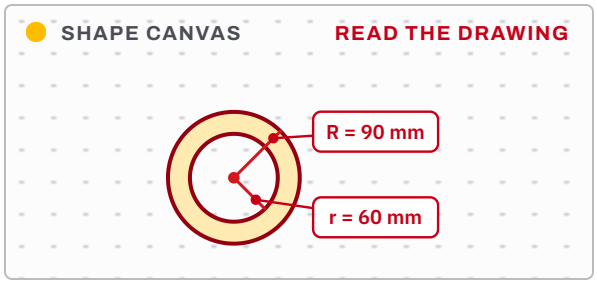
5 Find the area of the disc to the nearest mm².



WORKING

Answer _____

6 Find the area of the washer to two decimal places.



WORKING

Answer _____

WORKSHEET 10B

Break it down

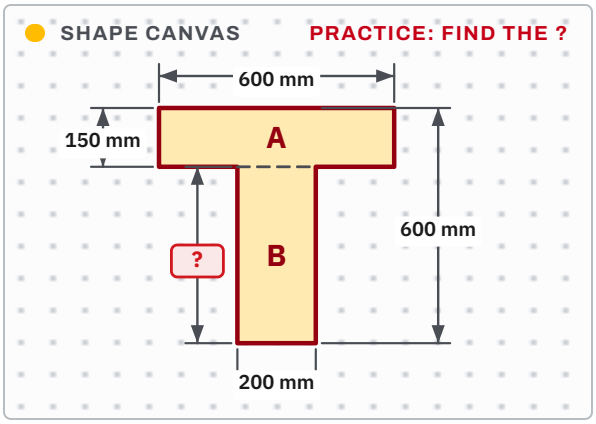
Split each part into basic shapes A, B, C

Name _____

Date _____

Find any ? first. Then find the area of each piece and add, or subtract a hole. Give answers to two decimal places where π is used.

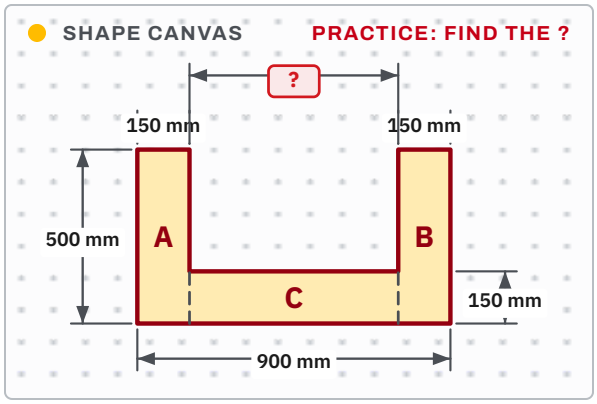
1 Find the hidden height of B, then the area of the T-plate.



WORKING

Answer _____

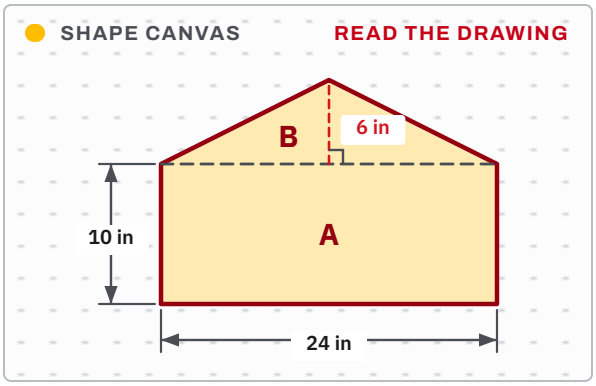
2 Find the hidden width, then the area of the U-shaped plate.



WORKING

Answer _____

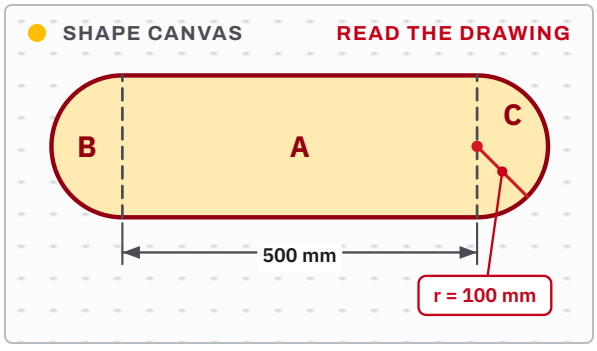
3 Find the area of the gusset in in², then in ft² to two decimal places.



WORKING

Answer _____

4 Find the area of the slotted plate. B and C are half circles.



WORKING

Answer _____

WORKSHEET 10C

Unfold it

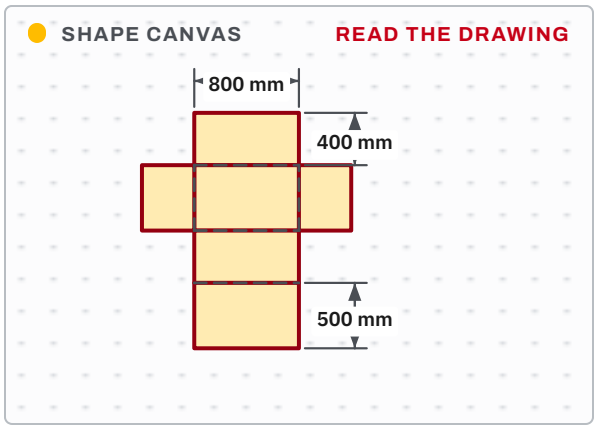
Solids and their nets

Name _____

Date _____

Find the area of each piece of the net, then add. Give answers to two decimal places where π is used.

1 This is the net of a closed box. Find the surface area of the box in mm² and in m².



WORKING

Answer _____

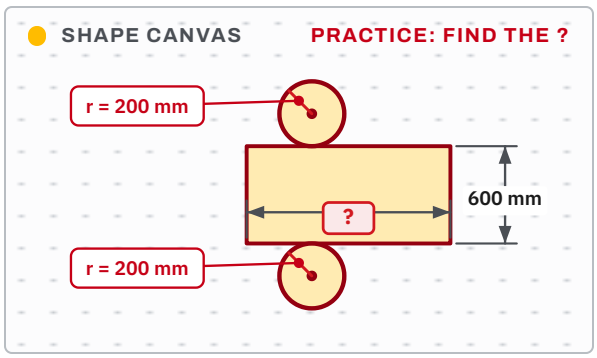
2 Change one value: a disc has a radius of 100 mm. The radius is doubled to 200 mm. Find both areas to two decimal places. How many times bigger is the new area?



WORKING

Answer _____

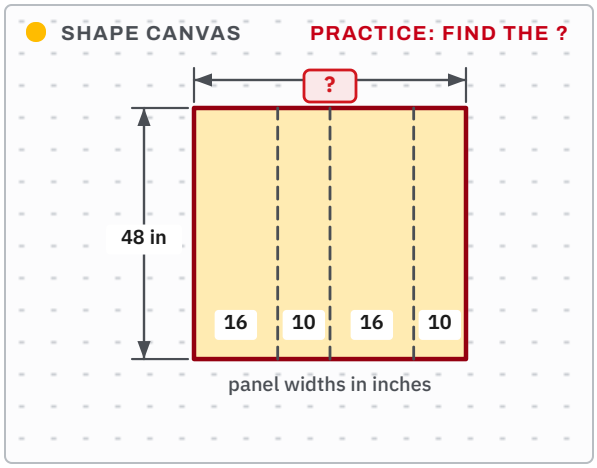
3 This is the net of a closed cylinder with radius 200 mm and height 600 mm. Find the hidden stretch-out to two decimal places, then the surface area.



WORKING

Answer _____

4 The duct is 16 in by 10 in and open at both ends. Find the hidden width of its net, then the sheet area in ft² to two decimal places.



WORKING

Answer _____

WORKSHEET 10D

Materials takeoff

Area, openings, waste, whole units and cost

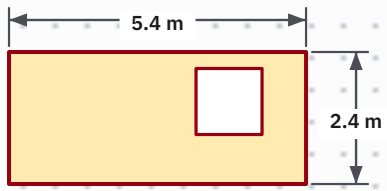
Name _____

Date _____

Sheet sizes, coverage, waste and prices are given in each problem. Show the net area, the units before rounding, the whole units you buy and the cost.

1 Drywall sheets are 1.2 m by 2.4 m. How many sheets cover the wall?

SHAPE CANVAS READ THE DRAWING

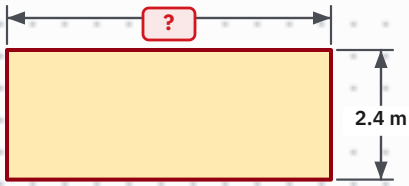


WORKING

Answer _____

2 The wall has no openings and took exactly 5 sheets of 1.2 m by 2.4 m drywall. Find the hidden length.

SHAPE CANVAS PRACTICE: FIND THE ?

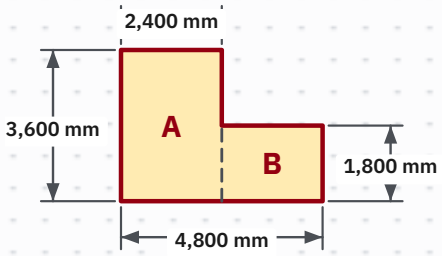


WORKING

Answer _____

3 Tile the L-shaped floor. Allow 10% waste. A box covers 1.08 m² and costs $42.00. How many boxes, and what is the cost?

SHAPE CANVAS READ THE DRAWING

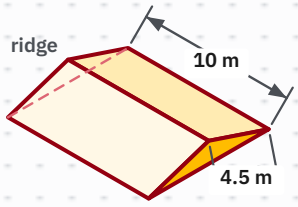


WORKING

Answer _____

4 Shingle both sides of the roof. One bundle covers 3.1 m² and costs $36.00. How many bundles, and what is the cost?

SHAPE CANVAS READ THE DRAWING

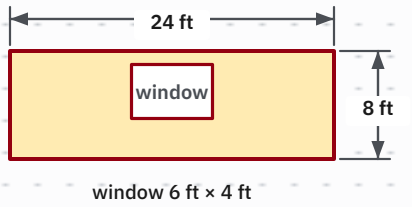


WORKING

Answer _____

5 Sheathing comes in 4 ft by 8 ft sheets at $27.40. How many sheets, and what is the cost?

SHAPE CANVAS READ THE DRAWING



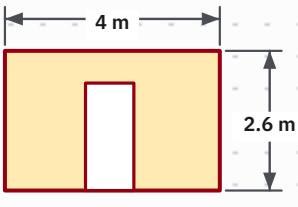
window 6 ft × 4 ft

WORKING

Answer _____

6 The wall gets two coats of paint. One litre covers 9 m², and paint is sold in 1 L tins. How many tins?

SHAPE CANVAS READ THE DRAWING



door 0.9 m × 2 m

WORKING

Answer _____

CHAPTER

11

Volume and Capacity

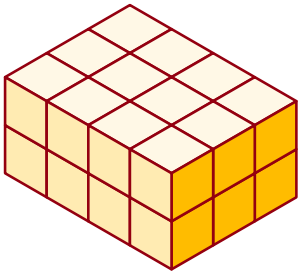
Volume is the amount of space a solid takes up or holds. You need it to size a tank, to know how much water fills a pipe run before a pressure test, and to work out what a hopper holds. This chapter counts cubic units, builds the volume formulas for prisms, cylinders, cones, pyramids and spheres, and changes volumes into litres and US gallons.

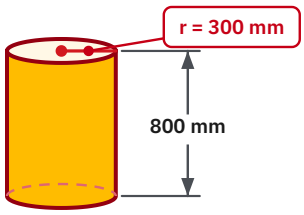
In this chapter you will

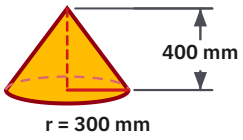
- Explain volume and use cubic units
- Find the volume of prisms with V = end area × length
- Find the volume of cylinders, tanks and pipes
- Find the volume of cones, pyramids and spheres
- Change volumes into litres and US gallons
- Break a composite solid into basic solids

Key words

- volume
- cubic unit
- prism
- cylinder
- cone and pyramid
- sphere
- capacity







11.1 Volume and cubic units

Volume is the amount of space a solid takes up or holds. Length uses one direction and area uses two. Volume uses three: length, width and height.

Volume is counted in cubes. A **cubic unit** is a cube 1 unit long on each edge. A cube 1 cm on each edge is 1 cubic centimetre, written 1 cm³. The small 3 shows that three lengths were multiplied. The cubic units in this book are mm³, cm³ and m³, and in³ and ft³.

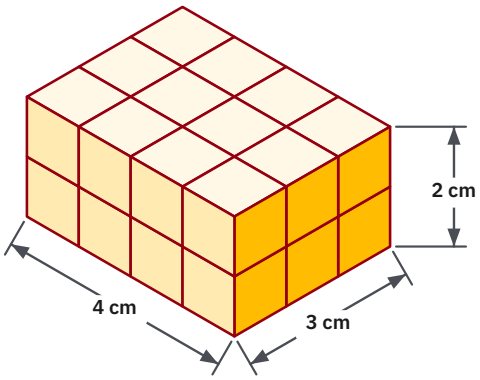


Figure 11.1 One layer holds 4 × 3 = 12 cubes of 1 cm³. Two layers hold 12 × 2 = 24 cm³.

Changing between cubic units uses the length factor three times. A cubic centimetre is 10 mm × 10 mm × 10 mm = 1,000 mm³. A cubic metre is 1,000 mm × 1,000 mm × 1,000 mm = 1,000,000,000 mm³. A cubic foot is 12 in × 12 in × 12 in = 1,728 in³.

THREE FACTORS, NOT ONE

To change mm³ to m³, divide by 1,000,000,000 (1,000 three times). Dividing by 1,000 only changes a length. Many people find it easier to change each length to metres first and then multiply.

TRY IT 11.1

1. A block of 1 cm cubes is 5 long, 2 wide and 3 high. How many cm³?
2. How many mm³ are in 1 cm³?
3. How many in³ are in 2 ft³?
4. Which is a volume: 8 m, 8 m² or 8 m³?

Answers are in the Answer Key.

11.2 Prisms

A **prism** is a solid with two identical, parallel ends joined by flat sides. Cut a slice anywhere along it and the slice is the same shape as the end. A box, a triangular trough and a length of angle bar are all prisms.

Think of the end as one layer. The volume is the area of that layer times how many layers there are, which is the length.

VOLUME OF A PRISM

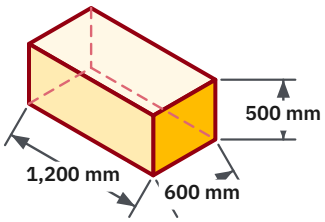
$$V = \text{end area} \times \text{length}$$

$$\text{Box: } V = l \times w \times h$$

Find the end area with the formulas from Chapter 10. All lengths in the same unit.

Example 11.1 A rectangular tank

A tank is 1,200 mm long, 600 mm wide and 500 mm high inside. Find its volume in mm³ and in m³.



SOLUTION

1. $V = l \times w \times h = 1{,}200 \times 600 \times 500 = 360{,}000{,}000$ mm³.
2. In metres the tank is 1.2 m by 0.6 m by 0.5 m.
3. $V = 1.2 \times 0.6 \times 0.5 = 0.36$ m³.

CALCULATOR

TI-30Xa

mm³

1200 × 600 × 500 =

Display 360000000.

m³

1.2 × 0.6 × 0.5 =

Display 0.36

Sharp EL-531XT

mm³

1200 × 600 × 500 =

Display 360’000’000.

m³

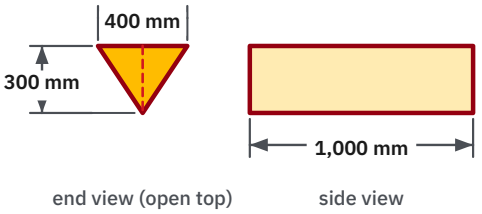
1.2 × 0.6 × 0.5 =

Display 0.36

Answer 360,000,000 mm³ = 0.36 m³

Example 11.2 A triangular trough

A trough is a triangular prism. Its end is a triangle with a base of 400 mm and a height of 300 mm. The trough is 1,000 mm long. Find the volume in m³.



SOLUTION

1. End area: 400 × 300 ÷ 2 = 60,000 mm².
2. Volume: 60,000 × 1,000 = 60,000,000 mm³.
3. In m³: 60,000,000 ÷ 1,000,000,000 = 0.06 m³.

CALCULATOR

TI-30Xa

400 × 300 ÷ 2 × 1000 ÷ 1000000000 =

Display 0.06

Sharp EL-531XT

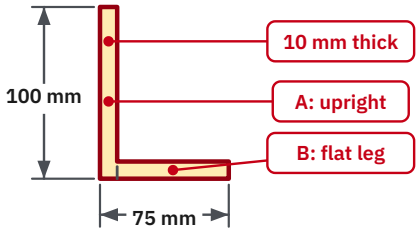
400 × 300 ÷ 2 × 1000 ÷ 1000000000 =

Display 0.06

Answer 0.06 m³

Example 11.3 Break down the end of an angle bar

An angle bar has legs of 100 mm and 75 mm, is 10 mm thick and 300 mm long. Find the volume of steel in cm³.



SOLUTION

1. Break the L-shaped end into two rectangles. Upright: 100 × 10 = 1,000 mm². Flat leg without the corner: (75 − 10) × 10 = 650 mm².
2. End area: 1,650 mm². Volume: 1,650 × 300 = 495,000 mm³.
3. In cm³ (1 cm³ = 1,000 mm³): 495,000 ÷ 1,000 = 495 cm³.

CALCULATOR

TI-30Xa

1650 × 300 ÷ 1000 =

Display 495.

Sharp EL-531XT

1650 × 300 ÷ 1000 =

Display 495.

Answer 495 cm³

TRY IT 11.2

1. Box 800 mm by 500 mm by 300 mm. Volume in m^3?
2. Box 24 in by 18 in by 12 in. Volume in in^3 and ft^3?
3. Triangular prism: end base 0.5 m, end height 0.4 m, length 3 m. Volume?
4. A flat bar 50 mm by 10 mm, 2,000 mm long. Volume in cm^3?

Answers are in the Answer Key.

11.3 Capacity: litres and US gallons

Capacity is how much a container holds. Liquids are measured in litres (L) or in gallons. The litre is tied to the cubic units: a cube 10 cm (100 mm) on each edge holds exactly 1 L.

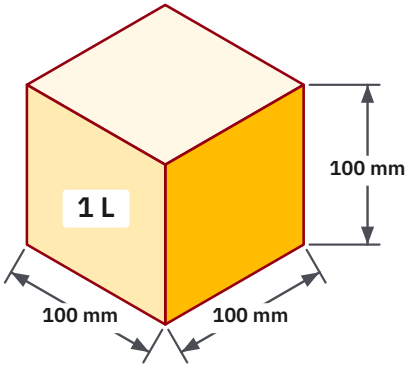


Figure 11.2 A cube 100 mm on each edge: 10 cm × 10 cm × 10 cm = 1,000 cm^3 = 1 L.

Volume	Capacity	Exact or rounded	Source
1,000 cm^3	1 L	exact	SI definition of the litre
1,000,000 mm^3	1 L	exact	100 mm × 100 mm × 100 mm
1 m^3	1,000 L	exact	SI definition of the litre
231 in^3	1 US gal	exact	NIST SP 1038
1 US gal	3.785411784 L	exact	231 × 16.387064 cm^3 (NIST)

Table 11.1 Capacity factors. The gallon factors are the NIST SP 1038 factors used in Chapter 7.

Example 11.4 Litres in a tank

How many litres does the 1,200 mm by 600 mm by 500 mm tank from Example 11.1 hold?

SOLUTION

1. Its volume is 360,000,000 mm³, which is 0.36 m³.
2. Method 1: 1 L = 1,000,000 mm³, so 360,000,000 ÷ 1,000,000 = 360 L.
3. Method 2: 1 m³ = 1,000 L, so 0.36 × 1,000 = 360 L.

CALCULATOR

TI-30Xa

360000000 ÷ 1000000 =

Display 360.

Sharp EL-531XT

360000000 ÷ 1000000 =

Display 360.

Answer 360 L

Example 11.5 A tank in US gallons and litres

A US-made tank is 24 in by 18 in by 12 in inside. Find its capacity in US gallons and in litres, to two decimal places.

SOLUTION

1. Volume: 24 × 18 × 12 = 5,184 in³.
2. US gallons: 1 US gal = 231 in³, so 5,184 ÷ 231 = 22.44 US gal.
3. Litres: multiply the full gallon value by 3.785411784: 84.95 L.

CALCULATOR

TI-30Xa

US gallons

24 × 18 × 12 ÷ 231 =

Display 22.44155844

Litres

24 × 18 × 12 ÷ 231 × 3.785411784 =

Display 84.95053978

Sharp EL-531XT

US gallons

24 × 18 × 12 ÷ 231 =

Display 22.44155844

Litres

24 × 18 × 12 ÷ 231 × 3.785411784 =

Display 84.95053978

Answer 22.44 US gal = 84.95 L

TWO DIFFERENT GALLONS

The US gallon is defined as 231 in³, which is 3.785411784 L (NIST SP 1038). The imperial gallon is defined as 4.54609 L (Weights and Measures Act). Check which gallon a label means. This book uses US gallons.

Section 11.10 works out tank capacities in litres, US gallons and imperial gallons side by side.

TRY IT 11.3

1. Write 2.5 m³ in litres.
2. A tank is 1,000 mm by 500 mm by 400 mm inside. Litres?
3. 50 US gal in litres, to two decimal places?
4. 200 L in US gallons, to two decimal places?
5. A box 12 in by 11 in by 7 in. US gallons?
6. How many litres in 1 ft³? Use 1 ft³ = 1,728 in³. Two decimal places.

Answers are in the Answer Key.

11.4 Cylinders: tanks and pipes

A **cylinder** has two equal circular ends joined by a curved side. It works like a prism: every slice is the same circle. So its volume is the circle area times the height (or length).

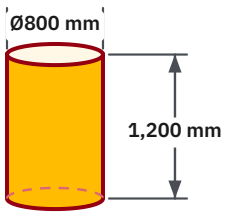
VOLUME OF A CYLINDER

$$V = \pi \times r^2 \times h$$

r = radius of the end, h = height or length. For a pipe, h is the length of the run.

Example 11.6 A vertical tank

A vertical tank is 800 mm across inside and 1,200 mm high. How many litres does it hold, to two decimal places?



SOLUTION

1. Radius: 800 ÷ 2 = 400 mm.
2. V = π × 400² × 1,200 = 603,185,789.49 mm³.
3. Litres: ÷ 1,000,000 gives 603.19 L.

CALCULATOR

TI-30Xa

π × 400 x² × 1200 ÷ 1000000 =

Display 603.1857895

Sharp EL-531XT

π × 400 x² × 1200 ÷ 1000000 =

Display 603.1857895

Answer 603.19 L

Example 11.7 Filling a pipe run

A pipe run is 30 m long and the pipe has an inside diameter of 102 mm. How many litres of water fill it, to two decimal places?

SOLUTION

1. Work in metres. Inside radius: 102 ÷ 2 = 51 mm = 0.051 m.
2. V = π × 0.051² × 30 = 0.24514 m³.
3. Litres: × 1,000 = 245.14 L.

CALCULATOR

TI-30Xa

π × 0.051 x² × 30 × 1000 =

Display 245.1384748

Sharp EL-531XT

π × 0.051 x² × 30 × 1000 =

Display 245.1384748

Answer 245.14 L

USE THE INSIDE DIAMETER

Capacity uses the inside diameter, because the water fills the inside. The outside diameter tells you how much space the pipe or tank takes up.

Inside diameter	Litres per metre	Compared with 25 mm
25 mm	0.49	1 time
50 mm	1.96	4 times
100 mm	7.85	16 times
200 mm	31.42	64 times

Table 11.2 Litres per metre of pipe, π × r² × 1 m × 1,000. Doubling the diameter holds 2 × 2 = 4 times as much.

TRY IT 11.4

1. Tank 600 mm across inside, 900 mm high. Litres to two decimal places?
2. Pipe, inside diameter 50 mm, 40 m long. Litres to two decimal places?
3. Tank 24 in across inside, 36 in tall. US gallons to two decimal places?
4. Pipe, inside diameter 150 mm. Litres per metre to two decimal places?

Answers are in the Answer Key.

11.5 Cones and pyramids

A **cone** has a circular base that narrows to a point. A **pyramid** has a flat base, often a square, and triangular sides that meet at a point. A cone holds one third of a cylinder with the same base and height, and a pyramid holds one third of a prism with the same base and height.

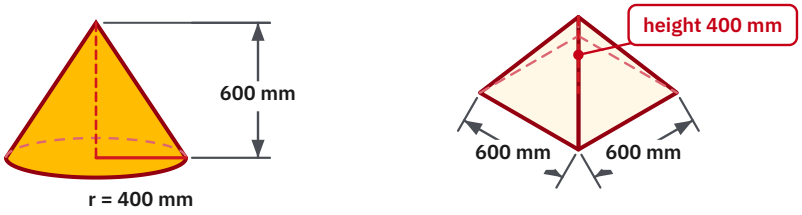


Figure 11.3 A cone and a square pyramid. Each height is the dashed line from the apex straight down to the centre of the base.

VOLUME OF A CONE AND A PYRAMID

$$V = \text{base area} \times h \div 3$$

$$\text{Cone: } V = \pi \times r^2 \times h \div 3$$

$$\text{Square pyramid: } V = b^2 \times h \div 3$$

h = height at right angles to the base, not the sloping side.

Example 11.8 A cone-shaped hopper

A cone-shaped hopper has a radius of 400 mm and a height of 600 mm. How many litres does it hold, to two decimal places?

SOLUTION

1. Base area: $\pi \times 400^2 = 502{,}654.82$ mm².
2. $V = 502{,}654.82 \times 600 \div 3 = 100{,}530{,}964.91$ mm³.
3. Litres: $\div 1{,}000{,}000 = 100.53$ L.

CALCULATOR

TI-30Xa

π × 400 x^2 × 600 ÷ 3 ÷ 1000000 =

Display 100.5309649

Sharp EL-531XT

π × 400 x^2 × 600 ÷ 3 ÷ 1000000 =

Display 100.5309649

Answer **100.53 L**

TRY IT 11.5

1. Cone, radius 300 mm, height 400 mm. Litres to two decimal places?
2. Square pyramid, base 600 mm, height 400 mm. Litres?
3. Square pyramid, base 12 in, height 10 in. Volume in in³?
4. A cone has a radius of 300 mm, a height of 400 mm and a sloping side of 500 mm. Which length goes into the formula as h?

Answers are in the Answer Key.

11.6 Spheres

A **sphere** is a perfectly round ball. Every point on its surface is the same distance, the radius, from the centre. Half a sphere is a hemisphere.

VOLUME OF A SPHERE

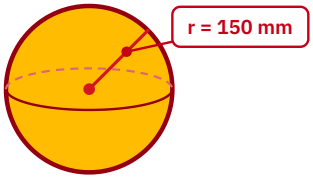
$$V = 4 \div 3 \times \pi \times r^3$$

Hemisphere: half of that

r = radius. r^3 means r × r × r. Use the x^3 key, or y^x 3.

Example 11.9 A spherical float

A spherical float has a radius of 150 mm. Find its volume in litres, to two decimal places.



SOLUTION

1. $r^3 = 150 \times 150 \times 150 = 3{,}375{,}000$.
2. $V = 4 \div 3 \times \pi \times 3{,}375{,}000 = 14{,}137{,}166.94$ mm³.
3. Litres: $\div 1{,}000{,}000 = 14.14$ L.

CALCULATOR

TI-30Xa

4 ÷ 3 × π × 150 2nd x³ ÷ 1000000 =

Display 14.13716694

Sharp EL-531XT

4 ÷ 3 × π × 150 x³ ÷ 1000000 =

Display 14.13716694

Answer **14.14 L**

TRY IT 11.6

1. Sphere, radius 100 mm. Litres to two decimal places?
2. Hemisphere, radius 300 mm. Litres to two decimal places?
3. Sphere, diameter 1 m. m³ to three decimal places?
4. Sphere, radius 6 in. Volume in in³ to two decimal places?

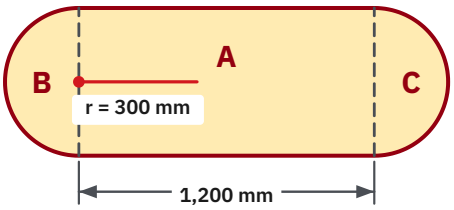
Answers are in the Answer Key.

11.7 Composite solids

Many tanks and hoppers are composite solids. Break the solid into basic solids, find each volume, and add. Take away the volume of any hollow part.

Example 11.10 A tank with round ends

A tank is a cylinder 1,200 mm long with a hemisphere on each end. The radius is 300 mm. How many litres does it hold, to two decimal places?



SOLUTION

1. The two hemispheres make one sphere. Cylinder: π × 300² × 1,200 = 339,292,006.59 mm³.
2. Sphere: 4 ÷ 3 × π × 300³ = 113,097,335.53 mm³.
3. Total: 452,389,342.12 mm³ = 452.39 L.

CALCULATOR

TI-30Xa

(π × 300 x^2 × 1200 + 4 ÷ 3 × π × 300 2nd x^3) ÷ 1000000 =

Display 452.3893421

Sharp EL-531XT

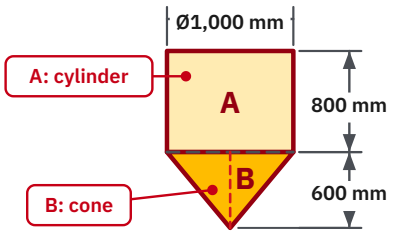
(π × 300 x^2 × 1200 + 4 ÷ 3 × π × 300 x^3) ÷ 1000000 =

Display 452.3893421

Answer 452.39 L

Example 11.11 A hopper

A hopper is a cylinder 1,000 mm across and 800 mm high, on top of a cone 600 mm high. Find the capacity in litres, to two decimal places.



SOLUTION

1. Radius: 1,000 ÷ 2 = 500 mm.
2. A, cylinder: π × 500² × 800 = 628,318,530.72 mm³.
3. B, cone: π × 500² × 600 ÷ 3 = 157,079,632.68 mm³.
4. A + B = 785,398,163.40 mm³ = 785.40 L.

CALCULATOR

TI-30Xa

(π × 500 x² × 800 + π × 500 x² × 600 ÷ 3) ÷ 1000000 =

Display 785.3981634

Sharp EL-531XT

(π × 500 x² × 800 + π × 500 x² × 600 ÷ 3) ÷ 1000000 =

Display 785.3981634

Answer **785.40 L**

CHANGE ONE VALUE

Double the radius of a cylinder and its volume becomes 2 × 2 = 4 times bigger, because r is squared. Double every length of a solid, and the volume becomes 2 × 2 × 2 = 8 times bigger.

TRY IT 11.7

1. Tube: outside diameter 60 mm, inside diameter 50 mm, 2,000 mm long. Volume of metal in cm³, to two decimal places?
2. A cylinder 400 mm across and 500 mm high sits on a cone 300 mm high with the same radius. Litres to two decimal places?
3. A rectangular tank 1,200 mm by 600 mm is filled to a depth of 350 mm. Litres?
4. A cylinder holds 20 L. The radius is doubled and the height kept. How many litres now?

Answers are in the Answer Key.

11.8 Concrete and steel

Concrete is ordered by volume. In Canada it is ordered in cubic metres, and some plans and suppliers still use cubic yards. Slabs, footings and walls are prisms, and round piers are cylinders, so the formulas in this chapter give the volume. Change every size to metres first.

A **cubic yard** is a cube 1 yard on each edge. A yard is exactly 0.9144 m (Weights and Measures Act), so 1 yd³ = 0.9144 × 0.9144 × 0.9144 = 0.764554857984 m³. A yard is also 3 ft, so 1 yd³ = 3 × 3 × 3 = 27 ft³.

CONCRETE VOLUMES

$$m^3 = \text{length} \times \text{width} \times \text{thickness, all in metres}$$

$$yd^3 = m^3 \div 0.9144^3$$

$$yd^3 = ft^3 \div 27$$

On the calculator, key ÷ 0.9144 x³ to change m³ into cubic yards.

Example 11.12 A garage slab

A garage slab is 7.2 m by 6 m and 100 mm thick. How much concrete is that in m³ and in cubic yards, to two decimal places?

SOLUTION

1. Thickness in metres: 100 mm = 0.1 m.
2. V = 7.2 × 6 × 0.1 = 4.32 m³.
3. Cubic yards: 4.32 ÷ 0.9144³ = 5.65 yd³.

CALCULATOR

TI-30Xa

m³

7.2 × 6 × 0.1 =

Display 4.32

yd³

4.32 ÷ 0.9144 2nd x³ =

Display 5.650346675

Sharp EL-531XT

m³

7.2 × 6 × 0.1 =

Display 4.32

yd³

4.32 ÷ 0.9144 x³ =

Display 5.650346675

Answer **4.32 m³ = 5.65 yd³**

The BC Building Code sets the smallest footing a house may have. Sentence 9.15.3.4.(1) says the minimum footing width or area must comply with Table 9.15.3.4. (page 9-120). For strip footings under exterior walls the table gives 250 mm for 1 floor supported, 350 mm for 2 floors and 450 mm for 3 floors. Under interior walls it gives 200 mm, 350 mm and 500 mm. Column footings, for columns spaced 3 m o.c., need 0.4 m², 0.75 m² and 1.0 m². The table applies unless Sentence (2) (supported joist spans over 4.9 m), Sentence (3) (gravel, sand or silt with a high water table) or Articles 9.15.3.5. to 9.15.3.7. change it.

Sentence 9.15.3.8.(1) (page 9-121) sets the thickness: not less than the greater of 100 mm or the width of the projection of the footing beyond the wall it supports. For a footing centred under a wall, the projection is (footing width − wall thickness) ÷ 2.

A **What the table sizes, Sentence (1)**

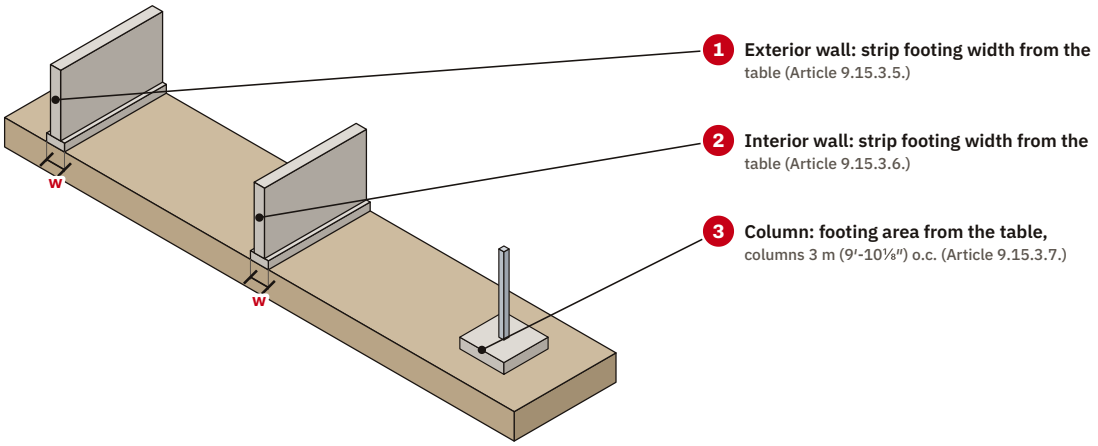


TABLE 9.15.3.4. MINIMUM FOOTING SIZES

Floors supported	Strip footing, exterior walls (2)	Strip footing, interior walls (3)	Column footing area, columns 3 m o.c. (1)
1	250 mm (9 13/16″)	200 mm (7⅞″)	0.4 m² (4.31 ft²)
2	350 mm (13¾″)	350 mm (13¾″)	0.75 m² (8.07 ft²)
3	450 mm (17 11/16″)	500 mm (19 11/16″)	1.0 m² (10.76 ft²)

(1) See Sentence 9.15.3.7.(1). (2) See Sentence 9.15.3.5.(1).
(3) See Sentence 9.15.3.6.(1).

Joist span over 4.9 m, Sentence (2)(b)

`W = w × [Σ sjs / (storeys × 4.9)]`

W = minimum footing width
w = width from the table (joists not over 4.9 m)
Σ sjs = sum of supported joist spans per storey (half spans each side for an interior wall)
storeys = number of storeys supported

Note A-9.15.3.4.(2) example

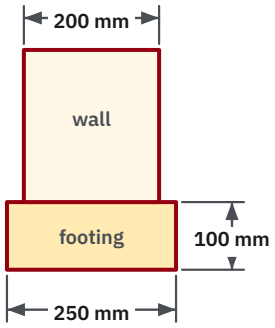
2-storey house, wood I-joists spanning 6 m:
sum1 = 6 + 6 = 12 m
sum2 = 4.9 × 2 = 9.8 m
ratio = 12 / 9.8 = 1.22
1.22 × 350 mm = 427 mm

Figure 11.4 Strip footings under exterior and interior walls, and a column footing, with Table 9.15.3.4. The plate shows the sizes. The code text is quoted above.

Source: BCBC 2024, Division B, Article 9.15.3.4. (page 9-120)

Example 11.13 Footings at the code minimum

A one-storey house with wood-frame floors has 200 mm concrete foundation walls. The strip footings under the exterior walls are centred under the walls, at the minimum width and thickness the code allows. They run 38 m in all. Find the footing size and the concrete in m³ and yd³.



SOLUTION

1. Width: 1 floor supported, exterior walls, Table 9.15.3.4.: 250 mm.
2. Projection: (250 – 200) ÷ 2 = 25 mm.
3. Thickness, Sentence 9.15.3.8.(1): the greater of 100 mm and 25 mm, so 100 mm.
4. Volume: 0.25 × 0.1 × 38 = 0.95 m³.
5. Cubic yards: 0.95 ÷ 0.9144³ = 1.24 yd³.

CALCULATOR

TI-30Xa

m³

0.25 × 0.1 × 38 =

Display 0.95

yd³

0.95 ÷ 0.9144 2nd x³ =

Display 1.242553088

Sharp EL-531XT

m³

0.25 × 0.1 × 38 =

Display 0.95

yd³

0.95 ÷ 0.9144 x³ =

Display 1.242553088

Answer **250 mm by 100 mm, 0.95 m³ = 1.24 yd³**

MINIMUM IS NOT THE DESIGN

The code sizes are the smallest a footing may be. The drawings for a job may call for larger footings. Take the sizes from the drawings, and use the code to check that they are not below the minimum.

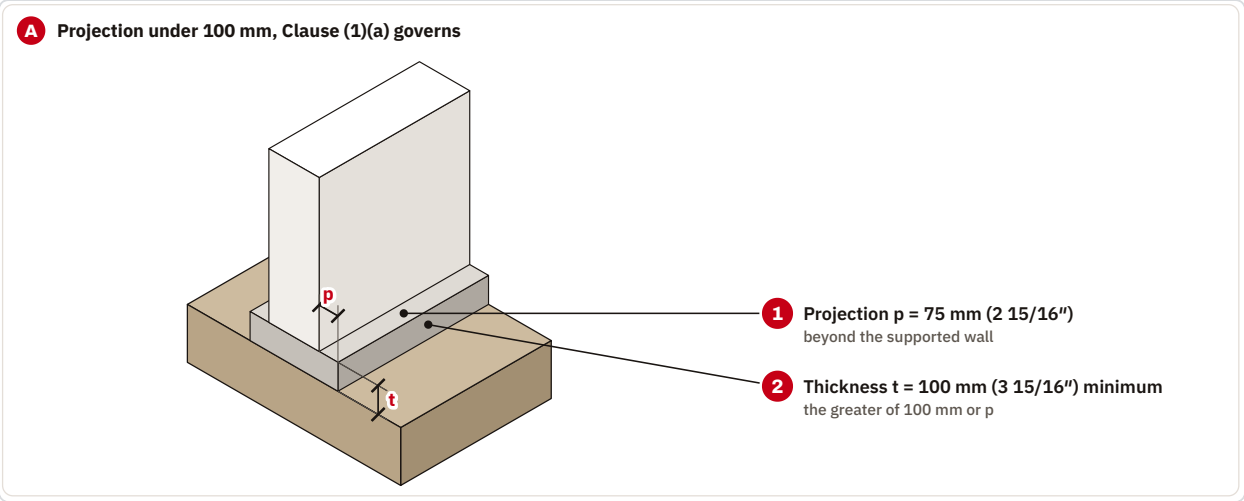


Figure 11.5 Panel A: the projection p is under 100 mm, so the footing is at least 100 mm thick.
Source: BCBC 2024, Division B, Article 9.15.3.8. (page 9-121)

Example 11.14 Footing and foundation wall

A two-storey house has exterior footings at the Table 9.15.3.4. minimum width, centred under 200 mm walls. The walls are 2.4 m high and the footing and the wall each run 40 m. Find the footing thickness, then the total concrete in m³ and yd³.

SOLUTION

1. Width: 2 floors supported, exterior walls: 350 mm. Projection: (350 – 200) ÷ 2 = 75 mm, so the thickness is the greater of 100 mm and 75 mm: 100 mm.
2. Footing: 0.35 × 0.1 × 40 = 1.40 m³.
3. Wall: 0.2 × 2.4 × 40 = 19.20 m³.
4. Total: 1.40 + 19.20 = 20.60 m³ = 26.94 yd³.
5. Sentence 9.15.4.6.(1) (page 9-127) also says exterior foundation walls must extend not less than 150 mm above finished ground level. Check on the drawings that the top of the wall is at least that high.

CALCULATOR

TI-30Xa

m³

0.35 × 0.1 × 40 + 0.2 × 2.4 × 40 =

Display 20.6

yd³

20.60 ÷ 0.9144 2nd x³ =

Display 26.94378276

Sharp EL-531XT

m³

0.35 × 0.1 × 40 + 0.2 × 2.4 × 40 =

Display 20.6

yd³

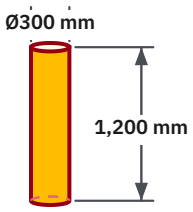
20.60 ÷ 0.9144 x³ =

Display 26.94378276

Answer **20.60 m³ = 26.94 yd³**

Example 11.15 Round piers

A deck sits on 8 round concrete piers. Each pier is 300 mm across and 1.2 m deep. How much concrete, in m³ and yd³, to two decimal places?



SOLUTION

1. Radius: 300 ÷ 2 = 150 mm = 0.15 m.
2. One pier: π × 0.15² × 1.2 = 0.0848 m³.
3. Eight piers: × 8 = 0.68 m³.
4. Cubic yards: ÷ 0.9144³ = 0.89 yd³.

CALCULATOR

TI-30Xa

m³

π × 0.15 x² × 1.2 × 8 =

Display 0.678584013

yd³

π × 0.15 x² × 1.2 × 8 ÷ 0.9144 2nd x³ =

Display 0.88755438

Sharp EL-531XT

m³

π × 0.15 x² × 1.2 × 8 =

Display 0.678584013

yd³

π × 0.15 x² × 1.2 × 8 ÷ 0.9144 x³ =

Display 0.88755438

Answer **0.68 m³ = 0.89 yd³**

A concrete floor on the ground sits on a layer of crushed stone or gravel. Sentence 9.16.4.3.(1) (page 9-130) says concrete slabs shall not be less than 75 mm thick, not counting any topping. Sentence 9.16.2.1.(1) (page 9-129) says not less than 100 mm of coarse clean granular material, with not more than 10% passing a 4 mm sieve, shall be placed beneath floors-on-ground. Sentence (2) lists the exceptions, such as slabs in garages, carports and accessory buildings.

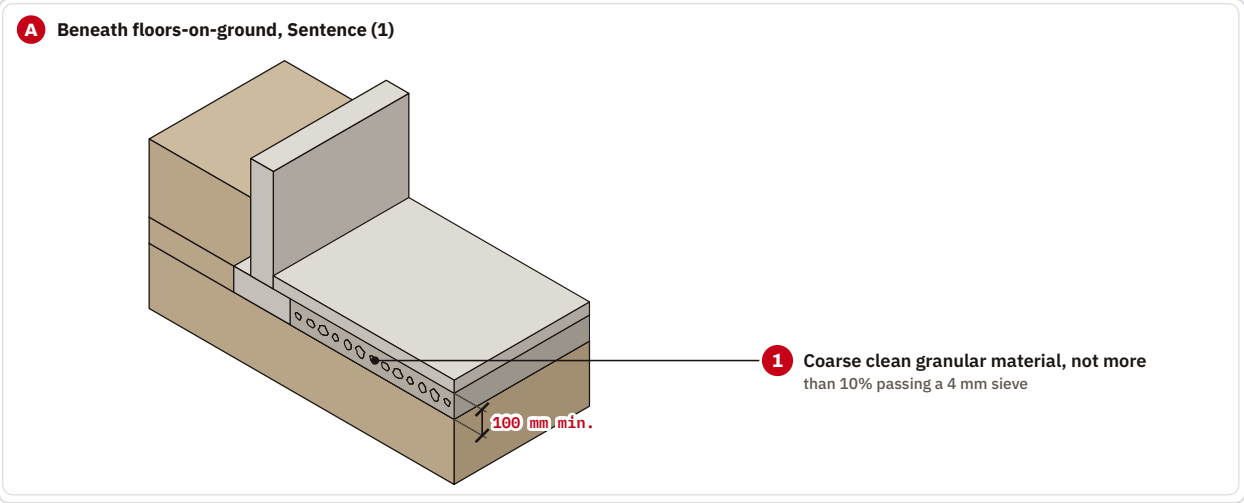
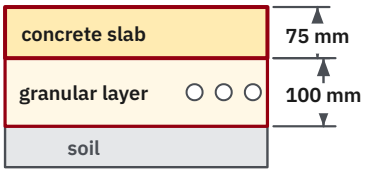


Figure 11.6 Panel A: the granular layer under a floor-on-ground, at least 100 mm deep.
Source: BCBC 2024, Division B, Article 9.16.2.1. (page 9-129)

Example 11.16 Slab and granular base

A basement floor is 9.6 m by 7.2 m. The slab and the granular layer under it are at the code minimum thicknesses. How much concrete and how much granular material, in m³?



SOLUTION

1. Slab: 75 mm = 0.075 m. 9.6 × 7.2 × 0.075 = 5.184 m³.
2. Granular layer: 100 mm = 0.1 m.
   9.6 × 7.2 × 0.1 = 6.912 m³.

CALCULATOR

TI-30Xa

Concrete

9.6 × 7.2 × 0.075 =

Display 5.184

Granular

9.6 × 7.2 × 0.1 =

Display 6.912

Sharp EL-531XT

Concrete

9.6 × 7.2 × 0.075 =

Display 5.184

Granular

9.6 × 7.2 × 0.1 =

Display 6.912

Answer **5.184 m³ of concrete, 6.912 m³ of granular material**

Example 11.17 Backfill around a footing

A trench 40 m long, 0.6 m wide and 0.5 m deep is dug for a footing. The footing is 350 mm wide and 100 mm thick, and the 200 mm wall on it rises from the top of the footing to the top of the trench and beyond. How much backfill is needed to fill the trench back to the top?

SOLUTION

1. Trench: 40 × 0.6 × 0.5 = 12 m³.
2. Footing: 0.35 × 0.1 × 40 = 1.4 m³.
3. Wall inside the trench: 0.2 × (0.5 − 0.1) × 40 = 3.2 m³.
4. Backfill: 12 − 1.4 − 3.2 = 7.4 m³.

CALCULATOR

TI-30Xa

40 × 0.6 × 0.5 − 0.35 × 0.1 × 40 − 0.2 × 0.4 × 40 =

Display 7.4

Sharp EL-531XT

40 × 0.6 × 0.5 − 0.35 × 0.1 × 40 − 0.2 × 0.4 × 40 =

Display 7.4

Answer **7.4 m³ of backfill**

Concrete and steel are heavy. A load on a truck, a lift with a crane or a load on a beam starts from the mass, and the mass is the volume times the **density**. Density is the mass of one cubic metre of the material, in kg/m³.

MASS FROM VOLUME

$$\text{mass (kg)} = \text{volume (m}^3\text{)} \times \text{density (kg/m}^3\text{)}$$

$$1\ \text{t (tonne)} = 1{,}000\ \text{kg}$$

Use the density the supplier or the drawings give. The values in Table 11.3 are for practice.

Material	Density	Source
Concrete, normal weight	2,240 to 2,400 kg/m³	FHWA, Portland Cement Concrete Materials (fhwa.dot.gov/pavement/pubs/013683.pdf), read Oct 20 2026
Steel	7,850 kg/m³	The Engineering ToolBox, Metals and Alloys: Densities (engineeringtoolbox.com), read Oct 20 2026, the source of the hub units module

Table 11.3 Densities used in this chapter. Concrete varies with its stone, air and water, so the source gives a range.

Example 11.18 Mass of the garage slab

Find the mass of the 4.32 m³ garage slab from Example 11.12, at the low and the high end of the concrete range in Table 11.3.

SOLUTION

1. Low end: 4.32 × 2,240 = 9,676.8 kg.
2. High end: 4.32 × 2,400 = 10,368 kg.
3. In tonnes: about 9.68 t to 10.37 t.

CALCULATOR

TI-30Xa

Low

4.32 × 2240 =

Display 9676.8

High

4.32 × 2400 =

Display 10368.

Sharp EL-531XT

Low

4.32 × 2240 =

Display 9’676.8

High

4.32 × 2400 =

Display 10’368.

Answer 9.68 t to 10.37 t

Example 11.19 Mass of a steel plate

A steel plate is 2,440 mm by 1,220 mm and 10 mm thick. Find its mass to two decimal places, using 7,850 kg/m³.

SOLUTION

1. In metres: 2.44 × 1.22 × 0.01 = 0.029768 m³.
2. Mass: 0.029768 × 7,850 = 233.68 kg.

CALCULATOR

TI-30Xa

2.44 × 1.22 × 0.01 × 7850 =

Display 233.6788

Sharp EL-531XT

2.44 × 1.22 × 0.01 × 7850 =

Display 233.6788

Answer **233.68 kg**

TRY IT 11.8

1. A sidewalk is 12 m long, 1.2 m wide and 100 mm thick. Concrete in m³ and yd³, to two decimal places?
2. A pour is 300 ft³. How many cubic yards, to two decimal places?
3. Six round piers, each 250 mm across and 1.5 m deep. Concrete in m³, to two decimal places?
4. A strip footing under an interior wall of a one-storey house, at the Table 9.15.3.4. minimum width, is 100 mm thick and 12 m long. Concrete in m³?
5. A steel round bar is 25 mm across and 6 m long. Mass at 7,850 kg/m³, to two decimal places?

Answers are in the Answer Key.

11.9 Board feet

Softwood and hardwood lumber is often priced by the **board foot**. One board foot is a piece 1 in thick, 12 in wide and 1 ft long, which is 144 in³, or 1/12 of a cubic foot. Prices are often given per thousand board feet, written MBF.

Lumber is named by its nominal size, such as 2 × 4. The Canadian Wood Council lists the actual surfaced dry (S-Dry) size of a 2 × 4 as 38 mm × 89 mm, or 1½ in × 3½ in. Board feet are tallied on the nominal size: a 1 × 12 that is 16 ft long is sold as 16 board feet, even though the dressed piece is smaller (Briggs, Forest Products Measurements and Conversion Factors, University of Washington).

BOARD FEET

$$\text{BF} = \text{thickness (in)} \times \text{width (in)} \times \text{length (ft)} \div 12$$

Thickness and width are the nominal sizes in inches. Length is in feet. MBF = 1,000 board feet.

Nominal size, in	S-Dry size, mm	S-Dry size, in
2 × 4	38 × 89	1½ × 3½
2 × 6	38 × 140	1½ × 5½
2 × 8	38 × 184	1½ × 7¼
2 × 10	38 × 235	1½ × 9¼
2 × 12	38 × 286	1½ × 11¼
4 × 4	89 × 89	3½ × 3½

Table 11.4 Nominal and actual surfaced dry lumber sizes, from the Canadian Wood Council table of dimension lumber sizes (cwc.ca/articles/lumber), read Oct 20 2026.

Example 11.20 A lumber order

An order is for 24 pieces of 2 × 6, each 12 ft long. The yard charges $1,150.00 per MBF. How many board feet, and what is the cost?

SOLUTION

1. One piece: 2 × 6 × 12 ÷ 12 = 12 BF.
2. 24 pieces: 12 × 24 = 288 BF.
3. Cost: 288 ÷ 1,000 × $1,150.00 = $331.20.

CALCULATOR

TI-30Xa

Board feet

2 × 6 × 12 ÷ 12 × 24 =

Display 288.

Cost

288 ÷ 1000 × 1150 =

Display 331.2

Sharp EL-531XT

Board feet

2 × 6 × 12 ÷ 12 × 24 =

Display 288.

Cost

288 ÷ 1000 × 1150 =

Display 331.2

Answer 288 BF, $331.20

Example 11.21 Nominal and actual

Find the board feet in one 2 × 4 that is 8 ft long. Then find how many board feet of wood the dressed piece holds, using its S-Dry size of 1½ in by 3½ in.

SOLUTION

1. Nominal: 2 × 4 × 8 ÷ 12 = 5.33 BF.
2. Actual: 1.5 × 3.5 × 8 ÷ 12 = 3.5 BF.
3. The piece is sold as 5.33 BF but holds 3.5 BF of wood.

CALCULATOR

TI-30Xa

Nominal

2 × 4 × 8 ÷ 12 =

Display 5.333333333

Actual

1.5 × 3.5 × 8 ÷ 12 =

Display 3.5

Sharp EL-531XT

Nominal

2 × 4 × 8 ÷ 12 =

Display 5.333333333

Actual

1.5 × 3.5 × 8 ÷ 12 =

Display 3.5

Answer **5.33 BF sold, 3.5 BF of wood**

TRY IT 11.9

1. Ten pieces of 2 × 10, each 16 ft long. Board feet, to two decimal places?
2. One 1 × 8 board, 10 ft long. Board feet, to two decimal places?
3. One 4 × 4 post, 8 ft long. Board feet, to two decimal places?
4. Forty pieces of 2 × 4, each 8 ft long, at $1,200 per MBF. Board feet and cost?

Answers are in the Answer Key.

11.10 Tank capacity in litres and gallons

A tank may be rated in litres, in US gallons or in imperial gallons. Find the inside volume, change it to litres, and then divide by the size of the gallon you need. The US gallon is 3.785411784 L (231 in³, NIST SP 1038). The imperial gallon is 4.54609 L (Weights and Measures Act). Both factors are exact.

CAPACITY IN THREE UNITS

$$L = mm^3 \div 1{,}000{,}000 = m^3 \times 1{,}000$$

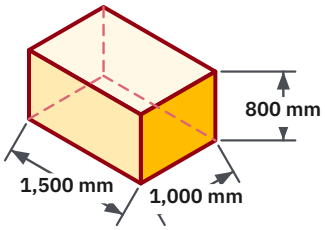
$$\text{US gal} = L \div 3.785411784$$

$$\text{imperial gal} = L \div 4.54609$$

Use the inside sizes of the tank. Keep the full litre value in the calculator before dividing.

Example 11.22 A rectangular tank

A tank is 1,500 mm by 1,000 mm by 800 mm inside. Find its capacity in litres, US gallons and imperial gallons, to two decimal places.



SOLUTION

1. Volume: 1,500 × 1,000 × 800 = 1,200,000,000 mm³ = 1,200 L.
2. US gallons: 1,200 ÷ 3.785411784 = 317.01 US gal.
3. Imperial gallons: 1,200 ÷ 4.54609 = 263.96 imperial gal.

CALCULATOR

TI-30Xa

US gal

1200 ÷ 3.785411784 =

Display 317.0064628

Imperial gal

1200 ÷ 4.54609 =

Display 263.963098

Sharp EL-531XT

US gal

1200 ÷ 3.785411784 =

Display 317.0064628

Imperial gal

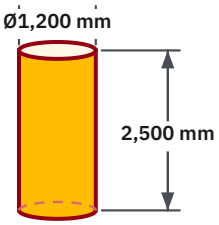
1200 ÷ 4.54609 =

Display 263.963098

Answer 1,200 L = 317.01 US gal = 263.96 imperial gal

Example 11.23 A round tank

A vertical tank is 1,200 mm across and 2,500 mm high inside. Find its capacity in litres, US gallons and imperial gallons, to two decimal places.



SOLUTION

1. Radius: 1,200 ÷ 2 = 600 mm = 0.6 m. Height 2.5 m.
2. Volume: π × 0.6² × 2.5 = 2.82743 m³ = 2,827.43 L.
3. US gallons: ÷ 3.785411784 = 746.93 US gal.
4. Imperial gallons: ÷ 4.54609 = 621.95 imperial gal.

CALCULATOR

TI-30Xa

Litres

π × 0.6 x² × 2.5 × 1000 =

Display 2827.433388

US gal

π × 0.6 x² × 2.5 × 1000 ÷ 3.785411784 =

Display 746.9288811

Imperial gal

π × 0.6 x² × 2.5 × 1000 ÷ 4.54609 =

Display 621.948397

Sharp EL-531XT

Litres

π × 0.6 x² × 2.5 × 1000 =

Display 2′827.433388

US gal

π × 0.6 x² × 2.5 × 1000 ÷ 3.785411784 =

Display 746.9288811

Imperial gal

π × 0.6 x² × 2.5 × 1000 ÷ 4.54609 =

Display 621.948397

Answer **2,827.43 L = 746.93 US gal = 621.95 imperial gal**

THE IMPERIAL GALLON IS BIGGER

One imperial gallon is 4.54609 ÷ 3.785411784 = 1.201 US gallons, so the same tank holds fewer imperial gallons than US gallons.

TRY IT 11.10

1. A tank holds 2,000 L. US gallons and imperial gallons, to two decimal places?
2. A tank is 48 in by 24 in by 30 in inside. US gallons, litres and imperial gallons, to two decimal places?
3. A tank is 900 mm across and 1,500 mm high inside. Litres, US gallons and imperial gallons, to two decimal places?
4. A boiler manual says 100 imperial gal. How many US gallons, to two decimal places?

Answers are in the Answer Key.

WORKSHEET 11A

Find the volume

Practice mode: one drawing hides a dimension behind a red ?

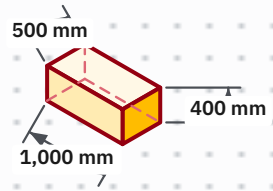
Name _____

Date _____

Write the formula and the numbers. Round to two decimal places where π is used.

1 Volume in m³? Capacity in litres?

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

2 Volume of the cube in mm³ and in litres?

SHAPE CANVAS — READ THE DRAWING

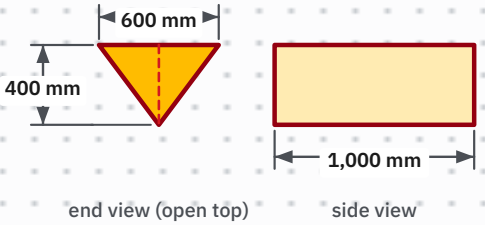


WORKING

Answer _____

3 Capacity of the trough in litres?

SHAPE CANVAS — READ THE DRAWING

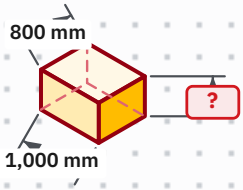


WORKING

Answer _____

4 The tank holds 480 L. Find the ?

SHAPE CANVAS — PRACTICE: FIND THE ?

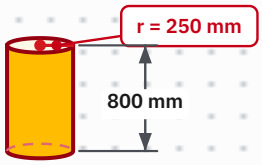


WORKING

Answer _____

5 Capacity of the cylinder in litres?

SHAPE CANVAS — READ THE DRAWING

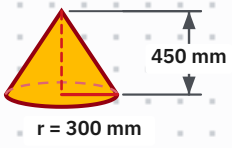


WORKING

Answer _____

6 Capacity of the cone in litres?

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

WORKSHEET 11B

Tanks and hoppers

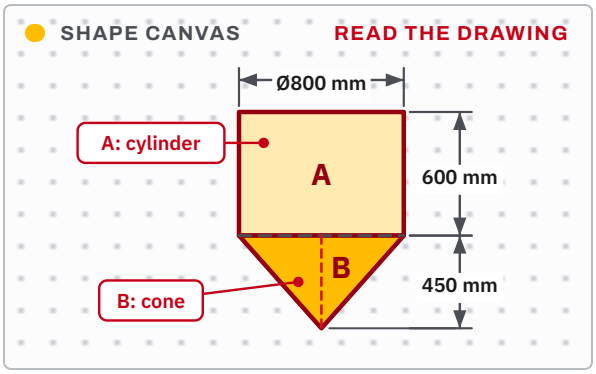
Break each solid into basic solids

Name _____

Date _____

Find each part, then add. Give answers to two decimal places.

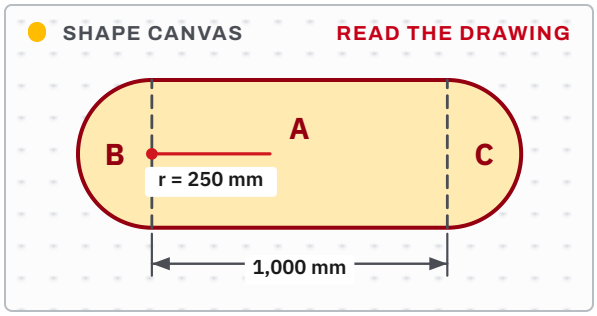
1 Find the capacity of the hopper in litres.



WORKING

Answer _____

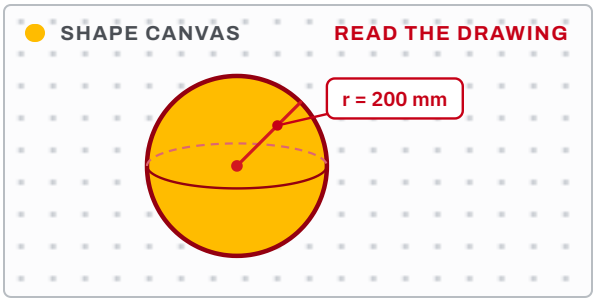
2 The tank has a hemisphere on each end. Find its capacity in litres.



WORKING

Answer _____

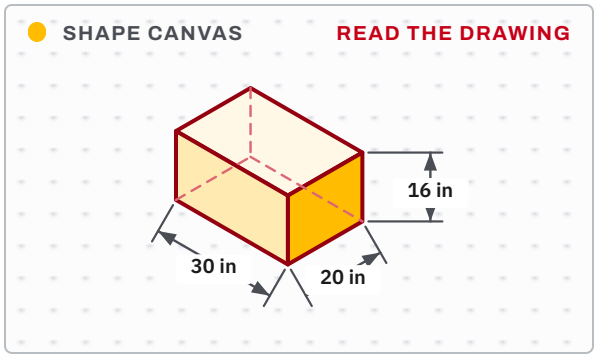
3 Find the capacity of the spherical tank in litres.



WORKING

Answer _____

4 A US-made tank is 30 in by 20 in by 16 in inside. Find its capacity in US gallons.



WORKING

Answer _____

WORKSHEET 11C

Change one value

Unfold it, fold it back, and see what changes

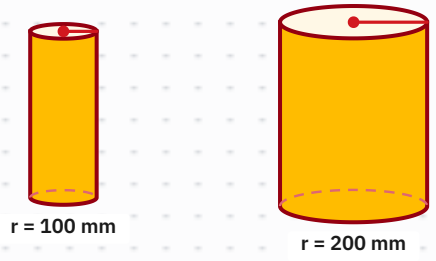
Name _____

Date _____

Give answers to two decimal places. For each pair, say how many times bigger the second answer is.

1 Both cylinders are 500 mm high. The radius doubles from 100 mm to 200 mm. Find both capacities in litres.

SHAPE CANVAS READ THE DRAWING



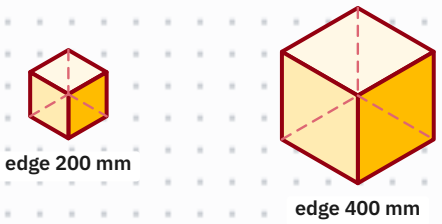
Both are 500 mm high.

WORKING

Answer _____

2 A cube edge doubles from 200 mm to 400 mm. Find both capacities in litres.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

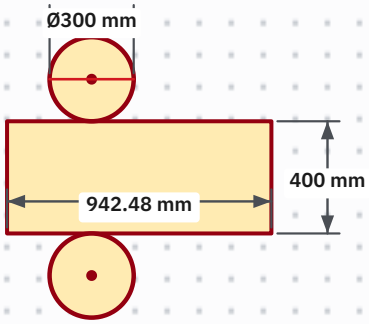
3 A 20 m pipe run has an inside diameter of 50 mm. A second run of the same length is 100 mm inside. How many litres fills each?

WORKING

Answer _____

4 This net folds up into a closed cylinder. Find the capacity of the cylinder in litres.

SHAPE CANVAS READ THE DRAWING



WORKING

Answer _____

CHAPTER

12

Angles and Right Triangles

Square corners hold a shop together. A frame that is out of square will not sit flat, a brace cut to the wrong length will not fit, and a bolt circle laid out at the wrong angle will not line up with its flange.

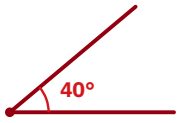
This chapter starts with angles: how they are measured in degrees and how they are named. Then it moves to the right triangle and the Pythagorean theorem, the rule that lets you find a missing side, check a corner for square with the 3-4-5 method and work out the diagonal of a frame.

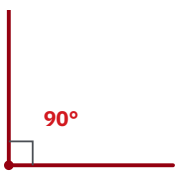
In this chapter you will

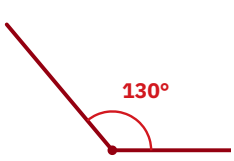
- Measure and name angles in degrees
- Tell acute, right, obtuse and straight angles apart
- Find a missing angle on a line, around a point and in a triangle
- Name the legs and the hypotenuse of a right triangle
- Use the Pythagorean theorem to find a missing side
- Check a corner for square with the 3-4-5 method and with diagonals

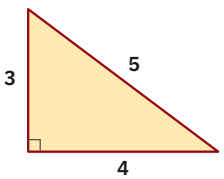
Key words

- angle
- degree
- right angle
- acute angle
- obtuse angle
- leg
- hypotenuse
- Pythagorean theorem









12.1 Angles and degrees

An **angle** is the amount of turn between two lines that meet at a point. The point is called the vertex, and the two lines are the arms of the angle.

Angles are measured in **degrees**. One full turn is 360 degrees, written 360°. Half a turn is 180°, and a quarter turn is 90°.

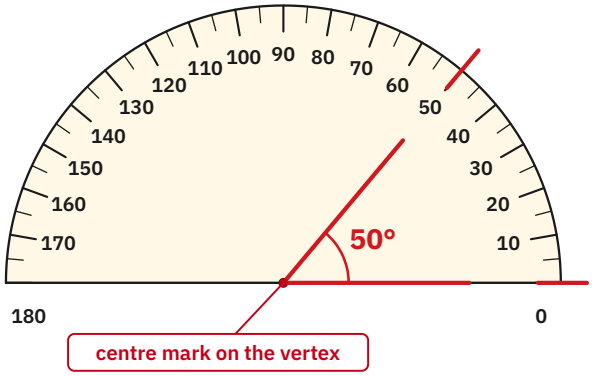


Figure 12.1 A protractor measures an angle. Put the centre mark on the vertex, line one arm up with 0 and read where the other arm crosses the scale. This angle is 50°.

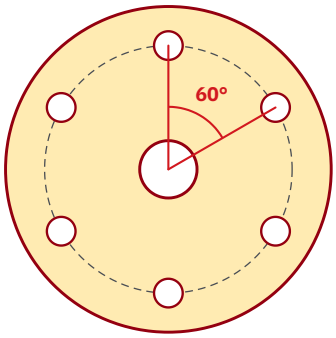
LONG ARMS DO NOT MAKE A BIGGER ANGLE

The size of an angle is the amount of turn, not the length of the arms. A 50° angle drawn with 20 mm arms is the same angle as one drawn with 2 m arms.

Equal parts of a turn come up in layout work all the time. Holes spaced evenly around a circle, such as the bolt holes in a flange, split the full 360° turn into equal angles.

Example 12.1 Angle between bolt holes

Six bolt holes are spaced evenly around a circle. What is the angle between two holes that are next to each other, measured at the centre?



SOLUTION

1. A full turn is 360°.
2. Six equal spaces share the turn: 360° ÷ 6 = 60°.

Answer **60° between holes.**

TRY IT 12.1

1. How many degrees are in half a turn?
2. How many degrees are in one eighth of a turn?
3. Eight bolt holes are evenly spaced on a circle. Angle between neighbours?
4. Five holes are evenly spaced on a circle. Angle between neighbours?
5. Twelve holes are evenly spaced on a circle. Angle between neighbours?
6. What fraction of a full turn is 270°?

Answers are in the Answer Key.

12.2 Types of angles

Angles are named by their size. The most important one in the shop is the **right angle**, exactly 90°. A right angle is a square corner, and drawings mark it with a small square instead of an arc.

An **acute angle** is less than 90°. An **obtuse angle** is more than 90° and less than 180°. A straight angle is exactly 180°: the two arms make a straight line.

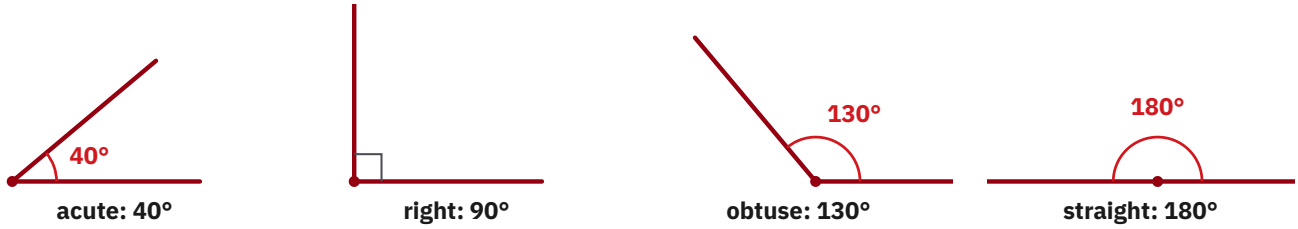
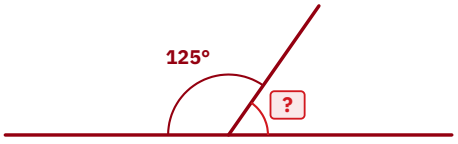


Figure 12.2 The four kinds of angle, drawn at their true size.

Two facts let you find a missing angle without a protractor. Angles that sit side by side on a straight line add to 180°. Angles that go all the way around a point add to 360°. For example, if three lines meet at a point and two of the angles are 110° and 95°, the third angle is 360° − 110° − 95° = 155°.

Example 12.2 A bent strap

A flat strap is bent so that one side of the bend makes an angle of 125° with the straight part, as shown. Find the angle on the other side of the bend.



SOLUTION

1. The two angles sit on a straight line, so they add to 180°.
2. Missing angle = 180° − 125° = 55°.

Answer **55°, an acute angle.**

Angles that add to 90° or to 180°, the equal angles across from each other where two lines cross, and the angles made where a line crosses two parallel lines are in Section 12.7, Angle pairs and parallel lines.

TRY IT 12.2

1. Name each angle: 35°, 90°, 150°, 89°, 180°.
2. Two angles on a straight line. One is 72°. Find the other.
3. Two angles on a straight line. One is 135°. Find the other.
4. Three angles around a point: 120°, 135° and ?. Find the ?
5. Four equal angles go around a point. How big is each?
6. Is 91° acute, right or obtuse?

Answers are in the Answer Key.

12.3 Angles in a triangle

The three angles inside any triangle add to 180°. You can see why by tearing the three corners off a paper triangle and laying them side by side. They fit together into a straight line.



Figure 12.3 The corners of a 50°, 60°, 70° triangle (left) placed side by side make a straight angle: 50° + 60° + 70° = 180°.

ANGLES IN A TRIANGLE

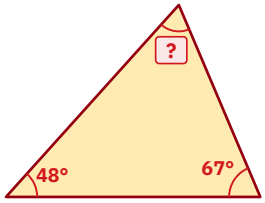
$$\angle A + \angle B + \angle C = 180°$$

In a right triangle: the two acute angles add to 90°

∠ means "angle". A right triangle already uses 90° of the 180°, so the other two angles share the other 90°.

Example 12.3 The third angle of a gusset

A triangular gusset has angles of 48° and 67°. Find the third angle.



SOLUTION

1. The three angles add to 180°.
2. Third angle = 180° − 48° − 67° = 65°.

Answer 65°

A triangle with two equal sides also has two equal angles. They are the angles at the ends of the third side. If the angle between the equal sides is 40°, the other two angles share 180° − 40° = 140°, so each one is 140° ÷ 2 = 70°.

TRY IT 12.3

1. A triangle has angles of 35° and 85°. Find the third angle.
2. A triangle has angles of 90° and 27°. Find the third angle.
3. One acute angle of a right triangle is 45°. Find the other.
4. Can a triangle have two right angles? Why?
5. A gusset has two equal sides with 50° between them. Find the other two angles.
6. All three angles of a triangle are equal. How big is each?

Answers are in the Answer Key.

12.4 Right triangles and the Pythagorean theorem

A right triangle has one right angle. Its sides have special names. The two sides that make the right angle are the **legs**. The side across from the right angle is the **hypotenuse**.

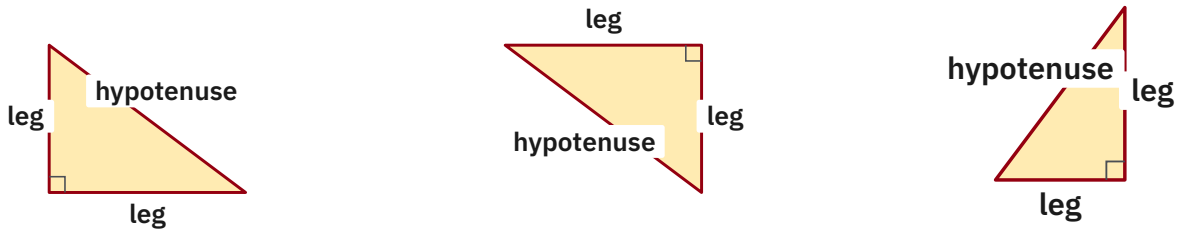


Figure 12.4 Three right triangles turned different ways. The hypotenuse is always across from the small square that marks the right angle, and it is always the longest side.

The **Pythagorean theorem** links the three sides of every right triangle. Square each leg and add the results. The total is the **square** of the hypotenuse.

Figure 12.5 shows it with real squares. A right triangle with legs of 3 and 4 units has a hypotenuse of 5 units. The squares on the legs hold 9 and 16 small squares. The square on the hypotenuse holds 25, and 9 + 16 = 25.

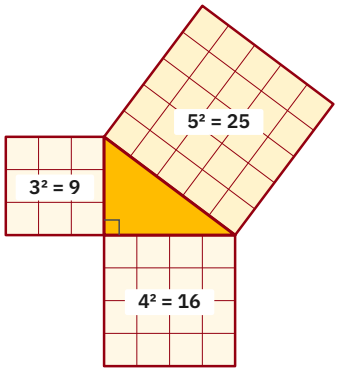


Figure 12.5 The squares on the sides of a 3-4-5 right triangle: 9 + 16 = 25.

PYTHAGOREAN THEOREM

$$a^2 + b^2 = c^2$$

Find the hypotenuse: $c = \sqrt{(a^2 + b^2)}$

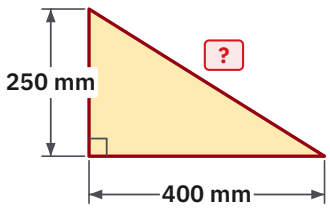
Find a leg: $a = \sqrt{(c^2 - b^2)}$

a and b are the legs, c is the hypotenuse. The second and third lines are the theorem **transposed** (Chapter 8) to make c or a the subject.

1. Sketch the triangle and mark the right angle.
2. Label the hypotenuse c and the legs a and b.
3. To find the hypotenuse, square both legs, add, then take the **square root**.
4. To find a leg, square the hypotenuse and the known leg, subtract, then take the square root.
5. Check: the hypotenuse must be the longest side.

Example 12.4 Length of a gusset edge

A triangular gusset has legs of 400 mm and 250 mm. Find the length of the sloped edge (the hypotenuse) to two decimal places.



SOLUTION

1. $c^2 = 400^2 + 250^2 = 160{,}000 + 62{,}500 = 222{,}500$.
2. $c = \sqrt{222{,}500}$.

CALCULATOR

TI-30Xa

(400 x^2 + 250 x^2) √x

Display 471.6990566

Sharp EL-531XT

√ (400 x^2 + 250 x^2) =

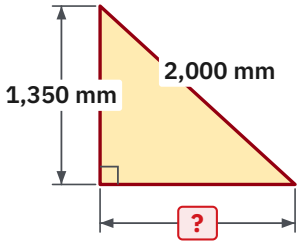
Display 471.6990566

The brackets make the calculator add the squares before it takes the square root. The Sharp puts the √ key first and the brackets after it.

Answer **471.70 mm**

Example 12.5 Finding a leg

A brace 2,000 mm long runs from the floor to a column. Its top end is 1,350 mm above the floor. How far from the column is its bottom end? Give the answer to two decimal places.



SOLUTION

1. The brace is the hypotenuse: c = 2,000. The known leg is a = 1,350.
2. $b^2 = 2{,}000^2 - 1{,}350^2 = 4{,}000{,}000 - 1{,}822{,}500 = 2{,}177{,}500$.
3. $b = \sqrt{2{,}177{,}500}$.

CALCULATOR

TI-30Xa

(2000 x^2 − 1350 x^2) √x

Display 1475.635456

Sharp EL-531XT

√ (2000 x^2 − 1350 x^2) =

Display 1’475.635456

Answer **1,475.64 mm. Check: it is shorter than the 2,000 mm brace.**

ADD TO FIND THE HYPOTENUSE, SUBTRACT TO FIND A LEG

If you add the squares when you are looking for a leg, your answer will be longer than the hypotenuse. That cannot happen. A leg is always shorter than the hypotenuse.

Example 12.6 Feet and inches

A right triangle has legs of 2′ 8″ and 1′ 8″. Find the hypotenuse to the nearest 1/16″, in feet and inches.

SOLUTION

1. Change the legs to inches: 2′ 8″ = 2 × 12 + 8 = 32″ and 1′ 8″ = 1 × 12 + 8 = 20″.
2. $c^2 = 32^2 + 20^2 = 1{,}024 + 400 = 1{,}424$.
3. $c = \sqrt{1{,}424} = 37.73592$″ (the display shows more digits).
4. Change the decimal part to sixteenths: 0.73592 × 16 = 11.77, which rounds to 12 sixteenths.
5. 12/16 = $\frac{3}{4}$ in lowest terms.

CALCULATOR

TI-30Xa

(32 x² + 20 x²) √x

Display 37.73592453

Sharp EL-531XT

√ (32 x² + 20 x²) =

Display 37.73592453

Answer $37\frac{3}{4}$″, which is 3′ $1\frac{3}{4}$″.

TRY IT 12.4

1. Legs 600 mm and 800 mm. Find the hypotenuse.
2. Legs 1,500 mm and 900 mm. Find the hypotenuse to two decimal places.
3. Hypotenuse 1,300 mm, one leg 500 mm. Find the other leg.
4. Hypotenuse 2.5 m, one leg 1.8 m. Find the other leg to two decimal places.
5. Legs 9″ and 12″. Find the hypotenuse.
6. Legs 2′ 0″ and 1′ 3″. Find the hypotenuse to the nearest 1/16″.

Answers are in the Answer Key.

12.5 The 3-4-5 check for square

The theorem also works backwards. If the sides of a triangle fit $a^2 + b^2 = c^2$, the corner between a and b is a right angle. The smallest whole numbers that fit are 3, 4 and 5: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$.

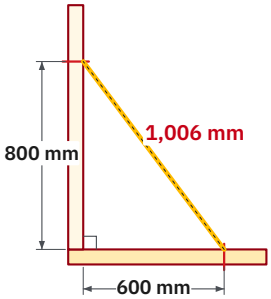
Any multiple of 3, 4 and 5 works too, in any unit. Pick the largest set that fits on the work. A reading error of 1 mm matters less on a 1,000 mm diagonal than on a 500 mm one.

1. From the corner, measure and mark 3 units along one edge.
2. From the same corner, measure and mark 4 units along the other edge.
3. Measure the straight distance between the two marks.

4 If it is exactly 5 units, the corner is square. If it is longer, the corner is open past 90°. If it is shorter, the corner is closed to less than 90°.

Example 12.7 Is the frame square?

You mark 600 mm along one side of a frame corner and 800 mm along the other. The tape reads 1,006 mm between the marks. Is the corner square?



SOLUTION

1. For a square corner: $600^2 + 800^2 = 360{,}000 + 640{,}000 = 1{,}000{,}000$, and $\sqrt{1{,}000{,}000} = 1{,}000$ mm.
2. The tape reads 1,006 mm, which is 6 mm too long.
3. A diagonal that is too long means the corner is opened past 90°.

Answer **No. Close the corner until the marks are 1,000 mm apart.**

Leg a	Leg b	Diagonal c	Check: $a^2 + b^2 = c^2$
300 mm	400 mm	500 mm	$90{,}000 + 160{,}000 = 250{,}000 = 500^2$
600 mm	800 mm	1,000 mm	$360{,}000 + 640{,}000 = 1{,}000{,}000 = 1{,}000^2$
3 ft	4 ft	5 ft	$9 + 16 = 25 = 5^2$
6 ft	8 ft	10 ft	$36 + 64 = 100 = 10^2$

Table 12.1 Multiples of 3-4-5 in metric and imperial units.

TRY IT 12.5

1. Marks at 300 mm and 400 mm. What should the diagonal be?
2. Marks at 9 ft and 12 ft. What should the diagonal be?
3. Marks at 900 mm and 1,200 mm. The diagonal reads 1,494 mm. Is the corner open or closed?
4. Marks at 1.2 m and 1.6 m. What should the diagonal be?
5. Marks at 15″ and 20″. What should the diagonal be?
6. Do 5, 12 and 13 make a right triangle? Show the check.

Answers are in the Answer Key.

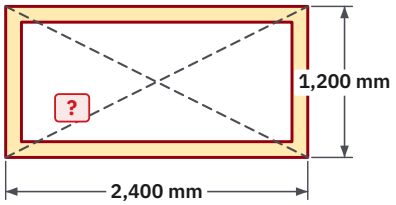
12.6 Diagonals of rectangles

A diagonal joins opposite corners of a rectangle. It cuts the rectangle into two right triangles, so its length comes from the Pythagorean theorem: $d = \sqrt{(l^2 + w^2)}$.

Diagonals are the fastest check for a whole frame. Measure both diagonals corner to corner. If the opposite sides are equal and the two diagonals are equal, every corner is 90°.

Example 12.8 Diagonal of a base frame

A base frame is 2,400 mm long and 1,200 mm wide. What should each diagonal measure, to the nearest millimetre?



SOLUTION

1. $d^2 = 2{,}400^2 + 1{,}200^2 =$
   $5{,}760{,}000 + 1{,}440{,}000 = 7{,}200{,}000.$
2. $d = \sqrt{7{,}200{,}000} = 2{,}683.28$ mm.

CALCULATOR

TI-30Xa

(2400 x² + 1200 x²) √x

Display 2683.281573

Sharp EL-531XT

√ (2400 x² + 1200 x²) =

Display 2′683.281573

Answer 2,683 mm

SQUARING A FRAME

When the two diagonals of a frame are different, the two corners at the ends of the long diagonal are less than 90°. Push those two corners toward each other and measure again until both diagonals match.

TRY IT 12.6

1. A rectangle is 1,600 mm by 1,200 mm. Find the diagonal.
2. A frame is 1,800 mm by 750 mm. Find the diagonal.
3. A frame is 3,000 mm by 1,500 mm. Find the diagonal to the nearest millimetre.
4. A square has 300 mm sides. Find the diagonal to two decimal places. Hint: both legs are 300 mm.
5. A frame measures 2,013 mm on one diagonal and 2,020 mm on the other. Is it square?
6. A door opening is 36″ by 80″. Find the diagonal to the nearest 1/16″.

Answers are in the Answer Key.

12.7 Angle pairs and parallel lines

Some pairs of angles come up so often that they have names. Two angles that add to 90° are **complementary angles**. The two acute angles of a right triangle are complementary. Two angles that add to 180° are **supplementary angles**. The two angles at the bend of the strap in Example 12.2 are supplementary: 125° + 55° = 180°.

When two straight lines cross, they make four angles. The angles across from each other are **vertically opposite angles**, and they are always equal. Each angle and the angle beside it sit on a straight line, so they are supplementary.

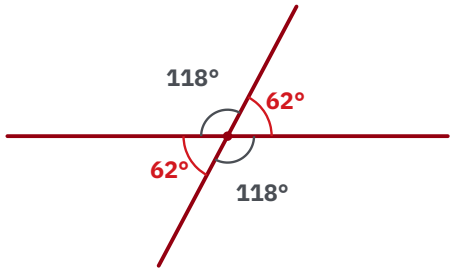


Figure 12.6 Two lines cross. The angles across from each other are equal: 62° and 62°, 118° and 118°. Any two angles side by side add to 180°.

Two lines are **parallel** when they run in the same direction and stay the same distance apart, like the two rails of a gate or the top and bottom plates of a wall. A line that crosses two or more lines is a **transversal**. Where a transversal crosses two parallel lines, it makes the same four angles at each crossing.

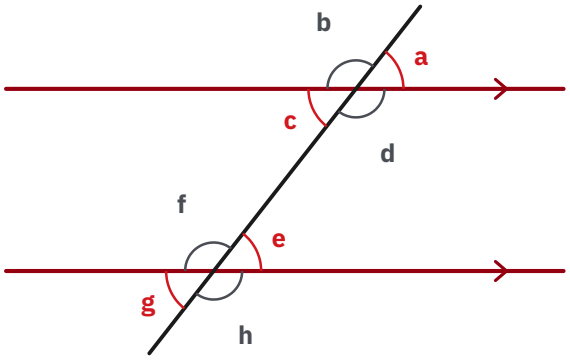


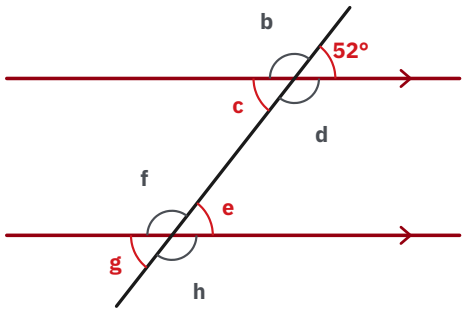
Figure 12.7 A transversal crosses two parallel lines and makes eight angles, a to h. The arrow marks show that the two lines are parallel.

Angle pair	Where the angles are	Rule	In Figure 12.7
Complementary	any two angles	add to 90°	the two acute angles of a right triangle
Supplementary	any two angles	add to 180°	a and b, a and d
Vertically opposite	across from each other at a crossing	equal	a and c, b and d
Corresponding	in the same position at each crossing	equal	a and e, d and h
Alternate	between the parallel lines, on opposite sides of the transversal	equal	c and e, d and f
Co-interior	between the parallel lines, on the same side of the transversal	add to 180°	d and e, c and f

Table 12.2 Angle pairs and their rules.

Example 12.9 Angles on a gate brace

A brace crosses the two parallel rails of a gate. At the top rail, angle a is 52°. Find angles b to h.



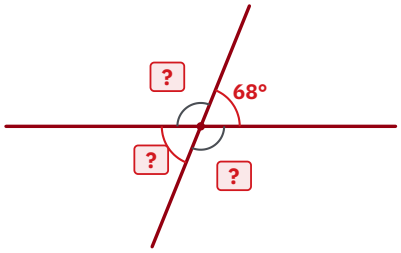
SOLUTION

1. b sits beside a on the straight rail: b = 180° − 52° = 128°.
2. c is opposite a, so c = 52°. d is opposite b, so d = 128°.
3. Corresponding angles are equal, so the bottom crossing repeats the top one: e = 52°, f = 128°, g = 52°, h = 128°.
4. Check with a co-interior pair: d + e = 128° + 52° = 180°.

Answer **a, c, e and g are 52°. b, d, f and h are 128°.**

Example 12.10 Crossed braces

Two braces cross like an X. One of the four angles where they cross is 68°. Find the other three.



SOLUTION

1. The angle beside 68° makes a straight line with it: 180° − 68° = 112°.
2. Vertically opposite angles are equal: 68° across from 68°, and 112° across from 112°.
3. Check: 68° + 112° + 68° + 112° = 360°, a full turn.

Answer **112°, 68° and 112°.**

TRY IT 12.7

1. Find the complement of 28°.
2. Find the supplement of 28°.
3. Two lines cross. One angle is 104°. Find the other three.
4. A transversal crosses two parallel lines. One of the eight angles is 41°. What sizes are the other seven?
5. Two alternate angles lie between two parallel lines. One is 73°. Find the other.
6. Two co-interior angles lie between two parallel lines. One is 115°. Find the other.

Answers are in the Answer Key.

12.8 Degrees, minutes and seconds

A degree can be split into smaller parts. One degree is 60 **minutes**, and one minute is 60 **seconds**. An angle written this way is in degrees, minutes and seconds, D°M′S for short. 36°52′12″ is read "36 degrees, 52 minutes, 12 seconds".

The marks ′ and ″ are the same marks used for feet and inches. After a degree sign they mean minutes and seconds. An angle can also be written in decimal degrees, such as 36.87°. The SIN, COS and TAN keys work in decimal degrees, so you need to change from one form to the other.

DEGREES, MINUTES AND SECONDS

1° = 60′ and 1′ = 60″, so 1° = 3,600″

decimal degrees = D + M ÷ 60 + S ÷ 3,600

minutes = decimal part of the degrees × 60

seconds = decimal part of the minutes × 60

D, M and S are the degrees, minutes and seconds.

Example 12.11 D°M′S to decimal degrees

A drawing gives an angle of 36°52′12″. Change it to decimal degrees.

SOLUTION

1. Minutes: 52 ÷ 60 = 0.866667.
2. Seconds: 12 ÷ 3,600 = 0.003333.
3. Add: 36 + 0.866667 + 0.003333 = 36.87°.

CALCULATOR

TI-30Xa

36.5212 | 2nd | DMS▸DD

Display: 36.87

Sharp EL-531XT

36 | D°M′S | 52 | D°M′S | 12 | = | 2ndF | ↔DEG

Display: 36.87

On the TI-30Xa, key the angle as 36.5212: the degrees, the decimal point, then two digits of minutes and two of seconds. 2nd DMS▸DD changes it to decimal degrees. On the Sharp, press D°M′S after the degrees and after the minutes. The Sharp shows the angle in D°M′S, and 2ndF ↔DEG switches the display to decimal degrees.

Answer 36.87°

Example 12.12 Decimal degrees to D°M′S

Change 47.37° to degrees, minutes and seconds.

SOLUTION

1. The whole number is the degrees: 47°.
2. Minutes: 0.37 × 60 = 22.2. The whole number is the minutes: 22′.
3. Seconds: 0.2 × 60 = 12″.

CALCULATOR

TI-30Xa

47.37 | 2nd | DD▸DMS

Display: 47°22'12"0

Sharp EL-531XT

47.37 | = | 2ndF | ↔DEG

Display: 47°22′12″

On the TI-30Xa, 2nd DD▸DMS shows the angle in D°M′S while you look at it. The next calculation goes on in decimal degrees. On the Sharp, 2ndF ↔DEG switches the display between decimal degrees and D°M′S.

Answer 47°22′12″

KEYING ANGLES IN D°M′S

TI-30Xa: key D.MMSS, with two digits for the minutes and two for the seconds, using 0s where needed. 48°5′3″ is keyed 48.0503. Change a D°M′S value with 2nd DMS▸DD before you use it in a calculation.

Sharp EL-531XT: key the degrees, D°M′S, the minutes, D°M′S, then the seconds. The Sharp adds and subtracts D°M′S values as they are and shows the result in D°M′S.

Sources: TI-30Xa Guidebook, page 7 (DMS), and Sharp EL-531XT Operation Manual, Calculation Examples [10].

Example 12.13 Adding angles

Add 27°45′30″ + 18°32′45″.

SOLUTION

1. Add each column: 45°, 77′ and 75″.
2. 75″ is more than 60″: 75″ = 1′ 15″. Carry 1 to the minutes: 78′.
3. 78′ = 1° 18′. Carry 1 to the degrees: 46°.

CALCULATOR

TI-30Xa

27.4530 | 2nd | DMS▸DD | + | 18.3245

2nd | DMS▸DD | = | 2nd | DD▸DMS

Display: 46°18'15"0

Sharp EL-531XT

27 | D°M′S | 45 | D°M′S | 30 | + | 18

D°M′S | 32 | D°M′S | 45 | =

Display: 46°18′15″

The TI-30Xa adds in decimal degrees, and 2nd DD▸DMS shows the sum in D°M′S. The Sharp adds the D°M′S values and shows the sum in D°M′S.

Answer 46°18′15″

Example 12.14 Subtracting from 90°

Find 90° − 37°25′15″, the complement of 37°25′15″.

SOLUTION

1. Write 90° as 89°59′60″: borrow 1° = 60′, then 1′ = 60″.
2. Subtract each column: 89 − 37 = 52°, 59 − 25 = 34′, 60 − 15 = 45″.

CALCULATOR

TI-30Xa

90 | − | 37.2515 | 2nd | DMS▸DD | = | 2nd | DD▸DMS

Display: 52°34'45"0

Sharp EL-531XT

90 | − | 37 | D°M′S | 25 | D°M′S | 15 | = | 2ndF | ↔DEG

Display: 52°34′45″

Answer 52°34′45″

36.52° IS NOT 36°52′

Decimal degrees and minutes are not the same thing. 36°52′ = 36 + 52 ÷ 60 = 36.8667°, but 36.52° = 36°31′12″. Always change one form to the other before you compare or calculate.

TRY IT 12.8

1. Change 25°30′ to decimal degrees.
2. Change 12°15′36″ to decimal degrees.
3. Change 63.75° to D°M′S.
4. Change 18.125° to D°M′S.
5. Add 14°38′50″ + 22°45′25″.
6. Find the supplement of 72°41′36″.

Answers are in the Answer Key.

12.9 Stair geometry

A straight stair is a row of small right triangles that make one large right triangle. Each step has a **unit rise**, the height of one riser, and a **unit run**, the depth of one tread measured from nosing to nosing. The whole flight has a **total rise** from floor to floor and a **total run** along the floor. The stringer, the board that carries the treads, slopes along the long side.

Chapter 8 shows the stair formulas: how many risers a stair needs, the unit rise and the total run. This section puts those numbers on a drawing, checks them against the BC Building Code and finds the stringer line with the Pythagorean theorem.

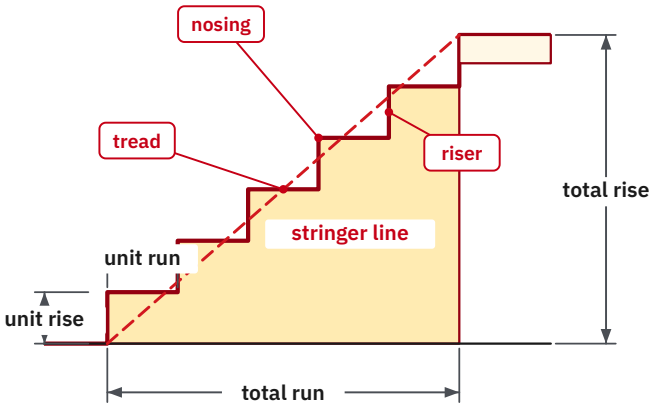


Figure 12.8 A straight flight with 6 risers and 5 treads. The upper floor is the last step, so this flight has one tread fewer than it has risers. The dashed stringer line runs from the bottom corner to the top corner.

STAIR GEOMETRY

$$\text{treads} = \text{risers} - 1 \text{ (a flight from floor to floor)}$$

$$\text{total rise} = \text{risers} \times \text{unit rise}$$

$$\text{total run} = \text{treads} \times \text{unit run}$$

$$\text{stringer line} = \sqrt{(\text{total rise}^2 + \text{total run}^2)}$$

The stringer line is the hypotenuse of the right triangle that has the total rise and the total run as its legs. The board for a stringer is longer than the line, to leave room for the cuts at each end.

What the BC Building Code says

The BC Building Code 2024 (BCBC), Division B, Part 9, limits the rise and run of stairs. The limits depend on the stair type. In the code, private stairs are "exterior and interior stairs that serve a. single dwelling units, b. houses with a secondary suite including their common spaces, or c. garages that serve houses described in Clause a) or b)". Public stairs are "all stairs not described as service stairs or private stairs" (Notes to Tables 9.8.4.1. and 9.8.4.2., page 9-31). The rise and run tables do not apply to "stairs serving areas only used as service rooms or service spaces".

What is limited	Private stair	Public stair	Where in the BCBC 2024, Division B
Rise, maximum	200 mm	180 mm	Article 9.8.4.1, Sentence (1), Table 9.8.4.1, page 9-31
Rise, minimum	125 mm	125 mm	Article 9.8.4.1, Sentence (1), Table 9.8.4.1, page 9-31
Run, maximum	355 mm	No limit	Article 9.8.4.2, Sentence (1), Table 9.8.4.2, page 9-31
Run, minimum	255 mm	280 mm	Article 9.8.4.2, Sentence (1), Table 9.8.4.2, page 9-31
Depth of a rectangular tread	not less than its run and not more than its run plus 25 mm	the same	Article 9.8.4.2, Sentence (2), page 9-31
Risers in one flight	5 mm between adjacent treads or landings, 10 mm between the tallest and shortest	the same	Article 9.8.4.4, Sentence (1), page 9-32
Runs of rectangular treads in one flight	5 mm between adjacent treads, 10 mm between the deepest and shallowest	the same	Article 9.8.4.4, Sentence (3), page 9-32

Table 12.3 Stair limits quoted from the BCBC 2024. The rise "is measured as the vertical nosing-to-nosing distance" (Sentence 9.8.4.1.(1)).

Clear height over a stair is measured "vertically, over the clear width of the stair, from a straight line tangent to the tread and landing nosings to the lowest point above". It "shall not be less than 2 050 mm", and over stairs "serving a single dwelling unit or a house with a secondary suite including their common spaces" it "shall not be less than 1 950 mm" (Article 9.8.2.2, Sentences (1) to (3), page 9-30).

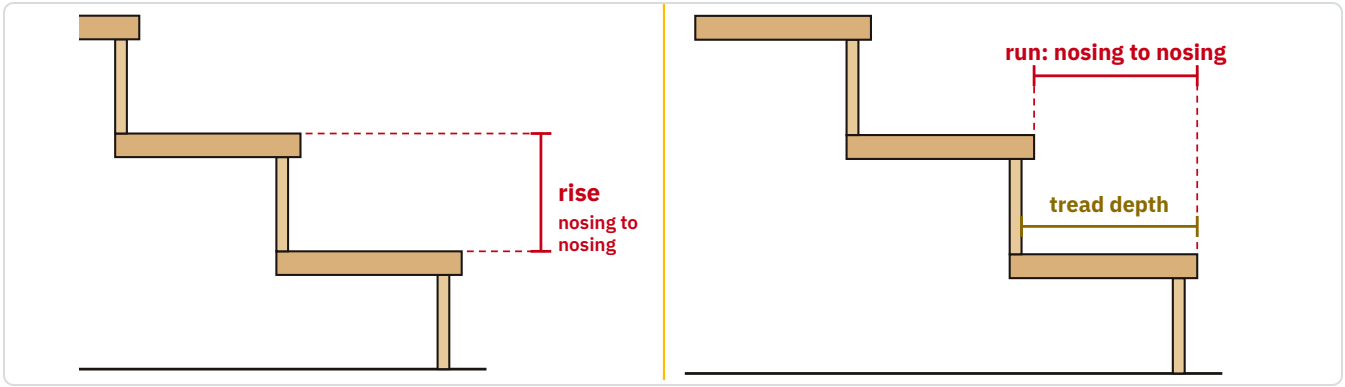
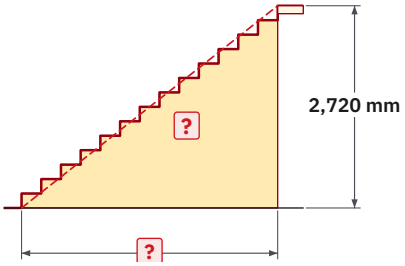


Figure 12.9 Rise (left) and run (right) are both measured from nosing to nosing. Tread depth (right) is measured from the nosing to the riser, so it is a little more than the run.

Source: BCBC 2024, Division B, Article 9.8.4.1. (page 9-30), and BCBC 2024, Division B, Article 9.8.4.2. (page 9-31)

Example 12.15 Laying out a stair

A stair in a house runs from one floor to the next. The total rise is 2,720 mm. Use the private stair maximum rise of 200 mm to find the number of risers and the unit rise. The treads have a unit run of 265 mm. Check both against Table 12.3, then find the total run and the stringer line, to two decimal places.



SOLUTION

1. Risers (Chapter 8): 2,720 ÷ 200 = 13.6. Round up to 14 risers, so no riser is taller than 200 mm.
2. Unit rise = 2,720 ÷ 14 = 194.29 mm. It is between 125 mm and 200 mm.
3. Treads = 14 − 1 = 13. The unit run of 265 mm is between 255 mm and 355 mm.
4. Total run = 13 × 265 = 3,445 mm.
5. Stringer line = √(2,720² + 3,445²).

CALCULATOR

TI-30Xa

Unit rise

2720 ÷ 14 =

Display 194.2857143

Stringer line

(2720 x² + 3445 x²) √x

Display 4389.353597

Sharp EL-531XT

Unit rise

2720 ÷ 14 =

Display 194.2857143

Stringer line

√ (2720 x² + 3445 x²) =

Display 4'389.353597

Answer **14 risers of 194.29 mm and 13 treads. Total run 3,445 mm, stringer line 4,389.35 mm.**

Example 12.16 A deck stair in inches

A deck stair has 6 risers of $7\frac{1}{4}$″ and 5 treads with a unit run of $10\frac{1}{2}$″. Find the total rise, the total run and the stringer line to the nearest 1/16″. Then change the unit rise and run to millimetres (1 in = 25.4 mm, Chapter 7) and check them against the private stair limits.

SOLUTION

1. Total rise = 6 × 7.25 = 43.5″. Total run = 5 × 10.5 = 52.5″.
2. Stringer line = √(43.5² + 52.5²) = 68.180″, which is $68\frac{3}{16}$″ or 5′ $8\frac{3}{16}$″.
3. Unit rise: 7.25 × 25.4 = 184.15 mm, between 125 mm and 200 mm. Unit run: 10.5 × 25.4 = 266.70 mm, between 255 mm and 355 mm.

CALCULATOR

TI-30Xa

(43.5 x² + 52.5 x²) √x

Display 68.17990906

Sharp EL-531XT

√ (43.5 x² + 52.5 x²) =

Display 68.17990906

Answer **Total rise 43.5″, total run 52.5″, stringer line 5′ $8\frac{3}{16}$″. Both step sizes are inside the private stair limits.**

Example 12.17 Checking the risers of a flight

The risers in one flight measure 191, 194, 189, 195, 192, 190 mm, from the bottom up. Does the flight meet Article 9.8.4.4, Sentence (1)?

SOLUTION

1. Differences between neighbours: 3, 5, 6, 3, 2 mm. The largest is 6 mm, more than the 5 mm allowed between adjacent treads or landings.
2. Tallest minus shortest: 195 − 189 = 6 mm. That is inside the 10 mm allowed in a flight.

Answer **No. The flight passes the 10 mm check but fails the 5 mm check between the 189 mm and 195 mm risers.**

TRY IT 12.9

1. A stair has 12 risers of 185 mm and 11 treads with a 270 mm unit run. Find the total rise, the total run and the stringer line to two decimal places.
2. Is a unit rise of 205 mm allowed on a private stair? On a public stair?
3. A public stair has a unit run of 275 mm. Does it meet Table 9.8.4.2?
4. A rectangular tread has a run of 260 mm. What tread depths does Sentence 9.8.4.2.(2) allow?
5. Measured straight up from the nosing line, a beam is 2,000 mm above a stair. Is that enough over a stair serving a single dwelling unit? Over a public stair?
6. A stair has 4 risers of 7½″ and 3 treads of 11″. Find the stringer line to the nearest 1/16″.

Answers are in the Answer Key.

12.10 Rafters and squaring a layout

A gable roof is two right triangles back to back. In this section the **run** of a rafter is the level distance it covers, from the wall line to the centre line of the ridge, and the **rise** is how far it climbs over that run. The rafter line length is the hypotenuse. When the ridge is in the middle, the run is half the span.

Roof slope is often written as rise in 12: the number of units the roof rises for every 12 units of run. A 5 in 12 roof rises 5 mm for every 12 mm of run, or 5″ for every foot. Chapter 13 finds the angle of the roof from the same two numbers.

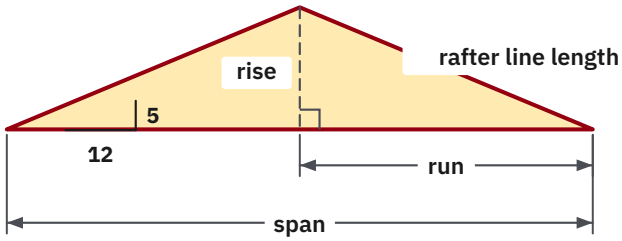


Figure 12.10 A gable roof with a 5 in 12 slope. The rafter line length is the hypotenuse of the triangle with the run and the rise as legs. The small triangle on the left shows the slope: 5 up for every 12 across.

RAFTER LINE LENGTH

$$\text{rise} = \text{run} \times (\text{rise in } 12) \div 12$$

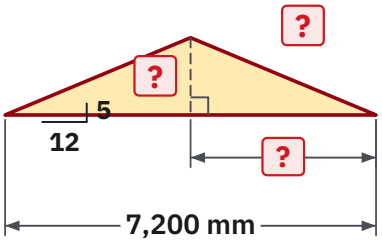
$$\text{rafter line length} = \sqrt{(\text{run}^2 + \text{rise}^2)}$$

$$\text{length per foot of run} = \sqrt{(12^2 + (\text{rise in } 12)^2)}, \text{ in inches}$$

Multiply the length per foot of run by the run in feet to get the rafter line length in inches.

Example 12.18 Rafter line length in millimetres

A building is 7,200 mm wide with the ridge in the middle. The roof slope is 5 in 12. Find the run, the rise and the rafter line length.



SOLUTION

1. Run = 7,200 ÷ 2 = 3,600 mm.
2. Rise = 3,600 × 5 ÷ 12 = 1,500 mm.
3. Rafter line length = √(3,600² + 1,500²).

CALCULATOR

TI-30Xa

(3600 x² + 1500 x²) √x

Display 3900.

Sharp EL-531XT

√ (3600 x² + 1500 x²) =

Display 3’900.

Answer **Run 3,600 mm, rise 1,500 mm, rafter line length 3,900 mm. The sides are 12, 5 and 13 times 300 mm.**

Example 12.19 Rafter line length in feet and inches

A 6 in 12 roof has a rafter run of 11′ 6″. Find the rafter line length to the nearest 1/16″, two ways.

SOLUTION

1. Run = 11 × 12 + 6 = 138″. Rise = 138 × 6 ÷ 12 = 69″.
2. Way 1: √(138² + 69²) = 154.289″.
3. Way 2: length per foot of run = √(12² + 6²) = 13.416″. The run is 11.5 ft, and 11.5 × 13.416 = 154.284″. The rounded constant makes the last digit differ a little.

CALCULATOR

TI-30Xa

Way 1

(138 x² + 69 x²) √x

Display 154.2886904

Length per foot of run

(12 x² + 6 x²) √x

Display 13.41640787

Sharp EL-531XT

Way 1

√ (138 x² + 69 x²) =

Display 154.2886904

Length per foot of run

√ (12 x² + 6 x²) =

Display 13.41640786

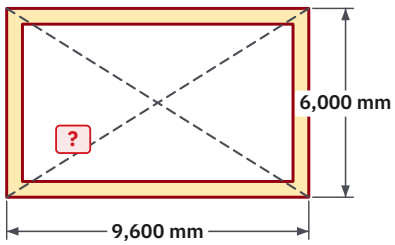
Answer $154\frac{5}{16}$″, which is 12′ $10\frac{5}{16}$″.

Squaring a layout

The outline of a building, a deck or a wall is laid out square with the methods of Sections 12.5 and 12.6. Use the largest multiple of 3, 4 and 5 that fits at a corner, then check that both diagonals match.

Example 12.20 Squaring a foundation layout

A foundation layout is 9,600 mm by 6,000 mm. (a) What 3-4-5 marks would you use at a corner, with 1,200 mm as the unit? (b) What should each diagonal measure, to the nearest millimetre? (c) The diagonals measure 11,332 mm and 11,309 mm. Is the layout square?



SOLUTION

1. (a) 3 × 1,200 = 3,600 mm along one side and 4 × 1,200 = 4,800 mm along the other. The marks should be 5 × 1,200 = 6,000 mm apart.
2. (b) d = √(9,600² + 6,000²) = 11,320.78 mm.
3. (c) The diagonals differ by 11,332 − 11,309 = 23 mm, so the layout is not square.

CALCULATOR

TI-30Xa

(9600 x² + 6000 x²) √x

Display 11320.77736

Sharp EL-531XT

√ (9600 x² + 6000 x²) =

Display 11′320.77736

Answer **(a) 3,600 mm and 4,800 mm, 6,000 mm apart. (b) 11,321 mm. (c) No. Push the two corners at the ends of the 11,332 mm diagonal toward each other until both diagonals read 11,321 mm.**

TRY IT 12.10

1. A roof has a 4 in 12 slope and a run of 3,000 mm. Find the rise and the rafter line length to two decimal places.
2. A building is 9,000 mm wide with the ridge in the middle and an 8 in 12 slope. Find the run, the rise and the rafter line length to two decimal places.
3. A 12 in 12 roof has a run of 10′. Find the rafter line length to the nearest 1/16″.
4. Find the length per foot of run for a 7 in 12 roof, to three decimal places.
5. A deck frame is 4,800 mm by 3,600 mm. What should each diagonal measure?
6. A wall layout is 20′ by 12′. What should the diagonal be, to the nearest 1/16″?

Answers are in the Answer Key.

WORKSHEET 12A

Angles on the canvas

Each drawing shows the angles at their true size. Find every red ?

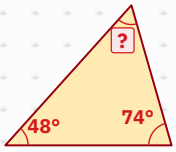
Name _____

Date _____

Use the angle facts from this chapter: a straight line is 180°, a full turn is 360° and the angles of a triangle add to 180°. Write the fact you used in the working box.

1 Find the third angle of the triangle.

SHAPE CANVAS PRACTICE: FIND THE ?

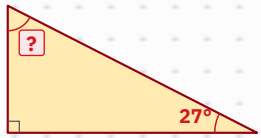


WORKING

Answer _____

2 Find the missing angle of the right triangle.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

3 A strap is bent as shown. Find the ?

SHAPE CANVAS PRACTICE: FIND THE ?

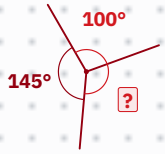


WORKING

Answer _____

4 Three lines meet at a point. Find the ?

SHAPE CANVAS PRACTICE: FIND THE ?

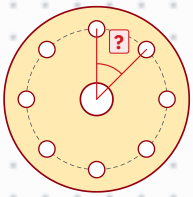


WORKING

Answer _____

5 Eight holes are evenly spaced. Find the angle between neighbours.

SHAPE CANVAS PRACTICE: FIND THE ?

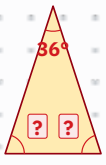


WORKING

Answer _____

6 The gusset has two equal sides. Find each ?

SHAPE CANVAS PRACTICE: FIND THE ?

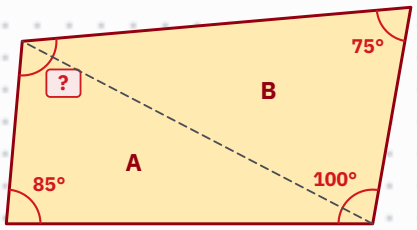


WORKING

Answer _____

7 Break it down. The dashed line splits the plate into triangles A and B. How many degrees do the four angles of the plate add to? Find the ?

SHAPE CANVAS — PRACTICE: FIND THE ?

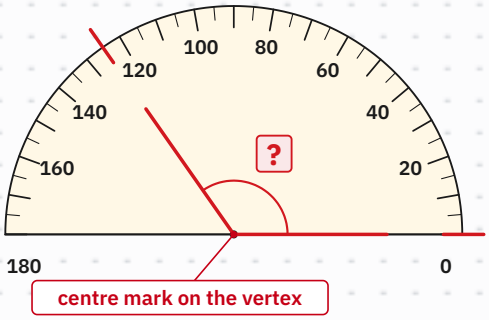


WORKING

Answer _____

8 Read the protractor. How big is the angle, and what type of angle is it?

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

9 Change one value. The plate in problem 5 is made with 10 holes instead of 8, still evenly spaced. What is the angle between neighbours now? With 12 holes? Do more holes give a bigger or a smaller angle?

WORKING

Answer _____

WORKSHEET 12B

Pythagoras on the canvas

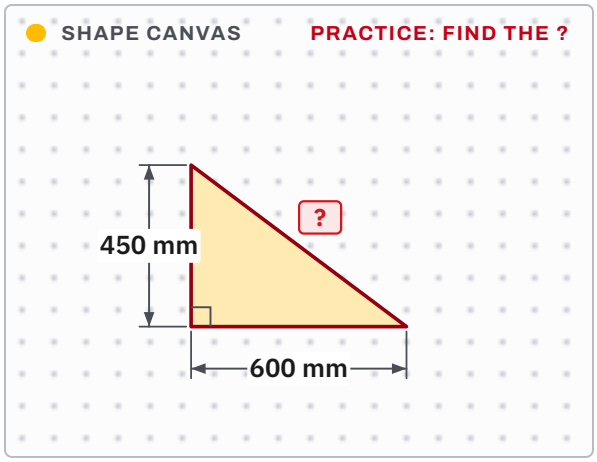
Every triangle has a right angle. Find every red ?

Name _____

Date _____

Label the hypotenuse c before you start. Add the squares to find the hypotenuse. Subtract to find a leg. Check that the hypotenuse is the longest side.

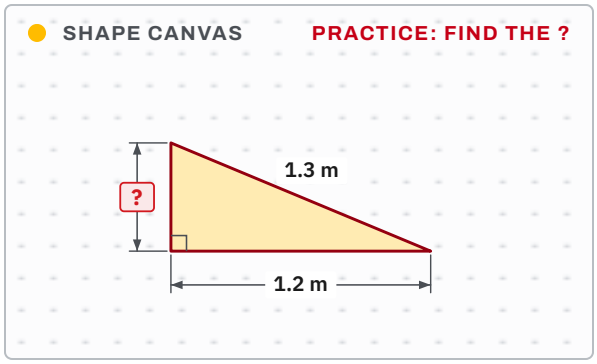
1 Find the hypotenuse.



WORKING

Answer _____

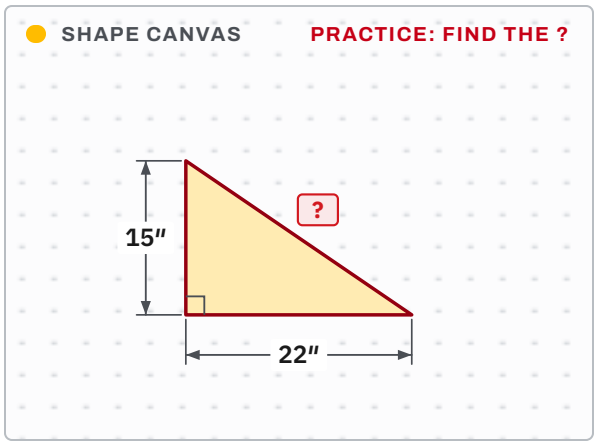
2 Find the missing leg.



WORKING

Answer _____

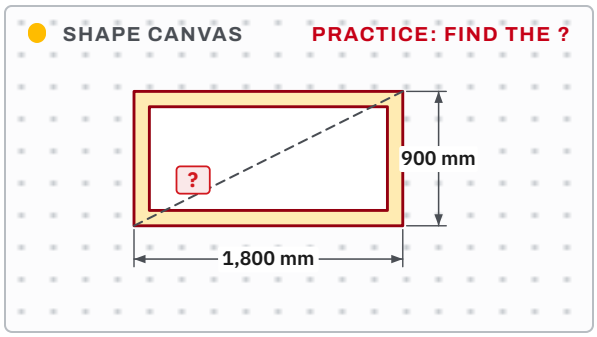
3 Find the hypotenuse to the nearest 1/16″.



WORKING

Answer _____

4 Find the diagonal of the frame to two decimal places.

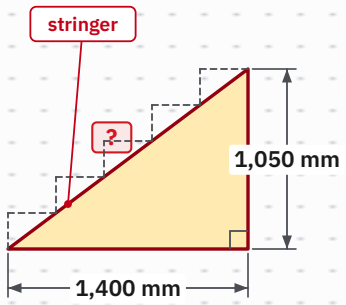


WORKING

Answer _____

5 A stair has a total rise of 1,050 mm and a total run of 1,400 mm. How long is the stringer from the bottom corner to the top corner?

SHAPE CANVAS — PRACTICE: FIND THE ?

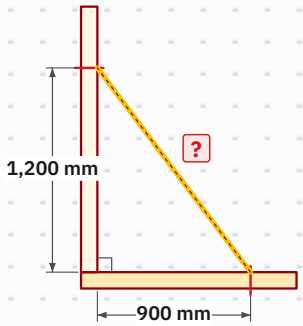


WORKING

Answer _____

6 You mark 900 mm and 1,200 mm from a corner. The tape reads 1,507 mm between the marks. What should it read? Is the corner open or closed?

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

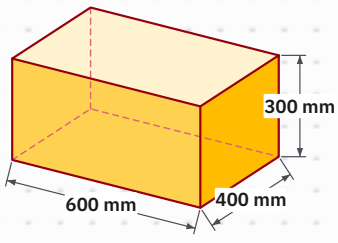
7 Change one value. A right triangle has legs of 300 mm and 400 mm. Find the hypotenuse. Then double both legs. What happens to the hypotenuse?

WORKING

Answer _____

8 Break it down. A box is 600 mm long, 400 mm wide and 300 mm high. First find the diagonal across the bottom. Then use it as a leg to find the longest straight rod that fits inside, corner to corner. Give the answer to two decimal places.

SHAPE CANVAS — READ THE DRAWING



WORKING

Answer _____

CHAPTER

13

Trigonometry

The Pythagorean theorem finds a side when you know two sides. On the job you often know one side and an angle instead: a brace set at 35°, a ramp that must not be steeper than a given slope, a pipe offset made with 45° fittings.

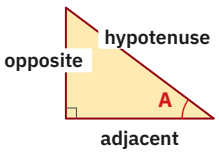
Trigonometry links the angles of a right triangle to its sides. With three ratios, sine, cosine and tangent, and the SIN, COS and TAN keys on your calculator, you can find any side or any angle of a right triangle.

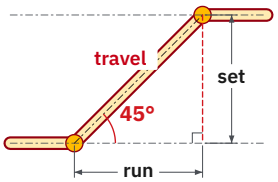
In this chapter you will

- Name the opposite side, adjacent side and hypotenuse from a given angle
- Write the sine, cosine and tangent ratios
- Use SIN, COS and TAN in DEG mode on both calculators
- Find a missing side from one side and one angle
- Find a missing angle with the inverse keys
- Solve slopes, ramps, pipe offsets, bevels and mitres

Key words

- trigonometry
- opposite side
- adjacent side
- sine
- cosine
- tangent
- inverse trig function
- offset





13.1 Naming the sides

Trigonometry is the math that links the angles of a right triangle to the lengths of its sides. It always starts from one of the two acute angles. From that angle, each side gets a name.

- The **hypotenuse** is across from the right angle. It is the longest side.
- The **opposite side** is across from the angle you are working with. It does not touch that angle.
- The **adjacent side** is next to the angle. It is the side that touches the angle and is not the hypotenuse.



Figure 13.1 The same triangle named from angle A (left) and from angle B (right). The opposite and adjacent sides swap. The hypotenuse stays the same.

MARK THE ANGLE FIRST

Put your finger on the angle you are working with. The side your finger cannot touch is the opposite side. Of the two sides that touch it, the longer one is the hypotenuse and the other one is the adjacent side.

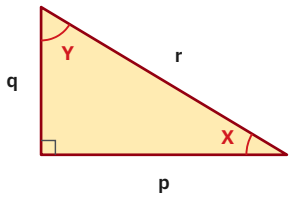


Figure 13.2 A right triangle with sides p, q and r and acute angles X and Y.

TRY IT 13.1

1. In Figure 13.2, which side is opposite angle X?
2. Which side is adjacent to angle X?
3. Which side is the hypotenuse?
4. Which side is opposite angle Y?
5. Which side is adjacent to angle Y?
6. Angle X is 31°. How big is angle Y?

Answers are in the Answer Key.

13.2 Sine, cosine and tangent

Draw any right triangle with a 30° angle, large or small. Divide the opposite side by the hypotenuse and you always get 0.5. The ratio of two sides depends only on the angle, not on the size of the triangle. That is why one angle and one side are enough to find the rest.

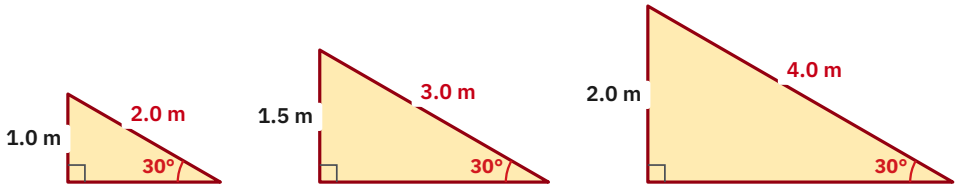


Figure 13.3 Three 30° triangles. In each one, opposite ÷ hypotenuse = 1 ÷ 2 = 1.5 ÷ 3 = 2 ÷ 4 = 0.5.

Each pair of sides gives a ratio with its own name. The **sine** (sin) of an angle is opposite ÷ hypotenuse. The **cosine** (cos) is adjacent ÷ hypotenuse. The **tangent** (tan) is opposite ÷ adjacent.

THE THREE TRIG RATIOS

$$\sin A = \text{opposite} \div \text{hypotenuse}$$

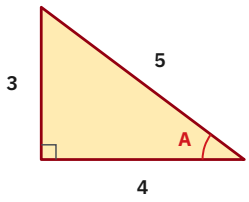
$$\cos A = \text{adjacent} \div \text{hypotenuse}$$

$$\tan A = \text{opposite} \div \text{adjacent}$$

A memory aid: SOH CAH TOA. S = O ÷ H, C = A ÷ H, T = O ÷ A.

Example 13.1 The ratios of a 3-4-5 triangle

A right triangle has sides 3, 4 and 5. Angle A is opposite the side of 3. Write sin A, cos A and tan A as fractions and as decimals.



SOLUTION

1. From A: opposite = 3, adjacent = 4, hypotenuse = 5.
2. sin A = 3 ÷ 5 = 0.6.
3. cos A = 4 ÷ 5 = 0.8.
4. tan A = 3 ÷ 4 = 0.75.

Answer **sin A = 3/5 = 0.6, cos A = 4/5 = 0.8, tan A = 3/4 = 0.75.**

The trig keys and DEG mode

Angles in this book are in degrees, so both calculators must be in degree mode. On the TI-30Xa, check that DEG shows in the display. If it does not, press DRG until it does. On the Sharp EL-531XT, DEG shows at the top of the display, and the DRG key steps through DEG, RAD and GRAD. Chapter 2 shows the keys.

The keys go in a different order. On the TI-30Xa, key the angle first, then SIN, COS or TAN. On the Sharp, press sin, cos or tan first, then the angle, then =.

CALCULATOR Check your mode: sin 30°

TI-30Xa

30 SIN

Display 0.5

Sharp EL-531XT

sin 30 =

Display 0.5

If sin 30° does not give 0.5, the calculator is not in degree mode. In radian mode the same keys give −0.988. Press DRG until DEG shows, then try again.

TRY IT 13.2

1. sin 40° to four decimal places
2. cos 40° to four decimal places
3. tan 40° to four decimal places
4. tan 45°
5. sin 60° and cos 30° to four decimal places. What do you notice?
6. A right triangle has opposite 5, adjacent 12, hypotenuse 13. Find sin A and tan A to four decimal places.

Answers are in the Answer Key.

13.3 Finding a side

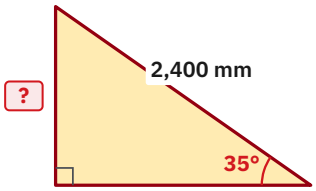
1. Mark the angle you know.
2. Name the side you know and the side you want: opposite, adjacent or hypotenuse.
3. Pick the ratio that uses those two sides: SOH, CAH or TOA.
4. Write the ratio with the numbers in it and **transpose** it for the side you want.
5. Check: the hypotenuse must be the longest side.

WHEN THE SIDE YOU WANT IS ON THE BOTTOM, DIVIDE

If the unknown side is the hypotenuse in SOH or CAH, or the adjacent side in TOA, it sits under the fraction bar. Then you divide the known side by sin, cos or tan instead of multiplying.

Example 13.2 Height of a brace (sine)

A brace 2,400 mm long is set at 35° to the floor. How high up the wall does it reach? Give the answer to two decimal places.



SOLUTION

1. From the 35° angle: the brace is the hypotenuse, the height is the opposite side.
2. Opposite and hypotenuse: SOH. sin 35° = opposite ÷ 2,400.
3. Multiply both sides by 2,400: opposite = 2,400 × sin 35°.

CALCULATOR

TI-30Xa

35 | SIN | × | 2400 | =

Display 1376.583447

Sharp EL-531XT

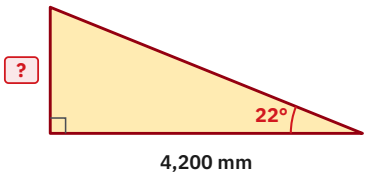
sin | 35 | × | 2400 | =

Display 1’376.583447

Answer 1,376.58 mm

Example 13.3 Rise of a truss (tangent)

The top chord of a truss rises at 22° over a half span of 4,200 mm. How high is the peak above the bottom chord? Give the answer to two decimal places.



SOLUTION

1. The half span is the adjacent side. The height is the opposite side.
2. Opposite and adjacent: TOA. opposite = 4,200 × tan 22°.

CALCULATOR

TI-30Xa

22 | TAN | × | 4200 | =

Display 1696.910149

Sharp EL-531XT

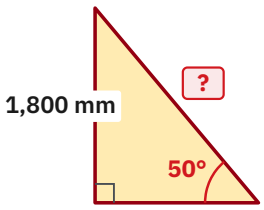
tan | 22 | × | 4200 | =

Display 1’696.910149

Answer 1,696.91 mm

Example 13.4 Length of a brace (divide)

A brace must reach 1,800 mm up a column and meet the floor at 50°. How long is the brace? Give the answer to two decimal places.



SOLUTION

1. The height is the opposite side. The brace is the hypotenuse.
2. SOH: sin 50° = 1,800 ÷ hypotenuse.
3. Transpose: hypotenuse = 1,800 ÷ sin 50°.

CALCULATOR

TI-30Xa

[1800] [÷] [50] [SIN] [=]

Display: 2349.733121

Sharp EL-531XT

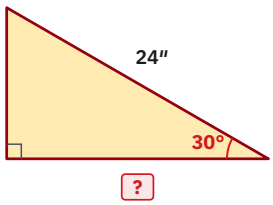
[1800] [÷] [sin] [50] [=]

Display: 2'349.733121

Answer **2,349.73 mm. Check: longer than the 1,800 mm height.**

Example 13.5 Inches to the nearest sixteenth

A 24″ gusset edge is set at 30° to the base. Find the base (the adjacent side) to the nearest 1/16″.



SOLUTION

1. CAH: adjacent = 24 × cos 30° = 20.78461″ (the display shows more digits).
2. Sixteenths: 0.78461 × 16 = 12.55, which rounds to 13.

CALCULATOR

TI-30Xa

[30] [COS] [×] [24] [=]

Display: 20.78460969

Sharp EL-531XT

[cos] [30] [×] [24] [=]

Display: 20.78460969

Answer $20\frac{13}{16}''$

TRY IT 13.3

1. Hypotenuse 800 mm, angle 25°. Find the opposite side to two decimal places.
2. Hypotenuse 1,500 mm, angle 40°. Find the adjacent side to two decimal places.
3. Adjacent 2,000 mm, angle 18°. Find the opposite side to two decimal places.
4. Opposite 600 mm, angle 35°. Find the hypotenuse to two decimal places.
5. Opposite 900 mm, angle 52°. Find the adjacent side to two decimal places.
6. Hypotenuse 30″, angle 40°. Find the opposite side to the nearest 1/16″.

Answers are in the Answer Key.

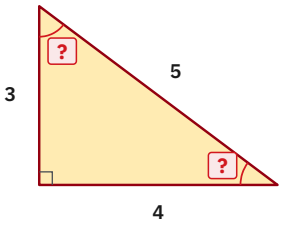
13.4 Finding an angle

When you know two sides, work out their ratio and then turn the ratio back into an angle. That is the job of an **inverse trig function**: SIN^{-1}, COS^{-1} and TAN^{-1}. They are the second functions of the SIN, COS and TAN keys.

On the TI-30Xa, key the ratio, then 2nd and SIN, COS or TAN. Put a division in brackets so it is worked out first. On the Sharp, press 2ndF and sin, cos or tan first, then the ratio, then =.

Example 13.6 The angles of a 3-4-5 triangle

Find both acute angles of a 3-4-5 triangle to two decimal places. Angle A is opposite the side of 3.



SOLUTION

1. From A, the opposite side is 3 and the adjacent side is 4: TOA.
2. $\tan A = 3 \div 4$, so $A = \tan^{-1}(3 \div 4) = 36.87°$.
3. The two acute angles add to 90°: $B = 90° - 36.87° = 53.13°$.

CALCULATOR

TI-30Xa

A from TOA

(3 ÷ 4) 2nd TAN⁻¹

Display 36.86989765

Check with SOH

(3 ÷ 5) 2nd SIN⁻¹

Display 36.86989765

Sharp EL-531XT

A from TOA

2ndF tan⁻¹ (3 ÷ 4) =

Display 36.86989765

Check with SOH

2ndF sin⁻¹ (3 ÷ 5) =

Display 36.86989765

Any ratio that uses two known sides gives the same angle. Here SOH, sin A = 3 ÷ 5, checks the answer.

Answer **A = 36.87°, B = 53.13°**

RATIO KEY OR ANGLE KEY?

SIN, COS and TAN turn an angle into a ratio. SIN^{-1}, COS^{-1} and TAN^{-1} turn a ratio into an angle. A sine or cosine ratio is always less than 1, because the hypotenuse is the longest side. If yours is more than 1, the sides are mixed up.

TRY IT 13.4

1. Opposite 500, adjacent 800. Find the angle to two decimal places.
2. Opposite 700, hypotenuse 1,000. Find the angle to two decimal places.
3. Adjacent 1,100, hypotenuse 1,400. Find the angle to two decimal places.
4. Opposite 9″, adjacent 12″. Find both acute angles to two decimal places.
5. sin A = 0.5. Find A.
6. tan A = 1. Find A.

Answers are in the Answer Key.

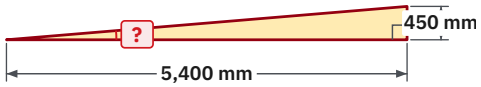
13.5 Slopes and ramps

A slope is a right triangle lying on its side. The rise is the opposite side, the run is the adjacent side and the length along the slope is the hypotenuse. The angle of the slope comes from the tangent: angle = $\tan^{-1}$(rise ÷ run).

Slopes are also given as a **percent** grade: rise ÷ run × 100. A slope written as 1 in 12 rises 1 unit for every 12 units of run.

Example 13.7 Angle and grade of a ramp

A ramp rises 450 mm over a run of 5,400 mm. Find its angle and its percent grade to two decimal places.



SOLUTION

1. Rise and run are opposite and adjacent: TOA.
2. Angle = $\tan^{-1}(450 \div 5{,}400)$.
3. Grade = $450 \div 5{,}400 \times 100 = 8.33\%$.

CALCULATOR

TI-30Xa

(450 ÷ 5400) 2nd TAN⁻¹

Display 4.763641691

Sharp EL-531XT

2ndF tan⁻¹ (450 ÷ 5400) =

Display 4.763641691

Answer **Angle 4.76°, grade 8.33%. The drawing is to scale: a slope this gentle looks almost flat.**

1 IN 12

450 ÷ 5,400 is the same ratio as 1 ÷ 12. So a slope of 1 in 12 has the same angle, 4.76°.

Example 13.8 Fall of a drain line

A drain line runs 12,000 mm at a grade of 2%. How far does it fall, and what is its angle to two decimal places?

SOLUTION

1. Grade = fall ÷ run × 100, so fall = run × grade ÷ 100 = 12,000 × 2 ÷ 100 = 240 mm.
2. Angle = $\tan^{-1}(2 \div 100) = \tan^{-1}(0.02) = 1.15°$. Key it the same way as the ramp angle.

Answer **Fall 240 mm, angle 1.15°.**

Inches per foot

A slope can also be given as the rise in inches for every foot (12″) of run, such as $\frac{1}{4}$″ per foot. Divide the inches by 12 to get the ratio rise ÷ run. $\frac{1}{4}$″ per foot is 0.25 ÷ 12, the same as 1 in 48, which is also written 1:48. As a grade it is 0.25 ÷ 12 × 100 = 2.08%, and its angle is $\tan^{-1}(0.25 \div 12) = 1.19°$.

A roof that rises 5″ per foot is a 5 in 12 roof. Its ratio is 5 ÷ 12, which is 1 in 2.4, and its angle is $\tan^{-1}(5 \div 12) = 22.62°$. Section 13.8 works with roof slopes in this form.

ONE SLOPE, FIVE WAYS

$\frac{1}{4}$″ per foot = 1 in 48 = 1:48 = 2.08% grade = 1.19°. The first three are ratios of rise to run, the grade is the ratio × 100 and the angle comes from TAN^{-1} of the ratio.

TRY IT 13.5

1. Rise 300 mm, run 4,000 mm. Find the angle and the grade to two decimal places.
2. A slope of 1 in 20. Find the angle to two decimal places.
3. A 1% grade over a 15,000 mm run. How far does it fall?
4. A ramp is 3,000 mm long, measured along the slope. It slopes at 4°. Find the rise to two decimal places.
5. A roof rises 6″ for every 12″ of run. Find its angle to two decimal places.
6. A ramp must rise 600 mm at an angle of 3°. How long is the run, to the nearest millimetre?

Answers are in the Answer Key.

13.6 Pipe offsets

An **offset** moves a pipe over to a new line that is parallel to the first one. In this book an offset has two fittings of the same angle, and the three measurements of an offset are the sides of a right triangle. In this book, set, run and travel are measured along the centreline of the pipe.

- **Set:** how far the pipe moves over, square to the two parallel lines. It is the side opposite the fitting angle.
- **Run:** the distance along the line of the pipe that the offset takes up. It is the side adjacent to the fitting angle.
- **Travel:** the length along the centreline of the angled part, from one fitting to the other. It is the hypotenuse.

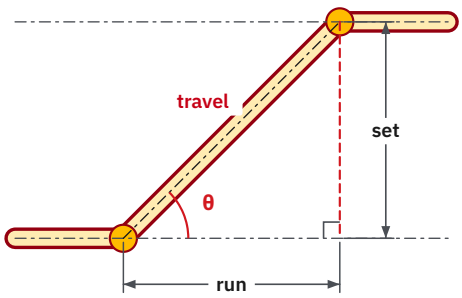


Figure 13.4 A 45° offset. Set, run and travel form a right triangle. The fitting angle θ (theta) is at the first fitting.

OFFSETS

$$\text{travel} = \text{set} \div \sin\theta$$

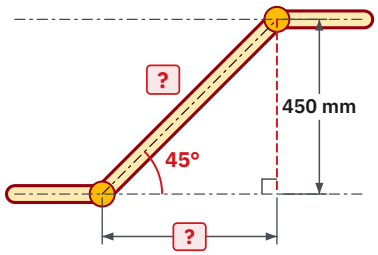
$$\text{run} = \text{set} \div \tan\theta$$

$$\text{set} = \text{travel} \times \sin\theta$$

θ (theta) is the fitting angle. These are SOH and TOA with the set as the opposite side.

Example 13.9 A 45° offset

A line must move over 450 mm using 45° fittings. Find the travel and the run, to two decimal places.



SOLUTION

1. Travel = 450 ÷ sin 45°.
2. Run = 450 ÷ tan 45°. Since tan 45° = 1, the run equals the set: 450.00 mm.

CALCULATOR

TI-30Xa

450 ÷ 45 SIN =

Display 636.3961031

Sharp EL-531XT

450 ÷ sin 45 =

Display 636.3961031

Answer **Travel 636.40 mm, run 450.00 mm.**

Example 13.10 A 60° offset in inches

A line must move over 14″ using 60° fittings. Find the travel and the run to the nearest 1/16″.

SOLUTION

1. Travel = 14 ÷ sin 60° = 16.166″.
2. Run = 14 ÷ tan 60° = 8.083″.
3. Change each decimal part to sixteenths and round.

CALCULATOR

TI-30Xa

Travel

14 ÷ 60 SIN =

Display 16.16580754

Run

14 ÷ 60 TAN =

Display 8.082903769

Sharp EL-531XT

Travel

14 ÷ sin 60 =

Display 16.16580754

Run

14 ÷ tan 60 =

Display 8.082903769

Answer Travel $16\frac{3}{16}$″, run $8\frac{1}{16}$″.

Where offset constants come from

Travel = set ÷ sin θ is the same as set × (1 ÷ sin θ). The number 1 ÷ sin θ depends only on the fitting angle, so it can be worked out once and kept in a table. The same goes for 1 ÷ tan θ for the run. Every one of these numbers comes from the calculator:

Fitting angle θ	Travel constant 1 ÷ sin θ	Run constant 1 ÷ tan θ
60°	1.155	0.577
45°	1.414	1.000
30°	2.000	1.732
22.5°	2.613	2.414
11.25°	5.126	5.027

Table 13.1 Offset constants worked out from sine and tangent, to three decimal places.

CALCULATOR The 45° travel constant: 1 ÷ sin 45°

TI-30Xa

45 | SIN | 1/x

Display 1.414213562

Sharp EL-531XT

(| sin | 45 |) | 2ndF | x^{-1} | =

Display 1.414213562

The 1/x key (2ndF x^{-1} on the Sharp) divides 1 by the number. For the run constant, use TAN instead of SIN.

A ROUNDED CONSTANT GIVES A ROUNDED ANSWER

With the constant 1.414, the 45° travel above is 450 × 1.414 = 636.30 mm. Dividing by sin 45° on the calculator gives 636.40 mm. The difference grows with the set, so use the calculator value when you can. To find the fitting angle from a set and a run, use $\theta = \tan^{-1}(\text{set} \div \text{run})$.

TRY IT 13.6

1. 45° fittings, set 600 mm. Find the travel to two decimal places.
2. 60° fittings, set 400 mm. Find the travel and run to two decimal places.
3. 22.5° fittings, set 250 mm. Find the travel to two decimal places.
4. 45° fittings, set 10″. Find the travel to the nearest 1/16″.
5. Travel 900 mm with 30° fittings. Find the set.
6. Set 350 mm, run 350 mm. What fitting angle?

Answers are in the Answer Key.

13.7 Bevel and mitre angles

A mitre joint joins two pieces at a corner. Each piece is cut at an angle so that the joint line splits the corner angle in half. For a square corner (90°), each piece is cut at 45° to its edge.

A frame with equal sides and equal corners turns 360° in all, so each corner turns 360° ÷ n, where n is the number of sides. The corner angle inside the frame is 180° − 360° ÷ n, and each piece is cut at half of that to its edge.

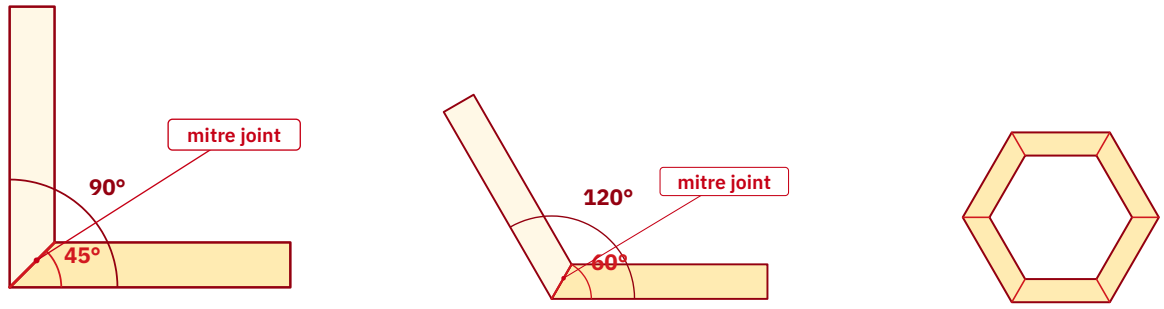


Figure 13.5 Mitre joints for a square corner (90°, cut at 45°) and for a six-sided frame (120°, cut at 60°).

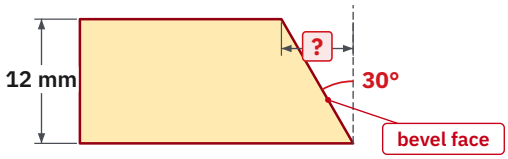
Sides n	Corner angle 180° – 360° ÷ n	Cut angle to the edge (half the corner)	Angle away from a square cut
4	90°	45°	45°
6	120°	60°	30°
8	135°	67.5°	22.5°
12	150°	75°	15°

Table 13.2 Mitre angles for frames with equal sides.

A bevel is an angled face cut along the edge of a plate. In this book the bevel angle is measured from the square edge. Through the full thickness t, the bevel is t × tan(angle) wide at the top. Two plates bevelled at the same angle and butted together open a groove of twice the bevel angle.

Example 13.11 Width of a bevel

A 12 mm plate is bevelled at 30° through its full thickness. How wide is the bevel at the top, to two decimal places?



SOLUTION

1. The thickness is the side adjacent to the bevel angle. The width is the opposite side: TOA.
2. Width = 12 × tan 30°.

CALCULATOR

TI-30Xa

12 × 30 TAN =

Display 6.92820323

Sharp EL-531XT

12 × tan 30 =

Display 6.92820323

Answer **6.93 mm. Two plates bevelled like this open a 60° groove. To find a bevel angle from the thickness and the width, use $\tan^{-1}$(width ÷ thickness).**

TRY IT 13.7

1. A frame has 5 equal sides. Find the corner angle and the cut angle.
2. Each corner of a frame is 150°. How many sides does it have?
3. A 20 mm plate is bevelled at 30° through its thickness. Bevel width to two decimal places?
4. A 16 mm plate has a 9 mm wide bevel. Find the bevel angle to two decimal places.
5. Two plates are each bevelled at 35°. What is the groove angle?
6. A square frame corner is cut at 45°. A cut of 46° is made by mistake on both pieces. What corner angle do the pieces make?

Answers are in the Answer Key.

13.8 Roofs, stairs and ramps

Carpentry slopes are right triangles too. A roof slope written as rise in 12, a stair step with its unit rise and unit run, and a ramp written as 1 in 10 all give a rise and a run. The angle is $\tan^{-1}$(rise ÷ run), the same as for the ramps in Section 13.5. Section 12.10 finds rafter lengths from the same numbers.

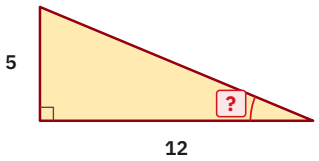
Roof slope

Rise in 12	As 1 in …	Percent grade	Angle $\tan^{-1}$(rise ÷ 12)
2 in 12	1 in 6	16.67%	9.46°
3 in 12	1 in 4	25.00%	14.04°
4 in 12	1 in 3	33.33%	18.43°
5 in 12	1 in 2.4	41.67%	22.62°
6 in 12	1 in 2	50.00%	26.57°
8 in 12	1 in 1.5	66.67%	33.69°
10 in 12	1 in 1.2	83.33%	39.81°
12 in 12	1 in 1	100.00%	45.00°

Table 13.3 Roof slopes worked out with the calculator, to two decimal places.

Example 13.12 Angle of a 5 in 12 roof

Find the angle of a 5 in 12 roof to two decimal places, and in degrees, minutes and seconds to the nearest second (Section 12.8).



SOLUTION

1. Rise 5 and run 12 are the opposite and adjacent sides: TOA.
2. Angle = $\tan^{-1}(5 \div 12) = 22.62°$.
3. In D°M′S: 22°37′12″.

CALCULATOR

TI-30Xa

Angle

(5 ÷ 12) 2nd TAN⁻¹

Display 22.61986495

In D°M′S

(5 ÷ 12) 2nd TAN⁻¹ 2nd DD▸DMS

Display 22°37'11"5

Sharp EL-531XT

Angle

2ndF tan⁻¹ (5 ÷ 12) =

Display 22.61986495

In D°M′S

2ndF tan⁻¹ (5 ÷ 12) = 2ndF ↔DEG

Display 22°37′11.5138″

Answer 22.62°, which is 22°37′12″ to the nearest second.

Example 13.13 A slope in inches per foot

A drain line slopes $\frac{1}{4}$″ per foot. Write the slope as 1 in …, as a percent grade and as an angle to two decimal places. How far does it fall over a 30 ft run?

SOLUTION

1. Ratio: 0.25 ÷ 12. Turn it over: 12 ÷ 0.25 = 48, so the slope is 1 in 48.
2. Grade = 0.25 ÷ 12 × 100 = 2.08%.
3. Angle = $\tan^{-1}(0.25 \div 12)$ = 1.19°.
4. Fall = 30 ft × 0.25″ per foot = 7.5″.

CALCULATOR

TI-30Xa

Angle

(| 0.25 | ÷ | 12 |) | 2nd | TAN⁻¹

Display 1.193489424

Grade

0.25 | ÷ | 12 | × | 100 | =

Display 2.083333333

Sharp EL-531XT

Angle

2ndF | tan⁻¹ | (| 0.25 | ÷ | 12 |) | =

Display 1.193489424

Grade

0.25 | ÷ | 12 | × | 100 | =

Display 2.083333333

Answer **1 in 48, 2.08%, 1.19°. It falls 7.5″ ($7\frac{1}{2}$″) over 30 ft.**

The BC Building Code 2024 (BCBC), Division B, sets the slopes that roof coverings may be put on. Sentence 9.26.3.1.(1) says the slopes "shall conform to Table 9.26.3.1." In that table (page 9-242), asphalt shingles have a minimum slope of "1 in 3" for normal application and "1 in 6" for low slope application, with "no limit" as the maximum. Wood shingles have a minimum of "1 in 4" and cedar shakes "1 in 3".

TABLE 9.26.3.1. ROOFING TYPES AND SLOPE LIMITS
Forming Part of Sentence 9.26.3.1.(1)

Type of Roofing	Minimum Slope	Maximum Slope
Asphalt Shingles		
Low slope application	1 in 6	no limit
Normal application	1 in 3	no limit

Figure 13.6 The asphalt shingle rows of Table 9.26.3.1. A slope of 1 in 3 rises 1 unit for every 3 units of run, the same as 4 in 12.

Source: BCBC 2024, Division B, Article 9.26.3.1. (page 9-241)

Example 13.14 Shingles on a 3 in 12 roof

Can asphalt shingles go on a 3 in 12 roof in normal application? In low slope application? Compare the angles.

SOLUTION

1. 3 in 12 is 3 ÷ 12 = 1 in 4. Its angle is $\tan^{-1}(3 \div 12) = 14.04°$.
2. Normal application needs at least 1 in 3 = 4 in 12, an angle of $\tan^{-1}(1 \div 3) = 18.43°$. The roof is flatter than that.
3. Low slope application needs at least 1 in 6 = 2 in 12, an angle of $\tan^{-1}(1 \div 6) = 9.46°$. The roof is steeper than that.

Answer **Not in normal application. Yes in low slope application.**

Roof slope also changes how much venting a roof space needs. Sentence 9.19.1.2.(1) says "the unobstructed vent area shall be not less than 1/300 of the insulated ceiling area." Sentence 9.19.1.2.(2) says "Where the roof slope is less than 1 in 6 or in roofs that are constructed with roof joists, the unobstructed vent area shall be not less than 1/150 of the insulated ceiling area" (page 9-136).

WORKED EXAMPLE: 120 M² (1 292 FT²) CEILING

Sentence (1): 120 ÷ 300 = 0.40 m² (4.3 ft²)
of unobstructed vent area

Clause (3)(b): at least 25% at the top,
0.10 m² (1.1 ft²) or more

Clause (3)(c): at least 25% at the bottom,
0.10 m² (1.1 ft²) or more

Sentence (2): slope less than 1 in 6, or roof
joists: 120 ÷ 150 = 0.80 m² (8.6 ft²)

Sentence (5): vents comply with CAN3-A93-M,
"Natural Airflow Ventilators for Buildings"

Figure 13.7 The worked example on the 9.19.1.2 plate: the same ceiling needs twice the vent area when Sentence (2) applies.
Source: BCBC 2024, Division B, Article 9.19.1.2. (page 9-136)

Example 13.15 Vent area for a low roof

A roof framed with trusses has a slope of 1.5 in 12. The insulated ceiling below it is 96 m². Which Sentence of Article 9.19.1.2 applies, and what is the least vent area?

SOLUTION

1. 1.5 in 12 is 1.5 ÷ 12 = 1 in 8. 1 in 6 is 2 in 12. The roof is flatter, so its slope is less than 1 in 6.
2. Sentence (2) applies: vent area = 96 ÷ 150 = 0.64 m².
3. For comparison, a roof of 1 in 6 or steeper over the same ceiling needs 96 ÷ 300 = 0.32 m² (Sentence (1)).

Answer **Sentence (2). At least 0.64 m² of unobstructed vent area.**

Stair angle

Each step of a stair is a small right triangle. The stair angle is $\tan^{-1}$(unit rise ÷ unit run). The largest rise with the smallest run allowed by Tables 9.8.4.1 and 9.8.4.2 (Section 12.9) gives the steepest stair those two limits allow: $\tan^{-1}(200 \div 255) = 38.11°$ for a private stair and $\tan^{-1}(180 \div 280) = 32.74°$ for a public stair.

Example 13.16 Angle of a stair

The stair in Example 12.15 has a unit rise of 194.29 mm (2,720 ÷ 14) and a unit run of 265 mm. Find the stair angle to two decimal places.

SOLUTION

1. Angle = $\tan^{-1}$(unit rise ÷ unit run) = $\tan^{-1}(194.29 \div 265)$.
2. Key 2,720 ÷ 14 in the brackets so the calculator keeps every digit of the unit rise.

CALCULATOR

TI-30Xa

(2720 ÷ 14 ÷ 265) 2nd TAN^{-1}

Display 36.24714196

Sharp EL-531XT

2ndF tan^{-1} (2720 ÷ 14 ÷ 265) =

Display 36.24714196

Answer 36.25°. It is less steep than 38.11°, the steepest the private stair limits allow.

Ramp slope

Sentence 9.8.5.4.(1) says "The slope of ramps shall be not more than a) 1 in 10 for exterior ramps, b) 1 in 10 for interior ramps serving residential occupancies, c) 1 in 6 for industrial occupancies, and d) 1 in 8 for all other occupancies" (pages 9-33 and 9-34).

Ramp	Clause	Steepest slope	Percent grade	Angle
Exterior ramps	(a)	1 in 10	10.00%	5.71°
Interior ramps serving residential occupancies	(b)	1 in 10	10.00%	5.71°
Industrial occupancies	(c)	1 in 6	16.67%	9.46°
All other occupancies	(d)	1 in 8	12.50%	7.13°

Table 13.4 The ramp limits of Sentence 9.8.5.4.(1) as grades and angles.

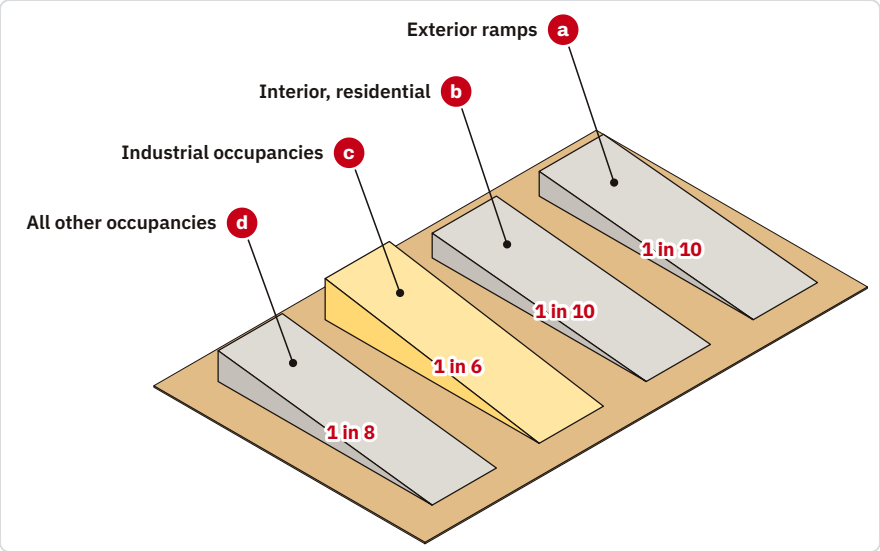
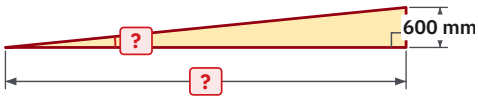


Figure 13.8 Four ramps, each at the steepest slope Sentence 9.8.5.4.(1) allows for its use.

Source: BCBC 2024, Division B, Article 9.8.5.4. (page 9-33)

Example 13.17 Shortest exterior ramp

An exterior ramp must rise 600 mm. Find the shortest run Clause 9.8.5.4.(1)(a) allows, the length along the slope to two decimal places and the angle.



SOLUTION

1. 1 in 10 means the run is at least 10 times the rise:
   600 × 10 = 6,000 mm.
2. Length along the slope = $\sqrt{(6{,}000^2 + 600^2)} = 6{,}029.93$ mm.
3. Angle = $\tan^{-1}(1 \div 10) = 5.71°$.

CALCULATOR

TI-30Xa

Length along the slope

(6000 x² + 600 x²) √x

Display 6029.925373

Angle

(1 ÷ 10) 2nd TAN⁻¹

Display 5.710593138

Sharp EL-531XT

Length along the slope

√ (6000 x² + 600 x²) =

Display 6'029.925373

Angle

2ndF tan⁻¹ (1 ÷ 10) =

Display 5.710593137

Answer **Run 6,000 mm, length 6,029.93 mm, angle 5.71°.**

Rafter cuts

A rafter sits at the roof angle θ. Two of its cuts are plumb or level when the rafter is in place. The plumb cut at the ridge is vertical. The birdsmouth over the wall has a level seat cut and a plumb heel cut. Because the rafter edge slopes at θ, the seat cut meets the rafter edge at θ, and a plumb cut meets the rafter edge at 90° − θ. A square cut across the rafter is 90° to the edge, so a plumb cut is θ away from square.

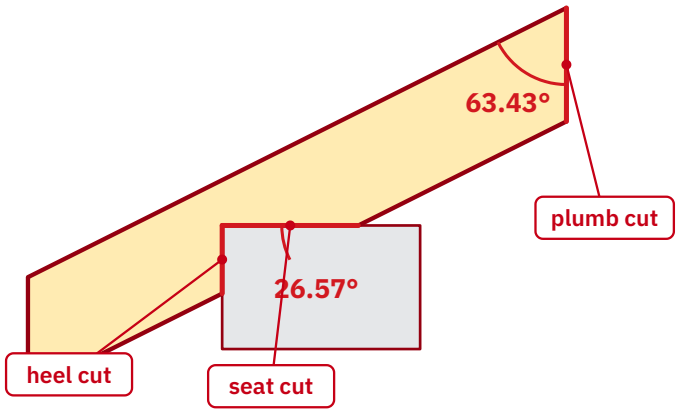


Figure 13.9 A 6 in 12 rafter with a plumb cut at the ridge end and a birdsmouth over the wall plate. The roof angle is 26.57°.

Example 13.18 Cut angles for a 6 in 12 rafter

Find the roof angle of a 6 in 12 roof, the angle between the seat cut and the rafter edge, and the angle between the plumb cut and the rafter edge, to two decimal places.

SOLUTION

1. Roof angle $\theta = \tan^{-1}(6 \div 12) = 26.57°$.
2. The seat cut is level, so it meets the rafter edge at $\theta = 26.57°$.
3. The plumb cut meets the rafter edge at $90° - 26.57° = 63.43°$. It is 26.57° away from a square cut.

CALCULATOR

TI-30Xa

(6 ÷ 12) 2nd TAN⁻¹

Display 26.56505118

Sharp EL-531XT

2ndF tan⁻¹ (6 ÷ 12) =

Display 26.56505118

Answer θ = 26.57°, seat cut 26.57°, plumb cut 63.43° to the rafter edge.

TRY IT 13.8

1. Find the angle of a 4 in 12 roof to two decimal places.
2. A roof is 7 in 12. Write it as 1 in … to two decimal places, and find its angle.
3. A drain slopes $\frac{1}{8}$″ per foot. Write the slope as 1 in … and as a percent grade to two decimal places.
4. Can asphalt shingles go on a 2.5 in 12 roof in normal application? In low slope application? Use Table 9.26.3.1.
5. An industrial ramp rises 450 mm at the steepest slope Clause 9.8.5.4.(1)(c) allows. Find the run and the angle.
6. A rafter is 8 in 12. Find the angle between the seat cut and the rafter edge, and between the plumb cut and the rafter edge.

Answers are in the Answer Key.

WORKSHEET 13A

Find the side

Each triangle is drawn at its true angle. Find every red ?

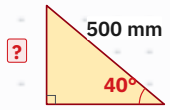
Name _____

Date _____

Mark the angle, name the sides, pick SOH, CAH or TOA, then multiply or divide. Give metric answers to two decimal places and inch answers to the nearest 1/16″.

1 Find the opposite side.

SHAPE CANVAS PRACTICE: FIND THE ?

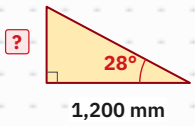


WORKING

Answer _____

2 Find the height.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

3 Find the base.

SHAPE CANVAS PRACTICE: FIND THE ?

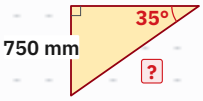


WORKING

Answer _____

4 Find the hypotenuse.

SHAPE CANVAS PRACTICE: FIND THE ?

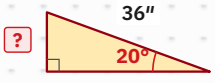


WORKING

Answer _____

5 Find the opposite side to the nearest 1/16″.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

6 Find the hypotenuse to the nearest 1/16″.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

WORKSHEET 13B

Find the angle

Use the inverse keys. Find every red ?

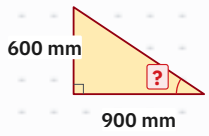
Name _____

Date _____

Name the two sides you know, pick the ratio, then use SIN^{-1}, COS^{-1} or TAN^{-1}. Give angles and percent grades to two decimal places. Check: the two acute angles add to 90°.

1 Find the angle.

SHAPE CANVAS PRACTICE: FIND THE ?

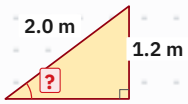


WORKING

Answer _____

2 Find the angle.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

3 Find the angle.

SHAPE CANVAS PRACTICE: FIND THE ?

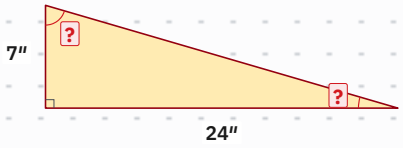


WORKING

Answer _____

4 Find both acute angles.

SHAPE CANVAS PRACTICE: FIND THE ?

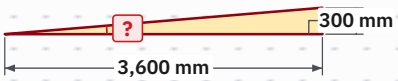


WORKING

Answer _____

5 A ramp rises 300 mm over a run of 3,600 mm. Find its angle and its percent grade.

SHAPE CANVAS PRACTICE: FIND THE ?

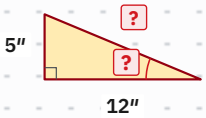


WORKING

Answer _____

6 A roof rises 5″ for every 12″ of run. Find the roof angle and the sloped length.

SHAPE CANVAS PRACTICE: FIND THE ?



WORKING

Answer _____

WORKSHEET 13C

Shop triangles

Offsets, bevels and mitres. Find every red ?

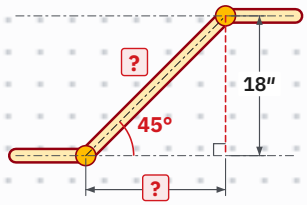
Name _____

Date _____

Find the right triangle in each job first. Set, run and travel are opposite, adjacent and hypotenuse from the fitting angle.

1 A 45° offset has a set of 18″. Find the travel to the nearest 1/16″ and the run.

SHAPE CANVAS PRACTICE: FIND THE ?

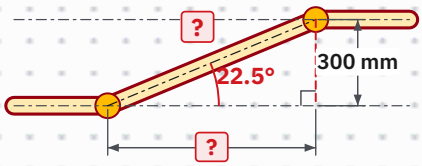


WORKING

Answer _____

2 A 22.5° offset has a set of 300 mm. Find the travel and the run to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

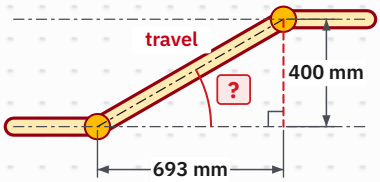


WORKING

Answer _____

3 An offset has a set of 400 mm and a run of 693 mm. Find the fitting angle to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

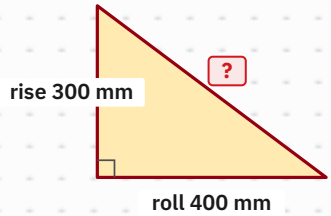


WORKING

Answer _____

4 Break it down. A rolling offset rises 300 mm and rolls 400 mm. First find the true offset (the set) with the Pythagorean theorem. Then find the travel with 45° fittings. Give both to two decimal places.

SHAPE CANVAS PRACTICE: FIND THE ?

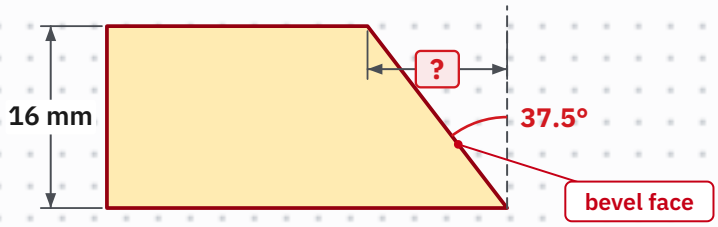


WORKING

Answer _____

5 A 16 mm plate is bevelled at 37.5° through its thickness. Find the bevel width to two decimal places.

SHAPE CANVAS — PRACTICE: FIND THE ?

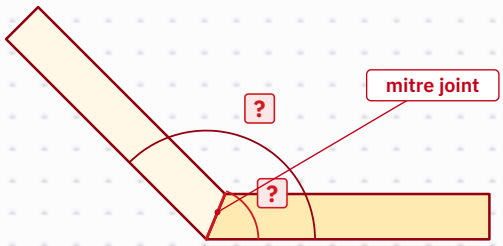


WORKING

Answer _____

6 An eight-sided frame. Find the corner angle and the cut angle of each piece.

SHAPE CANVAS — PRACTICE: FIND THE ?



WORKING

Answer _____

7 Change one value. A ramp is 3,000 mm long, measured along the slope, and slopes at 6°. Find its rise to two decimal places. Then double the angle to 12°. Does the rise double too?

WORKING

Answer _____

APPENDIX

A

Kitchen Math

A cook uses the same math as the rest of the shop. A recipe that feeds 12 has to feed 45. A recipe written in cups has to be measured in millilitres. A case of carrots loses weight when it is peeled, so the cook has to buy more than the plate needs.

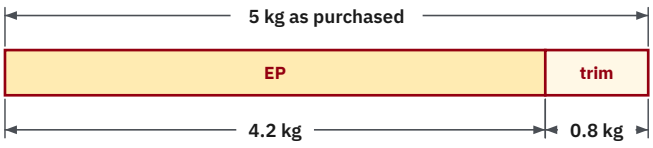
This appendix covers three kitchen skills: scaling a recipe with a scaling factor, changing between cooking measures, and food yield percent. They use the fractions, decimals and percents from Chapters 4, 5 and 6.

In this appendix you will

- Find a scaling factor and use it to make a recipe larger or smaller
- Change cups and spoons to millilitres, and pounds and ounces to kilograms and grams
- Find a yield percent from a yield test
- Find the edible portion, the amount to buy and the cost per kilogram after trimming

Key words

- yield
- scaling factor
- as purchased (AP)
- edible portion (EP)
- yield percent



A.1 Scaling a recipe

The **yield** of a recipe is the amount it makes, such as 12 portions or 3 kg. To make a different amount, every ingredient is multiplied by the same number, the **scaling factor**.

The scaling factor is the new yield divided by the old yield. A factor greater than 1 makes the recipe larger. A factor less than 1 makes it smaller. The yields must be in the same unit: portions with portions, grams with grams.

SCALING A RECIPE

Scaling factor = new yield ÷ old yield

New amount = old amount × scaling factor

New yield over old yield. Never the other way around.

Example A.1 Soup for a bigger crew

A soup recipe makes 12 portions. The kitchen needs 45 portions. Scale the recipe.

SOLUTION

1. Scaling factor = new yield ÷ old yield = 45 ÷ 12 = 3.75.
2. Multiply every ingredient by 3.75. For the stock: 2.4 L × 3.75 = 9 L.

Ingredient	For 12 portions	× 3.75	For 45 portions
Chicken stock	2.4 L		9 L
Diced onion	600 g		2,250 g
Vegetable oil	30 mL		112.5 mL
Salt	10 mL		37.5 mL

CALCULATOR

TI-30Xa

45 ÷ 12 =

Display 3.75

Sharp EL-531XT

45 ÷ 12 =

Display 3.75

The scaling factor is **3.75**.

Answer **9 L of stock, 2,250 g of onion, 112.5 mL of oil and 37.5 mL of salt.**

Sometimes the portion size changes too. Then work out each yield as a total amount first: number of portions × portion size.

Example A.2 More portions, bigger portions

A rice recipe makes 20 portions of 150 g and uses 1.2 kg of dry rice. The kitchen needs 64 portions of 200 g. How much dry rice is needed?

SOLUTION

1. Old yield: 20 × 150 g = 3,000 g.
2. New yield: 64 × 200 g = 12,800 g.
3. Scaling factor = 12,800 ÷ 3,000 = 4.2667 (rounded here, kept in full on the calculator).
4. Rice: 1.2 kg × the factor = 5.12 kg.

CALCULATOR

TI-30Xa

12800 ÷ 3000 × 1.2 =

Display 5.12

Sharp EL-531XT

12800 ÷ 3000 × 1.2 =

Display 5.12

Divide and multiply in one run, so the full factor is used: **5.12 kg**.

Answer **5.12 kg of dry rice.**

Example A.3 Scaling down

A biscuit recipe makes 40 biscuits with 1,600 g of flour and 1.2 L of milk. Make 15 biscuits.

SOLUTION

1. Scaling factor = 15 ÷ 40 = 0.375. It is less than 1, so the recipe gets smaller.
2. Flour: 1,600 g × 0.375 = 600 g.
3. Milk: 1.2 L × 0.375 = 0.45 L = 450 mL.

CALCULATOR

TI-30Xa

15 ÷ 40 =

Display 0.375

Sharp EL-531XT

15 ÷ 40 =

Display 0.375

Answer **600 g of flour and 450 mL of milk.**

NEW OVER OLD

The factor is new yield ÷ old yield. If you divide the other way, a recipe you meant to make larger comes out smaller. Check: more portions needs a factor greater than 1.

ROUND ONLY WHAT YOU MEASURE

Keep the full scaling factor on the calculator and round only the amount you will weigh or measure. Measure dry goods by mass when you can. It is more exact than cups.

TRY IT A.1

1. A recipe makes 8 portions. Find the scaling factor to make 30 portions.
2. A recipe makes 50 portions. Find the scaling factor to make 20 portions.
3. A sauce for 16 portions uses 1.8 L of tomatoes. How much for 40 portions?
4. A stew for 24 portions uses 3.6 kg of beef. How much for 18 portions?
5. A recipe makes 30 portions of 120 g. Find the scaling factor to make 45 portions of 160 g.
6. A dough uses 900 g of flour for 36 rolls. How much flour for 120 rolls?

Answers are in the Answer Key.

A.2 Cooking measures

Canadian kitchens measure liquids in millilitres and litres, and many recipes still give cups and spoons. Table A.1 is the conversion list Health Canada gives with Canada's food guide. It uses a metric cup of 250 mL, a tablespoon (Tbsp) of 15 mL and a teaspoon (tsp) of 5 mL.

Metric	Cups and spoons	Metric	Cups and spoons
750 mL	3 cups	60 mL	$\frac{1}{4}$ cup
500 mL	2 cups	45 mL	3 Tbsp
375 mL	$1\frac{1}{2}$ cups	30 mL	2 Tbsp
310 mL	$1\frac{1}{4}$ cups	15 mL	1 Tbsp
250 mL	1 cup	10 mL	2 tsp
180 mL	$\frac{3}{4}$ cup	7.5 mL	$\frac{1}{2}$ Tbsp
150 mL	$\frac{2}{3}$ cup	5 mL	1 tsp
125 mL	$\frac{1}{2}$ cup	2 mL	$\frac{1}{2}$ tsp
80 mL	$\frac{1}{3}$ cup	1 mL	$\frac{1}{4}$ tsp

Table A.1 Metric and cup and spoon measures. Source: Health Canada, Canada's food guide, "Measure ingredients like a pro", canada.ca (page dated 2026-03-31, read October 19, 2026).

Some rows are rounded to a size that is easy to measure. A quarter of 250 mL is 62.5 mL, and the list gives 60 mL. Three tablespoons is 3 × 15 mL = 45 mL, which is exact.

One tablespoon is 3 teaspoons: 15 mL = 3 × 5 mL. To change teaspoons to tablespoons, divide by 3.

Example A.4 Cups and spoons to millilitres

A recipe calls for $2\frac{1}{2}$ cups of stock and 3 Tbsp of oil. Measure them in millilitres with a 250 mL cup and a 15 mL spoon.

SOLUTION

1. Stock: 2.5 × 250 mL = 625 mL.
2. Oil: 3 × 15 mL = 45 mL. Table A.1 lists 45 mL for 3 Tbsp too.

CALCULATOR

TI-30Xa

2.5 × 250 =

Display 625.

Sharp EL-531XT

2.5 × 250 =

Display 625.

Answer **625 mL of stock and 45 mL of oil.**

Example A.5 Teaspoons after scaling

After scaling, a spice rub needs 12 tsp of paprika. How many tablespoons is that, and how many millilitres?

SOLUTION

1. 3 tsp = 1 Tbsp, so 12 tsp ÷ 3 = 4 Tbsp.
2. 4 × 15 mL = 60 mL. Check: 12 × 5 mL = 60 mL.

Answer **4 Tbsp, or 60 mL.**

A recipe from the United States uses the U.S. cup. In the United States the cup, tablespoon and teaspoon are defined as 8, $\frac{1}{2}$ and $\frac{1}{6}$ U.S. fluid ounce, and 1 U.S. fluid ounce is 29.573 53 mL (NIST SP 1038, Section 5.1.5 and its note 19). So a U.S. cup is 8 × 29.573 53 mL = 236.59 mL, a little less than the 250 mL metric cup. The same note gives 250 mL, 15 mL and 5 mL as the practical metric equivalents.

Meat and produce are often priced by the pound. One pound is exactly 0.453 592 37 kg, and one ounce is $\frac{1}{16}$ of a pound (Weights and Measures Act, Schedule II). So one ounce is 0.453 592 37 ÷ 16 × 1,000 = 28.35 g.

Example A.6 Pounds and ounces to kilograms

A recipe from an old card uses 5 lb of potatoes and 12 oz of butter. Change both to metric. Round to the nearest gram.

SOLUTION

1. Potatoes: 5 × 0.453 592 37 kg = 2.268 kg.
2. Butter: 12 oz = 12 ÷ 16 lb = 0.75 lb. 0.75 × 0.453 592 37 kg = 0.340 kg = 340 g.

CALCULATOR

TI-30Xa

Potatoes, kg

5 × 0.45359237 =

Display 2.26796185

Butter, kg

12 ÷ 16 × 0.45359237 =

Display 0.340194277

Sharp EL-531XT

Potatoes, kg

5 × 0.45359237 =

Display 2.26796185

Butter, kg

12 ÷ 16 × 0.45359237 =

Display 0.340194277

Answer **2.268 kg of potatoes and 340 g of butter.**

TWO KINDS OF CUP

A metric cup is 250 mL. A U.S. cup is about 236.6 mL. For a large batch the difference adds up: 40 cups is 10,000 mL with metric cups and about 9,464 mL with U.S. cups. Use the cup the recipe was written for.

TRY IT A.2

1. Change $1\frac{1}{2}$ cups to millilitres, using a 250 mL cup.
2. Change 4 Tbsp to millilitres.
3. Change 18 tsp to tablespoons.
4. How many 250 mL cups are in 2 L of milk?
5. Change 3 lb of ground beef to kilograms. Round to three decimal places.
6. Change 8 oz of cheese to grams. Round to the nearest gram.

Answers are in the Answer Key.

A.3 Food yield percent

Food is bought whole and loses weight when it is trimmed, peeled, boned or cored. The amount bought is the **as purchased** amount, written AP. What is left to cook or serve is the **edible portion**, written EP.

The **yield percent** is the EP as a percent of the AP. A kitchen finds it with a yield test: weigh the food as purchased, trim it, and weigh what is left.

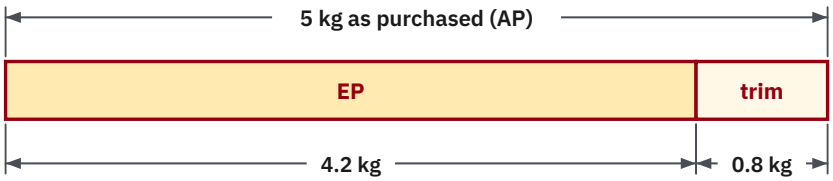


Figure A.1 A yield test on a case of carrots. The edible portion (EP) and the trim add up to the amount as purchased (AP).

FOOD YIELD

$$\text{Yield percent} = \text{EP} \div \text{AP} \times 100$$

$$\text{EP} = \text{AP} \times \text{yield percent} \div 100$$

$$\text{AP} = \text{EP} \div (\text{yield percent} \div 100)$$

EP and AP in the same unit. To find how much to buy, divide by the yield as a decimal.

Example A.7 A yield test

A cook peels and trims 5 kg of carrots and is left with 4.2 kg. What is the yield percent?

SOLUTION

1. Yield percent = EP ÷ AP × 100 = 4.2 ÷ 5 × 100 = 84%.
2. The trim was 5 − 4.2 = 0.8 kg, which is 16% of the AP.

CALCULATOR

TI-30Xa

4.2 ÷ 5 × 100 =

Display 84.

Sharp EL-531XT

4.2 ÷ 5 × 100 =

Display 84.

Answer 84%

Example A.8 Finding the edible portion

The kitchen's yield test on whole squash gave 70%. How many kilograms of squash are ready to cook from 12 kg as purchased?

SOLUTION

1. EP = AP × yield percent ÷ 100 = 12 × 70 ÷ 100 = 8.4 kg.

CALCULATOR

TI-30Xa

12 × 70 ÷ 100 =

Display 8.4

Sharp EL-531XT

12 × 70 ÷ 100 =

Display 8.4

Answer **8.4 kg**

Example A.9 How much to buy

A recipe needs 9 kg of trimmed onions. The kitchen's yield test on onions gave 88%. How many kilograms should be bought? Onions come in 1 kg bags.

SOLUTION

1. AP = EP ÷ (yield percent ÷ 100) = 9 ÷ 0.88 = 10.227 kg.
2. Check: 10.227 × 0.88 is about 9 kg.
3. Bags hold whole kilograms, so round up: 11 bags.

CALCULATOR

TI-30Xa

9 ÷ 0.88 =

Display 10.22727273

Sharp EL-531XT

9 ÷ 0.88 =

Display 10.22727273

Answer **10.227 kg, so buy 11 kg (11 bags).**

DIVIDE TO FIND WHAT TO BUY

To find the AP, divide by the yield as a decimal. Multiplying gives a smaller number and the kitchen runs short. The AP is always more than the EP.

Example A.10 Cost of the edible portion

Carrots cost $2.40 per kilogram as purchased. The yield is 80%. What does one kilogram of trimmed carrots cost?

SOLUTION

1. Only 80% of each kilogram bought can be used, so a usable kilogram costs more.
2. EP cost = AP price ÷ (yield percent ÷ 100) = $2.40 ÷ 0.8 = $3.00.

CALCULATOR

TI-30Xa

2.4 ÷ 0.8 =

Display 3.

Sharp EL-531XT

2.4 ÷ 0.8 =

Display 3.

Answer **$3.00 per kilogram of trimmed carrots.**

YIELD TESTS ARE YOUR OWN NUMBERS

Yield depends on the food, its size and quality, and how it is trimmed. A kitchen keeps its own yield records from its own tests. The yield percents in this appendix are given as data for the problems.

TRY IT A.3

1. A 6.4 kg case of potatoes gives 5.12 kg after peeling. Find the yield percent.
2. A 3.5 kg pork loin gives 2.8 kg after trimming. Find the yield percent.
3. The yield is 85%. Find the EP from 20 kg as purchased.
4. The yield is 60%. How much must be bought to get 4.5 kg of EP?
5. The yield is 75%. How much must be bought to get 10 kg of EP? Round to two decimal places.
6. Celery costs $3.00 per kilogram as purchased and the yield is 75%. Find the cost per kilogram of EP.

Answers are in the Answer Key.

WORKSHEET A1

Kitchen math on the canvas

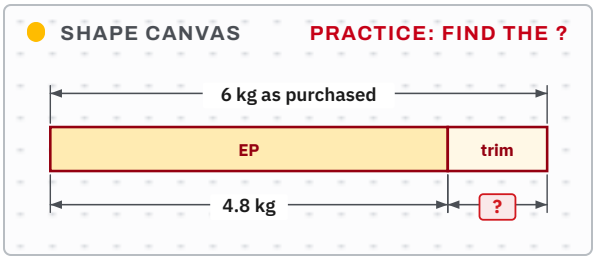
Yield bars, a recipe card and measures

Name _____

Date _____

Write the formula you use, then the numbers. Keep the full scaling factor on the calculator.

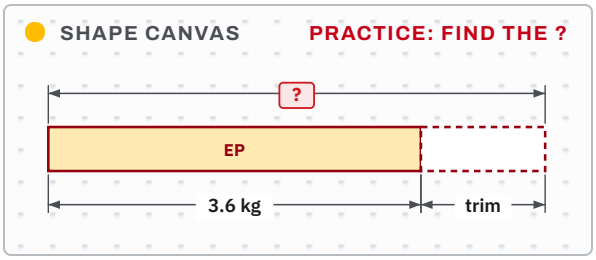
1 A yield test on onions. Find the mass of the trim and the yield percent.



WORKING

Answer _____

2 The yield percent is 75%. Find the amount as purchased.



WORKING

Answer _____

3 This chili recipe makes 10 portions. Scale it to 24 portions. Give each new amount.

Chili, 10 portions	Amount	Chili, 10 portions	Amount
Ground beef	1.5 kg	Crushed tomatoes	1.2 L
Cooked beans	800 g	Chili powder	30 mL

WORKING

Answer _____

4 Change one value: the chili in Problem 3 is made for 6 portions instead. How much ground beef is needed?

WORKING

Answer _____

5 A dressing needs 4 Tbsp and 2 tsp of vinegar. Give the amount in millilitres and in teaspoons.

WORKING

Answer _____

6 Green beans cost $3.60 per kilogram as purchased. The yield is 90%. Find the cost per kilogram of EP.

WORKING

Answer _____

7 A recipe makes 40 portions of 180 g. Find the scaling factor to make 25 portions of 240 g. Is the new batch larger or smaller?

WORKING

Answer _____

Formula sheet

Order of operations

Brackets, Exponents, Division and Multiplication (left to right), Addition and Subtraction (left to right)	**BEDMAS**

Perimeter and distance around

Rectangle	$P = 2 \times (l + w)$
Any shape with straight sides	**P = sum of the sides**
Circle	$C = \pi \times D$
Diameter and radius	$D = 2 \times r$

Area

Rectangle	$A = l \times w$
Square	$A = s^2$
Parallelogram	$A = b \times h$
Triangle	$A = b \times h \div 2$
Trapezoid	$A = (a + b) \times h \div 2$
Circle	$A = \pi \times r^2$
Ring (annulus)	$A = \pi \times (R^2 - r^2)$
Curved side of a cylinder	$A = \pi \times D \times h$

Volume

Rectangular prism	$V = l \times w \times h$
Any prism or cylinder	**V = base area × h**
Cylinder	$V = \pi \times r^2 \times h$
Cone	$V = \pi \times r^2 \times h \div 3$
Pyramid	**V = base area × h ÷ 3**
Sphere	$V = 4 \div 3 \times \pi \times r^3$

Right triangles

Pythagorean theorem	$c^2 = a^2 + b^2$
Angles of a triangle	$A + B + C = 180°$
Sine	$\sin \theta$ = **opposite ÷ hypotenuse**
Cosine	$\cos \theta$ = **adjacent ÷ hypotenuse**
Tangent	$\tan \theta$ = **opposite ÷ adjacent**

Percent and averages

Percent of a number	**part = percent ÷ 100 × whole**
Percent	**percent = part ÷ whole × 100**
Average	**sum ÷ how many**

Kitchen math (Appendix A)

Scaling factor	**new yield ÷ old yield**
New amount	**old amount × scaling factor**
Yield percent	**EP ÷ AP × 100**
Edible portion	**EP = AP × yield percent ÷ 100**
Amount to buy	**AP = EP ÷ (yield percent ÷ 100)**

Conversion factors

Exact factors are true by definition. Rounded factors are the exact value cut short. Keep the full exact factor on the calculator and round only the final answer.

From	To	Exact or rounded	Source
1 km	1,000 m	Exact	Weights and Measures Act
1 m	100 cm	Exact	Weights and Measures Act
1 m	1,000 mm	Exact	Weights and Measures Act
1 cm	10 mm	Exact	Weights and Measures Act
1 L	1,000 mL	Exact	Weights and Measures Act
1 kg	1,000 g	Exact	Weights and Measures Act
1 t (tonne)	1,000 kg	Exact	Weights and Measures Act
1 in	25.4 mm	Exact	NIST SP 811
1 ft	12 in = 304.8 mm = 0.3048 m	Exact	NIST SP 811
1 yd	0.9144 m	Exact	Weights and Measures Act
1/16 in	1.5875 mm	Exact	NIST SP 811
1 mm	0.03937 in	Rounded (0.039 370 078 7…)	NPC 2025 conversion tables
1 m	3.281 ft	Rounded (3.280 839 895…)	NPC 2025 conversion tables
1 m²	1,000,000 mm²	Exact	Weights and Measures Act
1 m²	10,000 cm²	Exact	Weights and Measures Act
1 ft²	144 in²	Exact	NIST SP 811
1 in²	645.16 mm²	Exact	NIST SP 811
1 ft²	0.092 903 04 m²	Exact	NIST SP 811
1 m²	10.76 ft²	Rounded (10.763 910 4…)	NPC 2025 conversion tables

1 m^3	1,000 L	Exact	Weights and Measures Act
1 L	1,000,000 mm^3	Exact	Weights and Measures Act
1 ft^3	1,728 in^3	Exact	NIST SP 811
1 ft^3	28.316 846 592 L	Exact	NIST SP 811
1 ft^3	7.48 US gal	Rounded (7.480 519 48… (1,728 ÷ 231))	NIST SP 1038
1 US gal	231 in^3 = 3.785 411 784 L	Exact	NIST SP 1038
1 imperial gal	4.546 09 L	Exact	Weights and Measures Act
1 imperial gal	1.201 US gal	Rounded (1.200 949 9…)	NIST SP 811
1 L	0.2200 imperial gal	Rounded (0.219 969 2…)	NPC 2025 conversion tables
1 psi	6.894 757 kPa	Rounded (6.894 757 293 168… (from the exact pound, inch and standard gravity))	NIST SP 811
1 kPa	0.1450 psi	Rounded (0.145 037 7…)	NPC 2025 conversion tables
1 bar	100 kPa	Exact	NIST SP 811
°F	1.8 × °C + 32	Exact	NIST SP 811
°C	(°F – 32) ÷ 1.8	Exact	NIST SP 811
1 lb	0.453 592 37 kg	Exact	Weights and Measures Act
1 kg	2.205 lb	Rounded (2.204 622 6…)	NPC 2025 conversion tables

Sources: Weights and Measures Act (Canada), Schedules I and II. NIST Special Publication 811, Guide for the Use of the International System of Units, Appendix B.8. NIST Special Publication 1038, The International System of Units (SI): Conversion Factors for General Use (2006), including the U.S. gallon of 231 cubic inches. National Plumbing Code of Canada 2025, conversion factor table (PDF page 245).

Glossary

acute angle
An angle less than 90°. Chapter 12

adjacent side
In a right triangle, the side next to the angle you are working with that is not the hypotenuse. Chapter 13

angle
The amount of turn between two lines that meet at a point. Angles are measured in degrees. Chapter 12

arc
A part of the circumference of a circle. Chapter 9

area
The amount of flat surface inside a shape, measured in square units. Chapter 10

as purchased (AP)
The amount of a food as it is bought, before any trimming, peeling or boning. Written AP. Appendix A

average
The sum of a set of numbers divided by how many numbers there are. Also called the mean. Chapter 1

base
The side of a shape that a height is measured from. Any side can be used as the base. Chapter 10

BEDMAS
A memory aid for the order of operations: Brackets, Exponents, Division and Multiplication (left to right), Addition and Subtraction (left to right). Chapter 3

board foot
A volume of lumber 1 in thick, 12 in wide and 1 ft long: 144 in^3, or 1/12 of a cubic foot. Chapter 11

brackets
The symbols ( ) and [ ] that group part of a calculation so that it is worked out first. Chapter 3

capacity
How much a container holds, usually given in litres (L) or gallons. Chapter 11

circumference
The distance around a circle. $C = \pi \times D$. Chapter 9

common denominator
A denominator that two or more fractions share. For 3/8 and 5/16, 16 is a common denominator. Chapter 4

complementary angles
Two angles that add to 90°. Chapter 12

composite shape
A shape made of two or more basic shapes, such as rectangles, triangles and circles. Chapter 10

cone
A solid with a circular base that narrows to a point (the apex). Chapter 11

constant
A value in a formula that is always the same, such as π or the 2 in $P = 2 \times (l + w)$. Chapter 8

conversion factor
A number that changes a measurement from one unit to another, such as 25.4 to change inches to millimetres. Chapter 7

cosine
In a right triangle, the adjacent side divided by the hypotenuse. Written cos. Chapter 13

cubic unit
A cube that is 1 unit long on each edge, such as 1 mm^3, 1 cm^3, 1 m^3, 1 in^3 or 1 ft^3. Volume is counted in cubic units. Chapter 11

cubic yard
A cube 1 yard (3 ft) on each edge. 1 yd^3 = 27 ft^3 = 0.9144^3 m^3, about 0.765 m^3. Chapter 11

cylinder
A solid with two equal, parallel circular ends joined by a curved side. Chapter 11

D.A.L.
The way the Sharp EL-531XT takes keys: in the same order the calculation is written. The line is worked out when you press =. Chapter 2

decimal
A number written with a decimal point, such as 2.375. The digits after the point are tenths, hundredths, thousandths and so on. Chapter 5

decimal feet
Feet written as a decimal number, such as 12.35 ft, instead of feet and inches. 0.01 ft is 0.12 in. Chapter 7

decimal inch
An inch measurement written as a decimal, such as 0.375 in for 3/8 in. Chapter 5

decimal place
One position after the decimal point. 2.375 has three decimal places. Chapter 5

decimal point
The dot that separates the whole number part of a decimal from the part that is less than one. Chapter 5

degrees
The unit for measuring angles. One full turn is 360 degrees, written 360°. Chapter 12

denominator
The bottom number of a fraction. It tells how many equal parts the whole is split into. Chapter 4

density
The mass of one cubic unit of a material, such as kilograms per cubic metre (kg/m³). Chapter 11

diagonals
The straight lines joining opposite corners of a shape with four sides. Chapter 10

diameter
The distance across a circle through its centre. D = 2 × r. Chapter 9

difference
The answer to a subtraction. Chapter 1

digits
The ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 used to write numbers. Chapter 1

display
The screen of the calculator. It shows the number you are keying in and the answer. Chapter 2

edible portion (EP)
The amount of a food left to cook or serve after trimming, peeling or boning. Written EP. Appendix A

elapsed time
The amount of time that passes between a start time and an end time. Chapter 1

equation
A statement that two quantities are equal, written with an equals sign, such as x + 250 = 900. Chapter 8

equivalent fractions
Fractions that are written differently but have the same value, such as 1/2, 2/4 and 8/16. Chapter 4

estimate
A quick, rounded answer used to check that an exact answer makes sense. Chapter 1

exponent
A small raised number that says how many of that number are multiplied together. In 5^3 the exponent is 3: 5^3 = 5 × 5 × 5 = 125. Chapter 3

expression
A group of numbers, letters and operation signs with no equals sign, such as x + 250 or 3 × l. Chapter 8

factors
Numbers that are multiplied together. Chapter 1

formula
A rule written with letters and symbols that shows how to work out one quantity from others, such as A = l × w. Chapter 8

fraction
A number that names part of a whole, written as one whole number over another, such as 3/8. Chapter 4

gross pay
Pay before anything is taken off: hours worked × the hourly rate, plus any overtime. Chapter 6

height
The distance from the base to the opposite corner or side, measured at right angles (90°) to the base. Chapter 10

hypotenuse
The side of a right triangle across from the right angle. It is always the longest side. Chapter 12

imperial system
The system of units based on the inch, foot, yard, pound and gallon. Canada and the United States define these units from the metric units. Chapter 7

improper fraction
A fraction whose numerator is equal to or bigger than its denominator, such as 11/4. It is worth one whole or more. Chapter 4

integer
A whole number, zero, or a whole number below zero: ... −3, −2, −1, 0, 1, 2, 3 ... Chapter 3

inverse operation
An operation that undoes another. Subtracting undoes adding, dividing undoes multiplying, and a square root undoes squaring. Chapter 8

inverse proportion
A relationship where one amount goes up as the other goes down, so their product stays the same. Twice the workers take half the time. Chapter 6

inverse trig function
A function that turns a trig ratio back into an angle: SIN^{-1}, COS^{-1} and TAN^{-1}. Chapter 13

kerf
The width of material removed by a saw cut. Chapter 1

key sequence
The list of keys you press, in order, to do a calculation. Chapter 2

legs
The two sides of a right triangle that form the right angle. Chapter 12

lowest terms
A fraction is in lowest terms when no whole number bigger than 1 divides evenly into both its numerator and its denominator. Chapter 4

median
The middle number when a set of numbers is put in order from smallest to largest. With an even count, it is halfway between the two middle numbers. Chapter 1

memory
A place in the calculator that holds a number until you need it again. STO stores the number and RCL recalls it. Chapter 2

metric system
A system of units based on powers of ten, such as the millimetre, metre and kilometre. Chapter 7

minutes of angle
One sixtieth of a degree, written ′. 60′ = 1°. Chapter 12

mixed number
A whole number and a proper fraction written together, such as 2 3/4. Chapter 4

mode
The number that comes up most often in a set of numbers. A set can have no mode or more than one. Chapter 1

net
A flat pattern that folds or rolls up to make a solid. Each face of the solid is one piece of the net. Chapter 10

net pay
Pay after deductions such as income tax and pension contributions are taken off the gross pay. Chapter 6

number line
A straight line with evenly spaced numbers on it: zero in the middle, positive numbers to the right and negative numbers to the left. Chapter 3

numerator
The top number of a fraction. It tells how many of the equal parts you have. Chapter 4

obtuse angle
An angle more than 90° and less than 180°. Chapter 12

offset
A change in the line of a pipe that moves it over to a new line parallel to the first. In this book an offset has two fittings of the same angle. Chapter 13

on centre
Measured from the centre of one framing member to the centre of the next. Written o.c., as in 400 mm o.c. Chapter 1

opposite side
In a right triangle, the side across from the angle you are working with. Chapter 13

order of operations
The agreed order for working out a calculation that has more than one operation in it. Chapter 3

parallel lines
Lines that run in the same direction and stay the same distance apart. They never meet. Chapter 12

percent
A number out of 100. 25% means 25 out of every 100, which is 0.25 or $\frac{1}{4}$. Chapter 6

perfect square
A whole number that is the square of a whole number, such as 49 = 7^2. Chapter 3

perimeter
The distance all the way around the outside of a flat shape. Chapter 9

pi
The number you get when you divide the circumference of any circle by its diameter, about 3.14159. Its symbol is π. Chapter 9

place value
The value a digit has because of its position in a number. In 12,450 the 2 is worth 2,000. Chapter 1

prefix
Letters in front of a unit name that multiply the unit by a power of ten, such as kilo (1,000) and milli ($\frac{1}{1{,}000}$). Chapter 7

prism
A solid with two identical, parallel ends joined by flat sides. Every slice parallel to the ends is the same shape. Chapter 11

product
The answer to a multiplication. Chapter 1

proper fraction
A fraction whose numerator is smaller than its denominator, such as 5/8. It is worth less than one whole. Chapter 4

proportion
A statement that two ratios are equal, such as $\frac{1}{4} = \frac{5}{20}$. Chapter 6

pyramid
A solid with a flat base, often a square, and triangular sides that meet at a point (the apex). Chapter 11

Pythagorean theorem
In a right triangle, the square of the hypotenuse equals the sum of the squares of the two legs: $a^2 + b^2 = c^2$. Chapter 12

quadrilateral
Any flat shape with four straight sides, such as a square, rectangle, parallelogram, rhombus, trapezoid or kite. Chapter 10

quotient
The answer to a division. Chapter 1

radius
The distance from the centre of a circle to its edge. Its letter is r. Chapter 9

rate
A ratio that compares two amounts in different units, such as litres per minute or dollars per metre. Chapter 6

ratio
A comparison of two amounts by division, written 3 : 1, 3 to 1 or $\frac{3}{1}$. Chapter 6

reciprocal
The fraction turned upside down. The reciprocal of 3/4 is 4/3. A number times its reciprocal is 1. Chapter 4

remainder
The amount left over when a number does not divide evenly. Chapter 1

right angle
An angle of exactly 90°, a quarter turn. A square corner. Chapter 12

rounding
Replacing a number with a nearby number that is easier to work with, such as the nearest ten, hundred or thousand. Chapter 1

scale
The ratio of a length on a drawing to the real length. At 1 : 20, 1 mm on the drawing stands for 20 mm on the real part. Chapter 6

scaling factor
The number every amount in a recipe is multiplied by to change its yield: new yield ÷ old yield. Appendix A

scientific notation
A way to write a number as a number from 1 up to 10, times a power of ten, such as 6.4×10^6 for 6,400,000. Chapter 7

second function
The job printed above a key. Press 2nd (TI-30Xa) or 2ndF (Sharp) first, then the key. Chapter 2

seconds of angle
One sixtieth of a minute of angle, written ″. 60″ = 1′. Chapter 12

SI
The International System of Units, the modern metric system. Its base units include the metre, the kilogram and the second. Chapter 7

sine
In a right triangle, the opposite side divided by the hypotenuse. Written sin. Chapter 13

sphere
A perfectly round ball. Every point on its surface is the same distance (the radius) from its centre. Chapter 11

square
A number multiplied by itself. The square of 5, written 5^2, is $5 \times 5 = 25$. Chapter 3

square root
The number that, multiplied by itself, gives the number you started with. $\sqrt{25} = 5$ because $5 \times 5 = 25$. Chapter 3

square unit
A square that is 1 unit long on each side, such as 1 mm^2, 1 m^2, 1 in^2 or 1 ft^2. Area is counted in square units. Chapter 10

stretch-out
The length of flat material that rolls or wraps into a round shape. For a cylinder it is $\pi \times D$. Chapter 9

subject of a formula
The letter on its own on one side of the equals sign. The formula tells you how to work it out. Chapter 8

substitute
To replace a letter in a formula with its value. Chapter 8

sum
The answer to an addition. Chapter 1

supplementary angles
Two angles that add to 180°. Chapter 12

surface area
The total area of all the outside faces of a solid. Find it by adding the areas of the faces in its net. Chapter 10

takeoff
A list of the materials a job needs, worked out from the drawings: how many sheets, boxes, cans or pieces to buy, and what they cost. Chapter 10

tangent
In a right triangle, the opposite side divided by the adjacent side. Written tan. Chapter 13

transpose
To rearrange a formula so that a different letter becomes its subject. Chapter 8

transversal
A line that crosses two or more other lines. Chapter 12

trigonometry
The math that links the angles of a right triangle to the lengths of its sides. Chapter 13

unit
A fixed amount used to measure, such as the metre, the inch or the litre. A measurement is a number and a unit. Chapter 7

unit price
The cost of one unit of an item, such as the price per litre or per screw. Found by dividing the price by the number of units. Chapter 6

variable
A letter in a formula that stands for a quantity that can change, such as l for length. Chapter 8

vertically opposite angles
The angles across from each other where two straight lines cross. They are always equal. Chapter 12

volume
The amount of space a solid takes up or holds, measured in cubic units. Chapter 11

whole number
A number with no fraction or decimal part: 0, 1, 2, 3 and so on. Chapter 1

yield
The amount a recipe makes, given as a number of portions or as a total mass or volume. Appendix A

yield percent
The edible portion as a percent of the amount as purchased: EP ÷ AP × 100. Appendix A